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Hybrid Cloud on IBM Power11

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Hybrid Cloud on IBM Power11

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First Edition (September 2026)

This edition applies to:
IBM Power11 servers (S1122, S1124, E1150, E1180)
IBM Power10 / Power9 compatibility modes
IBM Power Systems Firmware (FW950+)
IBM PowerVM (Hypervisor)
Matrix Multiply Assist (MMA) – on-chip AI acceleration
IBM Spyre AI Accelerator
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Secure Partition Platform Engine (SPPE)
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Preface

As enterprises accelerate their digital transformation journeys, hybrid clouds have emerged as a strategic imperative enabling flexibility, scalability, and resilience across IT environments. This IBM® Redbooks® publication explains how IBM Power11™ systems support hybrid cloud deployments with enterprise-grade performance, security, and automation.

The book begins by defining hybrid cloud in the context of IBM Power11 and explores its strategic relevance for modern organizations. It introduces IBM Power Virtual Server (PowerVS) as a key enabler of hybrid architecture, allowing seamless workload mobility between on-premises and cloud environments. The book examines core principles in depth, including workload portability, zero downtime, intelligent resource management, and AI-ready infrastructure.

Subsequent chapters present reference architectures, integration patterns with public clouds, and deployment models that use Red Hat OpenShift, IBM watsonx®, and IBM Cloud Pak for Data. The book also covers hybrid consumption models, migration strategies, and real-world use cases that demonstrate the business value of hybrid cloud on Power11.

The book addresses security, compliance, and governance with a focus on workload isolation, encryption, and quantum-safe cryptography. Finally, the publication highlights client success stories and future trends, providing guidance for organizations seeking to modernize with confidence in a hybrid multi-cloud world.

This Redbooks publication is intended for IT architects, system administrators, and decision-makers planning, designing, or implementing hybrid cloud solutions on IBM Power11.

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Yuichi Tamagawa has been with IBM Japan since 2002, spending more than a decade in the former GTS organization, where he was responsible for designing, building, and operating Power Systems infrastructures and managing related projects for enterprise clients. Since 2014, he has worked as a Technical Specialist for IBM Cloud and has been deeply involved with IBM Power Virtual Server since its service launch in Japan in 2020. By using his experience in both on-premises and cloud environments, he actively engages in solution design and technical consulting to address client challenges, while also contributing through technical writing and speaking engagements.

Michael Turek

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1

Foundations: Hybrid cloud on IBM Power11

IBM delivers open, secure, and enterprise-ready solutions that enable businesses to run workloads anywhere, on-premises or in private or public cloud, while maintaining consistency, control, and compliance.

Cloud computing has significantly transformed the IT landscape, with the global cloud market projected to grow 15.1% annually from $626.4 billion in 2023 to $1,266.4 billion by 2028. Forecasts show that cloud-related profits will reach $2 trillion by 2030[1].

As organizations accelerate their digital transformation initiatives, the role of hybrid cloud becomes increasingly strategic. Businesses seek platforms that provide the agility of cloud with the reliability and performance of on-premises systems, especially for data-intensive, regulated, or mission-critical workloads. IBM Power11 meets these requirements by delivering consistent operations across environments, seamless integration with cloud services, and the flexibility to run AI, analytics, and enterprise applications wherever they deliver the greatest value.

This chapter covers the following topics:

- “Introduction to hybrid cloud on Power11”
- “IBM CIO hybrid by design cloud”
- “Enabling AI with hybrid cloud infrastructure”
- “Defining hybrid cloud in the context of IBM Power11”
- “Data protection, encryption, and secure workload isolation”
- “Core principles of hybrid on Power11”
- “IBM hybrid cloud solutions: Built for flexibility, security, and innovation”

[1] Morales Avalos, V.; Tunc, C. *Trust Evaluation Based on a Risk Assessment and Game Theory for Virtual Machines in a Cloud Environment.* In: 2025 10th International Conference on Fog and Mobile Edge Computing (FMEC). IEEE, 2025, p. 1. (This paper cites Markets & Markets and Goldman Sachs for the stated figures.)

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1.1 Introduction to hybrid cloud on Power11

IBM's corporate strategy is to use hybrid cloud, AI, and quantum technologies to deliver competitive advantage for enterprise clients worldwide.

The internet was designed to avoid single points of failure. If a network link failed, protocols routed around the outage to maintain service. As network connections multiplied and speeds increased, the internet scaled globally and expanded payloads from text to voice to video streams, avoiding concentration and the resulting outages.

The cloud is designed to be an open, standards-based utility that any enterprise can plug into like electricity from the power grid. The global marketplace evolved from traditional on-premises systems into a mix of hyperscale cloud workloads and on-premises platforms driven by marketing and consulting companies selling cloud-first solutions.

This evolution resulted in an unusual concentration: tens of thousands of company workloads colocated in only a few sites. This concentration created hyper-targets for rogue actors, drove up costs, increased security vulnerabilities, created vendor lock-in, and introduced threats to sovereignty and national security due to the concentration of data, workloads, and control over processes and AI capabilities. Instead of distributing cloud workloads, the industry concentrated them into a few locations, creating huge failure risks.

In today's digital world, enterprises across every sector face pressure to modernize, innovate, and secure their operations without disrupting mission-critical workloads. The market is evolving from cloud-first strategies on a single hyperscaler to hybrid cloud as the open, standards-based strategic enabler of transformation. Hybrid cloud offers the flexibility to deploy and optimize applications, which include AI solutions, across public, private, and on-premises environments to avoid vendor lock-in on proprietary infrastructure and concentration in a few locations.

The next stage is *sovereign hybrid cloud* to protect the safety, control and ownership of data, AI and Quantum applications for individual nations. This ensures competitive growth and the ability to operate independently in a networked world.

Sovereign cloud services use one of two approaches: services owned by in-country companies (not subject to the U.S. CLOUD Act) or hybrid models that keep sensitive assets on-premises or in private clouds in-country while using public cloud for less sensitive workloads.

IBM Power11 supports autonomous IT operations and hybrid cloud compatibility. All Power11 models include features for hybrid cloud integration, large-scale virtualization, and high availability for critical workloads. The architecture includes Reliability, Availability, and Serviceability (RAS) features that support up to 99.9999% availability[2], which equals less than 32 seconds of unplanned downtime per year. The platform enables planned maintenance without downtime using capabilities such as Live Partition Mobility (LPM), hot patches, and rolling firmware updates.

Power11 provides quantum-safe cryptographic capabilities from the secure boot foundation through the entire stack (Figure 1-1), supporting the migration to quantum-resistant algorithms mandated by the National Security Agency (NSA) and standardized by the National Institute of Standards and Technology (NIST).[3]

[2] https://www.redbooks.ibm.com/abstracts/sg248596.html

[3] https://www.nist.gov/news-events/news/2022/07/nist-announces-first-four-quantum-resistant-cryptographic-algorithms

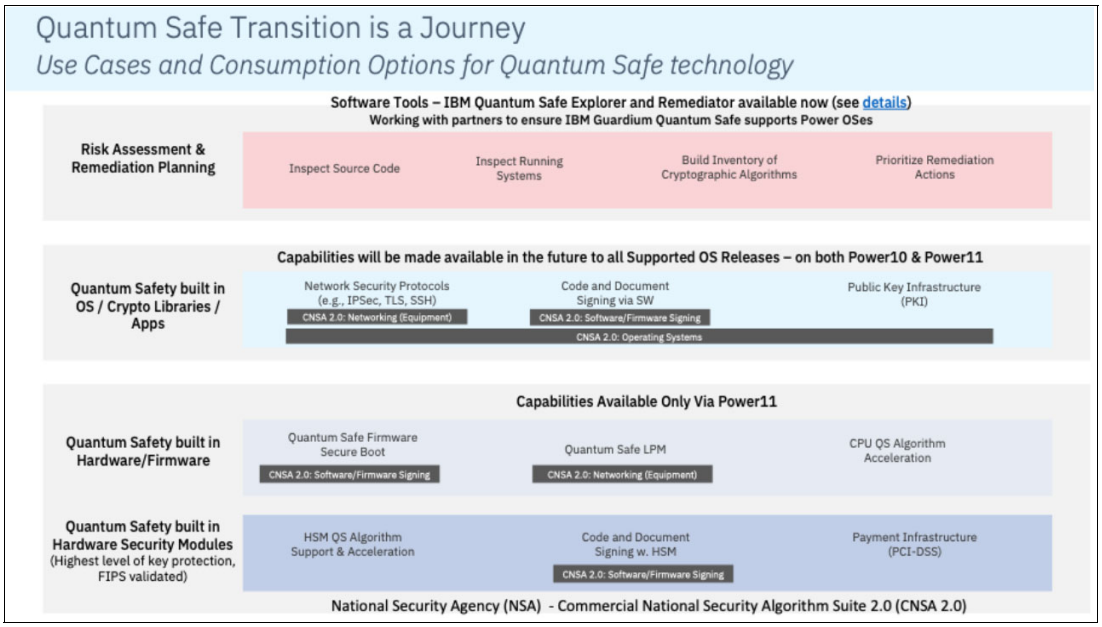


Figure 1-1 Quantum Safe transition diagram

The U.S. National Security Agency released the Commercial National Security Algorithm Suite 2.0 (CNSA 2.0)[4] requirements for National Security Systems (NSS). These requirements are recommended for the Department of Defense (DoD) and the Defense Industrial Base (DIB) in the United States and globally across Allied nations.

For electronic software signing and firmware signing, the NSA recommends the following timeline:

- Begin transitioning software and firmware signing immediately.
- Use CNSA 2.0 signing algorithms for new software and firmware by 2025.
- Transition deployed software and firmware that are not CNSA 1.0 compliant to CNSA 2.0-compliant algorithms by 2025.
- Transition all deployed software and firmware to CNSA 2.0-compliant signatures by 2030.

IBM also introduced the IBM Cyber Vault, which detects ransomware threats in under a minute and enables rapid recovery using immutable backup copies.

[4] https://media.defense.gov/2025/May/30/2003728741/-1/-1/0/CSA_CNSA_2.0_ALGORITHMS.PDF

Figure 1-2 shows the general information of IBM Power Cyber Vault.

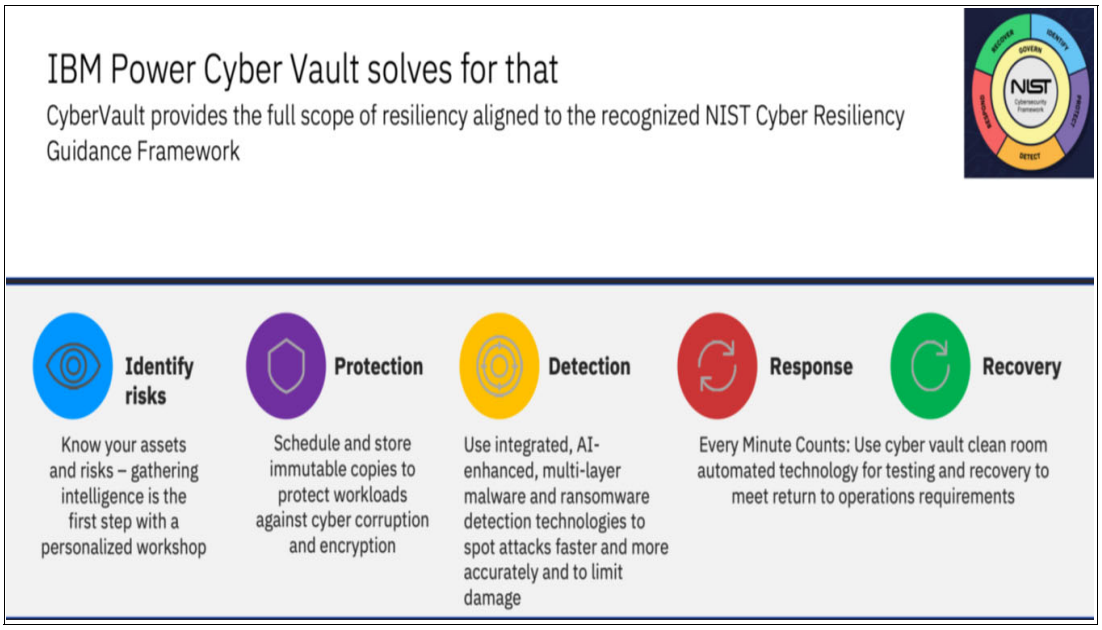


Figure 1-2 IBM Power Cyber Vault

1.2 IBM CIO hybrid by design cloud

IBM has built a world-class hybrid cloud platform by using Red Hat OpenShift in its cloud, on IBM Power, and on IBM Z mainframes. IBM achieved over $3.5 billion in productivity gains in the last two years, with $2 billion from AI and automation. IBM transformed its operating model to eliminate complexity, simplify end-to-end workflows, automate manual tasks, and embed AI on its hybrid cloud.

Central to this approach is the principle of testing IBM technologies internally before releasing them. IBM calls this the Client Zero approach.

1.2.1 IBM CIO Cirrus

Cirrus, IBM's internal hybrid cloud platform, uses Red Hat and IBM technologies. Modern and easy to use, Cirrus provides the speed, scale, security, and simplicity needed for IBM's digital business experiences to connect data, information, and people.

The IBM CIO Hybrid Cloud is the engine of digital transformation, delivering security-rich speed, scale, and simplicity to the IBM Chief Information Officer (CIO) organization. It provides capabilities not available when using exclusively public cloud or exclusively private cloud, giving IBM's digital application teams the velocity to focus on what matters most: IBM's digital business.

Cirrus is built on three principles:

1. Hybrid: One consistent experience across all platforms
2. Integrated: Common CI/CD pipeline and operating environment
3. Open: Open-source and standard solutions

Core to the *hybrid by design* approach is Red Hat OpenShift, which creates a common CI/CD (Continuous Integration/Continuous Deployment or Delivery) pipeline that works on x86, IBM Power (IBM AIX®, IBM i, and Linux), and IBM Z environments. This approach is shown in Figure 1-3.

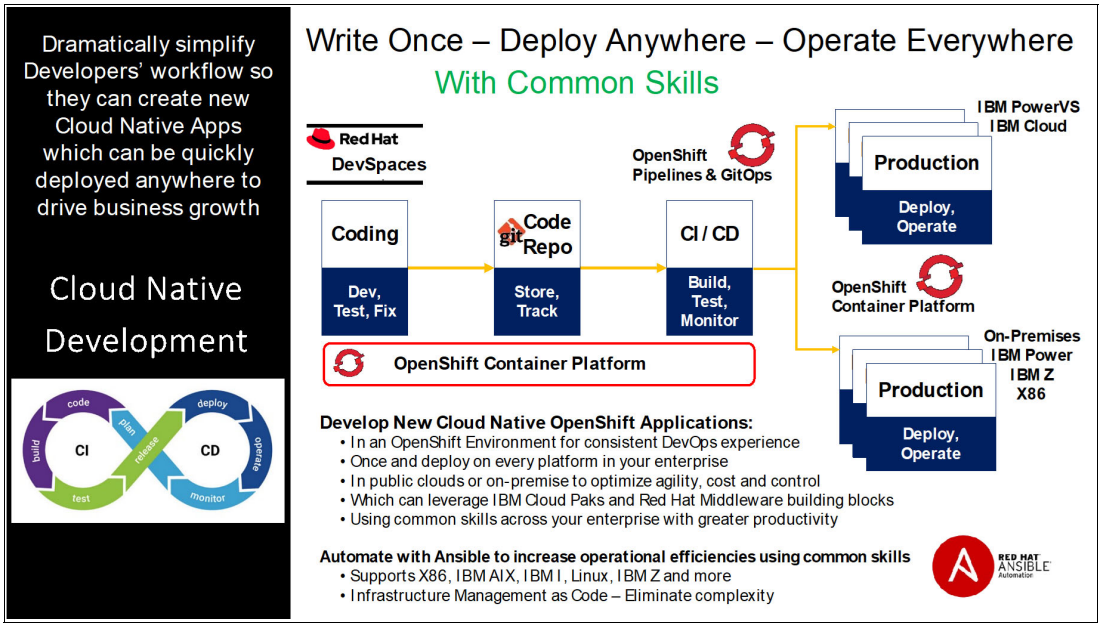


Figure 1-3 IBM CIO Cirrus goals

IBM is a company with more than 250,000 employees and annual revenues that exceed $50 billion. Thousands of internal applications manage all aspects of the company's diverse portfolio, including applications for chip design, sales, marketing, management, accounting, customer support, and many other functions. Some of these applications are as old as the computer industry. Others support the latest developments in quantum computing.

IBM's application teams faced difficult problems:

- Managing over 5,000 applications and 74 data centers
- Supporting 250,000+ users in over 170 countries
- Controlling hosting and operational costs
- Keeping current with the latest security patches
- Meeting availability and disaster recovery objectives
- Satisfying corporate security requirements
- Addressing minor changes that took months to implement

IBM's Client Zero success enabled the company to use a consolidated hybrid cloud supporting AI and automation solutions to transform enterprise productivity.

Platform engineering for IBM CIO Hybrid Cloud at scale

The engineering team created a single umbrella hybrid cloud platform called the CIO Hybrid Cloud, which is a hosting environment for both containerized applications and VM-based applications across a heterogeneous set of architectures. Red Hat OpenShift on IBM Cloud enabled the team to host identical environments on-premises and in the cloud. This is shown in Figure 1-4 on page 6.

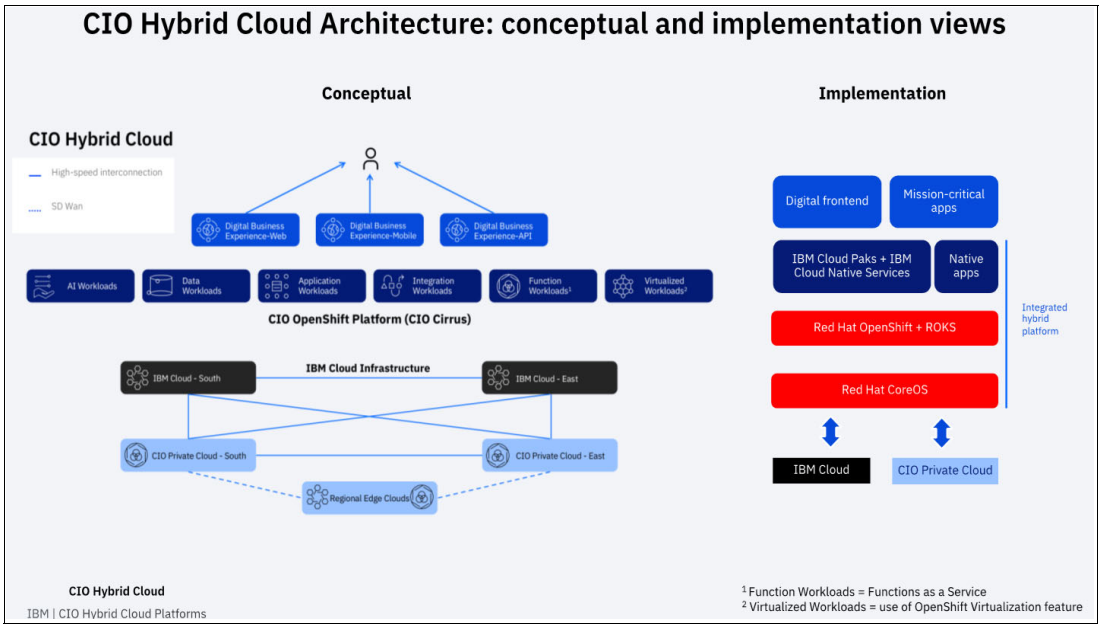


Figure 1-4 CIO Hybrid Cloud architecture

With increasing business requirements around security, resiliency, and cost optimization, IBM needed a hybrid cloud platform in which developers can focus on creating enterprise applications for their stakeholders. The CIO Hybrid Cloud platform meets these requirements simply by hosting applications on the platform. The IBM Cloud services integrated into the platform help balance application innovation with guardrails in areas such as databases, observability, and storage.

Figure 1-5 shows the hybrid cloud implementation.

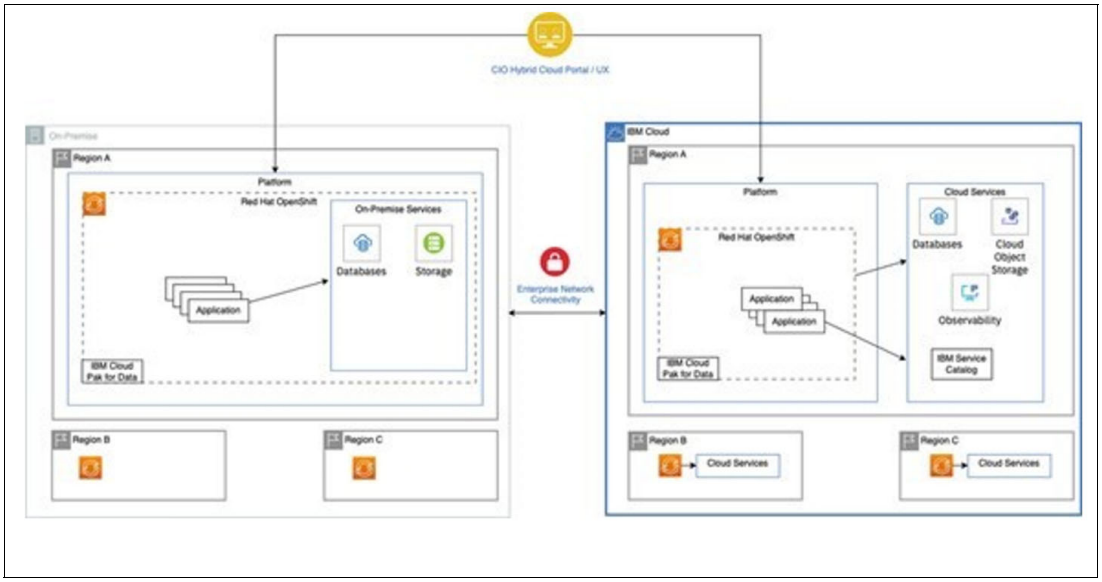


Figure 1-5 Hybrid cloud implementation

The expansion of OpenShift-based Cirrus to Power was part of the Intelligent Workload Placement initiative, a strategy that allows OpenShift-deployed applications to run transparently on multiple hardware architectures, including x86, Power, and Z. This strategy

also accelerates application modernization from monolithic applications to microservices and other modern development and deployment techniques.

Red Hat OpenShift on Power gained traction around 2017–2018, with IBM and Red Hat collaborating closely to enable the platform. In 2019, Red Hat announced extended update support (EUS) for Red Hat OpenShift on Power, allowing clients to stay on a specific version for a longer period with enterprise support.

In late 2019, IBM released PowerVS, an Infrastructure-as-a-Service (IaaS) solution that provides simplified automation and faster onboarding for clients. One year later, Red Hat OpenShift 4.6 introduced several new capabilities for Power, such as support for compute nodes with up to 512 threads and enhanced storage options. Red Hat OpenShift 4.6 also became available on IBM Z, IBM Power, and x86 platforms simultaneously, reinforcing the positioning of Red Hat OpenShift as a truly cross-platform solution.

The IBM CIO Power Virtualization Service delivers end-to-end virtualization services on IBM Power Systems, maximizing availability, standardization, and automation for IBM customers. The CIO Power Virtualization Service team manages and supports all on-premises virtualized Power Systems infrastructure across CIO data centers. The team provides the following capabilities:

- Ongoing management, support, and configuration of the global Power Systems infrastructure, including VIOS, HMCs, PowerVC, storage, network VLANs, and zone architecture
- Industry-leading best practices modeled on proven methods from IBM Consulting®, with knowledge transfer to Application Hosting Service Squads
- Consolidated management support processes with standardized tooling and monitoring
- Proactive creation of automated solutions to support the global IBM Power environment more efficiently
- Use of new and existing technologies to achieve no outages for planned maintenance, and reduce the scope, impact, and duration of unplanned outages
- IBM Security® processes to ensure health and compliance of Power Systems infrastructure, including tracking and addressing all security vulnerabilities
- Collaboration and open communication between Physical and Automation squads

The IBM Power Squad team delivered a Red Hat OpenShift cluster on Power infrastructure integrated with Cirrus within one quarter. This integration gave Cirrus the capability to move workloads from public cloud to private cloud and modernize applications currently running on Power, such as AIX and RHEL.

Benefits gained

Red Hat OpenShift on IBM Power optimizes infrastructure costs by reducing the number of servers without impacting performance. Figure 1-6 on page 8 show the TCO impact of using IBM Power compared to x86 servers with Red Hat OpenShift.

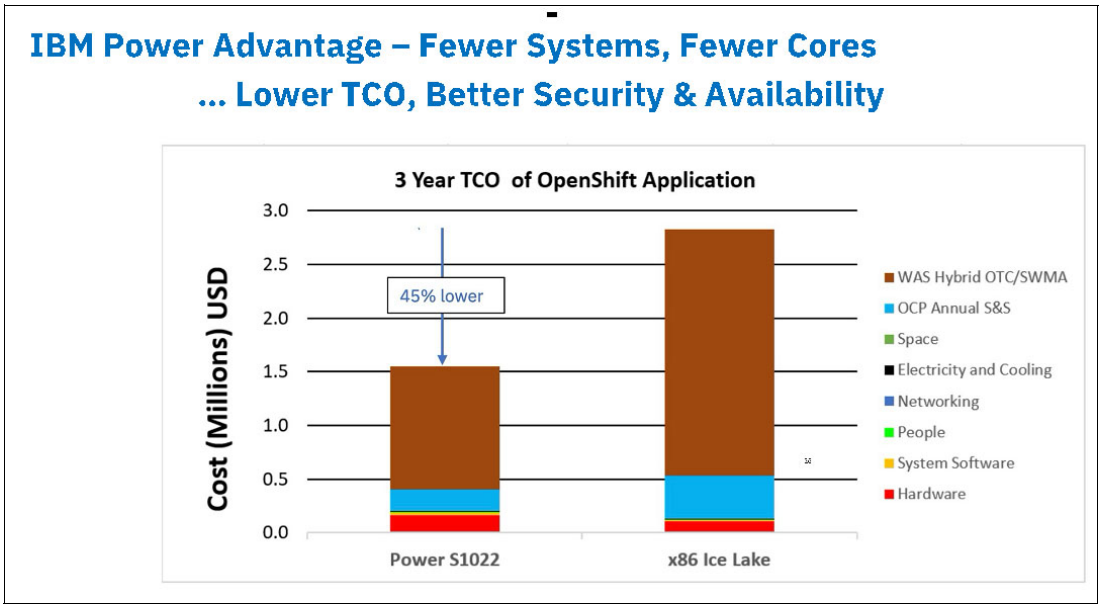


Figure 1-6 TCO Benefits of Power versus x86 for Red Hat OpenShift

The CIO Hybrid Cloud platform delivered significant results:

- Over 2,000 application components onboarded, including custom-built applications, application integrations, and data components
- A reduction in DevOps hours spent on operations, allowing application teams to focus on business value
- Cost savings for hosting containerized workloads compared to legacy hosting

Application teams now meet corporate objectives in the following areas:

- Speed: Application changes that took months can be delivered daily, and security patches are delivered rapidly.
- Security and availability: Highly available secure applications are the default rather than afterthoughts.
- Disaster recovery: Built-in disaster recovery support replaces the previous issues with implementation, testing, and associated costs

The agility of the IBM hybrid cloud allows the platform to scale elastically based on demand, eliminating the challenging capital expenditures and operational requirements to increase the scale and capabilities of the previous private cloud solution. The platform and applications use IBM Cloud Catalog services, which provide security-rich resources with low operational overhead and consumption-based pricing. Data and applications are located on-premises or in the cloud based on business requirements. Cloud data residency can be regional to meet regulatory requirements.

Continuous platform improvement relies on data-driven decisions with measurable results. The platform stores real-time application data in a data warehouse for platform and application analysis. As applications require updates over time, platform metrics inform required changes. When applications reach end of life, you can assess the impact of retiring them. Application owners view metrics through a common interface to understand performance and troubleshoot problems. The IBM Cloud Pak for Data is a central component of the data warehouse.

The platform analyzes underlying performance and measures the impact of changes. The platform also reports derived information to application teams, including compliance posture, criticality, and software currency. The IBM Cloud Security and Compliance Center, IBM Cloud Log Analysis, and IBM Cloud Activity Tracker support the observability strategy, along with IBM partners.

The platform applies AI-based analysis on the data warehouse. Applications can use vertical scaling of hosting resources through the IBM Turbonomic® solution. Intelligent workload placement allows applications to migrate for optimal cost and performance on x86, IBM Power, or IBM Z platforms. You can achieve up to 43% lower cloud costs, greater efficiency, and improved resource usage at scale. For more information, see Cloud performance optimization.

Application success by default

The CIO Hybrid Cloud hosts applications distributed across private and public clouds. Each application environment starts with built-in company business rules. All data centers reproduce a common application runtime environment that allows applications to run locally and handle regional disruptions. Applications run on compute and storage that reflect requirements and use cloud catalog services with consumption-based pricing.

The hybrid cloud containerized environment has reduced hosting expenses dramatically. Containerized workloads have replaced traditional virtual machines. Application-specific operational load has been consolidated into a common platform, allowing teams to focus on business value while reducing operational expenses.

The IBM CIO Hybrid Cloud, which is combined with application modernization and migration, delivers the following capabilities:

- End-to-end observability
- Consistent operations with speed, scale, and security

Platform evolution continues. Compliance requirements continue to evolve, and application performance demands have increased with AI adoption. The platform can further optimize application placement in private or public clouds by analyzing real-time application and cost data.

1.3 Enabling AI with hybrid cloud infrastructure

Generative AI is reshaping enterprise IT, placing unprecedented demands on infrastructure. AI workloads require massive computational throughput, rapid data access, and secure environments that scale across diverse platforms. As organizations deploy AI across their operations, hybrid cloud architecture provides the foundation for flexibility, scalability, and governance.

AI represents the next evolution in enterprise computing, following virtualization, cloud-native development, and hybrid cloud adoption. Today's AI models demand up to 100x more model parameters, 7-times faster threat response, and 10-times growth in AI-generated data. These demands require infrastructure that supports real-time inferencing, secure data handling, and seamless integration across environments.

Despite this urgency, only 16% of technology leaders feel confident their current infrastructure can support generative AI, and 71% of CEOs recognize that digital infrastructure is key to scaling innovation. This gap highlights the importance of intentional hybrid cloud design.

A hybrid-by-design approach integrates on-premises systems, edge computing, and multiple cloud platforms into a unified operating model. This approach contrasts with hybrid-by-default environments, which often result from ad hoc deployments and lead to siloed operations, inefficiencies, and constrained AI capabilities.

The CIO Organization envisioned a hybrid-by-design cloud platform, which is a single platform on which digital business application components run with full observability, transparency, and cost-optimized performance.

Matt Lyteson, CIO, VP Technology Platforms Transformation at IBM states *"The choice was clear: We must leverage IBM's products and technology at enterprise-scale. For us this meant using IBM's hybrid cloud technology to build an intelligent application platform to run internal applications, integrations, digital workflows and data components,"*.[5]

Figure 1-7 shows the IBM CIO vision for hybrid cloud

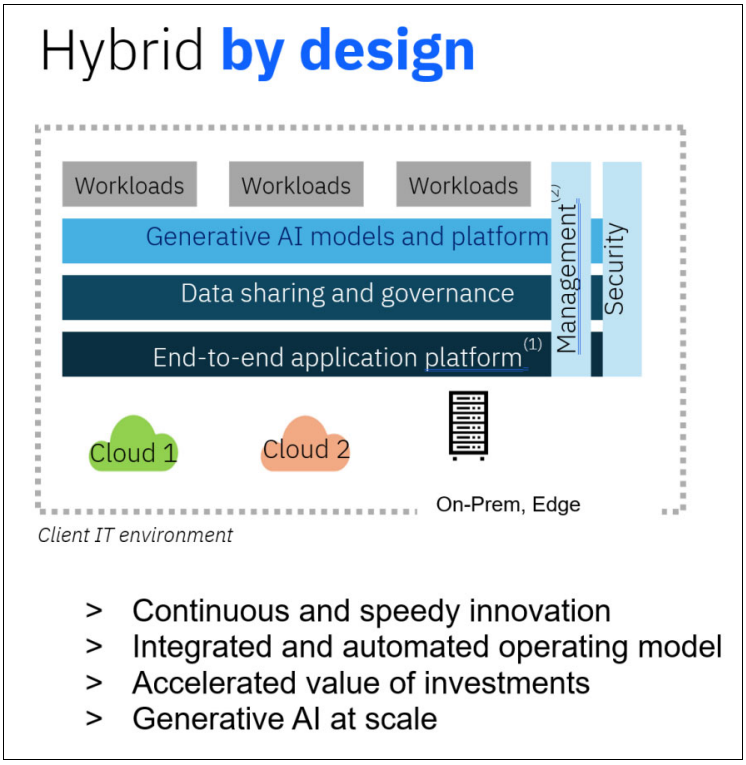


Figure 1-7 Hybrid by design concepts

IBM and Red Hat advocate for hybrid cloud architectures that are open, secure, and optimized for AI. The following key technologies support this architecture:

- Red Hat OpenShift for container orchestration across hybrid environments
- IBM Infrastructure (Z, Power, Cloud, Storage) purpose-built for AI workload.
- watsonx AI and data platform for model training, inferencing, and governance

These technologies enable consistent workload portability, automated deployment pipelines, and secure data governance across environments.

- Optimizing for AI Workloads.

[5] https://www.ibm.com/case-studies/ibm-cio-organization-hybrid-cloud

IBM's hybrid cloud infrastructure meets the unique needs of AI:

- IBM Z delivers encrypted, low-latency inferencing for transactional workloads.
- IBM Power offers in-core acceleration and high resiliency for industry-specific AI.
- IBM Cloud supports GPU-based compute and watsonx services for scalable AI.
- IBM Storage provides high-performance, software-defined storage for AI and HPC.

Red Hat complements this infrastructure with a platform-centric approach that streamlines developer productivity, accelerates AI adoption, and ensures consistency across hybrid environments.

Client examples demonstrate hybrid cloud impact for AI:

- Healthcare: AI-powered pathology analysis improves diagnostic accuracy.
- Finance: IBM Z shortens code understanding time by 80% for developers.
- Sports: AI models transform tennis data into real-time insights.
- Semiconductors: Hybrid cloud accelerates chip design and verification.

Figure 1-8 shows an example of the hybrid cloud deployment architecture.

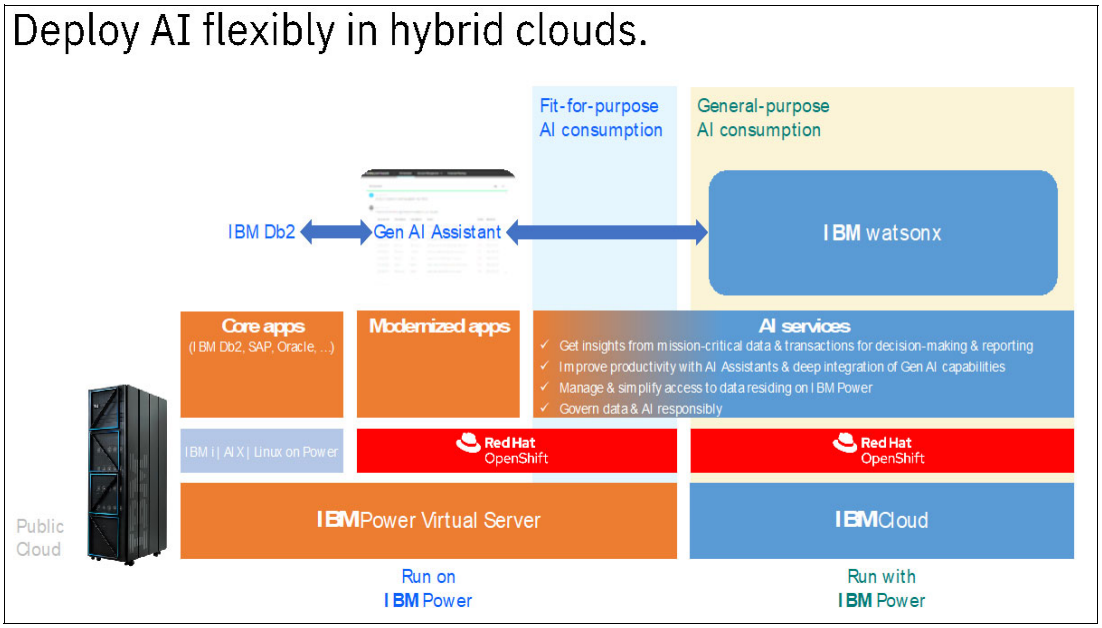


Figure 1-8 Deploy AI flexibility

1.4 Defining hybrid cloud in the context of IBM Power11

Hybrid cloud is an IT architecture that combines on-premises systems (or private cloud) with public cloud services through common orchestration tools. Clients can use IBM Power11 servers for enterprise workloads onsite and in private clouds and connect them with public cloud resources and services. IBM enables this architecture through OpenShift, the IBM Cloud Management Console, and solutions such as IBM Power Virtual Server on IBM Cloud and IBM PowerVS™ Private Cloud for client locations.

1.4.1 IBM offerings

Power11 systems are available both on-premises and in IBM Cloud as PowerVS. IBM offers the same PowerVS pods used in IBM Public Cloud locations to clients for Private Cloud services at client sites. These PowerVS Private Cloud pods (Power11 servers, Flash storage, networking, and cloud management software) are available at client sites with pricing based on a minimum monthly fee or actual usage, whichever is higher.

In a hybrid scenario, organizations keep sensitive, sovereign, or performance-intensive applications on their on-premises Power11 servers or PowerVS Private pods while using PowerVS for other workloads or cloud services for backup, disaster recovery, burst capacity, and modernization initiatives. This approach combines the stability and control of traditional IBM Power systems with the flexibility and scalability of cloud.

IBM Power11 Power Virtual Server Hybrid Cloud Package is a fixed-price offering that combines Power11 servers with cloud resources and support services. It includes a PowerVS subscription, deployment assistance, and advanced cloud support sized to match the selected server model (S1122, S1124, E1150, E1180). The package offers onboarding in five days and delivers a Power11 plus Power Virtual Server hybrid cloud solution.

Figure 1-9 illustrates deployment models for IBM Power Systems and IBM Power Virtual Server in different environments:

- On-premises private cloud:
  - Uses physical IBM Power servers located in the organization's data center
  - Ideal for workloads requiring full control over hardware and infrastructure
- Private cloud (IBM Power Virtual Server Private):
  - Virtualized IBM Power resources hosted in a private cloud environment
  - Offers flexibility and scalability while maintaining isolation for security and compliance
- Public cloud (IBM Power Virtual Server):
  - IBM Power Virtual Server hosted in IBM Cloud data centers
  - Enables hybrid cloud strategies, workload portability, and cost optimization



Figure 1-9 Compare Power11 on different cloud

1.4.2 Use cases

Hybrid cloud delivers value for Power11 in the following key use cases:

Accelerate application modernization: Developers can modernize legacy AIX and IBM i applications by integrating with cloud-native services or container platforms (such as Red Hat OpenShift on Power) without complete replatforming.

Strengthen business resiliency: Cloud-based backup and disaster recovery complement on-premises Power11 systems. For example, a production workload on Power11 can regularly replicate data to PowerVS for a warm DR standby, improving recovery time and preventing downtime during local outages.

Optimize and simplify the data center: Hybrid cloud enables dynamic scaling and workload balancing. Noncritical or transient workloads can run in the cloud to free up on-premises capacity, and cloud management tools simplify operations through unified management.

Figure 1-10 illustrates the hybrid value of Power 11.

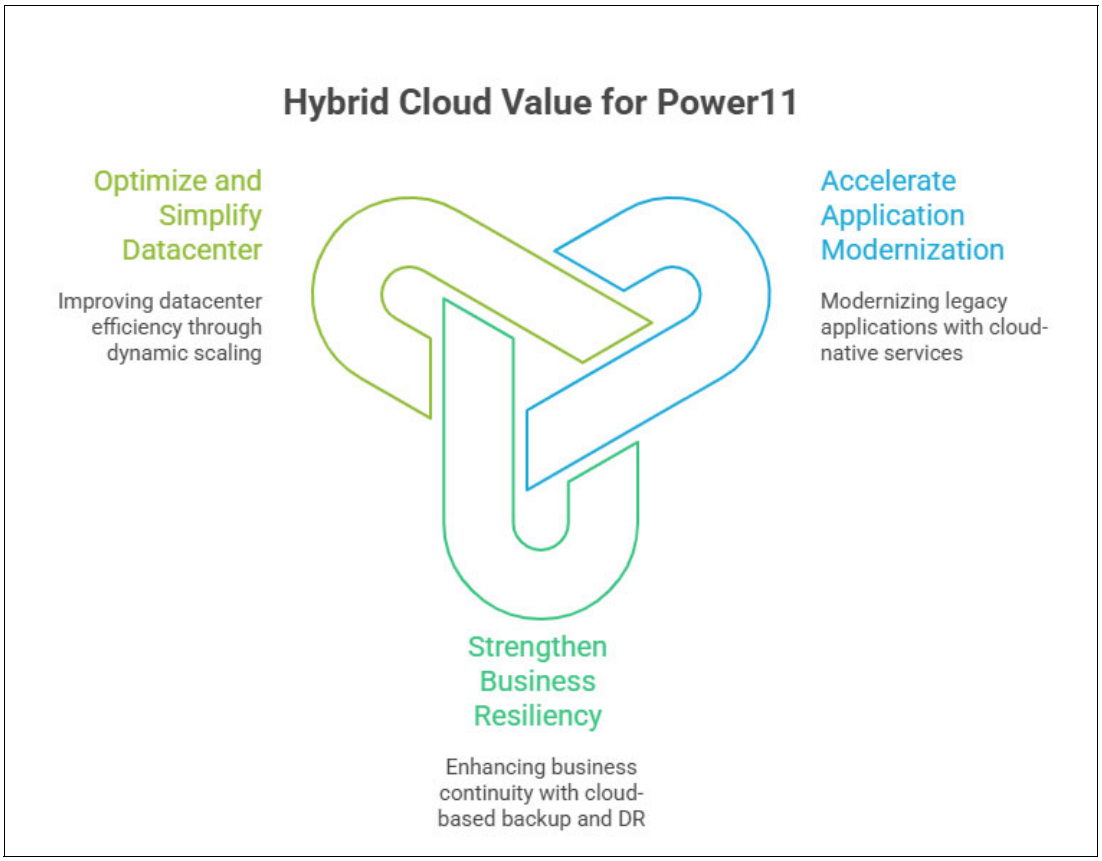


Figure 1-10 Hybrid cloud value for Power11.

IBM's approach to hybrid cloud with Power11 focuses on reducing complexity for clients. The hybrid package includes services that assist with configuring secure connections, such as VPN or IBM Cloud Transit Gateway, between on-premises infrastructure and IBM Cloud. After establishing the connection, Power11 partitions (supporting IBM i, AIX, or Linux) communicate with cloud services or migrate selected workloads to PowerVS.

Live Partition Mobility is a key capability of Power11 that enables active workloads to move between physical systems without interruption. When combined with hybrid cloud

connectivity, LPM establishes the foundation for future cross-site workload mobility. Even when LPM is primarily used within a local environment, such as during hardware maintenance, its integration with hybrid cloud services ensures that zero-downtime principles extend to supporting operations, including cloud-based backup and restore workflows.

With hybrid cloud on Power11, enterprises can confidently run core business applications on a highly reliable Power11 server in their own data center while extending capabilities to the cloud as needed. IBM provides the tools and services to make this extension straightforward.

The Power11 platform's full-stack integration (including support for container orchestration such as Red Hat OpenShift and AI services such as watsonx) helps ensure that hybrid cloud deployments are unified by consistent tools and open standards rather than siloed. With Power11, organizations can modernize at their own pace, using cloud for agility and innovation without abandoning the performance, security, and investment of their on-premises Power Systems.

1.5 Data protection, encryption, and secure workload isolation

A hybrid cloud environment requires robust security and data protection across both on-premises and cloud components. IBM Power11 addresses these requirements with a multi-layered security architecture that introduces enhancements in data encryption, cryptographic resilience, and workload isolation to protect sensitive workloads in any location.

This section explains how Power11 protects data at rest and in transit, uses advanced encryption with quantum-safe algorithms for future-proof security, and isolates workloads to prevent unauthorized access or interference.

1.5.1 Encryption and data protection

Power11 and PowerVS incorporate comprehensive encryption features to protect data at rest, in memory, and in transit. PowerVS encrypts all data stored in IBM Cloud, and network connections between on-premises Power11 systems and the cloud are secured through VPN or encrypted tunnels.

The platform uses modern encryption standards such as AES 256, accelerated by on-chip cryptographic engines. On IBM Power Virtual Server, cloud storage and disk data can be encrypted with keys from IBM Cloud Key Protect or external key managers.

Data traveling between the Power11 environment and IBM Cloud uses secure protocols (TLS, IPsec) to help ensure confidentiality and integrity. These measures align with Zero Trust security principles by ensuring that data remains protected at all times, even on shared or external networks.

Identity and access management is also integrated. Power11 systems and PowerVS support robust IAM controls, including integration with IBM Security Verify for single sign-on and multi-factor authentication. This integration ensures that the same strict authentication and authorization policies apply whether a workload runs locally or in the cloud, reducing the risk of breaches from compromised credentials.

An administrator accessing a Power11 LPAR or a PowerVS instance passes through the same level of identity verification, and fine-grained role-based access control limits their actions on each system. These controls enforce a consistent security posture across hybrid cloud components, which is especially important for industries with strict regulatory requirements on data privacy and access logging.

A hybrid cloud environment requires robust security and data protection across both on-premises and cloud components. IBM Power11 addresses these requirements with a multi-layered security architecture that introduces enhancements in data encryption, cryptographic resilience, and workload isolation to protect sensitive workloads in any location.

This section explains how Power11 protects data at rest and in transit, uses advanced encryption with quantum-safe algorithms for future-proof security, and isolates workloads to prevent unauthorized access or interference.

1.5.2 Memory encryption

Power11 includes memory encryption and other hardware-based data protection features. Building on the Power10 generation, Power11 implements *memory-to-core encryption*, which means data is automatically encrypted as it moves between memory and the processor caches. This guards against any snooping of the memory bus or installation of illicit hardware devices trying to capture sensitive information such as cryptographic keys or personal data while in use.

The memory encryption is transparent to applications and does not require modifications to software. Also, the memory of Power11 architecture uses DDR5 memory with an Open Memory Interface and includes integrity checks and redundancy to prevent data corruption or tampering in memory.

An important enhancement in Power11 is the extension of Secure Boot beyond just the processor firmware to also cover memory DIMM firmware. Each Power11 DDR5 DIMM module has an embedded controller that undergoes its own secure boot process: a hardware-anchored *Secure ROM* verifies the DIMM's firmware using digital signatures before the memory is trusted by the system.

If the memory firmware fails verification or shows any sign of tampering, the system automatically *unconfigures* that memory module (isolating it from use) to maintain overall system integrity. This level of granular hardware validation ensures that data residing in memory is not only encrypted, but also that the memory hardware itself has not been maliciously modified.

1.5.3 Cryptographic resilience and quantum-safe encryption

Power11 introduces quantum-safe cryptography to address security challenges from quantum computing. IBM incorporates post-quantum algorithms into critical system components because traditional algorithms such as RSA and ECC might become vulnerable to quantum attacks.

Secure firmware in Power11 uses a dual-signature approach that combines CRYSTALS-Dilithium (a lattice-based, quantum-resistant signature algorithm) with Elliptic Curve Digital Signature Algorithm (ECDSA) to verify boot code. This approach ensures that even if ECDSA is compromised by future quantum computers, Dilithium signatures maintain firmware authenticity.

IBM also uses CRYSTALS-Kyber, a quantum-safe key encapsulation mechanism, for securing data in transit. These algorithms align with CNSA 2.0 (Commercial National Security Algorithm Suite 2.0), the NSA's updated standard for protecting systems against classical and quantum threats, and are recommended by NIST and ETSI.

By adopting these algorithms early, Power11 provides long-term confidentiality for data. Data protected today remains secure even when quantum computing becomes mainstream.

Figure 1-11 illustrates the core cryptographic components of CNSA 2.0 (Commercial National Security Algorithm Suite 2.0), which is the NSA's updated standard for securing National Security Systems against classical and quantum threats.

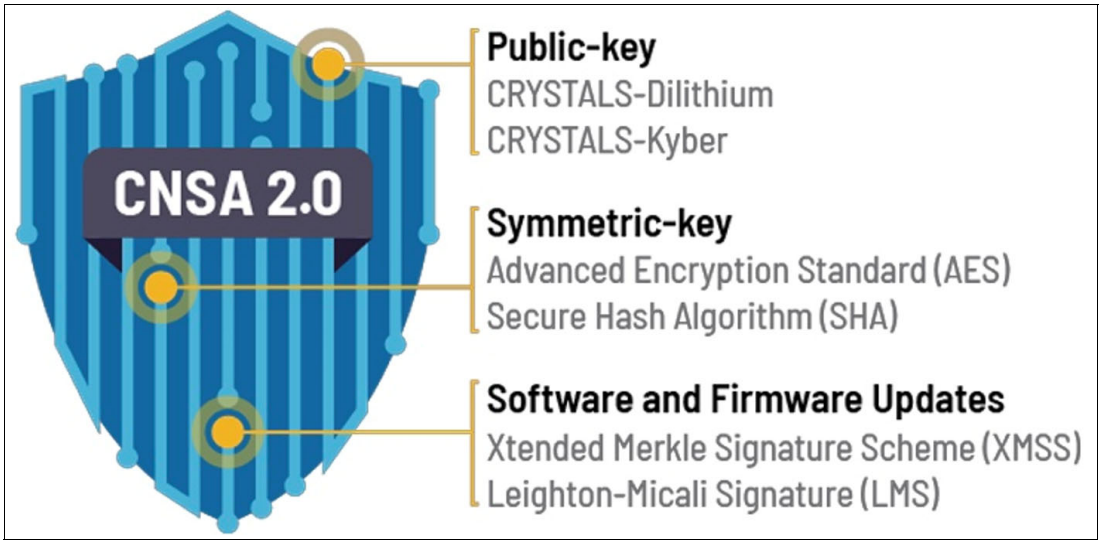


Figure 1-11 CNSA 2.0

Practical example of quantum-safe

A practical demonstration of the quantum-safe strategy in Power11 is its approach to securing Live Partition Mobility (LPM) operations within a hybrid cloud environment. With LPM, a running partition (a virtual machine) moves from one physical server to another without service interruption.

During LPM, the system's memory state and processor state transfer over the network, which is a sensitive process that must be encrypted. Power11 enhances LPM by using a hybrid key exchange that combines classical cryptography with quantum-safe cryptography to negotiate the encryption keys for migration.

During an LPM migration, Power11 generates a shared session key using both traditional algorithms (such as an Elliptic Curve Diffie-Hellman component authenticated by classical signatures) and quantum-safe algorithms (such as CRYSTALS-Kyber for key exchange and Dilithium for authentication).

Even if an attacker records the encrypted migration traffic today with the intent to decrypt it in the future (the harvest-now, decrypt-later attack), the quantum-resistant portion of the encryption ensures the data cannot be decrypted even with quantum computing power. IBM's implementation of this hybrid cryptography approach follows industry recommendations and provides a vital defense against emerging threats.

Power11 also employs quantum-safe methods in its Secure Boot process. The chain of trust that starts from hardware uses algorithms such as SHA3-512 hashing and Dilithium signatures for verifying firmware and operating system boot loaders.

This approach establishes a root of trust that is robust against both today's attackers and tomorrow's quantum adversaries. This cryptographic agility and resilience ensure that sensitive data (customer records, financial transactions) and cryptographic keys remain protected throughout the lifespan of the Power11 system, even as the threat landscape evolves.

IBM recommends that clients use a Bring Your Own Keys approach so that even IBM operations staff cannot decrypt your data, ensuring secure and sovereign data protection. If other organizations or governments obtained a copy of your quantum-resistant encrypted data and applications without the quantum keys, they cannot access your information.

1.5.4 Secure Boot and workload isolation

Protecting data encompasses both encryption and ensuring that only authorized code and workloads run on the system. IBM Power11 uses Secure Boot to verify the integrity of firmware, operating systems, and Hypervisor components every time the system boots.

Secure Boot on Power11 is anchored in hardware: an immutable boot ROM (the SROM) and one-time programmable fuses store IBM root cryptographic keys. During power-on, the system firmware (including the Power Hypervisor and system management firmware) is cryptographically verified using these keys before execution.

If any code shows tampering or is not signed by a trusted key, the Secure Boot process detects this and prevents its execution. This prevents low-level malware or rootkits from infiltrating the platform.

Power11 extends the Secure Boot process to multiple system components, including main processor firmware, adapter firmware, BMC (service processor) firmware, and memory modules.

Trusted Boot complements Secure Boot by measuring each software component's cryptographic hash and storing those measurements in a TPM or other secure store. This creates an attestation log that external systems or compliance auditors use to verify the platform's state. Together, Secure Boot and Trusted Boot ensure that the foundation for workloads is secure and trustworthy.

Secure workload isolation

With a trusted foundation in place, IBM Power11 provides secure workload isolation through its virtualization and partitioning features. Power Systems have a long history of strong logical partition (LPAR) isolation that is certified to separation standards. Previous Power Hypervisors have been evaluated to Common Criteria EAL4+ or higher.

On Power11, the Power Hypervisor isolates each LPAR (or VM) with dedicated or virtualized slices of processors, memory, and I/O. One partition cannot interfere with other partitions. The hardware enforces memory separation, and Hypervisor controls prevent one partition from reading or writing another's memory.

Power11 introduces Resource Groups and Trusted Execution features that further improve isolation and performance predictability for workloads, such as isolating certain cores or memory regions for high-priority VMs.

Additionally, Power11 introduces the Secure Partition Platform Engine (SPPE), a hardware mechanism that isolates the service processor (management controller) from the system's operational data paths. SPPE acts as a gatekeeper, preventing even the privileged service processor from directly accessing memory or registers of running partitions. This eliminates a potential attack vector and aligns with an internal Zero Trust approach. Even if the management firmware were compromised, it cannot bypass partition isolation to leak sensitive information.

These measures ensure that workloads on Power11, whether multiple tenants in a cloud environment or multiple departments in a private data center, remain strongly isolated from each other.

IBM Power Virtual Server (PowerVS) uses this isolation capability to provide secure multi-tenancy on the public cloud. The secure architecture of PowerVS ensures that one customer's VMs cannot access another customer's VMs, with strict controls preventing cross-tenant access or interference. Network segmentation and virtual firewalls in PowerVS further isolate workloads, and integration with cloud security services (such as IBM Cloud Monitoring and Compliance) provides continuous oversight.

Operational security

Power11 supports multiple operational security features to maintain a secure posture in hybrid environments. Power11 includes Cryptographic role-based access control (RBAC), enabling administrators to receive fine-grained roles that limit their security-sensitive operations, such as separate roles for managing keys versus managing VMs.

PowerSC, IBM's security and compliance tool for Power Systems, works with Power11 to automate security policy enforcement and compliance audits across hybrid cloud deployments. Features include trusted logging, security event forwarding to IBM QRadar® (SIEM), and integration with enterprise directory services (LDAP/AD) for unified identity across on-premises and cloud environments. These capabilities maintain hybrid cloud security.

Note: IBM Power11 delivers layered security for hybrid cloud workloads through encryption, quantum-safe cryptography, secure boot, and strict isolation without sacrificing performance or agility. Its next-generation cryptographic accelerators handle intensive encryption tasks with minimal impact, making always-on security practical for data replication across on-premises and cloud environments. Quantum-safe techniques enable organizations to future-proof their data protection strategies. Consistent security models between Power11 and PowerVS enable unified governance and compliance reporting throughout the hybrid environment.

1.6 Core principles of hybrid on Power11

The core principle of hybrid cloud is creating a unified IT environment that seamlessly connects on-premises infrastructure with public and private cloud resources. This integration enables organizations to place workloads where they perform best, balancing control, performance, and scalability. Hybrid cloud supports workload portability across environments through consistent operations and flexible deployment models, laying the foundation for achieving zero planned downtime through resilient architectures and dynamic resource management.

1.6.1 Workload portability (onprem and cloud)

Workload portability enables hybrid cloud flexibility by allowing applications and services to move seamlessly between on-premises environments and the cloud. With IBM Power Systems Virtual Server, organizations can replicate or migrate AIX, IBM i, and Linux on Power workloads without rearchitecting or refactoring. This capability ensures that businesses respond quickly to changing demands, optimize resource usage, and maintain consistent performance across platforms. PowerVS simplifies the transition between local and cloud

deployments by supporting common tooling and operating environments, helping IT teams maintain control while embracing cloud agility.

1.6.2 Zero downtime

As enterprises adopt hybrid cloud architectures to balance agility, scalability, and control, maintaining continuous availability across both on-premises and cloud environments becomes critical. IBM Power11 introduces a unified approach to Zero Planned Downtime (ZPD) that extends seamlessly into hybrid cloud deployments, enabling organizations to perform infrastructure maintenance without disrupting business operations. This includes updates to platform firmware, VIOS, I/O adapter firmware, and operating systems.

The Hardware Management Console (HMC) manages ZPD on Power11 through an automated update orchestration workflow. This workflow performs the following operations:

- Validates system readiness before initiating updates
- Downloads and stages fixes to minimize disruption
- Applies updates concurrently where supported by the platform
- Evacuates and returns logical partitions (LPARs) using Live Partition Mobility (LPM) when a disruptive step is required

This approach ensures that workloads remain online throughout the update process, significantly reducing operational downtime.

Note: IBM's ZPD capabilities for Power11 are based on internal validation. Although many update steps are nondisruptive, some might still require LPM. Administrators should verify their system topology against IBM's documented prerequisites before enabling automated update runs.

ZPD in hybrid cloud

Hybrid cloud environments combine the flexibility of public cloud with the control of private infrastructure. However, this complexity introduces challenges in maintaining uptime during updates, migrations, and scaling operations. IBM Power11 addresses these challenges by integrating ZPD capabilities across its supported operating systems (IBM i, AIX, and Linux) and enabling automated, nondisruptive maintenance workflows that span both physical and virtual infrastructure.

IBM i:

IBM i combines the operating system, database, and middleware into a single platform. This design simplifies operations and reduces risk, making IBM i a reliable foundation for mission-critical workloads.

ZPD capabilities in IBM i include the following features:

- Technology Refreshes for incremental enhancements
- Live Partition Mobility (LPM) for nondisruptive workload migration
- Migrate While Active to IBM Power Virtual Server for hybrid cloud flexibility

By using these features administrators can apply updates and perform migrations without impacting business operations.

AIX:

AIX, IBM's UNIX operating system, provides robust availability and security features for enterprise environments. Power11 enhances AIX® ZPD capabilities through the following technologies:

- Live Kernel Update (LKU) enables kernel patching without rebooting the system
- Live Library Update (LLU) applies updates to shared libraries while workloads remain active
- Live Partition Mobility (LPM) supports seamless migration of workloads during disruptive maintenance

Combined with HMC automation, AIX systems maintain near-continuous availability during both OS and platform-level updates.

Linux on Power:

Linux on Power supports multiple enterprise distributions, each offering live kernel patching technologies:

- kpatch for Red Hat Enterprise Linux (RHEL)
- Kernel Live Patching (KLP) for SUSE Linux Enterprise Server (SLES)
- Livepatch for Ubuntu

These technologies allow critical security and stability fixes to be applied without downtime. When integrated with Power11's HMC-driven automation and hybrid cloud mobility, Linux workloads fully participate in ZPD workflows.

Unified ZPD strategy with Power11

IBM i, AIX, and Linux form the foundation of a hybrid, AI-ready infrastructure. IBM Power11 unifies their maintenance strategies through the following capabilities:

- HMC-orchestrated automation
- AI-assisted operations through IBM Concert®
- Hybrid cloud mobility options

This integration makes ZPD a repeatable, scalable, and consistent process across diverse environments, reducing operational friction and ensuring continuous business availability.

1.7 IBM hybrid cloud solutions: Built for flexibility, security, and innovation

IBM supports enterprises in modernizing their IT environments and accelerating digital transformation through a comprehensive hybrid cloud strategy that unifies on-premises infrastructure, private cloud environments, and public cloud services into a single, cohesive operating model. This approach enables organizations to modernize at their own pace, maintain control over mission-critical workloads, and take advantage of cloud innovation where it delivers the most value. By integrating automation, security, resiliency, and AI-driven insights across this hybrid landscape, IBM provides a foundation that helps clients improve agility, reduce operational complexity, and innovate with confidence.

IBM’s flagship offerings help customers streamline operations, modernize applications, and accelerate their hybrid cloud journey:

- IBM Power Virtual Server (PowerVS)

Run AIX, IBM i, and Linux workloads in IBM Cloud with enterprise-grade security and performance. PowerVS enables hybrid deployments, workload portability, and disaster recovery across regions.

- Red Hat OpenShift on IBM Power

  A Kubernetes-based platform for containerized applications and AI workloads, delivering agility and scalability across hybrid environments.

- IBM Cloud Pak® Solutions

  Pre-integrated software packages for data, AI, automation, and security, which is built on Red Hat OpenShift. Examples include Cloud Pak for Data (AI and analytics) and Cloud Pak for Integration.

- IBM Storage for hybrid cloud

  Solutions like IBM FlashSystem® and Global Replication Services (GRS) provide secure, high-performance storage with multi-region replication for resilience.

- Automation and AIOps

  Tools such as IBM Turbonomic®, IBM Instana®, and Ansible Automation optimize workloads, automate failover, and help ensure operational efficiency across hybrid environments.

- Security & Compliance

  IBM integrates quantum-safe cryptography, identity management, and compliance frameworks to protect workloads across on-premises and cloud environments.

- Cost Optimization

  Offerings like Power Enterprise Pools 2.0 and Hybrid Cloud Packages help organizations scale resources dynamically and reduce costs.

For large-scale problems, IBM can provide large-scale solutions.

2

Network integration in reference architecture

This chapter outlines the reference architecture and integration strategies for building a secure, resilient, and scalable hybrid cloud on IBM Power. Connectivity forms the foundation of hybrid deployments, ensuring performance and compliance across on-premises and cloud environments. IBM Power Virtual Server networking options (primarily IBM Cloud Direct Link and VPN for VPC) provide the preferred methods for private, encrypted communication. The chapter covers multi-cloud integration, Red Hat OpenShift interoperability, and Data and AI platforms, supported by orchestration tools including Global Replication Services (GRS), Disaster Recovery Automation (DRA), and Security & Compliance Center for security posture management. These components form a unified architecture that delivers enterprise-grade resiliency and operational efficiency.

This chapter covers the following session:

- “Network connectivity for Power Virtual Server”
- “Private connectivity for IBM Virtual Private Cloud (VPC)”
- “Connectivity options and their relationship to Transit Gateway”
- “Network integration for RISE with SAP in IBM Power Virtual Server”
- “Integration with public clouds for IBM Power Systems”
- “Network design and address planning”

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2.1 Network connectivity for Power Virtual Server

Power Virtual Server networking delivers the secure, flexible, and high-performance connectivity required for modern hybrid cloud architectures. In environments where workloads span on-premises data centers, public cloud services, and cross-region deployments, reliable networking forms a foundational pillar for both operational resilience and security. PowerVS addresses these needs through a layered, cloud-native design anchored by three core components:

- Power Edge Router (PER) with Transit Gateway (TGW) for scalable hybrid connectivity
- Network Security Groups (NSGs) for fine-grained traffic control and workload isolation
- CRN-based metering for transparent tracking and governance of network resource consumption

These capabilities enable enterprises to integrate mission-critical Power workloads into their hybrid cloud strategies while maintaining consistent security policies, predictable performance, and the agility to adapt as business demands evolve.

Power Edge Router (PER), integrated with IBM Cloud Transit Gateway, forms the private routing fabric that interconnects PowerVS workspaces with VPCs, classic infrastructure, and on-premises environments. Using PER with IBM Cloud Transit Gateway keeps traffic on IBM's backbone and avoids public internet exposure.

Network Security Groups (NSGs) enable granular east-west segmentation within workspaces, allowing administrators to define explicit allow and deny rules and use Network Address Groups for simplified policy management.

Cloud Resource Name (CRN)-based metering provides visibility into resource usage and cost alignment, enabling tagging, grouping, and accurate chargeback models for network and security resources.

As of 2025, PowerVS workspaces are provisioned with Power Edge Router (PER) enabled by default. PER acts as the primary routing component, which is integrated with IBM Cloud Transit Gateway to provide secure, private connectivity between PowerVS, VPCs, and classic infrastructure. This design eliminates public internet exposure and simplifies hybrid cloud architectures.

For legacy workspaces created before 2023 that are not migrated to PER, connectivity is established using Cloud Connections. However, all examples and best practices in this chapter assume a PER-based environment.

2.1.1 IBM Power Edge Router and IBM TGW: Hybrid connectivity backbone

Effective hybrid cloud networking for IBM Power Virtual Server requires high-performance, secure, and simple connectivity. IBM Power Edge Router (PER) and IBM Cloud Transit Gateway form the foundation of this architecture, delivering high-performance, private, and scalable routing between PowerVS workspaces, Virtual Private Clouds (VPCs), and classic infrastructure. They provide the unified network fabric for running enterprise workloads seamlessly across IBM Cloud environments.

Power Edge Router (PER) is a high-performance routing solution for IBM Power Virtual Server (PowerVS) workspaces. It replaces Direct Link-based connectivity and connects directly to IBM Cloud MPLS backbone, providing 400 Gbps aggregate bandwidth per Performance Optimized Datacenter(POD).

For more information, see Getting started with the Power Edge Router.

Figure 2-1 presents the network architecture diagram, detailing the integration of the Power Edge Router (PER) into the IBM Cloud environment.

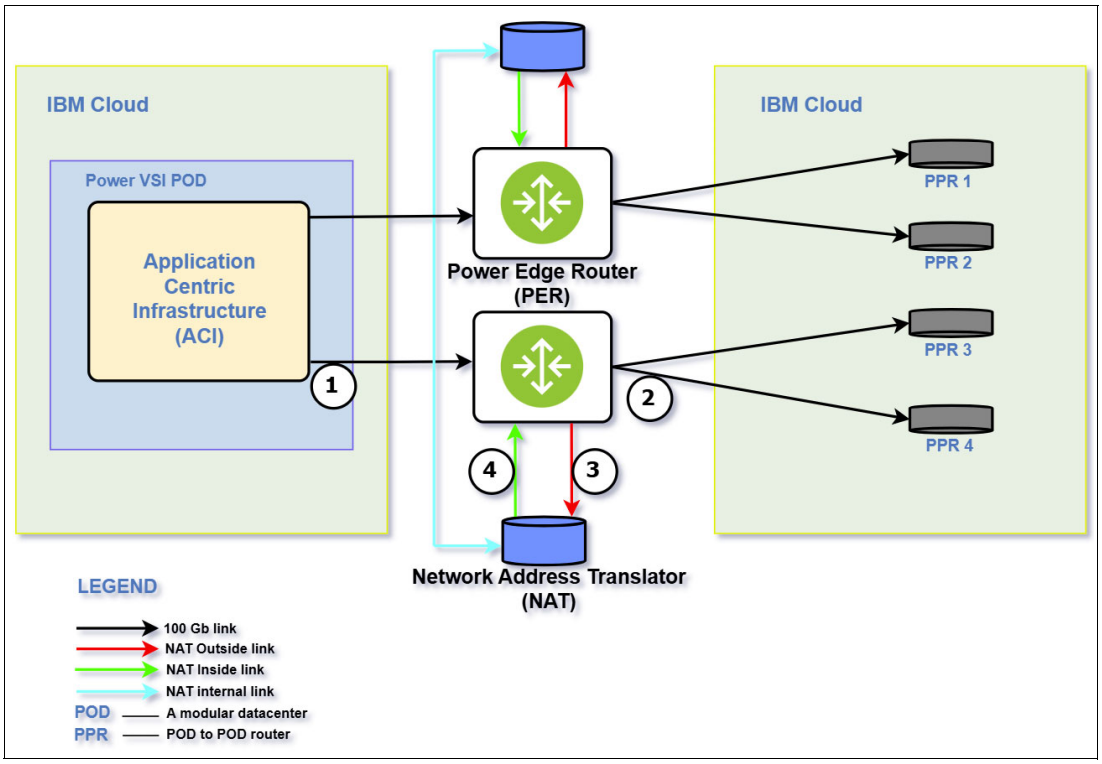


Figure 2-1 Power Edge Router network architecture diagram

IBM Cloud Transit Gateway (TGW) is a centralized network hub that interconnects IBM Cloud resources, including Virtual Private Clouds (VPCs), classic infrastructure, and Power Virtual Server (PowerVS) workspaces, over IBM's private backbone. Instead of relying on complex point-to-point links, TGW provides a unified routing fabric that simplifies network design and policy management. By keeping traffic within IBM's global network, TGW ensures private, in-transit data transfer between IBM Cloud data centers while delivering high availability and secure connectivity.

Organizations use TGW to support hybrid workloads and private data transfers while reducing administrative workload. When integrated with Power Edge Router (PER), TGW forms the core of the hybrid connectivity model for PowerVS, enabling seamless communication between PowerVS environments, VPCs, and classic infrastructure without exposing traffic to the public internet.

TGW acts as the central hub for routing across multiple VPCs, PowerVS workspaces, and VMware/Classic environments. TGW supports the following capabilities:

- Local and global routing for multi-region designs
- Zone-local pathing for optimized latency
- Centralized policy control for multi-VPC and cross-account connectivity

For more information about connectivity, see About IBM Cloud Transit Gateway. Figure 2-2 on page 26 illustrates interconnecting two or more VPCs within the same multi-zone region (MZR). Two VPCs in the same region connect by using a local Transit Gateway, enabling private, high-speed communication without relying on public internet paths.

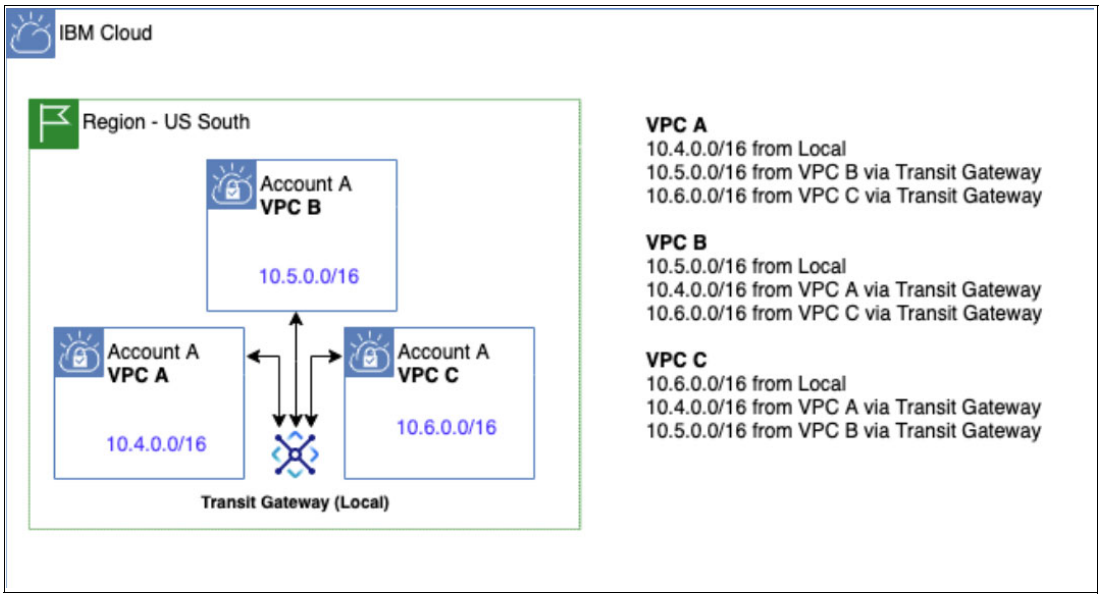


Figure 2-2 Multi-zone region architecture

2.1.2 East-west segmentation with network security groups

A network security group (NSG) enables administrators to define and enforce network traffic rules within an IBM Power Virtual Server workspace. NSGs provide granular control over east-west traffic by allowing or denying specific protocols and ports at the subnet or virtual machine (VM) network interface (NIC) level. This enhances security and simplifies traffic management between resources in the PowerVS environment.

Security rules in an NSG determine which inbound (ingress) traffic to permit or deny for its members. Members can include one or more NICs associated with subnets or VMs. All outbound (egress) traffic is allowed by default. Without NSGs, VMs within a workspace can freely exchange data regardless of subnet, requiring operating system-level controls for isolation. NSGs enable administrators to enforce network isolation and zero-trust principles directly at the infrastructure level.

Figure 2-3 on page 27 shows how Network Security Groups and Network Address Groups apply within a Power Virtual Server workspace and IBM Cloud VPC environment to control east-west traffic and enforce security policies.

NSGs in Power Virtual Server include the following key features:

- Explicit allow or deny rules for east-west traffic
- Protocol support for TCP, UDP, ICMP, or Any, with port-level granularity
- No additional cost; included with PowerVS workspaces
- No performance impact on network throughput or latency for NSG members
- Requirement for Power Edge Router (PER) and CRN-based metering in the region
- Network Address Group (NAG) integration for simplified rule management using external CIDR ranges, such as TGW-attached VPCs or private endpoints

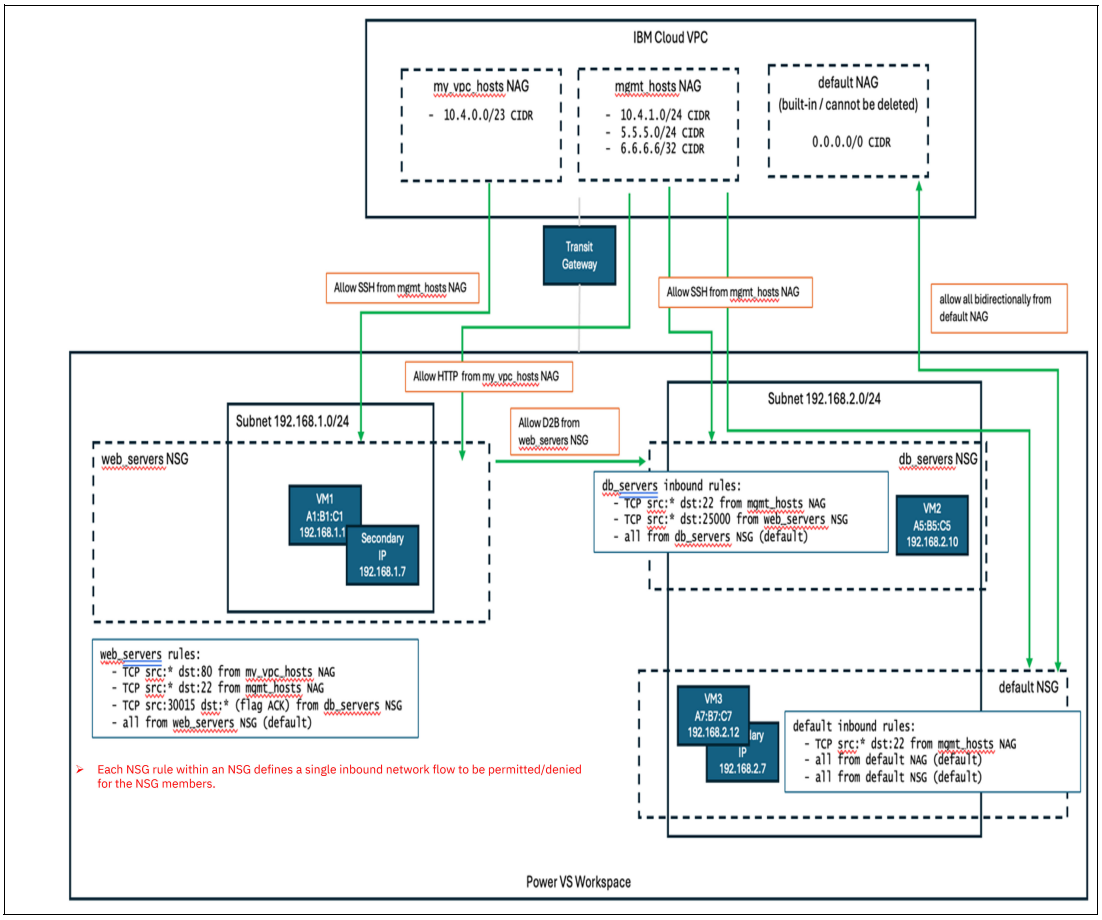


Figure 2-3 NSGs enforce east-west segmentation within a PowerVS workspace

NSGs strengthen workload protection, streamline policy management, and provide a more intuitive and scalable approach to securing east-west traffic within Power Virtual Server environments:

- Enhanced security and traffic control to prevent unauthorized access
- Reduced rule sprawl and improved scalability through NAGs
- A more intuitive network isolation compared to OS-level controls

Note: Use NSGs to segment workloads within a workspace, such as isolating application tiers or disaster recovery environments. Combine NSGs with NAGs to enforce consistent policies and implement zero-trust security models.

For more information, see Network security groups in IBM Power Virtual Server.

2.1.3 Visibility and cost alignment with CRN-based metering

IBM Cloud must consistently and uniquely identify resources such as virtual machines (VMs), storage volumes, snapshots, and dedicated hosts across services and regions. The Cloud Resource Name (CRN) provides this identification. CRNs are globally unique identifiers that enable precise resource management, billing, and automation.

A CRN is a structured string that encodes essential details about a resource, including its service, location, account, and instance information. CRNs enable unambiguous resource reference regardless of type or deployment region.

Figure 2-4 shows the PowerVS Cloud Resource Name (CRN) architecture for resource identification, grouping, and metering in IBM Cloud. CRNs apply within the PowerVS service hierarchy.

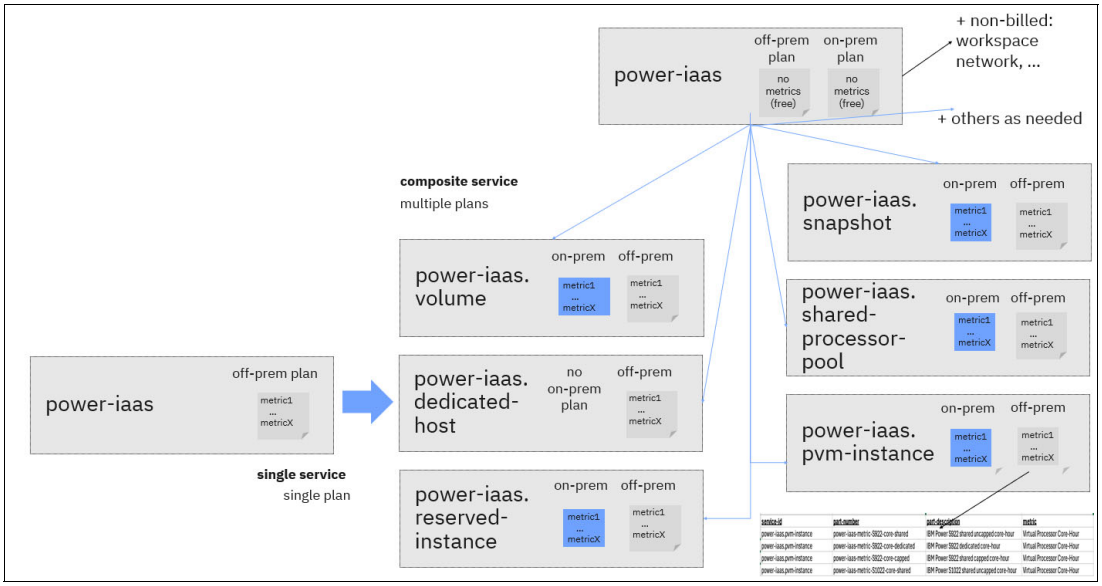


Figure 2-4 Cloud Resource Name diagram

CRN metering provides the following key capabilities:

- Granular usage visibility at the individual resource level
- Tagging and grouping for logical organization and filtering, enabling cost allocation by workload or project
- Enhanced resource groupings for virtual machines, shared processor pools, volumes, snapshots, and dedicated hosts

A CRN follows a standardized format:

`crn:v1:public:power-iaas:dal12:a/<tenant-id>:<instance-id>:pvm-instance:<UUID>`[1]

CRN-based metering introduces a major improvement in billing transparency for Power Virtual Server by shifting from aggregated workspace-level billing to precise, resource-level usage reporting. This change enables organizations to track consumption per virtual machine, volume, snapshot, or processor pool, apply accurate chargeback or showback models, and improve financial governance. CRNs paired with tagging provide granular and auditable cost allocation, and API integration enables automation tools to query, manage, and optimize resource usage with greater accuracy.

From workspace-level to resource-level billing

Before CRN-based metering, usage was aggregated at the workspace level, resulting in a single combined charge for all resources. This made it difficult to identify high-consuming resources, allocate costs accurately to projects or departments, or produce detailed chargeback and showback reports.

[1] https://cloud.ibm.com/docs/account?topic=account-crn

CRN-based metering itemizes billing by resource type, including VMs, volumes, snapshots, and processor pools. Each resource receives its own CRN, enabling precise usage tracking, detailed consumption reporting, and improved transparency for internal and external audits.

CRNs work with tags to group resources by workload, project, or business unit, enabling cost allocation models.

Billing transparency

Itemized billing improves invoice reconciliation through the following capabilities:

- Each line item corresponds to a specific CRN
- Charges can be validated against actual resource usage

This supports financial governance and compliance requirements.

Accurate modeling

CRN-based metering enables precise modeling of network and security costs. This supports the following use cases:

- Capacity planning
- Forecasting future expenses
- Optimizing resource allocation

Automation and integration

CRNs can be used in API calls and automation scripts to perform the following operations:

- Query usage for a specific resource
- Apply policies or lifecycle actions
- Integrate with cost management tools

2.2 Private connectivity for IBM Virtual Private Cloud (VPC)

Enterprises adopting hybrid cloud require secure, reliable, and high-performance private connectivity between on-premises data centers, IBM Cloud Virtual Private Cloud environments, and IBM Cloud services. This section describes patterns and implementation steps to establish private, encrypted connectivity by using IBM Cloud Transit Gateway (TGW), VPN for VPC (site-to-site), and Virtual Private Endpoints (VPEs). The section also covers routing constructs (ingress and egress routing tables), security group configuration, validation methods, and known limitations relevant to production designs.

An example hub-and-spoke topology with a simulated on-premises environment illustrates end-to-end private paths for application workloads and service endpoints.

Figure 2-5 on page 30 shows the reference architecture that is used throughout this section. Hybrid connectivity flows from the on-premises environment through a VPN to the Hub VPC, then across the Transit Gateway to the Spoke VPC with a VPE.

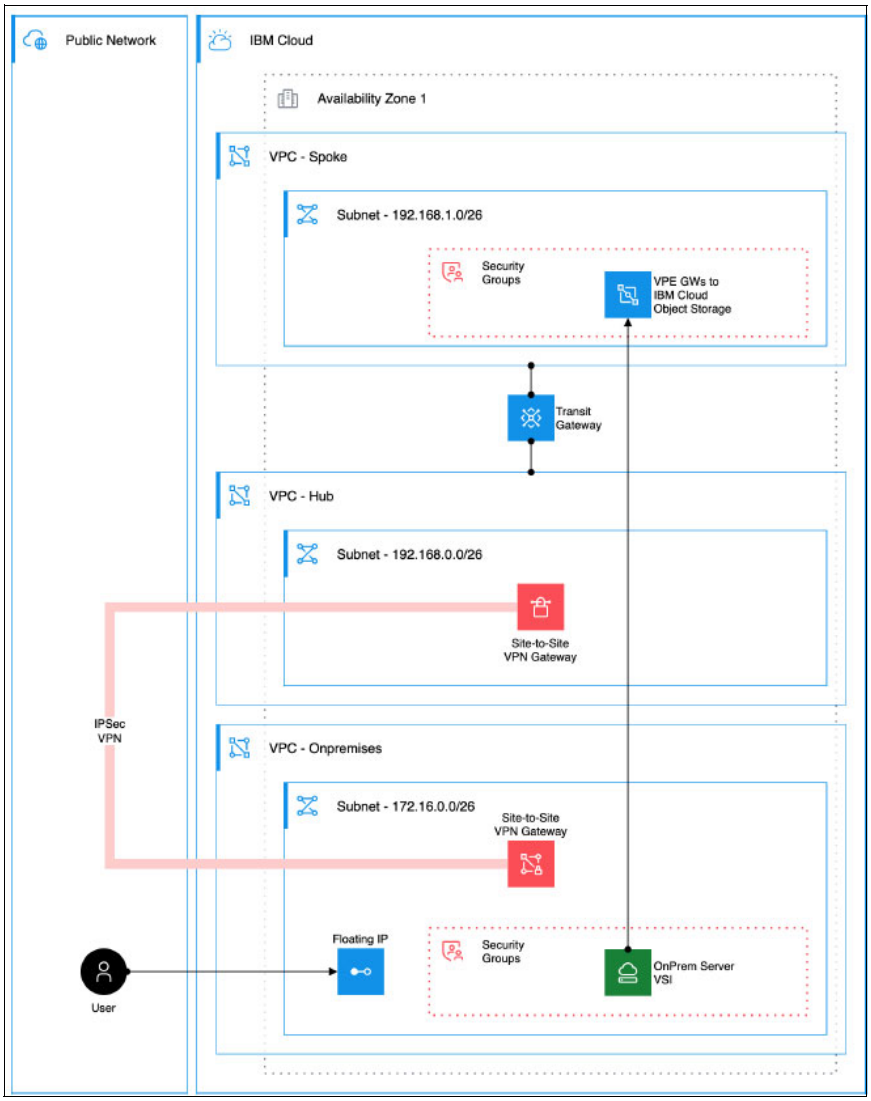


Figure 2-5 IBM Cloud network diagram

IBM Cloud VPC architecture uses Multi-Zone Regions (MZR) for high availability and disaster recovery. VPC architecture includes subnets, routing tables, ACLs, security groups, and private endpoints for secure and scalable network topologies.

Figure 2-6 on page 31 provides a comprehensive view of a Virtual Private Cloud deployment within the IBM Cloud environment, showing how components form a secure and scalable cloud infrastructure.

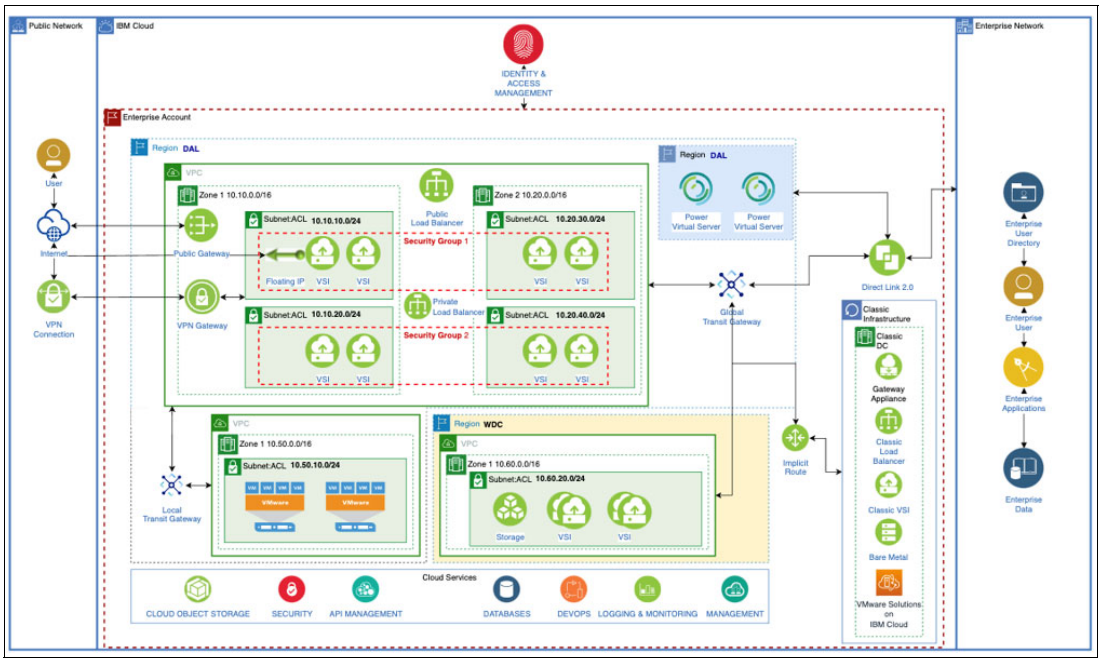


Figure 2-6 Virtual Private Cloud complete picture

IBM Cloud VPC architecture provides comprehensive capabilities that give organizations full control over network design, security posture, and hybrid connectivity options. IBM Cloud VPC architecture includes the following key capabilities:

- Full network control for defining IP ranges, subnets, routing tables, and firewall rules
- High availability and disaster recovery through Multi-Zone Regions with load balancers and replication
- Hybrid connectivity through Direct Link, VPN, and Transit Gateway for secure on-premises integration
- Private access to IBM services through Virtual Private Endpoints (VPE)
- Confidential computing to protect data at rest, in transit, and in use

IBM Cloud VPC provides additional capabilities to enchance security:

- Security and compliance through KYOK/BYOK encryption and FIPS 140-2 Level 4
- Performance with up to 200 Gbps networking throughput
- Developer support through REST APIs, CLI, and IBM Terraform®
- Cost efficiency through pay-as-you-go pricing and promotional offers

2.2.1 Key services and constructs

This design relies on several core IBM Cloud networking components that work together to provide secure, scalable, and structured hybrid connectivity. Each element plays a distinct role, from routing and inter-VPC connectivity to secure on-premises integration to private access to IBM services, ensuring that traffic flows predictably and remains fully within the private IBM Cloud network.

Transit Gateway

Transit Gateway (TGW) provides private interconnection between IBM Cloud VPCs and other IBM Cloud resources. It supports route propagation, centralized connectivity control, and high availability within a location.

Transit Gateway performs the following roles in this design:

- Learns VPC prefixes (/24 in this lab) from attached VPCs
- Advertises reachable networks between hub and spoke
- Acts as the distribution plane so that on-premises traffic, when terminated in the hub, can privately reach the spoke

VPN for VPC (site-to-site)

Site-to-site VPN for VPC establishes IPsec tunnels between an IBM Cloud VPC and an external peer. This session uses policy-based mode with pre-shared key (PSK) authentication and default IKEv2/IPsec policies.

VPN for VPC performs the following roles in this design:

- vpn-gw-hub (cloud side) terminates tunnels in the hub VPC.
- vpn-gw-onprem (simulated on-premises side) peers with the hub.
- Local and peer networks (policies) define which prefixes are permitted across the tunnel.

Virtual Private Endpoints

A Virtual Private Endpoints (VPE) exposes IBM Cloud services with private IPs inside a VPC, eliminating the need for public egress. VPEs are associated with security groups and VPC subnets.

VPE performs the following roles in this design:

- vpe-cos-1 in the spoke VPC provides private access to IBM Cloud Object Storage over the IBM Cloud network.
- On-premises clients reach the VPE privately through VPN, hub, TGW, and spoke.

VPC routing tables

VPC uses two routing table types to control path selection:

- Egress routing tables apply to subnets for outbound traffic from VMs toward other VPC resources, TGW, or the internet
- Ingress routing tables control incoming traffic from external sources such as TGW, Direct Link, or VPN gateways. Ingress tables must explicitly accept routes from these sources and can advertise routes back to TGW.

Design principles

- This architecture uses the following design principles to ensure secure, predictable, and well-governed hybrid connectivity:
- Hub and spoke: External connectivity terminates in a hub VPC to centralize inspection, policy, and shared services. TGW distributes to spokes.
- Explicit route control: Ingress tables in the hub VPC accept VPN routes and advertise reachability back to TGW for return paths.

- Least privilege security: VPE security groups are constrained to required CIDRs and ports, such as TCP 443 for Cloud Object Storage, and VSI SSH access is restricted to known sources.
- Deterministic testing: Each segment (VPN, hub routing, TGW propagation, spoke reachability, VPE access) is validated with discrete tools (ping, tcpdump, nc) before proceeding.

2.2.2 Implementation

This section provides a complete end-to-end implementation aligned with the reference architecture. The implementation follows a sequential approach, beginning with essential prerequisites, followed by provisioning the spoke VPE, and then building the hub-and-spoke VPC topology. Each stage ensures clarity, repeatability, and consistency with the overall design principles.

1. Verify prerequisites:

   The implementation requires the following prerequisites:

   – IBM Cloud account with permissions to create VPCs, subnets, TGW, VPN, VPE, VSIs, and security groups
   – Two resource groups or a single designated resource group
   – SSH key for VSI access
   – Chosen geography, region, and zone (examples use Dallas us-south, zone us-south-1)

2. Provision the spoke VPE (private service access):

   a. In the spoke VPC, create Virtual Private Endpoint Gateway vpe-cos-1.
   b. Select Cloud Object Storage as the service and choose the regional or global direct endpoint corresponding to the region.
   c. Place the endpoint in spoke-subnet-1 and associate security group sg-vpe (remove default SG if using a custom one).
   d. Record the VPE IP (for example, 192.168.1.x) for later validation.

   Note: VPEs expose service IPs inside the VPC. Clients from other networks require proper routing and SG permissions.

3. Create the hub and spoke VPCs:

   Hub VPC (vpc-hub):

   – Address prefix: 192.168.0.0/24
   – Subnet hub-subnet-1: 192.168.0.0/26
   – Default ACL: allow all (adjust per policy)
   – SG: allow ICMP/SSH as needed

   Spoke VPC (vpc-spoke):

   With spoke-subnet-1 (for example, 192.168.1.0/26) and the VPE

4. Attach hub and spoke to Transit Gateway

   – Create Transit Gateway tgw-lab-3 in the same location as the VPCs.
   – Add two VPC connections: vpc-hub and vpc-spoke.
   – After the TGW becomes Available, generate a route report and verify that it has learned both VPC /24 prefixes.

Note: TGW shows VPC address prefixes (/24) rather than individual subnet CIDRs (/26). This is expected behavior.

5. Build the simulated on-premises VPC and VPN
   - On-premises VPC (vpc-onprem):
     - Prefix: 172.16.0.0/24
     - Subnet onprem-subnet-1: 172.16.0.0/26
   - Site-to-site VPN gateways:
     - Create vpn-gw-hub in vpc-hub (policy-based; no connection yet).
     - Create vpn-gw-onprem in vpc-onprem (policy-based; no connection yet).
     - Record both public IPs.
6. Configure IPsec tunnel policies

   To establish a secure and reliable IPsec connection between the on-premises environment and the IBM Cloud VPC, the VPN configuration requires strong authentication, consistent policy settings, and correctly defined local and peer networks. The following steps bring both VPN gateways online and ensure a stable, encrypted tunnel:
   - Use a strong pre-shared key (PSK), ideally 20–40 characters with mixed-case, digits, and symbols.
   - The default IKEv2/IPsec policies are sufficient for this implementation and require no customization.
   - Create the VPN connection on each gateway, supplying the peer's public IP address, the shared PSK, and the appropriate local and peer CIDR lists.
   - Wait for both gateways to report an Active status before proceeding.

Table 2-1 shows the VPN configuration that defines which networks each gateway advertises (local networks) and which remote networks it expects to reach (peer networks).

Table 2-1 Define local and peer networks for each side

VPN gateway	Local networks	Peer networks
vpn-gw-hub	192.168.0.0/24, 192.168.1.0/24	172.16.0.0/24
vpn-gw-onprem	172.16.0.0/24	192.168.0.0/24, 192.168.1.0/24

2.2.3 Routing for private reachability

At this stage, pings from vpc-onprem to vpc-spoke might fail. The spoke can receive packets (visible in tcpdump), but return traffic cannot reach 172.16.0.0/24 because TGW has no path. An ingress routing table in the hub resolves this issue by performing the following operations:

- Accepts routes from the VPN gateway
- Sets Transit Gateway as a traffic source for ingress
- Advertises accepted routes back to TGW

Configure routing tables in the hub:

1. In vpc-hub, review the default egress table. The VPN service might insert an egress route toward on-premises based on tunnel policies.
2. Create a new ingress routing table hub-rt-ingress with the following settings:
   - Accepts routes from: VPN gateway
   - Traffic source: Transit gateway
   - Advertise to: Enabled
3. After creation, the VPN service automatically populates a route for 172.16.0.0/24 in hub-rt-ingress.
4. Regenerate the TGW route report. The 172.16.0.0/24 prefix now appears through vpc-hub.

   Return traffic can now traverse spoke → TGW → hub → VPN → on-premises.
5. Configure security controls

   Secure private connectivity between the simulated on-premises environment and IBM Cloud services requires appropriate security controls.

VPE security group

The Virtual Private Endpoint (VPE) security group (sg-vpe) must allow inbound traffic from the on-premises CIDR block 172.16.0.0/24 to the VPE IP address on TCP port 443 for Cloud Object Storage access. Specify an inbound rule with protocol TCP, port range 443, and source CIDR 172.16.0.0/24.

VSI access control

Restrict access to the on-premises Virtual Server Instance (VSI) by limiting SSH (port 22) to trusted IP addresses, such as a corporate egress IP or a specific workstation public IP. Default outbound traffic might remain open during testing, but tighten these policies for production environments.

DNS resolution

DNS resolution from the on-premises simulation to IBM Cloud services that use fully qualified domain names (FQDNs) is not supported by default. For lab validation, access services by using the VPE's private IP address or the direct endpoint IP obtained through tools such as nslookup. For production deployments, configure DNS forwarding or conditional resolvers to enable seamless name resolution across hybrid environments.

2.2.4 Validation and operations

After deploying the hybrid connectivity and security components, validate each segment of the design to ensure predictable behavior, proper routing, and secure access across the environment.

End-to-end validation

Perform the following validation steps:

1. VPN health:
   - Verify that both VPN connections are Active.
   - Confirm that Dead Peer Detection (DPD) defaults are in place.

2. TGW propagation:
   - Generate a route report.
   - Ensure that 192.168.0.0/24, 192.168.1.0/24, and 172.16.0.0/24 appear with correct attachments.
3. Spoke reachability:
   - From the on-premises VSI, ping the spoke VSI private or floating IP.
   - If traffic is unidirectional, run tcpdump -nne -i any net 172.16.0.0/12 on the spoke to confirm ingress.
4. VPE access:
   - From the on-premises VSI, test connectivity with nc -vz <vpe-cos-ip> 443.
   - If the connection is refused, add or adjust the sg-vpe inbound rule to allow TCP/443 from 172.16.0.0/24.
5. Observability

   Use the following tools for monitoring and troubleshooting:
   - Route reports (TGW): Snapshot of current learned and advertised prefixes
   - VPC routing tables: Verify accepted routes and advertisement flags
   - Security group logs: Review deny events if integrated with SIEM
   - Packet captures (tcpdump): Confirm traffic paths on VSIs

Limitations and design considerations

The architecture provides robust and secure hybrid connectivity, but several limitations and design considerations must be understood before moving to production:

- VPE and UDP through TGW

  VPEs do not currently support UDP traffic through TGW. For example, NTP to a VPE over TGW does not function. Use TCP-based service endpoints or alternative architectures for UDP requirements.
- DNS resolution from on-premises

  Without DNS integration, clients must use service or VPE IPs. For production, configure conditional forwarders or DNS resolvers within the hub or spoke to resolve IBM Cloud service FQDNs privately.
- Policy-based VPN scale

  Policy-based IPsec requires explicit CIDR policies. As prefixes grow, prefer route-based designs (when available) or centralize prefixes to reduce policy churn.
- Security hardening

  Apply the following security practices:
  - Restrict ACL and SG rules to required protocols and CIDRs only
  - Rotate PSKs and consider certificate-based authentication if supported

Table 2-2 on page 37 summarizes common troubleshooting scenarios and differentiates them by symptoms, likely causes, verification steps, and recommended remediation actions.

Table 2-2 Troubleshooting guide

Symptom	Likely cause	Verification	Remediation
One-way ping (on-premises -> spoke seen, reply lost)	Missing ingress route advertisement to TGW	TGW route report lacks 172.16.0.0/24	Create hub ingress table, accept from VPN GW. Enable Advertise to TGW
nc -vz <vpe-ip> 443 fails	sg-vpe lacks rule for on-premises CIDR	Inspect VPE SG inbound rules	Add inbound TCP/443 from 172.16.0.0/24
VPN stuck Down	PSK mismatch or peer IP wrong	VPN connection details on both gateways	Correct PSK/peer; re-create if needed
VPE is reachable by IP, not FQDN	DNS not configured across hybrid path	nslookup <FQDN> from on-premises VSI	Implement DNS forwarding/conditional resolvers
TGW report shows only /24s	Expected behavior (VPC prefix granularity)	Compare against subnet CIDRs	None; design with VPC prefix awareness

The following best practices maintain consistency, strengthen security, and ensure predictable routing behavior in hub-and-spoke network designs:

- Centralize ingress at the hub to simplify policy and inspections
- Document CIDR ownership and avoid overlaps across VPCs and on-premises environments
- Automate with Terraform® or CLI to maintain consistent and auditable configurations
- Use separate security groups for VPEs and workload NICs
- Apply least privilege principles to all security controls
- Regenerate TGW route reports regularly during changes to confirm propagation
- Aggregate CIDRs for policy-based VPNs to support growth and consider route-based alternatives where applicable

The following configuration summary consolidates the key components of the design:

- VPCs and prefixes:
  - vpc-hub: 192.168.0.0/24 →hub-subnet-1: 192.168.0.0/26
  - vpc-spoke: 192.168.1.0/24 → spoke-subnet-1: 192.168.1.0/26
  - vpc-onprem: 172.16.0.0/24 → onprem-subnet-1: 172.16.0.0/26
- Transit Gateway:

  tgw-lab-3: connections to vpc-hub and vpc-spoke
- VPN
  - Gateways: vpn-gw-hub<-> vpn-gw-onprem
  - Mode: Policy-based with IKEv2/IPsec defaults and PSK
  - Policies:
  - Cloud side:
    - Local: 192.168.0.0/24, 192.168.1.0/24;
    - Peer: 172.16.0.0/24

  - On-premises side:
    - Local: 172.16.0.0/24
    - Peer: 192.168.0.0/24, 192.168.1.0/24
- Routing:

  Hub ingress table (hub-rt-ingress): Accepts routes from VPN GW; advertises to TGW
- VPE
  - vpe-cos-1 in spoke-subnet-1
  - Security group: Allows TCP/443 from 172.16.0.0/24

Example 2-1 Operational validation commands

```
# On on-premises VSI: confirm egress path and reachability to spoke
ping <spoke-vsi-private-or-floating-ip>

# On spoke VSI: confirm ingress packets from on-premises
sudo tcpdump -nne -i any net 172.16.0.0/12

# Validate private service access via VPE (TCP-based)
nc -vz <vpe-cos-ip> 443

# Discover IBM Cloud service endpoint IP (from VSI)
nslookup s3.direct.<region>.cloud-object-storage.appdomain.cloud
```

Private connectivity on IBM Cloud VPC combines VPN, Transit Gateway, VPC routing tables, and VPEs to deliver ETE private paths across hybrid estates. The hub and spoke pattern centralizes control, and TGW provides scalable distribution. Correct use of ingress routing, route advertisement, and security groups is essential for symmetric traffic flow and secure service access. The lab steps in this session provide a repeatable blueprint you can adapt for production deployments.

2.3 Connectivity options and their relationship to Transit Gateway

IBM Power Virtual Server environments use different connectivity models based on their configuration and migration status. These options determine how workloads communicate with other IBM Cloud resources and whether traffic remains private or traverses the public internet.

2.3.1 PER with Transit Gateway

PER provides the entry point for Power Virtual Server workspaces to IBM's backbone, and Transit Gateway acts as the central routing hub. Through this integration, PER connects your Power Virtual Server workspace to Transit Gateway, and Transit Gateway dynamically routes traffic to other IBM Cloud resources such as VPCs, classic infrastructure, VMware environments, or other Power Virtual Server workspaces.

Transit Gateway enables communication between the following components:

- Power Virtual Server workspaces
- Virtual Private Clouds (VPCs) hosting x86 workloads
- Classic Infrastructure or VMware environments

All network flows remain on IBM's private network, avoiding public internet exposure. This design for hybrid and multi-cloud architectures provides secure, scalable, and high-performance connectivity.

Note: PER + TGW = Private, scalable, secure (preferred).

2.3.2 Cloud connections (legacy)

Before migration to PER, older Power Virtual Server workspaces use Cloud Connections to link to VPC or on-premises networks.

Cloud Connections use Direct Link for private connectivity or VPN over the public internet. However, this approach is less efficient and lacks the centralized routing and policy control available with Transit Gateway.

Note: Cloud Connections are transitional and might involve public paths if VPN is used.

2.3.3 Public connectivity

Public connectivity occurs when VPN over the internet is used instead of Direct Link or PER/Transit Gateway. IBM does not recommend this method for production workloads due to latency, security risks, and compliance concerns.

Note: Use the public internet only for fallback or temporary connectivity, not for production workloads.

2.3.4 IBM Cloud Direct Link

IBM Cloud Direct Link 2.0 Connect delivers secure, dedicated Layer 3 connectivity between enterprise on-premises environments and IBM Cloud, including VPC, Transit Gateway, and Power Virtual Server. By bypassing the public internet, Direct Link 2.0 Connect ensures consistent network performance, enhanced security, and greater reliability for hybrid cloud workloads.

Direct Link 2.0 Connect supports predictable throughput (50 Mbps to 10 Gbps), low jitter, and private routing. For high availability, deploy diverse links. Choose local or global routing based on your reachability requirements. When you provision Direct Link, select the same location as your Power Virtual Server workspace.

Direct Link is suitable for the following scenarios:

- Transfer of large volumes of data between on-premises systems and IBM Cloud
- Latency-sensitive applications that require consistent network performance
- Business-critical workloads where secure, private connectivity is mandatory

Based on your connectivity and capacity requirements, you can choose from multiple Direct Link options, such as Direct Link Dedicated (a physical port connection) or Direct Link Connect (connectivity through a service provider).

Many Power Virtual Server workspaces support Power Edge Router (PER). In these workspaces, you can establish the connection by using the following architecture:

On-premises-Carrier network → Direct Link 2.0 → Transit Gateway → Power Virtual Server

Figure 2-7 illustrates how BGP (Border Gateway Protocol) exchanges routing information between networks to ensure proper data paths.

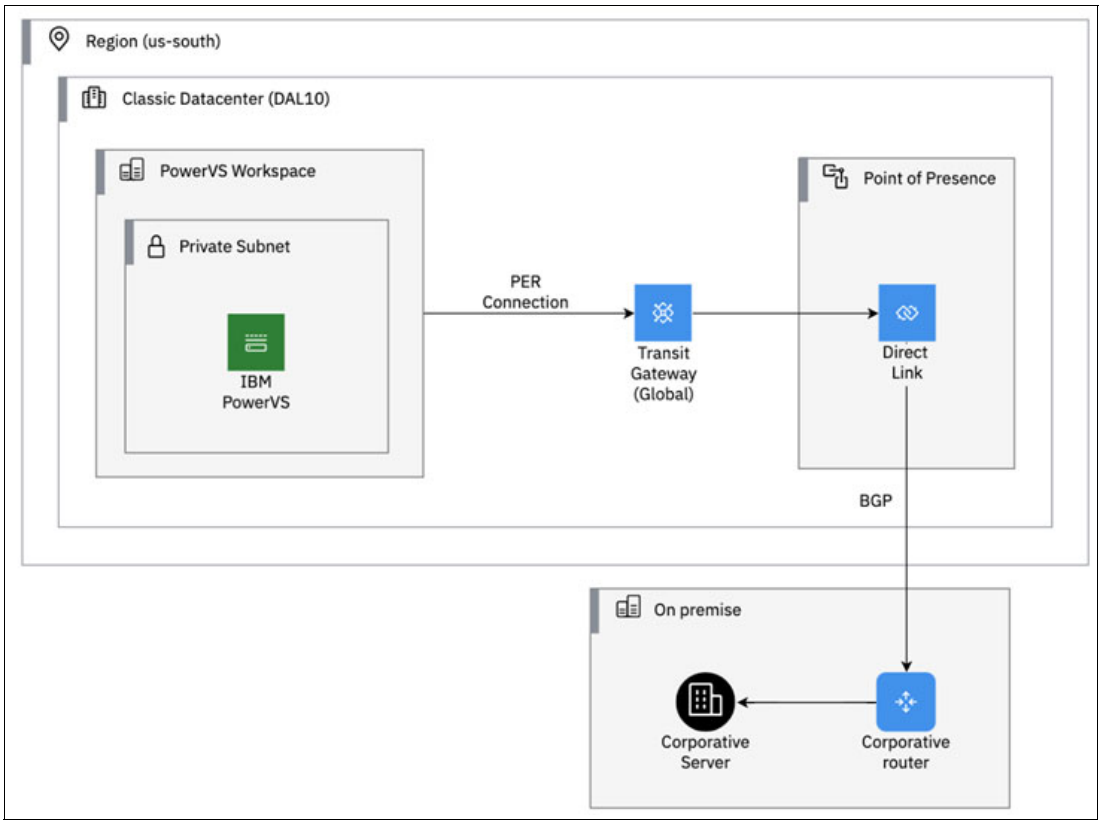


Figure 2-7 PER connection

Important: Effective July 1, 2025, IBM began applying metering charges to legacy Direct Link (DL 2.0) connections. These charges vary by port speed and can range from $100 to $5,000 USD per month[a]. To avoid these costs and benefit from improved performance, migrate to Power Edge Router (PER)-enabled workspaces[b].

a. https://cloud.ibm.com/status/announcement?query=cloud+connection
b. https://cloud.ibm.com/docs/power-iaas?topic=power-iaas-cloud-connections

2.3.5 IPsec VPN connectivity

Internet VPN Connectivity establishes a secure, encrypted tunnel over the public internet to connect enterprise networks or users with IBM Cloud and PowerVS resources. The following primary models provide connectivity:

- Site-to-Site VPN

  Provides secure connectivity between an on-premises network and IBM Cloud, enabling seamless integration of enterprise sites. Systems and applications communicate as if they were on the same private network.

  When private circuits are not an option, VPN for VPC offers encrypted internet-based connectivity. It can terminate into a transit VPC that peers with PowerVS through TGW, a common setup for quick-start environments and labs. Use non-overlapping CIDRs to prevent routing conflicts.

- Client-to-Site VPN

  Enables individual users or devices to connect securely to IBM Cloud/PowerVS resources over the internet. This approach is suitable for remote workers or administrators who require direct access to workloads in the cloud.

  Although internet VPNs are easy to deploy and cost-effective, they depend on public internet connectivity, which might introduce variability in latency, throughput, and availability compared to dedicated connections such as IBM Cloud Direct Link.

  Figure 2-8 shows a site-to-site VPN connection to PowerVS using VPN for VPC connectivity between your on-premises environment and IBM Cloud.

For more detailed instructions, see VPN connections.

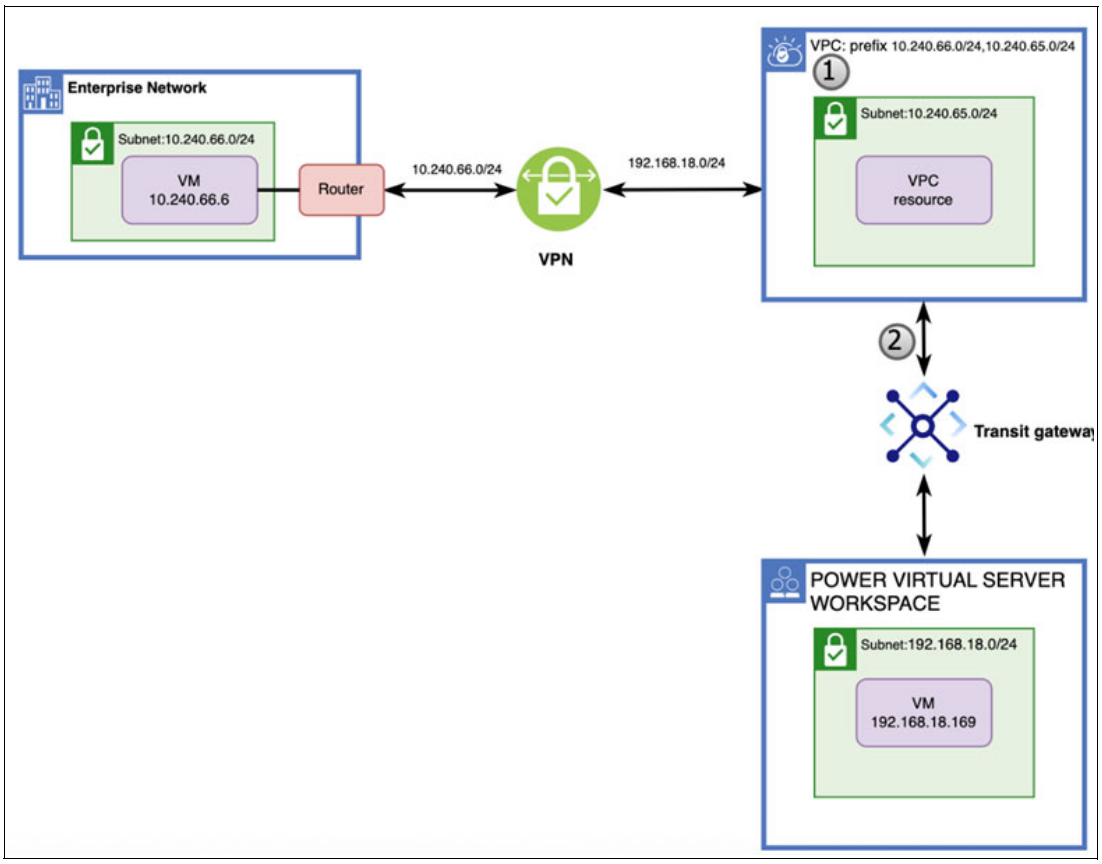


Figure 2-8 Site to site VPN diagram

For client-to-site VPN access, VPN for VPC (client-to-site) can be used. VPN for VPC supports OpenVPN protocol, so use client software that is compatible with OpenVPN.

Figure 2-9 illustrates how a user can securely connect to IBM Power Virtual Server using OpenVPN. The user installs an OpenVPN client on their computer and connects through the internet to a VPN server inside a Virtual Private Cloud (VPC). A Secrets Manager stores the login details, and a Transit Gateway links the VPC to the PowerVS environment. With this configuration, the user accesses virtual servers that are running systems like AIX and IBM i in the cloud.

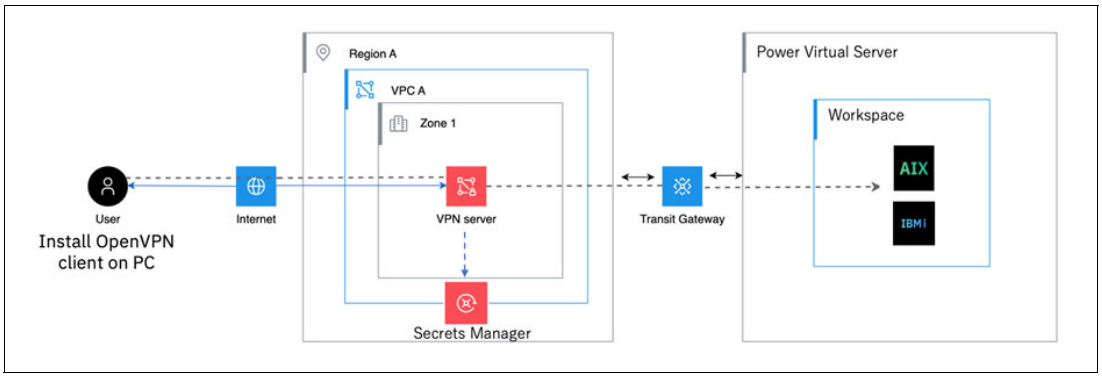


Figure 2-9 Client to site VPN access

2.3.6 Public connectivity

You can also access PowerVS over the internet. In such cases, give careful consideration to security.

Some use cases require exposing services hosted on PowerVS to the internet. Because PowerVS external IPs have fixed allowed ports and do not allow custom firewall configurations, use a VPC Floating IP instead. For load balancing and network address translation, Application Load Balancer for VPC (ALB) is the preferred option.

Security with public connectivity

When exposing services to the public, IBM Cloud Internet Services (CIS) can further enhance security and performance by providing DDoS protection, a Web Application Firewall (WAF), and caching capabilities. CIS functions as a reverse proxy within the overall communication flow: web access from clients is first received by CIS. Only requests that pass CIS security checks are forwarded to the Floating IP of the IBM Cloud VPC. VPC provides Security Groups to restrict traffic to accept only requests coming from CIS. As a result, only validated and secure requests reach the VPC and the PowerVS behind it.

Figure 2-10 on page 43 shows how IBM Cloud protects and connects workloads. Data from users travels through the internet to IBM Cloud Internet Services, which provides security features such as Web Application Firewall (WAF), DDoS protection, and performance tools such as Global Load Balancer (GLB) and Content Delivery Network (CDN). Traffic then moves through a floating IP into a Virtual Private Cloud (VPC) with security groups and an Application Load Balancer (ALB). A Transit Gateway links the VPC to Power Virtual Servers running systems such as AIX, IBM i, and Linux. This setup ensures secure, efficient routing and protection against threats.

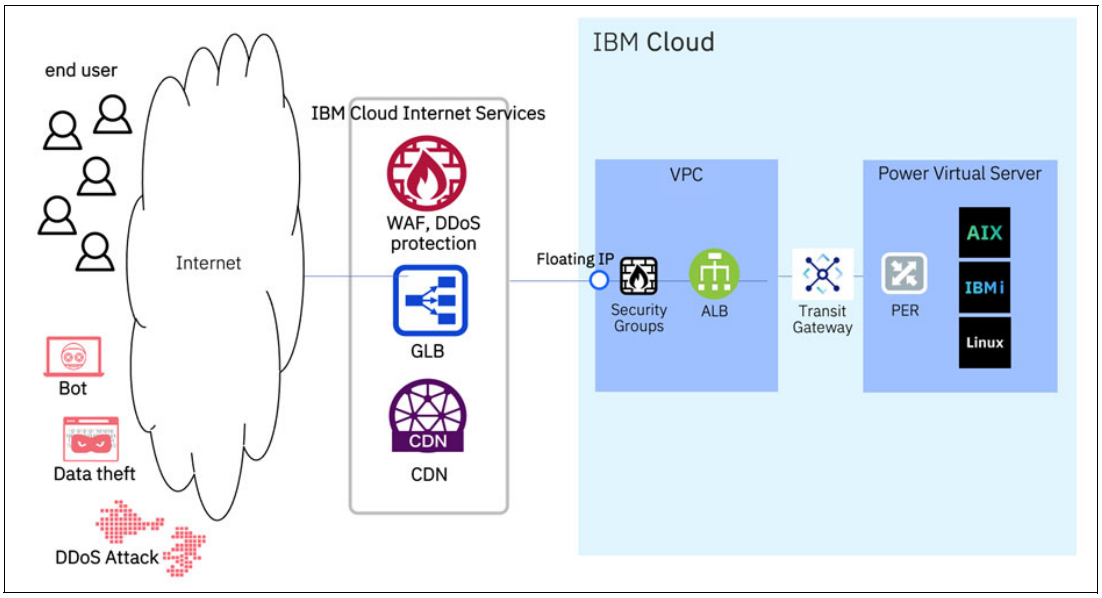


Figure 2-10 IBM Cloud protects and connects workloads.

Multi-workspace connectivity with Transit Gateway

To connect different PowerVS workspaces, use Transit Gateway. Transit Gateway is a service that relays communication between different networks, such as VPC, Classic Infrastructure, on-premises, and PowerVS. When connecting PowerVS workspaces located in different regions, the Transit Gateway must be of the Global type. This connection method enables easy data transfer between different workspaces.

Figure 2-11 illustrates how PowerVS environments in different regions can securely communicate over the IBM Cloud private network.

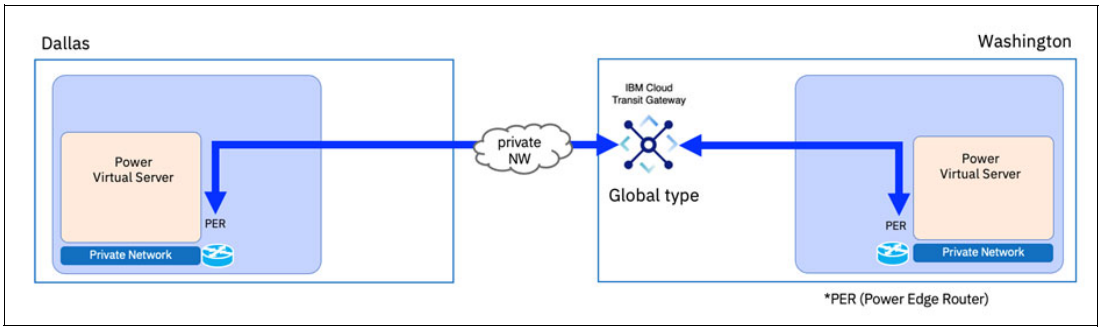


Figure 2-11 Connect from Dallas to Washington through private network

2.4 Network integration for RISE with SAP in IBM Power Virtual Server

With RISE with SAP on IBM Power Virtual Server, the SAP Enterprise Cloud Services (ECS) team manages the SAP systems. However, you are responsible for establishing network connectivity to RISE with SAP on Power Virtual Server. Consult with your SAP representative to understand the available options for connecting your on-premises networks and your IBM Cloud Account networks to the RISE with SAP on Power Virtual Server networks.

This document explains the concepts and best practices for a high-performance and secure solution to connect your networks with the RISE with SAP on Power Virtual Server networks.

In this document, the term *surround workload* defines a workload that integrates with your SAP system hosted in RISE with SAP on Power Virtual Server. These surround workloads can be non-RISE SAP or non-RISE non-SAP workloads. You can host these surround workloads in the following environments:

- On-premises: These surround workloads are typically loosely coupled workloads that do not require high bandwidth or low latency connections.
- Internet: Typically SaaS-based services.
- On another cloud: Your workloads or SaaS-based services.
- IBM Cloud: Your IBM Cloud workloads are hosted in one of the three IBM Cloud infrastructure environments: IBM Cloud Classic, IBM Cloud VPC, or IBM Power Virtual Server in an IBM data center.

Although your RISE with SAP on IBM Power Virtual Server workloads run in the SAP IBM Cloud Account, your IBM Cloud surround workloads run in your own IBM Cloud Account.

2.4.1 Virtual Private Cloud

A Virtual Private Cloud (VPC) provides network isolation and security in the IBM Cloud. A VPC can be a logical container for an enterprise division's resources (for example, marketing, development, or accounting) or a collection of resources operated by an enterprise team. You can connect VPCs to an enterprise's locations, to each other, and to VPCs in other IBM Cloud accounts.

With IBM Cloud Transit Gateway, you can define and control communication between resources hosted in the IBM Cloud infrastructure environments and on-premises networks. The IBM Cloud infrastructure environments include the following components:

- IBM Cloud VPC
- IBM Cloud Classic infrastructure
- Power Virtual Server workspaces

2.4.2 Transit Gateway

With an IBM Cloud Transit Gateway, you can connect the following resources:

- VPCs in your account or a different account.
- IBM Cloud classic infrastructure in your account or a different account.
- Direct Link in your account or a different account.
- IBM PowerVS in IBM data center in your account or a different account.
- Unbound Generic Routing Encapsulation (GRE) tunnels allow a connection between a zone and an endpoint in IBM Cloud Classic in your account or a different account.
- Redundant Generic Routing Encapsulation (GRE) tunnels allow a connection between two zones and an endpoint in IBM Cloud Classic or a VPC in your account or a different account.

2.4.3 RISE with SAP on IBM Power Virtual Server interconnectivity

RISE with SAP on IBM Power Virtual Server supports multiple connectivity configurations to meet diverse enterprise integration and networking requirements. These configurations enable secure, scalable, and high-performance communication between SAP workloads and external systems, including on-premises environments, other cloud services, and third-party applications.

Figure 2-12 illustrates the available interconnectivity options, including supported network topologies, integration points, and data flow paths for hybrid and multi-cloud infrastructures.

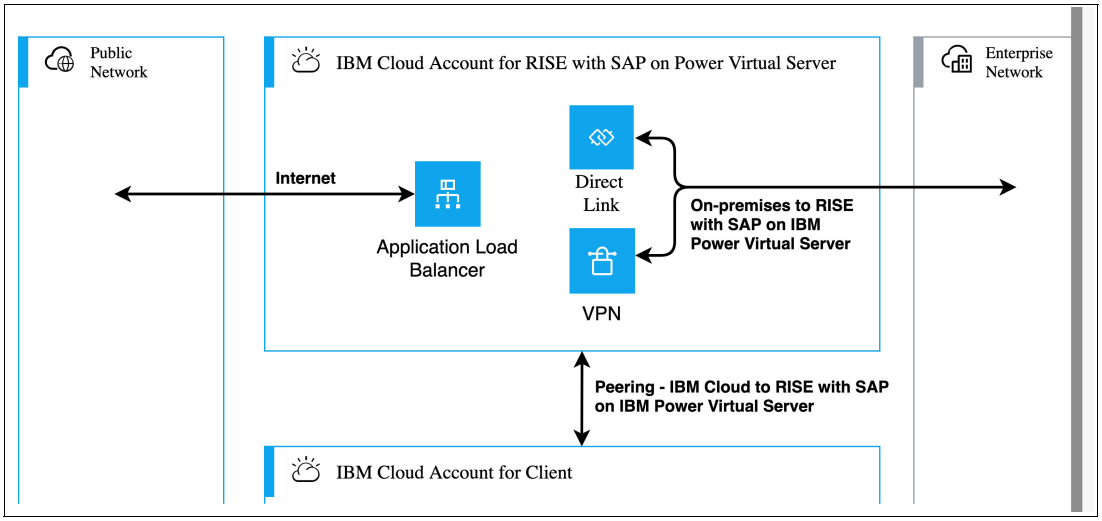


Figure 2-12 Connections for RISE with SAP

RISE with SAP on IBM Power Virtual Server supports the following connection types:

- On-premises to RISE with SAP on IBM Power Virtual Server

  Connects your users and workloads on-premises or at other non-IBM Cloud locations to the RISE with SAP on IBM Power Virtual Server service. You and SAP define this connectivity, which can be one of the following types:

  - Site-to-site IPsec VPN
  - Direct Link

- Internet to RISE with SAP on IBM Power Virtual Server provides application access from the internet.
- Peering - IBM Cloud to RISE with SAP on IBM Power Virtual Server

  Provides access between resources in your IBM Cloud account and your resources in RISE with SAP on IBM Power Virtual Server. A variant of this connectivity type uses your IBM Cloud account as a hub that connects your on-premises or external locations and your IBM Cloud resources, including your surround workloads, to RISE with SAP on IBM Power Virtual Server.

RISE with SAP on IBM PowerVS supports the following integration types:

- IBM Cloud Integration: Integrates RISE with SAP on IBM Power Virtual Server with your non-RISE SAP and non-RISE non-SAP workloads in IBM Cloud.
- Hybrid cloud integration: Integrates RISE with SAP on IBM Power Virtual Server with your non-RISE SAP and non-RISE non-SAP on-premises workloads.

- Multi-cloud integration: Integrates RISE with SAP on IBM Power Virtual Server with your non-RISE SAP and non-RISE non-SAP workloads hosted with other cloud providers.

Although your RISE with SAP on IBM Power Virtual Server workloads run in the SAP IBM Cloud Account, your IBM Cloud surround workloads run in your own IBM Cloud Account.

You can use the following connectivity patterns to peer your IBM Cloud surround workloads with your RISE with SAP on IBM Power Virtual Server workloads:

- IBM Cloud Transit Gateway
- IBM Cloud Site-to-Site VPN

Figure 2-13 illustrates the connectivity considerations.

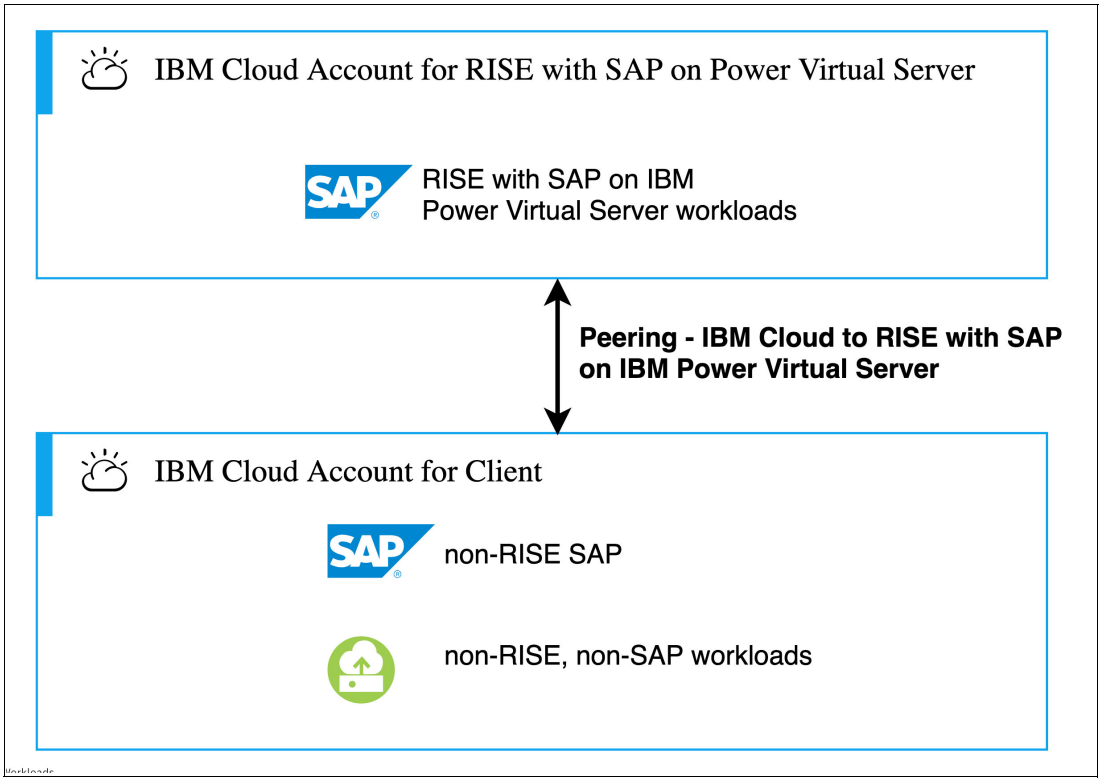


Figure 2-13 Cloud integration options

2.4.4 The IBM Cloud Transit Gateway pattern

With IBM Cloud Transit Gateway, you can define and control communication between resources hosted in the IBM Cloud infrastructure environments and on-premises networks. With an IBM Cloud Transit Gateway, you can connect the following resources:

- VPCs in your account or a different account.
- IBM Cloud classic infrastructure in your account or a different account.
- Direct Link in your account or a different account.
- IBM PowerVS in IBM data center in your account or a different account.
- Unbound Generic Routing Encapsulation (GRE) tunnels allow a connection between a zone and an endpoint in IBM Cloud Classic in your account or a different account.

- Redundant Generic Routing Encapsulation (GRE) tunnels allow a connection between two zones and an endpoint in IBM Cloud Classic or a VPC in your account or a different account.

The IBM Cloud Transit Gateway pattern includes the following subpatterns:

- IBM Cloud Transit Gateway in the SAP account
- IBM Cloud Transit Gateway in the client account

IBM Cloud Transit Gateway in the SAP account

This pattern uses the IBM Cloud Transit Gateway in the RISE with SAP on IBM Power Virtual Server IBM Cloud Account. Because this IBM Cloud Transit Gateway has the local routing option, it supports only your VPCs in the same region as the RISE with SAP on IBM Power Virtual Server VPC. Figure 2-14 shows the connectivity requirements.

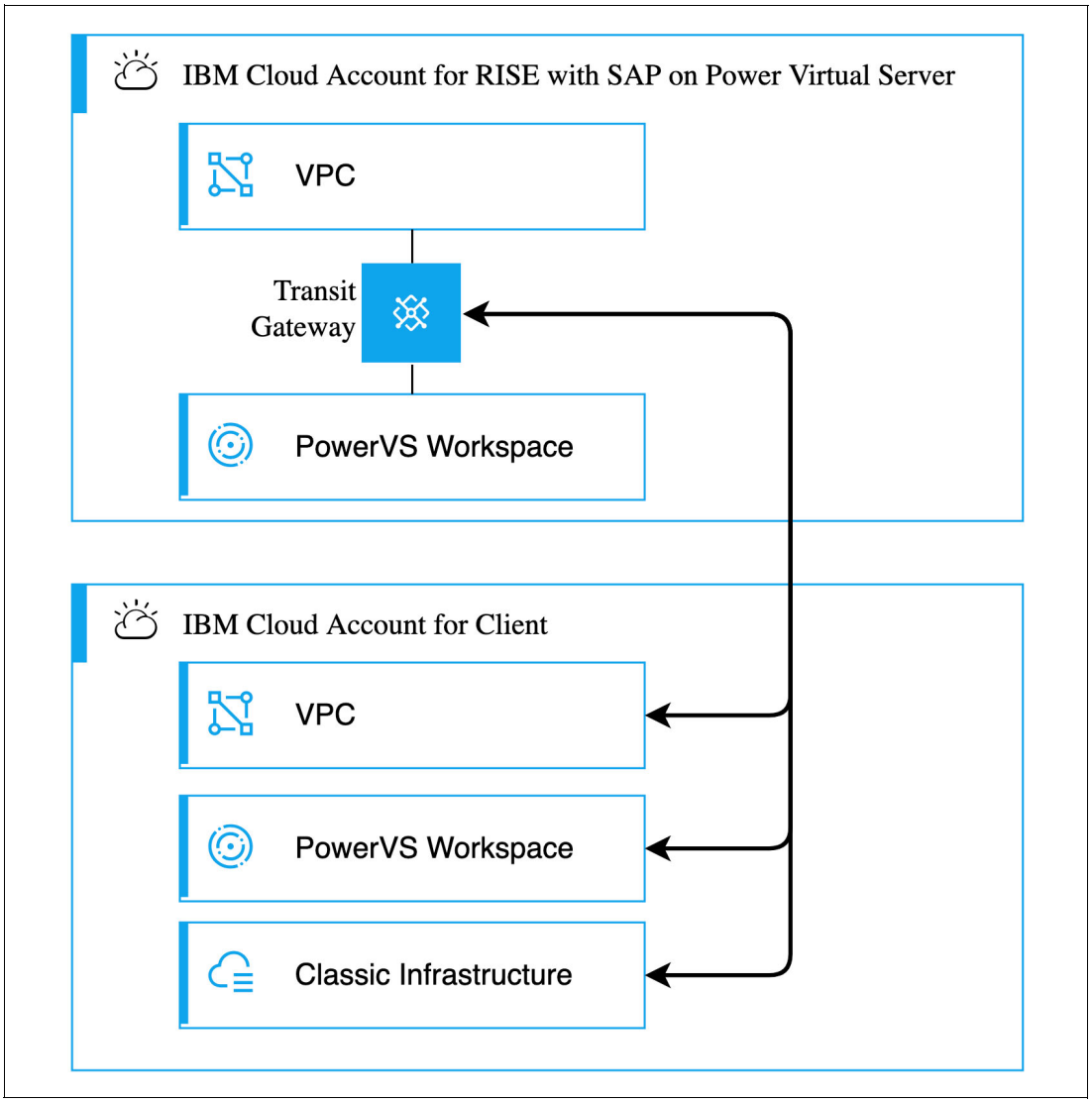


Figure 2-14 Connections for RISE with SAP

Perform the following steps:

1. Request the service from SAP and supply the following information:
   - VPC: your VPC CRN
   - Classic: Account ID
   - PowerVS: Power Systems Virtual Server CRN
2. SAP adds a new connection to their existing gateway to your designated VPC, Classic, or PowerVS environment.
3. Review and approve the connection request.

Figure 2-15 illustrates traffic flow between a Client's VPC and SAP resources in RISE with SAP on Power Virtual Server.

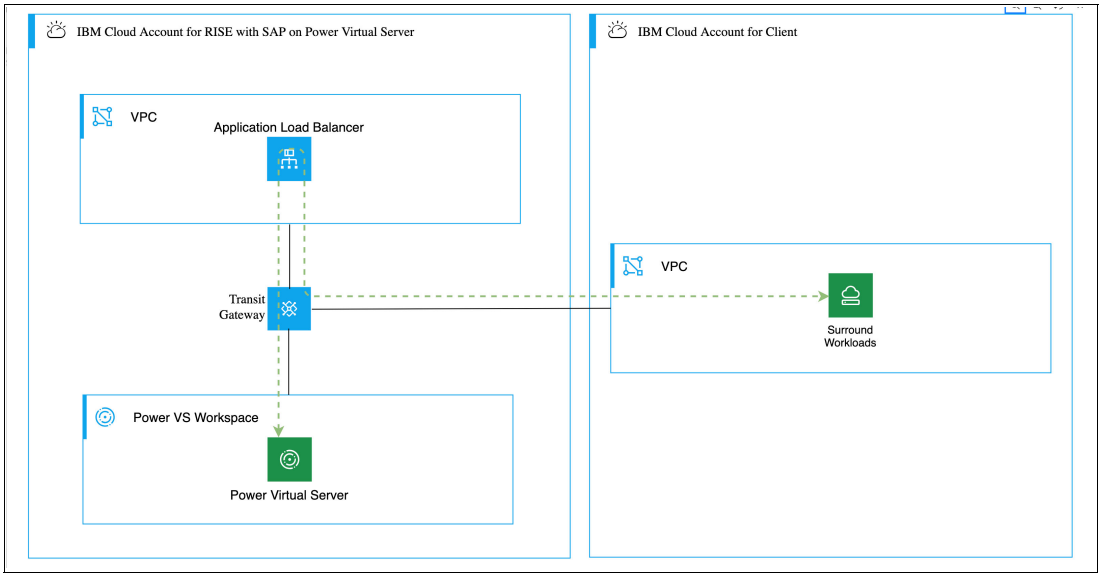


Figure 2-15 Traffic flow between client VPC and SAP resources

IBM Cloud Transit Gateway in the client account

To establish a connection with your RISE with SAP on IBM Power Virtual Server workload, use a new or existing IBM Cloud Transit Gateway in your account with a cross-account connection. When SAP approves the connection, your Transit Gateway connects to the VPC networks hosting your RISE with SAP on IBM Power Virtual Server workloads in the SAP IBM Cloud Enterprise Account.

Select an IBM Cloud Transit Gateway with local or global routing based on the location of your VPC and the RISE with SAP on IBM Power Virtual Server VPC. If the VPCs are in the same region, select local routing. If they are in different regions, select global routing. Figure 2-16 on page 49 shows the use of the Transit Gateway located in the client's account.

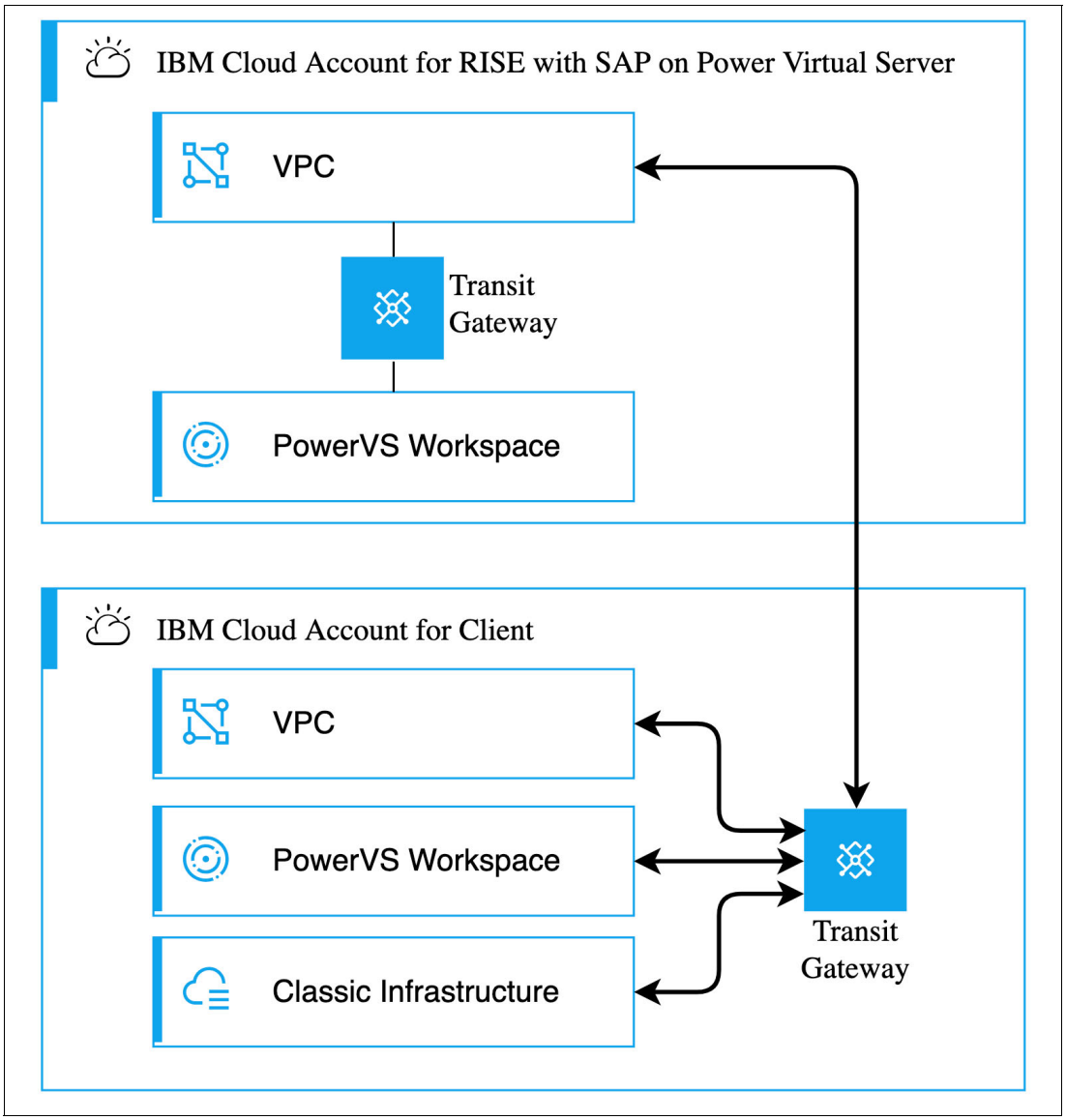


Figure 2-16 Client owned Transit Gateway

Perform the following steps:

1. Request SAP for the VPC CRN.
2. On a new or existing TGW, add a connection to a VPC in a different account and enter the VPC CRN supplied to you from SAP.
3. SAP reviews and approves the request.

Figure 2-17 illustrates traffic flow between a Client's VPC and SAP resources in RISE with SAP on Power Virtual Server.

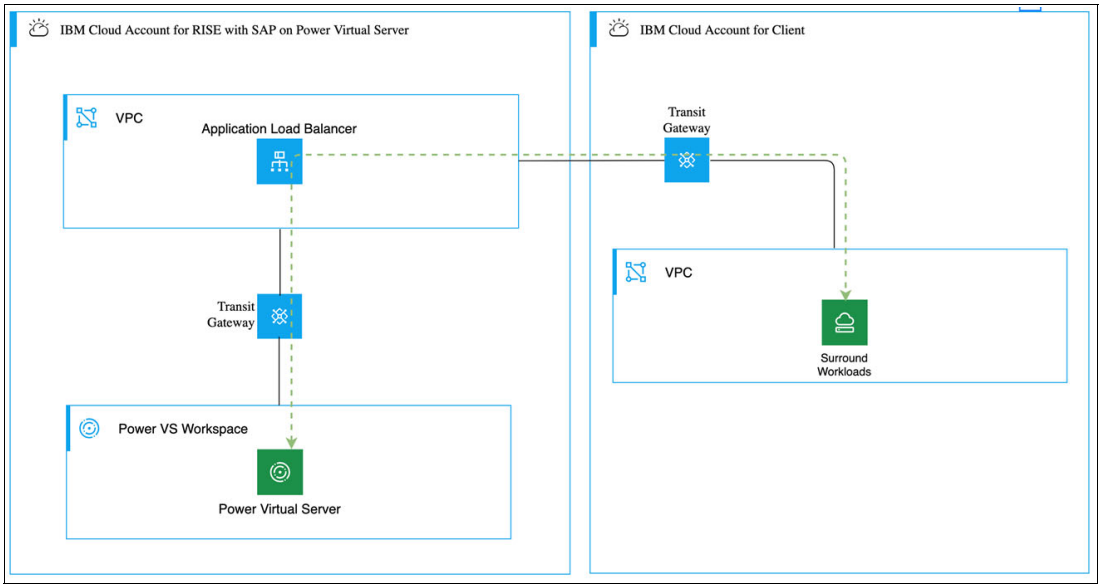


Figure 2-17 Traffic flow with client-owned Transit Gateway

2.4.5 IBM Cloud Site-to-Site VPN pattern

This pattern uses IBM Cloud VPN for VPC to securely connect your VPC to RISE with SAP on IBM Power Virtual Server. It uses either a static, route-based VPN or a policy-based VPN to establish an IPsec site-to-site tunnel between the VPCs. Figure 2-18 on page 51 shows the connections.

Perform the following steps:

1. Request the service from SAP.
2. SAP responds with their VPN endpoint details.
3. Create a site-to-site VPN and configure with the SAP provided details.
4. Provide SAP with your endpoint details.

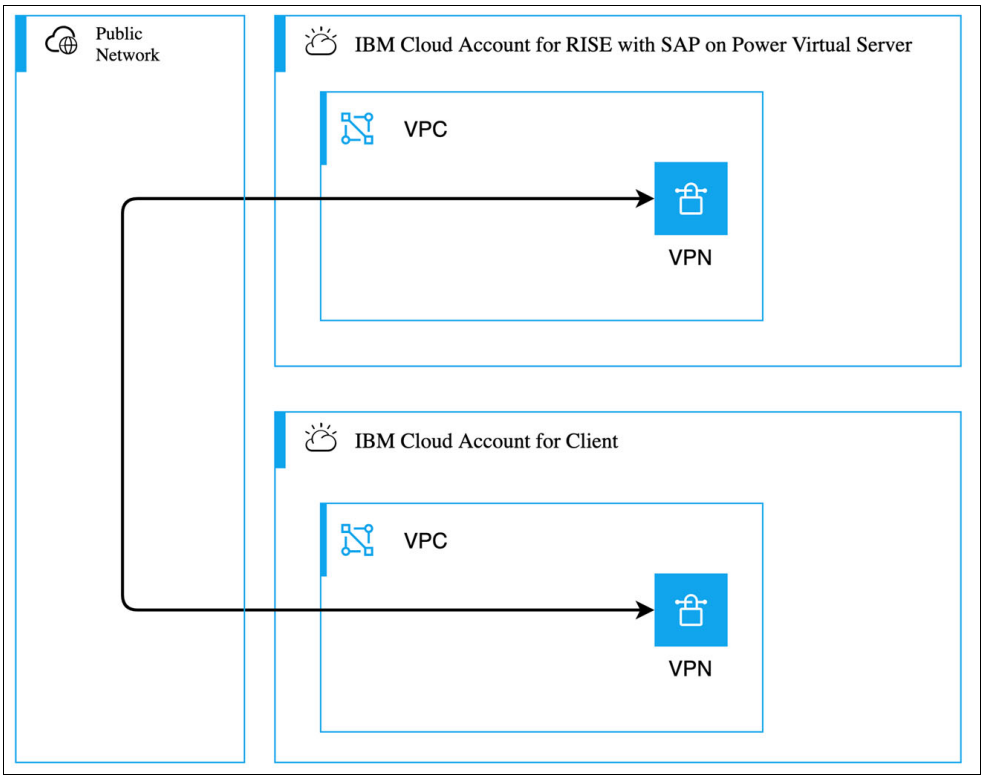


Figure 2-18 Connectivity using VPN

Hybrid cloud integration

This hybrid use case connects on-premises or external locations to your IBM Cloud Account and then connects your IBM Cloud Account to RISE with SAP on IBM Power Virtual Server, as shown in Figure 2-19.

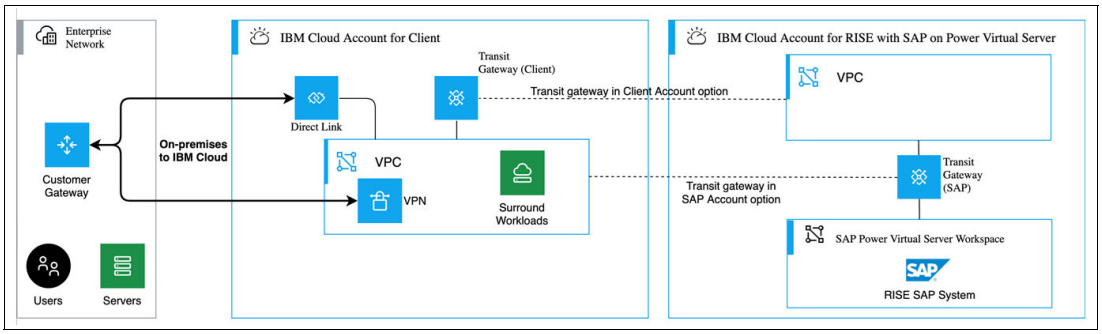


Figure 2-19 Hybrid cloud integration

This use case is useful if a client already has an IBM Cloud presence linked to their on-premises location and then adds the RISE with SAP on IBM Power Virtual Server service. This use case also applies to new client IBM Accounts as part of a move to IBM Cloud and RISE with SAP on IBM Power Virtual Server.

The connectivity between the Enterprise Network and the IBM Cloud Account for Client can be Direct Link, VPN, or both, depending on client requirements:

- VPN

  Enables access from your Enterprise Network to your IBM Cloud account with an IBM Cloud VPN for VPC site-to-site gateway. Traffic between IBM Cloud and your Enterprise Network is encrypted through an internet protocol security (IPsec) and transferred through a secure tunnel over the internet. This option is efficient and faster to implement compared to IBM Direct Link. IBM Cloud VPN for VPC provides a simple yet powerful solution for highly scalable and robust site-to-site VPN gateways. With this service, you can create site-to-site VPN tunnels for secure, encrypted connectivity. Connect from on-premises sites to IBM Cloud through a VPN gateway on an IBM Cloud VPC and a peer gateway on-premises. This service provides a mixture of industry-standard security and encryption options and support for Pre-Shared Key (PSK) authentication. This service also provides the ability to quickly add and remove VPN connections with the option to use pre-defined configurations.

- Direct Link

Use IBM Direct Link if you require higher throughput or a more consistent network experience than an internet-based connection. IBM Direct Link connects your on-premises resources to your cloud resources and offers more consistent, higher-throughput connectivity, keeping traffic within the IBM Cloud network. The following IBM Direct Link services are available:

  - Direct Link Dedicated: Terminates a single-tenant, fiber-based, cross-connect into your own IBM Cloud Private network connection. The supported speeds for the Direct Link Dedicated offering are 1 Gbps, 2 Gbps, 5 Gbps, and 10 Gbps.
  - Direct Link Connect: Uses a service provider to quickly establish and deliver connectivity to IBM Cloud. These service providers are already connected to the IBM Cloud network and use multi-tenant, high-capacity links known as network-to-network interfaces (NNI). Available speeds are based on your provider and provider's location. Partners include AT&T, Cologix, Console Connect, DE-CIX, Digital Realty, Equinix, Megaport, and Verizon SCI.

The peering between the IBM Cloud Account for Client and RISE with SAP on IBM Power Virtual Server can use either a client-provided Transit Gateway or the SAP Transit Gateway. You can also connect the IBM Cloud Account for Client to RISE with SAP on IBM Power Virtual Server using a VPN (not shown in the diagram). However, peering using a transit gateway is the preferred option.

Figure 2-19 on page 51 does not show the use of a Virtual Network Functions (VNF) appliance. VNF appliances are packaged third-party firewall offerings available on the IBM Catalog.

Enterprise hybrid pattern

Large organizations that use RISE with SAP on IBM Power Virtual Server often have complex networking and integration requirements. These enterprises typically run additional workloads such as analytics, data services, and business applications within their own IBM Cloud account.

The following diagrams show how to connect these environments in a hybrid cloud setup. In this scenario, a cross-account transit gateway links the client's IBM Cloud account to the SAP environment to enable secure communication between the two environments.

One common approach is the transit VPC pattern, which uses a central hub to manage traffic between different parts of the environment. This pattern includes the following components:

- The transit VPC acts as the hub
- Other environments such as the SAP VPC, Power Workspace, and additional VPCs connect as spokes
- All traffic between spokes flows through the hub for easier management and monitoring

An alternative is the edge VPC pattern, which focuses only on traffic entering or leaving the IBM Cloud account. Although both patterns might look similar in diagrams, they differ in how traffic is routed.

The diagrams show a Direct Link Dedicated connection to the enterprise network. You can replace this with Direct Link Connect or VPN, depending on your organization's requirements for bandwidth, security, and cost.

In some cases, SAP manages the transit gateway. After you initiate the peering process, SAP sends a connection request to your VPC. This approach simplifies setup and ensures secure integration between your environment and SAP systems. Figure 2-20 shows SAP ownership of the Transit Gateway.

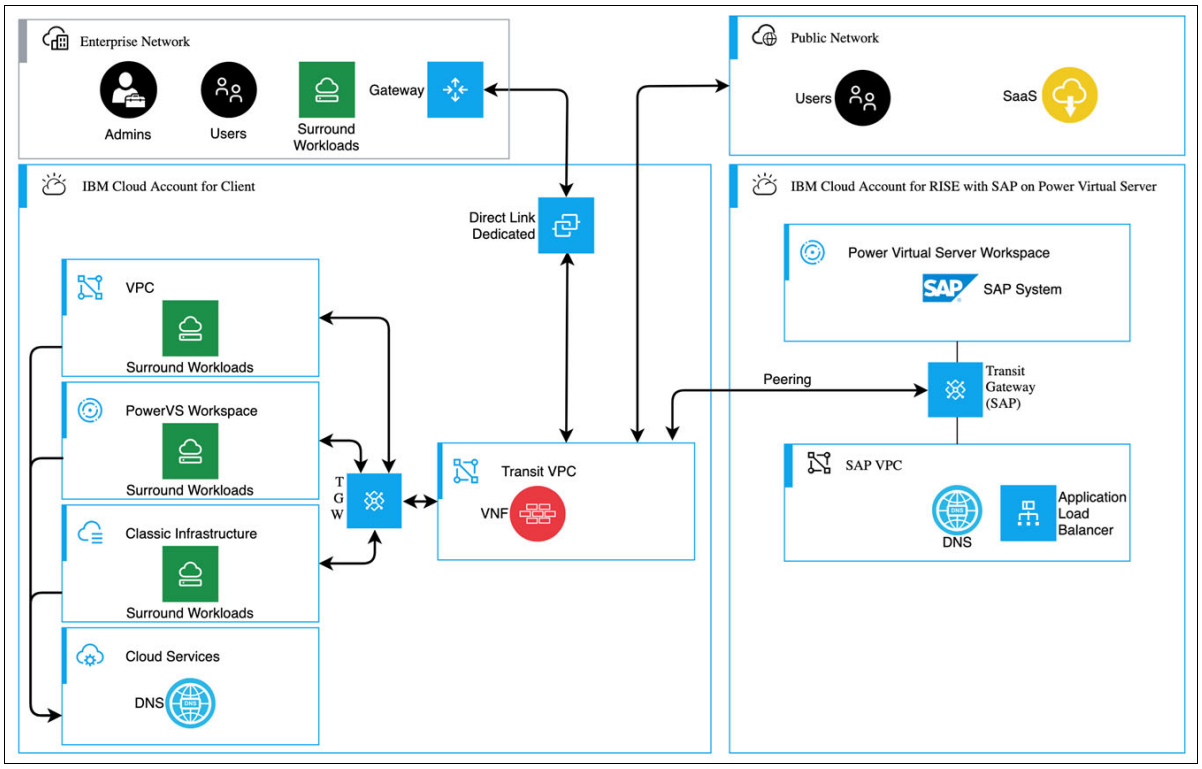


Figure 2-20 Enterprise cloud connection with SAP owned Transit Gateway

Figure 2-21 shows a scenario where the client owns the transit gateway.

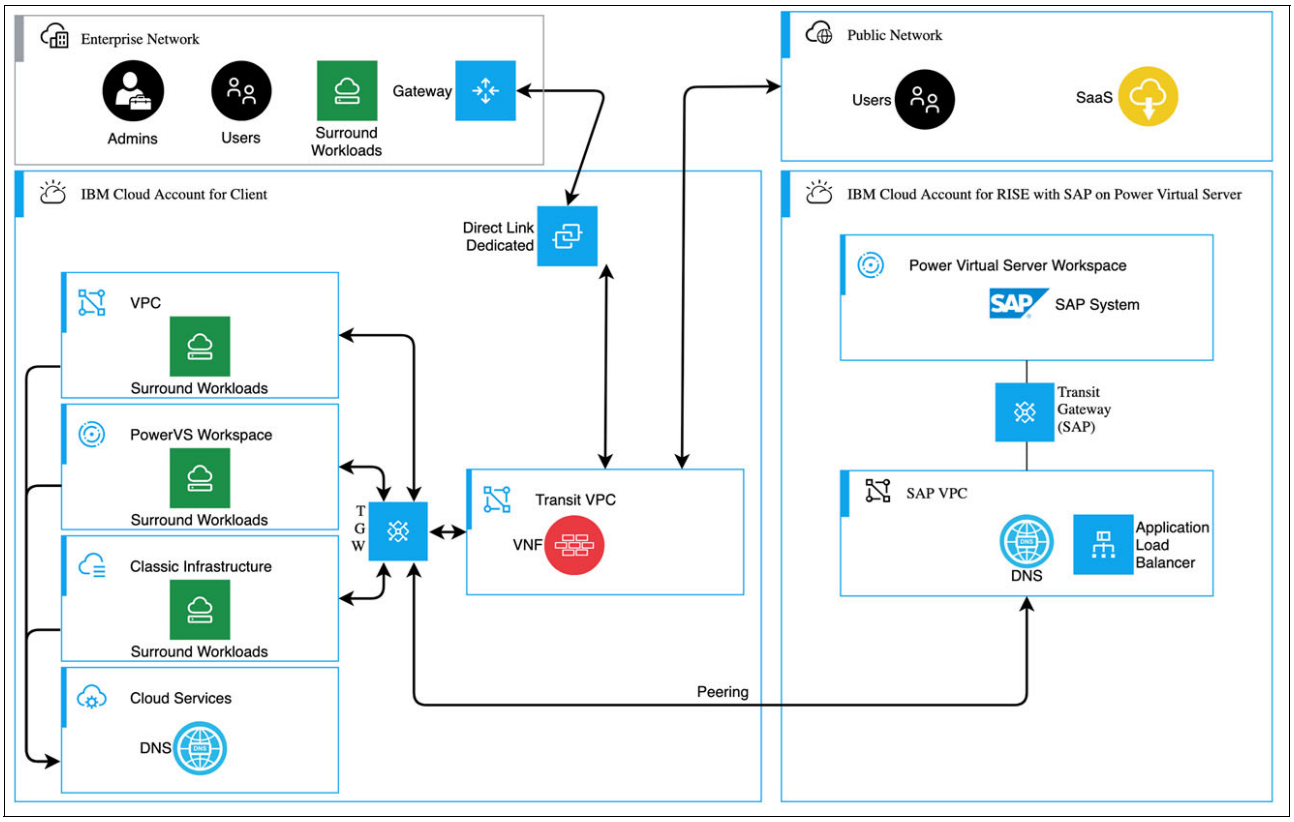


Figure 2-21 Enterprise cloud connection with client-owned Transit Gateway

Figure 2-22 shows a scenario where VPN is used to connect the two accounts.

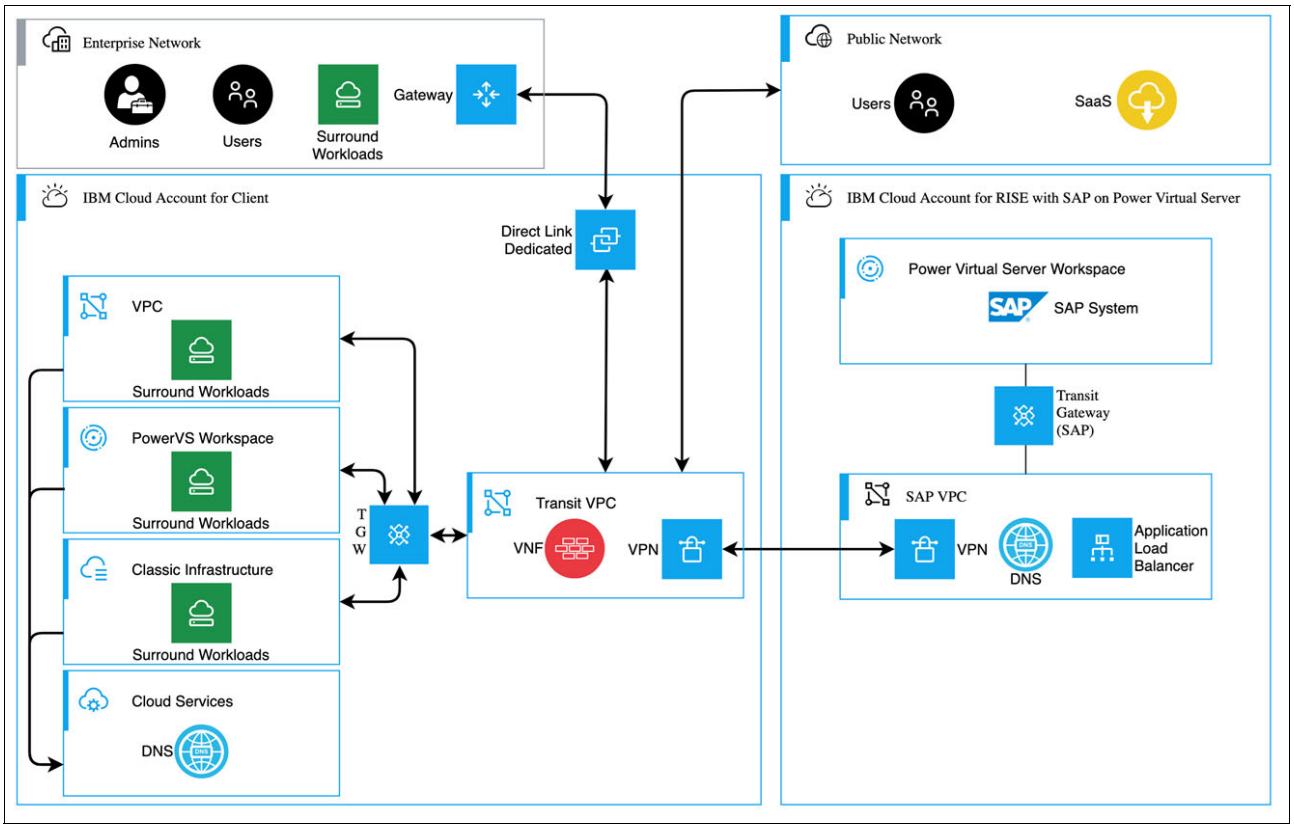


Figure 2-22 Enterprise cloud connection with VPN connection

Multi Cloud Integration

Multi-cloud is a cloud computing strategy in which enterprises use services from multiple public cloud providers simultaneously. Multi-cloud strategies integrate cloud services into a single environment through a combination of APIs, networking, and management tools. This integration allows enterprises to manage their cloud services centrally and to move workloads seamlessly between different cloud providers as needed.

Efficient multi-cloud implementation at scale requires distributed cloud exchange points that enable private, secure, low-latency communication among multiple clouds and SaaS services. IBM Cloud Direct Link Connect provides private access to the IBM Cloud infrastructure and any other clouds that are linked to your service provider. Many of these service providers offer cloud exchange services for creating multi-cloud connectivity and are already connected to IBM Cloud and other clouds through multi-tenant, high-capacity links known as network-to-network interfaces (NNI).

Figure 2-23 describes how IBM Cloud Direct Link Connect and your cloud exchange provider enable an enterprise's multi-cloud pattern.

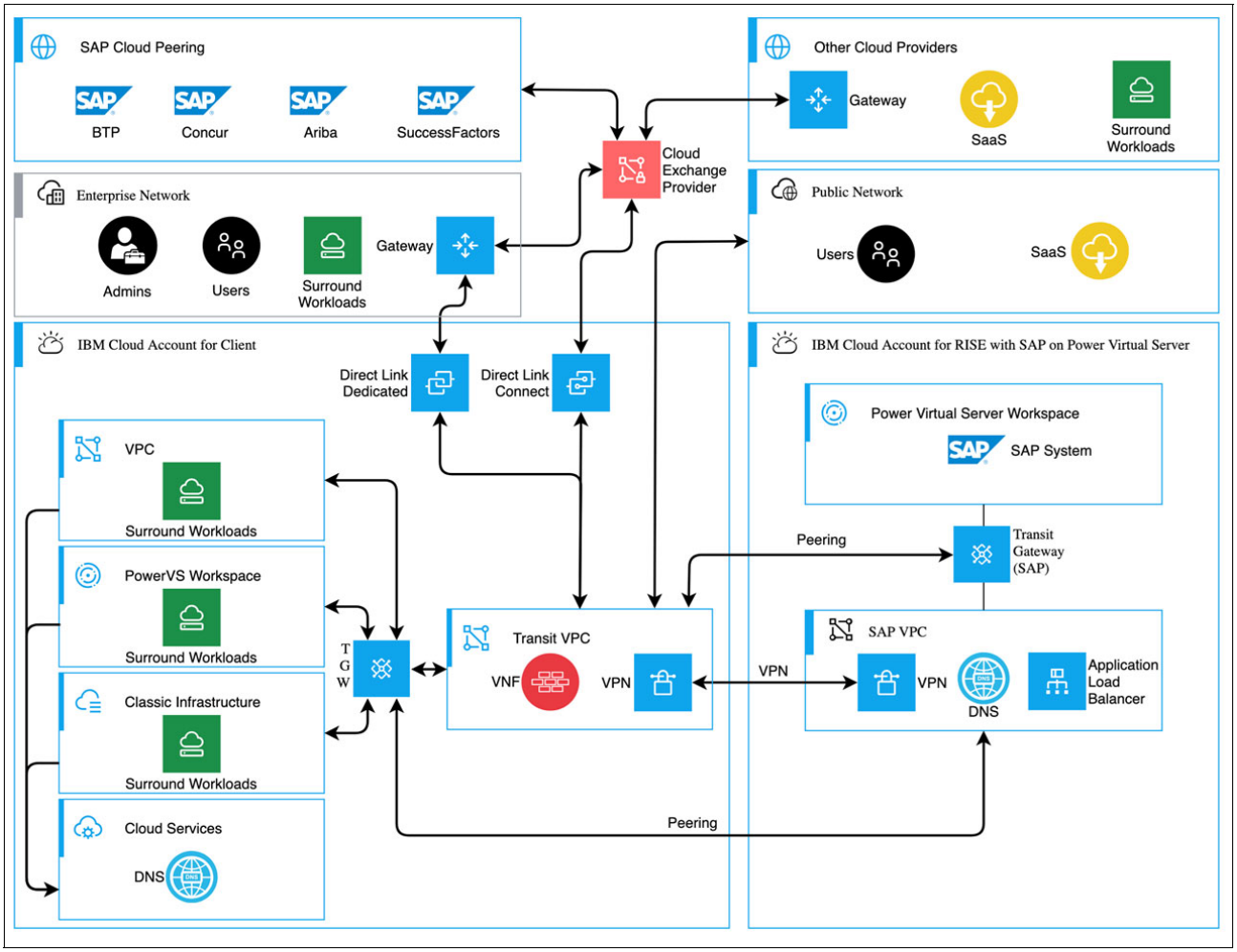


Figure 2-23 Multicloud connectivity

Figure 2-23 shows the following components:

- Your enterprise networks
- Your IT administrators
- Your internal users of the RISE with SAP, non-RISE SAP, and non-RISE non-SAP systems hosted in the multi-cloud environment
- Your surround workloads that remain in your locations
- A network gateway that connects your enterprise networks to external networks
- Cloud exchange provider: A client-selected partner who offers the required connections to your multi-cloud resources, including IBM.
- SAP Cloud Peering: A service that allows SAP customers to establish a direct, secure connection to SAP cloud services, such as SAP BTP, Concur, Ariba, and SuccessFactors. These SAP services, which are connected through a network of interconnection partners, bypass the public internet and provide high-speed, reliable connectivity for SAP applications. Many of these interconnection partners are also known as cloud exchange providers.

- Other cloud providers: Your cloud accounts in other cloud providers such as AWS, Azure, and GCP. These cloud providers host your additional surround workloads and SaaS products.
- Public networks: Internet-connected resources such as external users of the RISE with SAP, non-RISE SAP, and non-RISE non-SAP systems. SaaS products that are not accessible through the Cloud Exchange Provider must be accessed through the internet.
- IBM Cloud Account for client is an account that contains all your non-RISE SAP and non-RISE non-SAP surround workloads that you host in IBM Cloud. This account includes the following components:
  - IBM Cloud has three IaaS environments: VPC, Power Virtual Server, and Classic. Any of these IaaS environments can host your surround workloads.
  - You can also consume IBM Cloud Services such as DNS.
  - The pattern uses a transit VPC that enables a hub-and-spoke design where the hub is the transit VPC and the SAP VPC, your other VPCs, Power Workspace, and Classic infrastructure are the spokes. All spoke-to-spoke traffic passes through the hub. An alternative design is an edge VPC that receives only traffic leaving or entering your IBM Cloud Account.

    Diagrammatically, the two patterns look the same; the difference is in the routing configured in the VPC routing tables. The transit VPC hosts your selected Virtual Network Function (VNF) appliances. VNF appliances, such as routers and firewalls running on virtual server instances, can monitor, log, route, and filter the traffic through the transit VPC.
  - Integration of the different IBM Cloud IaaS environments, such as VPC, Classic, and PowerVS, is achieved by using the IBM Cloud Transit Gateway, which functions as a Border Gateway Protocol (BGP) router service. The IBM Cloud Transit Gateway can be local (limited to a specific region) or global (spanning multiple regions).
  - Network connectivity: The diagram shows Direct Link Dedicated connecting your enterprise networks to your transit VPC and Direct Link Connect connecting to your selected Cloud Exchange Provider. Alternatively, you can use Direct Link Connect for your enterprise network traffic.
  - Inbound internet access: Inbound internet access uses public application load balancers and optionally IBM Cloud Internet Services to provide the required security features. VNFs in high availability cannot be used for inbound internet traffic.
  - Outbound internet access: Outbound internet access uses a public gateway on the subnets in the transit VPC or Floating IPs attached to the VNFs, as described by the VNF type and deployment pattern.
- IBM Cloud account for RISE on Power Virtual Server: SAP's IBM Cloud account that hosts your RISE with SAP resources. The diagram shows three options to connect the IBM Cloud Account for Client and the IBM Cloud Account for RISE with SAP on Power Virtual Server: client-provided transit gateway, SAP-provided transit gateway, and VPN. Only one of these connections is required. Discuss your requirements with your SAP representative to determine which option satisfies your needs.

2.4.6 Virtual Network Functions

Virtual Network Functions (VNFs) are virtual appliances packaged as VPC virtual server instances. The third-party appliances interact with the IBM Software-Defined Networking (SDN) controller to offer an easier configuration of IBM VPC network features such as route tables and floating IPs. You can instantiate VNFs easily through IBM Cloud Schematics through the IBM Content catalog. You can achieve highly available, highly resilient VNF

deployments by using the routing mode feature of the IBM Cloud Network Load Balancer (NLB) for VPC. The following IBM Cloud VNF catalog offerings are available:

- Fortinet FortiGate Next-Generation Firewall
- Palo Alto VM-Series
- Check Point Security Gateway - CloudGuard Network Security Firewall with Threat Prevention
- Juniper vSRX
- F5 Big-IP

High availability VNF deployments have the following limitations:

- The VNF appliances must share one subnet with the Network Load Balancer (NLB)
- Routing public internet ingress traffic to a VNF is not supported
- Auto-scaling with the VNF is not supported

2.5 Integration with public clouds for IBM Power Systems

Modern enterprises increasingly adopt hybrid multicloud architectures to use the unique strengths of multiple cloud providers while maintaining critical workloads on-premises. For organizations running IBM Power Systems (AIX, IBM i, Linux on Power), integrating these systems into a multicloud strategy requires secure, high-performance, and resilient connectivity between the on-premises data center and public cloud environments.

Connecting on-premises IBM Power servers to public clouds such as IBM Cloud, AWS, Azure, GCP, and Oracle involves hybrid networking strategies that ensure low latency, high throughput, and strong security.

Consider the following factors when planning multicloud connectivity:

- Connectivity Options:
  - Direct Connectivity: Dedicated private links such as IBM Cloud Direct Link, AWS Direct Connect, and Azure ExpressRoute
  - VPN: IPsec tunnels for quick setup
  - SD-WAN: Dynamic routing and multi-cloud optimization
- Routing

  Use BGP for dynamic routing between on-premises and cloud
- Security

  Firewalls, ACLs, and encryption for data in transit
- Redundancy:

  Dual links and diverse paths for high availability

Table 2-3 Provider-Specific Connectivity

Cloud Provider	Service	Options	Key Steps
IBM Cloud	Direct Link	Dedicated or through partner (Equinix, Megaport)	Provision Direct Link -> Configure BGP -> Extend VLANs or IP routing

Cloud Provider	Service	Options	Key Steps
AWS	Direct Connect	Dedicated (1/10/100 Gbps) or Hosted	Create Direct Connect -> Set up VIF -> Configure BGP
Azure	ExpressRoute	Direct or through partner	Create ExpressRoute circuit -> Configure Private Peering -> Establish BGP
GCP	Interconnect	Dedicated or Partner	Reserve Interconnect -> Configure VLAN attachments
OCI	FastConnect	Colocation or Partner	Create FastConnect circuit -> Configure BGP -> Attach to VCN

Use the following best practices for IBM Power in Hybrid Multicloud:

- Use IBM PowerVS in IBM Cloud for native Power workloads, enabling elasticity without hardware refresh.
- Connect through IBM Cloud Direct Link for low-latency, private connectivity between on-premises and IBM Cloud.
- Use colocation exchange hubs (for example, Equinix Fabric, Megaport) to simplify multi-cloud interconnectivity.
- Implement cloud-native transit services (AWS TGW, Azure vWAN, GCP Cloud Router, OCI DRG) for scalable routing.
- Ensure redundancy and security with dual circuits, BGP-based failover, and optional IPsec overlays.
- Standardize IP addressing and VRFs to avoid overlaps and maintain isolation (Prod, Non-Prod, DR).

2.6 Network design and address planning

You can define your own IP address ranges when creating subnets. The CIDR information of these subnets is advertised through PER to the Transit Gateway, enabling external network routing to and from the PowerVS environment. The simplest and safest approach is to define a CIDR range that does not overlap with the on-premises network. However, if you want to migrate workloads to PowerVS without changing the on-premises IP addressing scheme, you can create a subnet with the same CIDR. In such cases, take care to avoid routing conflicts. If routing information to the same IP address range is advertised from both on-premises and PowerVS into your network, problems can occur. Therefore, until the switchover is complete, you must apply appropriate filtering to ensure that route advertisements from PowerVS are suppressed.

2.6.1 Network security

PowerVS provides Network Security Groups. This feature provides L4 Firewall functionality and is available in workspaces where PER is enabled. By default, Network Security Groups are disabled. You can enable this feature to make Network Security Groups available within the workspace, as shown in Figure 2-24.

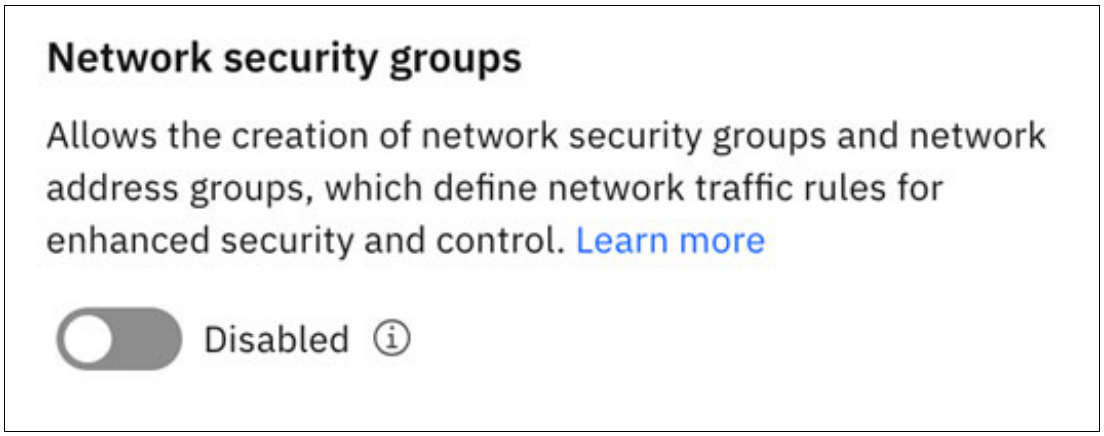


Figure 2-24 Network security group

Network Security Groups operate in a stateless manner, which means rules are defined for both requests and responses. As of October 2025, Network Security Groups control only inbound traffic (outbound traffic is not filtered).

If you require more advanced security capabilities, configure access to PowerVS through security appliances available in x86-based IaaS environments (VPC or Classic Infrastructure). In such cases, the routing configuration becomes more complex, for example, by establishing GRE tunnels.

Each PowerVS subnet has a property so that you can specify whether route advertisements are enabled. For subnets that you intend to use only within PowerVS, you can disable route advertisement to prevent the subnet from being advertised externally.

2.6.2 Disaster recovery deployment models in PowerVS

For mission-critical systems, a common pattern designates one region as the production site and another region as the disaster recovery (DR) site.

Figure 2-25 on page 61 illustrates a hybrid cloud architecture with production and DR sites in two regions. An IBM Cloud Transit Gateway connects both regions, ensuring secure and efficient communication. Each region uses Direct Links to connect to Points of Presence (PoPs) that link to a wide area network (WAN). A client site connects to the WAN for access to both production and DR environments. This design supports high availability and business continuity.

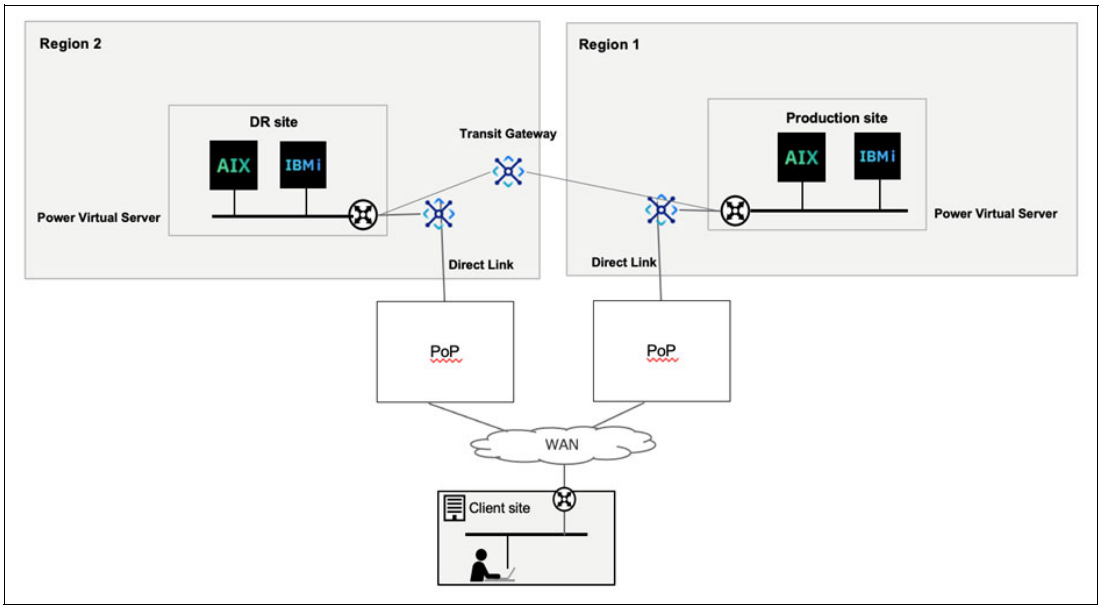


Figure 2-25 Disaster recovery

The approach differs depending on whether the IP addresses change between the production site and the DR site.

Approach 1: Different subnets for production and DR

The production site and DR site use different subnets. After a disaster, clients directly access the DR site using its IP address. For services accessed through FQDN, DNS record updates achieve failover. This approach simplifies implementation and minimizes the risk of routing conflicts.

Approach 2: Same subnet for production and DR

In some cases, the production site and DR site use the same subnet. This requires a more sophisticated configuration with two subnets in the DR site:

- One subnet uses the same CIDR as the production site and is active only during a disaster.
- The other subnet uses a different CIDR and is active during normal operations.
- DR site under normal conditions:
  - The first subnet (same CIDR) does not advertise routes.
  - The second subnet (different CIDR) advertises routes.
- LPARs in the DR site connect to both subnets, giving them two IP addresses.

  Clients normally access the DR site through the IP address of the second subnet.
- In the event of a disaster:
  - Disable route advertisements for the production site's subnet.
  - Enable route advertisements for the DR site's first subnet (same CIDR).

With this approach, clients continue using the same IP address during failover to the DR site.

2.6.3 Operations and monitoring

Use IBM Cloud Monitoring to monitor data transfer activity while using PowerVS.

From the Dashboard Library, selecting Power Virtual Server as shown in Figure 2-26.

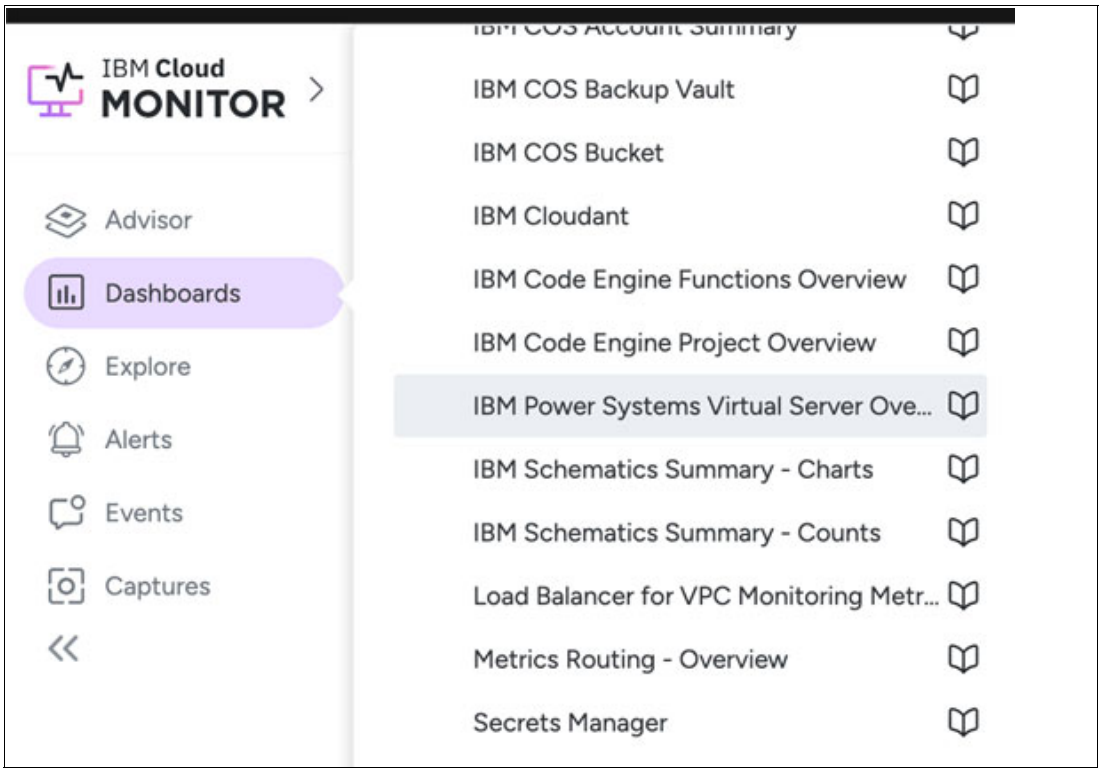


Figure 2-26 Navigate to the dashboard

Figure 2-27 shows graphs that display not only CPU and memory usage, but also inbound and outbound traffic per-server interface.

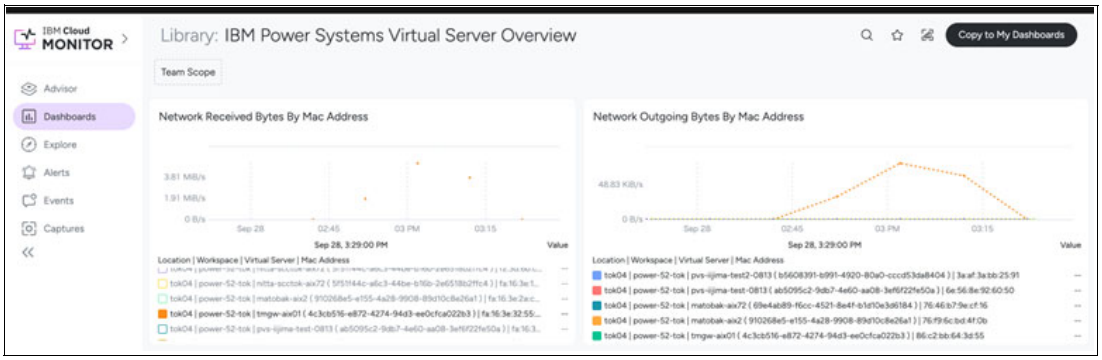


Figure 2-27 IBM Cloud Monitor dashboard

Also, you can view the ingress transfer rate of the Transit Gateway to check the overall network traffic for the PowerVS workspace as shown in Figure 2-28 on page 63.

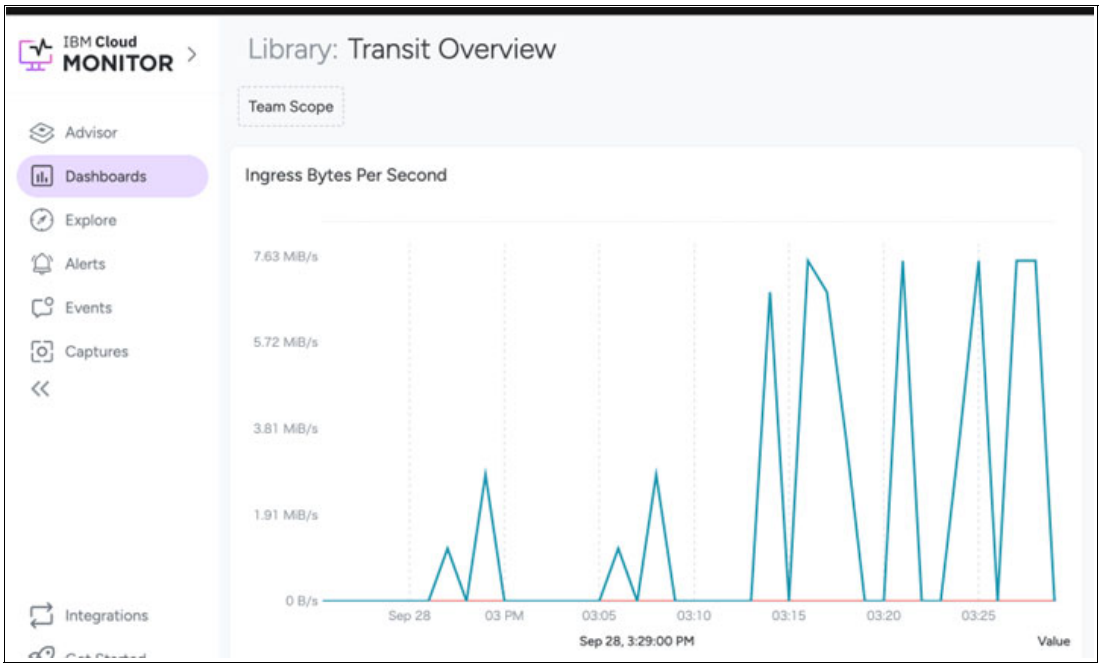


Figure 2-28 Transit overview

IBM Cloud automatically collects and visualizes these metrics when an IBM Cloud Monitoring instance with the Platform Metrics attribute enabled exists in the same region as the PowerVS workspace and Transit Gateway. This makes the service simple to use.

For more detailed information, you can rely on your own monitoring tools or operating system commands. IBM Cloud provides a built-in monitoring service for quick visibility.

For more information, see Monitoring platform metrics for IBM Power Virtual Server.

3

Hybrid cloud provisioning and automation

Hybrid cloud environments require flexibility for consuming, deploying, and scaling resources. This chapter explains capabilities that enable IBM Power Systems to operate seamlessly across on-premises and cloud infrastructures. The chapter begins with an overview of hybrid consumption and billing models, including Power Enterprise Pool 2.0 and the IBM Hybrid Cloud Package, which help organizations optimize costs and improve operational agility.

The chapter then explores deploying Red Hat OpenShift on IBM Power, covering supported deployment methods, multi-architecture compute, and AI acceleration features that enhance containerized workloads. It examines orchestration and optimization tools, with emphasis on integration models such as closed-loop automation for managing hybrid operations. Finally, the chapter outlines incremental adoption strategies that enable organizations to experiment with IBM Power Virtual Server (PowerVS) before committing to full-scale migration.

This chapter covers the following topics:

- Hybrid consumption and billing models
- Tooling for orchestration and optimization in hybrid cloud
- IBM Cloud Pak for Automation in hybrid cloud for IBM Power11
- Terraform deployment for IBM Cloud Pak for Data
- IBM Cloud catalog deployment

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3.1 Hybrid consumption and billing models

Businesses are moving toward flexible, pay-per-use consumption models to optimize costs and improve agility. IBM Power offers several options to support this trend:

- Shared Utility Capacity

  Shared Utility Capacity optimizes compute resource usage and reduces costs for handling peak or volatile demand across multiple Power servers while maintaining service-level agreements.

- Accelerating your journey to hybrid cloud

  IBM Power supports hybrid cloud adoption with common technology and consistent operations across on-premises and cloud environments. Pay-per-use capacity is available both on-premises and in IBM Cloud.

  This approach provides the following benefits:

  - Modernize applications and data across distributed environments
  - Optimize infrastructure for resiliency from edge to cloud to core
  - Secure regulated data and mitigate risks proactively

Figure 3-1 shows three layers of capacity. The bottom layer represents Permanent Capacity: resources that are purchased or leased and shared in a pool. The middle layer represents Metered Capacity: pay-per-use resources billed by the minute. The top layer represents Available Capacity: active standby resources ready when needed. These layers work together to provide a mix of fixed, flexible, and on-demand capacity.

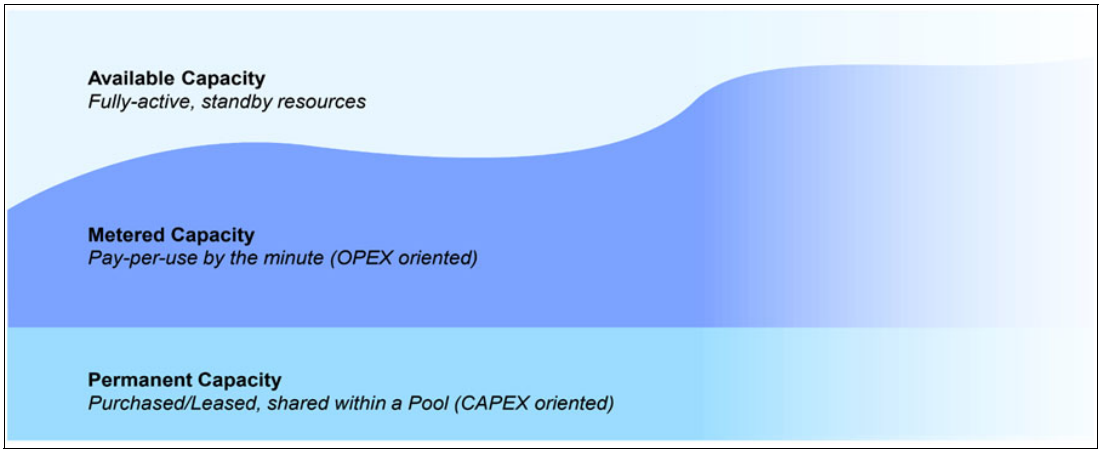


Figure 3-1 Three types of capacity in a resource management mode

3.1.1 Power Enterprise Pool 2.0 hybrid consumption

IBM introduced a pilot program for a Hybrid Consumption Model that is designed for clients who deploy a hybrid Power infrastructure across Power Enterprise Pools 2.0 (PEP 2.0) on-premises and PowerVS in IBM Cloud. This program also applies to clients with existing Power Enterprise Pools 2.0 who want their Metered Capacity usage that is billed together with PowerVS consumption on a single, consolidated invoice, with charges credited toward any PowerVS Enterprise Savings Plan commitment.

As part of this program, participating clients create a PowerVS Enterprise Savings Plan and link the Pool ID of their Power Enterprise Pool 2.0 to their IBM Cloud account. All charges for Metered Capacity consumption are invoiced in local country currency.

The pilot is open to clients in the following countries: US, Canada, United Kingdom, Ireland, Germany, Austria, Switzerland, Belgium, Netherlands, Luxembourg, Norway, Finland, Sweden, Denmark, France, Spain, and Italy.

Figure 3-2 shows pay-per-use capacity options for IBM Power environments.

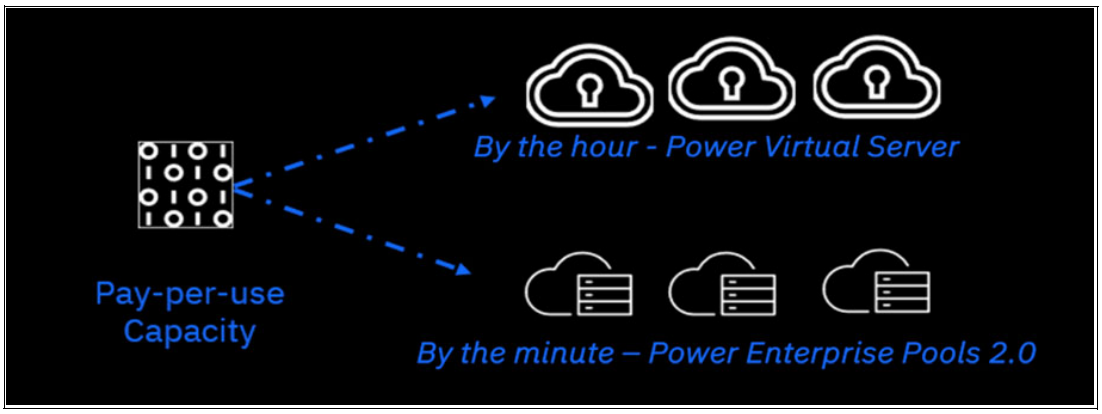


Figure 3-2 Pay per use offer in hybrid billing

The following capabilities highlight the flexibility and financial transparency of IBM's pay-per-use Power consumption model:

- Flexible rate and pace of deployment of Power pay-per-use capacity as required
- Single, consolidated invoice for pay-per-use compute consumption both on-premises (Pools 2.0) and in IBM Cloud
- Complete view of pay-per-use consumption across Pools and PowerVS and other Cloud usage Pools 2.0 Metered Capacity consumption invoiced alongside PowerVS and other Cloud elements and credited toward any PowerVS ESP Commitment

Creating a Power Enterprise Pool 2.0 with hybrid invoicing enables organizations to unify the billing and management of their metered capacity consumption across on-premises systems and IBM Cloud services. The process includes the following steps:

1. IBM Sales or a Business Partner submits an order for one or more systems configured for Power Enterprise Pools 2.0.
2. After the system is installed and visible in the Entitled Systems Support (ESS) inventory, you create your Pool ID through ESS.
3. During Pool ID creation, you select how your Metered Capacity consumption is charged: pre-paid Capacity Credits, Monthly Billing, or integrated as part of a PowerVS Enterprise Savings Plan.
4. If you select Monthly Billing or Enterprise Savings Plan, you select a valid system to associate with the Pool ID. The associated Business Partner handles the invoicing of any minutes of Metered Capacity consumption.
5. After you create the Pool ID, you start the Pool through the Cloud Management Console (CMC), add additional systems to it, and proceed with business-as-usual operations. The CMC begins monitoring and metering pool resources.
6. From that point forward, any minutes of Metered Capacity resource consumed above the Base of the Pool do not debit Capacity Credits but accrue and are invoiced each month through the selected Business Partner.

   You can view minutes of each Metered Capacity resource consumed in real time and historically in the CMC Enterprise Pools application.

3.1.2 Hybrid cloud package: Components and benefits

IBM offers a fixed-price hybrid cloud package with qualified Power11 server purchases.

Packaged services include everything needed to use hybrid cloud:

- PowerVS and IBM Cloud IaaS services
- PowerVS Build services by IBM Technology Expert Labs
- Cloud Advanced Support

Figure 3-3 highlights the comprehensive capabilities of IBM Power11 servers in hybrid cloud environments, emphasizing Power Virtual Server features such as VPN, Transit Gateway, and Secrets Manager IBM is offering a fixed priced, hybrid cloud package with qualified Power11 server purchases.

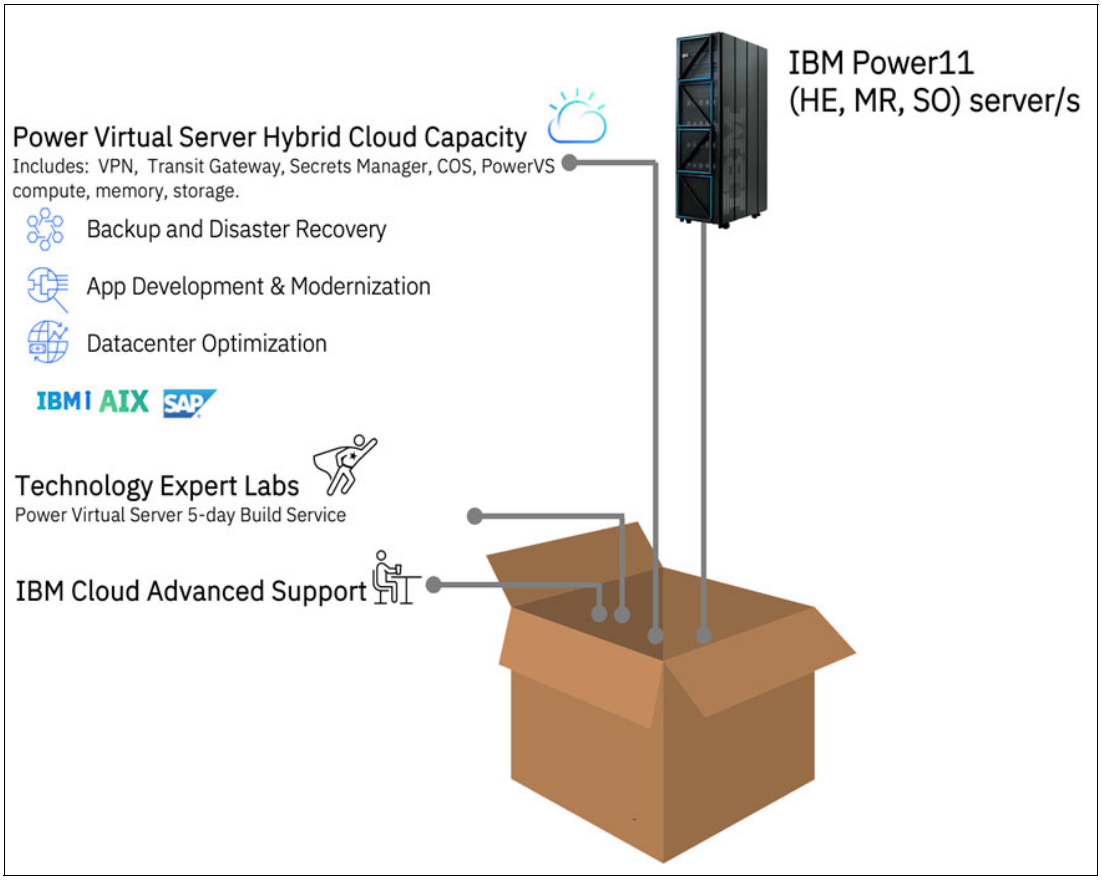


Figure 3-3 IBM Power11 servers and their associated services

The Hybrid Cloud Package comes in three sizes: small, medium, and large. You can add the package to a server configuration through the PVS1 feature. The package size is determined by the Power11 server model configured, such as S1122, S1124, E1150, or E1180 models.

The following rules and conditions apply to the IBM Hybrid Cloud Package (PVS1 feature) for Power11 server configurations:

- The package is intended for new clients to IBM Cloud by seeding PowerVS with Power11.
- The PVS1 package feature code can be used on a qualifying Power11 server only once per order and once per client. For example, if you purchase three servers, you can configure the PVS1 package with only one of them.

- The PVS1 Package is available only on new qualified Power11 orders.
- The PVS1 Package is not available on existing Power11 MES upgrade orders.
- You can apply the hybrid cloud capacity credits to any Power Virtual Server compute generation (IBM Power9®, Power10, or Power11).
- Provision your cloud account within 90 days of the qualifying IBM Power11 server order.

3.1.3 Ordering the IBM PowerVS Hybrid Cloud package

Perform the following steps to order the PowerVS Hybrid Cloud package by using e-config:

1. Open the e-config tool: IBM e-config on Cloud and select **Create Initial Order**.
   a. In the Brand field, select Power Systems & GTS/ISS.
   b. Use the Name field to name the configuration.
   c. Click **Create configuration** as shown in Figure 3-4.

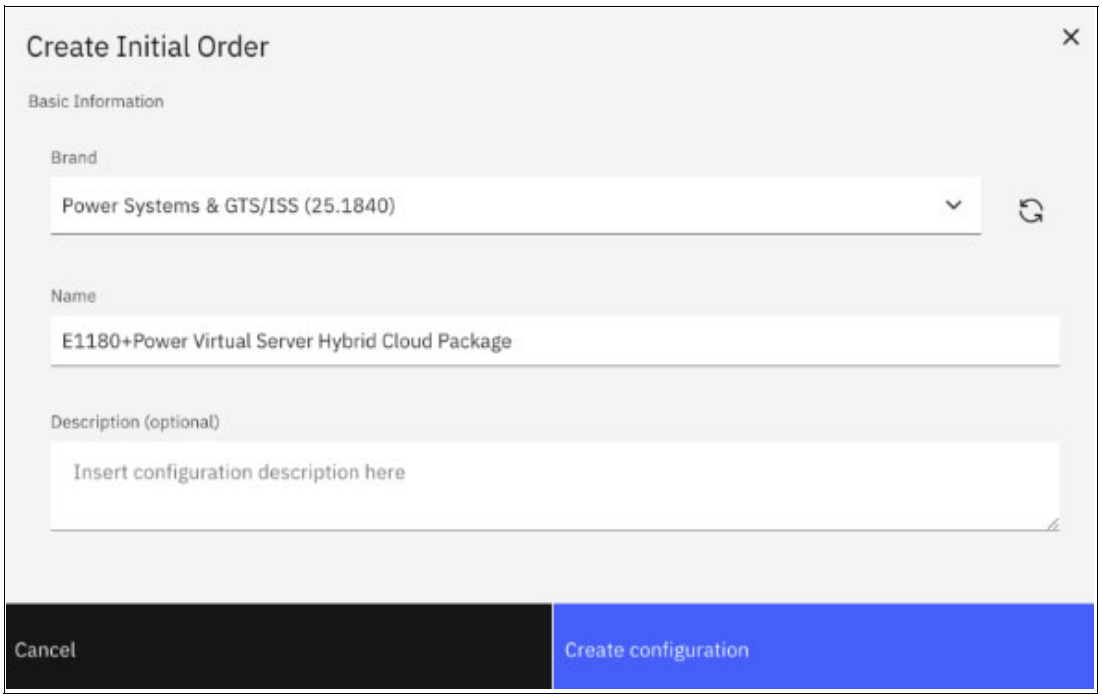


Figure 3-4 Create Initial Order

2. In Global Settings verify Geography, Country and CRAD (Figure 3-5 on page 70), and click **Finish**.

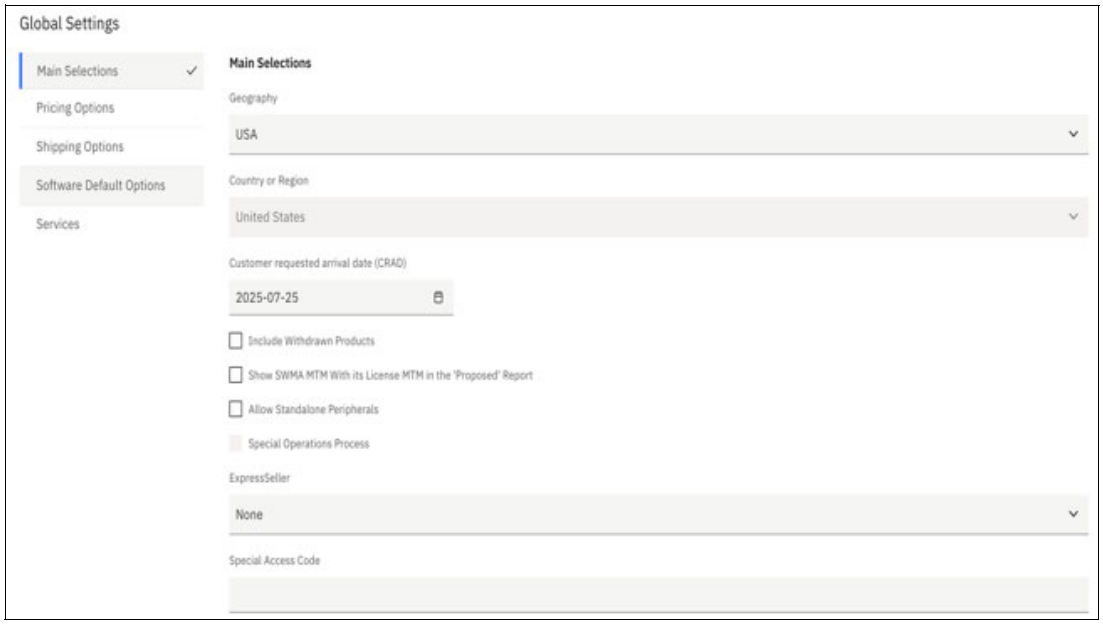


Figure 3-5 Global Settings

3. In the configuration list, click **Add Product** and then select **Hardware** → **Power11 Systems** as shown in Figure 3-6.

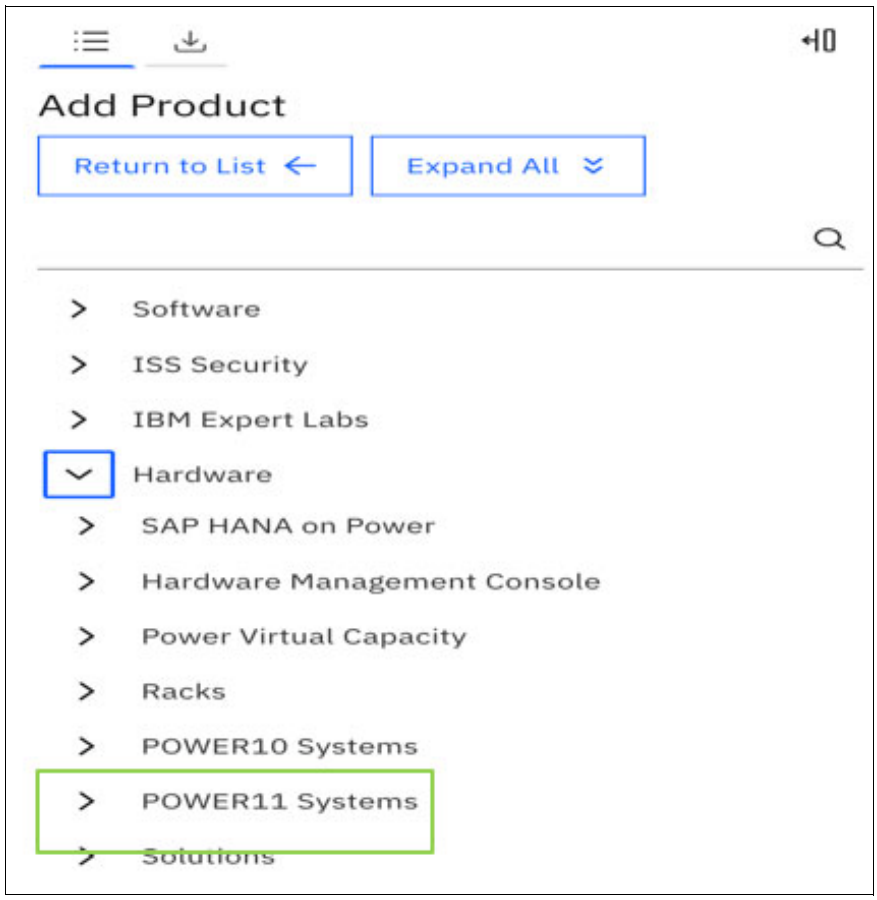


Figure 3-6 Add Product page

4. Select a qualified IBM Power11 system. For example, models such as S1122, S1124, E1150, or E1180.

   Ensure that the order is new and not an MES upgrade.

   In e-config, navigate to Power11 Systems and select a model, such as the E1180 shown in Figure 3-7.

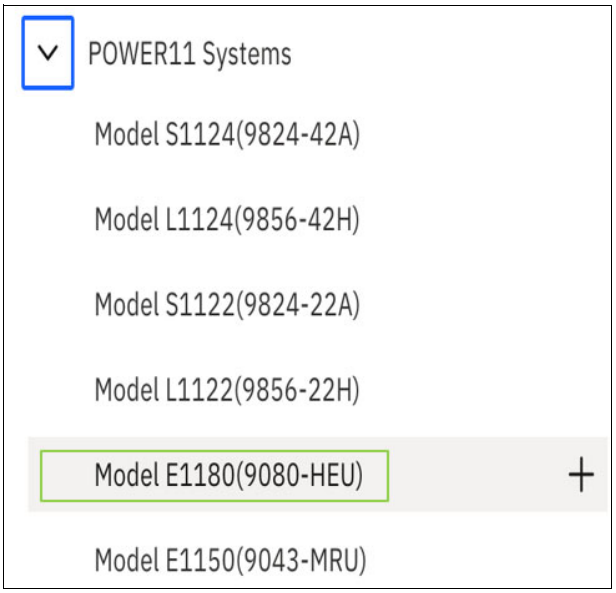


Figure 3-7 Power11 Systems

5. Add the PVS1 feature

   In e-config, locate the Hybrid Cloud Package feature code (PVS1). Add one PVS1 package per client. You can add one PVS1 package per order regardless of the number of servers purchased.

   Click the Edition section as shown in Figure 3-8 on page 72, In the drop-down there is an option to select (PVS1) - Power Virtual Server Hybrid Cloud Package. After PVS1 is selected, this results in the addition of 2 PIDs:

   - IBM Power Virtual Server Capacity – Small, Medium, or Large
   - IBM Cloud Advanced Support – Small, Medium, or Large

6. Review the package components and verify the included services:
   - PowerVS compute credits (usable across Power9, Power10, Power11 generations)
   - IBM Technology Expert Labs build services
   - Cloud Advanced Support
7. Confirm conditions:
   - Provision IBM Cloud account within 90 days of server order.
   - Package is one-time use per client.
   - Credits apply to PowerVS compute only.
8. Submit order:
   - Complete the configuration and submit through IBM Sales or Business Partner.
   - After installation, create a Pool ID in ESS and link to IBM Cloud account.

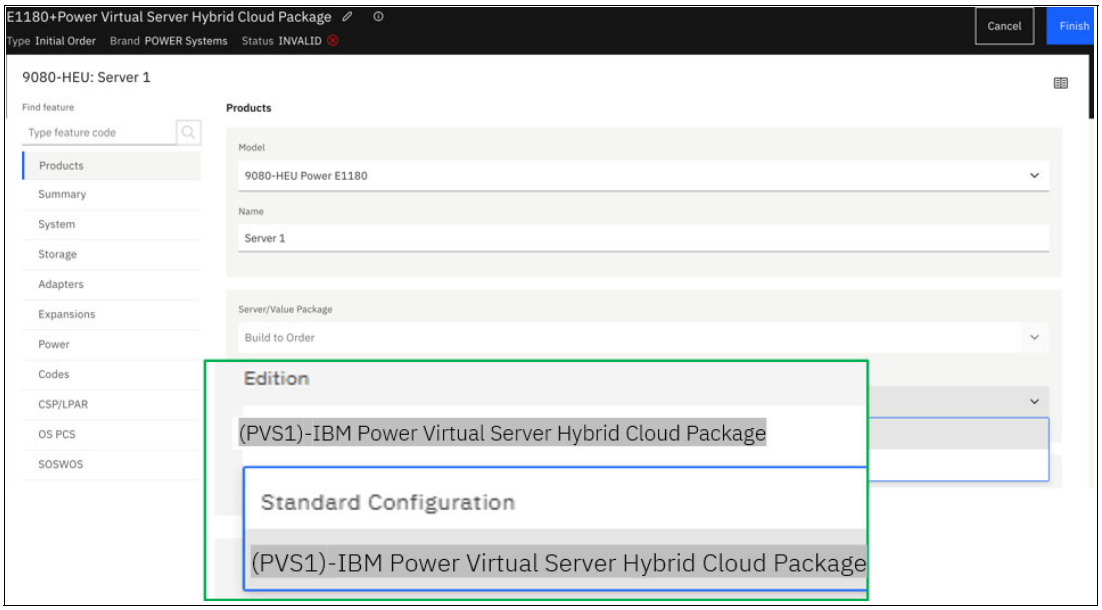


Figure 3-8 Select Cloud Package

3.1.4 IBM i software tiers (processor groups)

IBM i software tiers, also known as processor groups, are a licensing model that defines the resource limits and pricing structure for IBM i workloads running on either IBM Power Systems or IBM Power Virtual Server (PowerVS). The license is tied to either a physical server serial number or a Virtual Serial Number (VSN).

IBM i software tier breakdown

IBM i software tiers are categorized by physical servers or virtual machines based on the maximum number of virtual processors and memory they can use. These tiers align software licensing costs with workload size and performance needs.

Table 3-1 shows a breakdown of IBM i Software Tiers on-premises Power Servers.

Table 3-1 IBM i Software Tiers for on-premises Power Servers

Tier	Processor Group	Power11 Models	Typical Use Case
P10	Group 2	S1122	Medium workloads, production
P20 Standard Edition	Group 3	S1124/L1124	Large production workloads
P30 Enterprise Edition	Group 4	S1124/L1124/E1180	Enterprise-scale workloads

Table 3-2 shows a breakdown of IBM i Software Tiers for PowerVS.

Table 3-2 IBM i Software Tiers for PowerVS

Tier	Processor group	Max virtual processors	Max memory	Typical use case
P05	Group 1	1	64 GB	Small workloads, dev/test
P10	Group 2	4	1 TB	Medium workloads, production
P20	Group 3	12	1 TB	Large production workloads
P30	Group 4	No Limit	No Limit	Enterprise-scale workloads

Power Systems on-premises

On-premises, you can acquire IBM i software entitlement by using either the physical serial number of the server or a virtual serial number.

When you deploy a Logical Partition (LPAR), also known as a Virtual Machine (VM) when deployed with IBM Power Virtualization Center (PowerVC), you can tie the instance to either the physical server serial number or a virtual serial number.

Physical serial number

When using a physical serial number, migrating an LPAR from one physical server to another changes the serial number and impacts the IBM i operating system, Licensed Program Products (LPPs), and third-party software licenses. Movement to a different physical server using Dynamic LPAR (DLPAR) operations or redeployment impacts the licenses and requires manual re-entitlement.

Virtual serial number

A Virtual Serial Number (VSN) is a logical identifier assigned to a virtual IBM i instance running either on an IBM Power10 or Power11 Logical Partition (LPAR) or on the IBM Power Virtual Server (PowerVS) virtual server instance cloud environment. Like a physical serial number on traditional IBM Power Systems hardware, a VSN ties software entitlements, such as IBM i operating system licenses and licensed program products (LPPs), to a specific serial number. It supports subscription-based licensing and license portability.

On Power Systems with Power Systems Firmware version FW950 or later and Hardware Management Console (HMC) version 9.2.95 or later, you can assign a Virtual Serial Number (VSN) to the logical partition.

When you use a virtual serial number, migrating an LPAR from one physical server to another or to a system pool leaves the serial number unchanged and has no impact on software licensing.

Using virtual serial numbers on-premises provides the following benefits:

- Simplified license management across multiple systems
- Support for hybrid cloud models where workloads move between on-premises and PowerVS
- Dynamic scaling without relicensing
- Improved auditability and compliance tracking

When applicable, use the following best practices for using virtual serial numbers:

- Use consistent VSN naming conventions for traceability.
- Maintain a registry of VSNs and their associated entitlements.
- Integrate VSN management into LPAR provisioning workflows (for example, through HMC, Ansible, or CLI).
- Align VSNs with IBM i software tiers to ensure proper entitlement enforcement.

PowerVS virtual server instances

A Virtual Server Instance (VSI) in IBM PowerVS is a cloud-based virtual machine running on IBM PowerVS infrastructure. This is equivalent to an LPAR (a virtualized compute instance on a physical IBM Power System) or a virtual machine (VM) deployed by IBM PowerVC™.

When you deploy a VSI, you can tie the instance to either the physical server serial number or a virtual serial number. You select the IBM i software tier during provisioning.

Factors to consider when choosing a tier

Consider the following factors when selecting a tier:

- CPU and Memory Requirements: Match your workload's peak usage to the tier limits.
- License Cost Optimization: Use the lowest tier that meets your needs.
- Growth Forecast: If you expect rapid scaling, choose a tier that allows expansion.
- High Availability (HA): Larger tiers (P20/P30) support more robust HA configurations.
- Software Entitlements: Ensure your IBM i LPPs are compatible with the selected tier.

Physical serial number

If no virtual serial number is used, the IBM i software tier is tied to the model number of the server and the serial number of the system where it is deployed. The following table identifies the model number and software tier. Table 3-3 shows the available model number and software tiers when no virtual serial number is used.

Table 3-3 Model number with IBM i Software Tiers without VSN

Model	Tier	Core	Memory
S1122	P10	Up to 4	Up to 1 TB
E1180	P30	Up to max	Up to max

Virtual serial number

During deployment of a VSI with a virtual serial number, the IBM i software tier can be selected. The following Table 3-4 identifies the model number and software tiers available.

Table 3-4 Model number with IBM i Software Tiers with VSN

Model	Tier	Core	Memory
S1122	P05	Up to 1	Up to 64 GB
S1122/E1180	P10	Up to 4	Up to 1 TB
S1122/E1180	P20	Up to 12	Up to 1 TB
E1180	P30	Up to max	Up to max

When you use a virtual serial number, migrating a VSI from one physical server to another leaves the serial number unchanged and has no impact on software licensing.

Virtual serial number considerations

Key considerations when using Virtual Serial Numbers include the following tier characteristics:

- Tier selection is required when creating an IBM i VSI in PowerVS.
- Each tier has a corresponding license cost, with higher tiers costing more due to increased resource capacity.
- You can scale resources within a tier, but to exceed a tier's limits, you must change the tier, which might require powering off the VSI.
- Tiers are independent of physical hardware that they apply to virtual instances regardless of the underlying Power system (such as Power10, Power11).

Figure 3-9 shows the VSN licensing lifecycle.

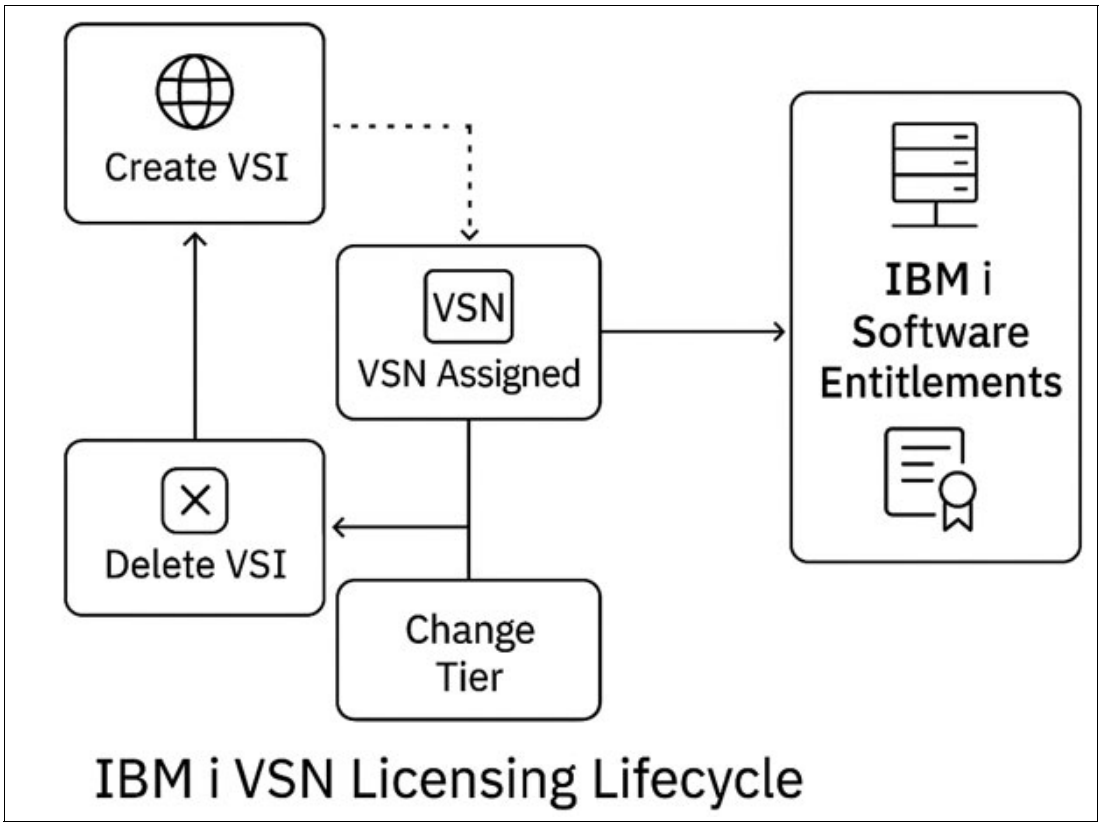


Figure 3-9 VSN for IBM i licensing

Virtual serial number benefits in PowerVS

Using a Virtual Serial Number (VSN) with IBM PowerVS offers several key benefits, especially for managing IBM i software entitlements in a cloud-native, scalable, and flexible environment:

- License portability:
  - VSNs allow IBM i software licenses (OS and LPPs) to decouple from physical hardware.
  - You can retain and reuse Virtual Server Networks across multiple virtual server instances, enabling license mobility.

- Simplified entitlement management:
  - VSNs can make it easier to track and manage IBM i entitlements dynamically in cloud environments.
  - You avoid the complexity of binding licenses to physical serial numbers or specific hardware.
- Support for subscription licensing:
  - IBM i in PowerVS uses term-based subscription licensing, which is tied to VSNs.
  - This aligns with modern cloud consumption models and simplifies cost forecasting.
- Workload flexibility:
  - You can scale workloads within the limits of the selected IBM i software tier (P05–P30) without relicensing.
  - VSNs enable dynamic provisioning and automated deployment in CI/CD pipelines.
- Hybrid cloud enablement:
  - VSNs support hybrid cloud strategies, allowing workloads to move between on-premises and PowerVS while maintaining license continuity.
  - This is especially useful for disaster recovery, test/dev, and migration scenarios.
- Audit and compliance readiness:
  - VSNs provide a clear and auditable trail of license usage.
  - VSNs help organizations stay compliant with IBM licensing policies.

3.2 Tooling for orchestration and optimization in hybrid cloud

Hybrid cloud adoption introduces significant challenges. Fragmented visibility across multiple environments complicates monitoring. Resource inefficiencies result from overprovisioning or underutilization. Unpredictable costs stem from complex pricing models and lack of financial governance. Complex orchestration is needed to coordinate workloads across heterogeneous platforms.

IBM Power Virtual Server addresses these issues through several key capabilities. It delivers rapid provisioning within minutes and dynamic resource scaling through DLPAR. Flexible storage tiers and automation through API and CLI support scheduled operations. PowerVS integrates seamlessly with IBM Cloud services for monitoring and orchestration, supports processor compatibility modes for legacy workloads, and provides enterprise-grade security with global compliance certifications. Combined with consumption-based pricing, immediate access to new hardware generations like Power11, and hybrid disaster recovery capabilities, PowerVS enables organizations to overcome operational complexity, optimize costs, and modernize their IT landscape while maintaining the reliability and performance of mission-critical systems.

This section explores how IBM and open-source tools along with AI-driven capabilities enable organizations to automate workflows, optimize resources, and maintain financial transparency and includes the following objectives:

- Understand the role of orchestration and optimization in hybrid cloud.
- Learn about IBM Instana, IBM Turbonomic, IBM Apptio®, and complementary tools.
- See how AI-driven insights improve performance and cost efficiency.

3.2.1 Hybrid cloud component

Table 3-5 describes the components in the IBM ecosystem.

Table 3-5 IBM hybrid cloud ecosystem components

Component	Role in the IBM Ecosystem
IBM Cloud Pak for Automation	Orchestrates workflows and automates business processes. It digitizes and manages complex tasks across hybrid environments, reducing manual effort and improving efficiency.
IBM Turbonomic	Provides AI-driven Application Resource Optimization (ARO). It continuously analyzes resource consumption and automatically adjusts resources to ensure performance while controlling costs.
IBM Instana	Offers observability and performance monitoring for hybrid workloads. Instana automatically discovers and monitors dynamic, cloud-native applications across all environments.
Ansible Automation Platform	Delivers Infrastructure as Code (IaC) for hybrid deployments. It automates provisioning, configuration management, and application deployment across diverse infrastructures.
PowerVC / PowerVS APIs	Enables orchestration for IBM Power environments. These tools manage virtualized resources on IBM Power Systems (AIX, IBM i, Linux) and integrate with cloud management platforms.

Together, these tools form a unified strategy for managing hybrid cloud complexity through automation, AI-driven optimization, and deep observability.

3.2.2 Integration model: Closed-loop automation

IBM's integration approach is based on three principles: observability, optimization, and automation.

Observability feeds optimization

IBM Instana serves as the observability layer, delivering real-time, high-fidelity visibility into application performance, service dependencies, and infrastructure behavior across containers, VMs, and platforms

Instana automatically discovers application components and maps dependencies, providing full-stack observability across hybrid environments.

This continuous telemetry stream feeds IBM Turbonomic, enabling data-driven decisions based on actual application demand rather than static thresholds.

Optimization triggers automation

The optimization decisions of IBM Turbonomic trigger Ansible playbooks for automated infrastructure changes. PowerVC and PowerVS APIs extend automation to IBM Power Systems, ensuring that mission-critical workloads are included in the hybrid strategy.
IBM Cloud Pak for Automation orchestrates higher-level workflows, integrating insights from Instana and actions from Turbonomic into business processes.

Unified hybrid cloud support

This integrated configuration enables enterprises to perform the following tasks:

- Monitor all applications and infrastructure with Instana
- Optimize performance and cost across all environments with Turbonomic
- Automate responses and workflows using Ansible and Cloud Pak for Automation

The result is an autonomous, AI-driven operational model that proactively manages hybrid cloud complexity.

3.2.3 Integration with orchestration platforms

Hybrid cloud environments require a combination of orchestration and automation tools to manage deployments, configurations, and monitoring. The following components provide a cohesive orchestration strategy:

- Kubernetes: Handles container orchestration across hybrid environments, ensuring consistent deployment and scaling of cloud-native applications.
- Ansible automation platform: Manages configuration and deployment through Infrastructure as Code (IaC), enabling automated provisioning and operational consistency.
- Terraform: Provides infrastructure provisioning and orchestration across multiple clouds and on-premises environments, supporting repeatable and scalable deployments.
- IBM Cloud Pak for Automation: Coordinates workflows and business process automation, integrating orchestration actions into enterprise-level processes.
- Prometheus and Grafana: Deliver advanced monitoring and visualization dashboards, complementing observability tools for performance insights and operational health.

Figure 3-10 shows concept diagram for workflow

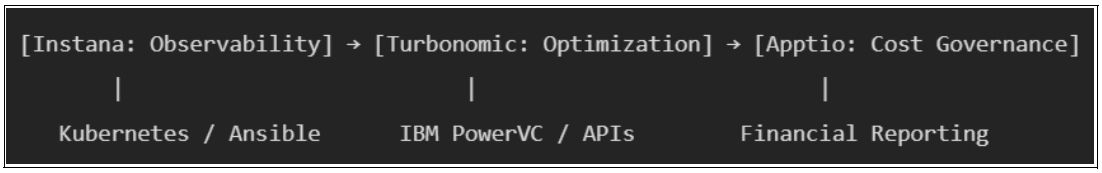


Figure 3-10 Integration workflow

3.3 IBM Cloud Pak for Automation in hybrid cloud for IBM Power11

Hybrid cloud environments demand agility, resilience, and intelligent automation to manage complex workloads across on-premises and cloud platforms. IBM Cloud Pak for Automation, when combined with IBM Power11 and Red Hat OpenShift on IBM Cloud (ROKS), provides a unified solution for automating business processes and IT operations. This integration enables enterprises to accelerate application delivery, optimize resource usage, and maintain security and compliance across heterogeneous environments.

Hybrid cloud environments on IBM Power11 use Red Hat OpenShift on IBM Cloud (ROKS) as the container orchestration layer. This integration provides a unified Kubernetes experience across public and private clouds, enabling organizations to modernize applications, adopt microservices architectures, and integrate AI-driven automation.

IBM managed operations with ROKS

IBM offers a fully managed Red Hat OpenShift experience through Red Hat OpenShift on IBM Cloud (ROKS). This service eliminates the operational burden of managing Kubernetes clusters, allowing organizations to focus on innovation and application development. IBM manages the entire lifecycle of Red Hat OpenShift clusters in the cloud, including the following aspects:

- Automated provisioning and configuration of compute, network, and storage resources
- High Availability (HA) with cross-zone configuration and automatic failure recovery
- Security patching and OS performance tuning for compliance and reliability
- Automatic scaling and capacity expansion with a single click
- Backups of core Red Hat OpenShift ETCD data for disaster recovery
- 24x7 global Site Reliability Engineering (SRE) support with deep expertise in IBM Cloud, Kubernetes, and Red Hat OpenShift

This managed approach delivers consistent performance, robust security, and seamless scalability, positioning Red Hat OpenShift on IBM Cloud as a foundation for hybrid cloud deployments on IBM Power11. Figure 3-11 illustrates the range of deployment and management options available, allowing clients to choose the level of control that best fits their operational needs.

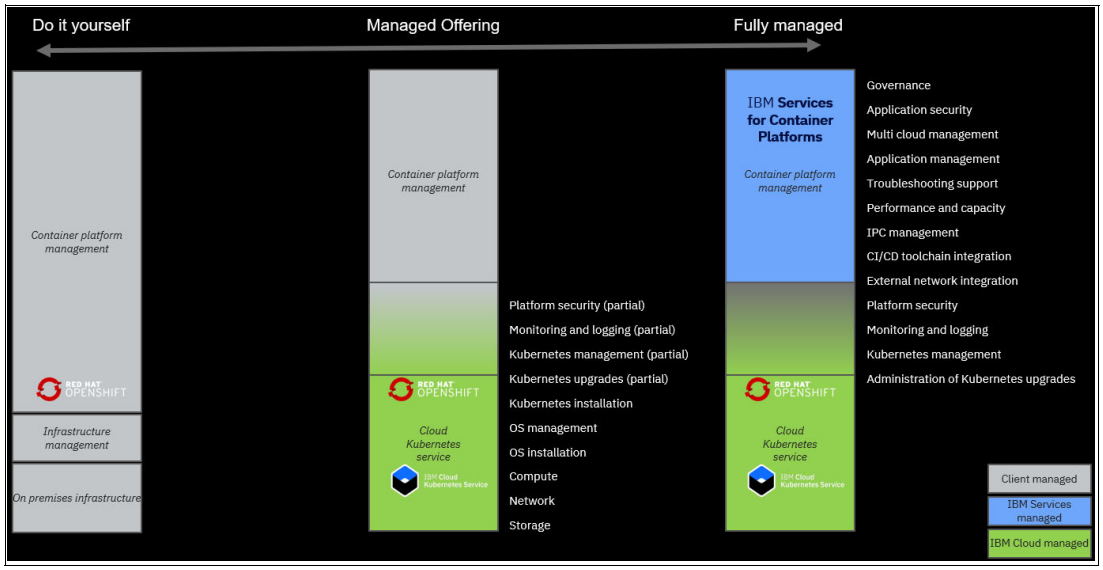


Figure 3-11 Deployment and management of Red Hat OpenShift offered

ROKS on IBM Cloud includes the following advanced features:

- AI Acceleration: Uses IBM Power11's AI capabilities for containerized AI workloads, enabling faster model training and inference.
- Multi-Architecture Compute: Supports heterogeneous environments (Power and x86) for flexibility and workload portability.
- Centralized Cluster Management: Uses Red Hat Advanced Cluster Management (ACM) for unified governance and policy enforcement.
- Operator Ecosystem: Automates platform functionality and extends Red Hat OpenShift capabilities through certified operators.

IBM Cloud Pak for Automation complements ROKS by providing the following suite of automation capabilities:

- Business Automation Workflow: Orchestrates processes across hybrid environments.
- Operational Decision Manager (ODM): Automates business rules and decisions.
- Robotic Process Automation (RPA): Handles repetitive task automation.
- Content Services: Manages documents and unstructured data.
- Process Mining: Identifies inefficiencies and optimizes workflows.

These capabilities deliver end-to-end process automation, reduce operational costs, and improve compliance.

Red Hat OpenShift on IBM Cloud simplifies cluster lifecycle management by providing the following capabilities:

- Built-in Security and Isolation: Delivers applications rapidly and securely.
- Access to 200+ IBM Cloud Services: Enables integration with AI, data, and automation tools.
- Global Availability: Six IBM regions and more than 35 data centers provide proximity and compliance for enterprise workloads.

3.3.1 Cloud Pak for Data

IBM Cloud Pak for Data (CP4D) is a cloud-native, fully managed data and AI platform built on Red Hat OpenShift. The platform provides a unified environment for data management and analytics, enabling organizations to make data-driven decisions and accelerate AI initiatives.

CP4D implements a data fabric architecture that connects siloed data across on-premises systems and multiple clouds without moving data. The platform automates data discovery and curation, enforces governance policies, and protects sensitive information across the organization.

CP4D enables organizations to build, run, automate, and manage AI models using trusted data. For managed deployments, Red Hat Site Reliability Engineers (SREs) handle installation, monitoring, maintenance, and upgrades, which reduces operational complexity.

IBM Cloud Pak for Data version 5.0 introduces innovations that simplify operations, improve compliance, and accelerate analytics. This release helps organizations implement a data fabric architecture for trusted, accessible, and AI-ready data.

Figure 3-12 on page 81 illustrates an IBM Cloud Pak for Data and Red Hat OpenShift hybrid cloud architecture management option.

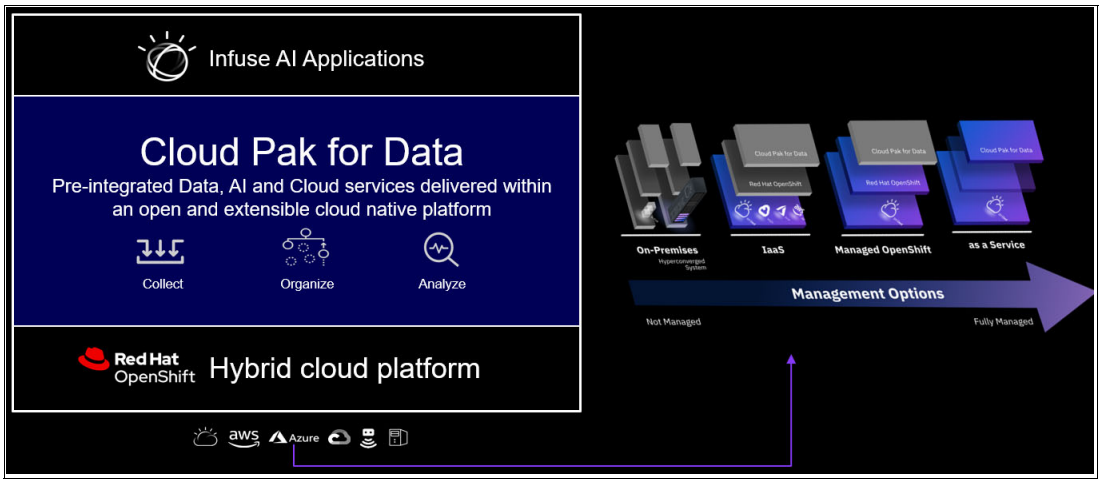


Figure 3-12 An Information Architecture for AI

IBM Cloud Pak for Data key capabilities

IBM Cloud Pak for Data includes the following key capabilities:

- Immersive experience

  Unifies IBM watsonx and Cloud Pak for Data on a single Red Hat OpenShift platform, eliminating integration complexity. Organizations can manage AI and data workloads together, reducing operational impact, and accelerating innovation.

- Remote data planes

  Brings compute to the data instead of moving data across regions or clouds. This capability minimizes transfer costs, ensures compliance with data sovereignty laws, and improves pipeline performance while consolidating multiple environments under one control plane.

- Connectivity

  Expands integration flexibility with 100+ certified connectors, including advanced support for Apache Iceberg, Delta Lake, and Milvus. Organizations can connect to modern data sources and implement new AI use cases.

Data fabric services:

CP4D provides the following data fabric services:

- Data integration: IBM DataStage® uses remote data planes for faster, compliant ETL processing.
- Data governance: Generative AI-powered Knowledge Catalog Cartridges automate metadata enrichment and compliance at scale.
- Relationship explorer: Provides a visual map of data relationships and governance flows, improving transparency and decision-making.

Data sharing

Data Product Hub transforms data sharing into a governed, enterprise-wide process. The hub accelerates collaboration by enabling reusable data products, reducing silos, and ensuring security and quality across domains.

IBM Cloud Pak for Data 5.0 delivers the following benefits:

- Breaks down data silos

- Ensures compliance and security
- Optimizes modern data workloads
- Scales analytics and AI with trusted, high-quality data

For more information, see IBM unveils Cloud Pak for Data 5.0.

3.3.2 Planning and cost estimation for IBM Cloud Pak for Data

IBM provides tools and trial options to help organizations plan and evaluate Cloud Pak for Data deployments effectively:

- IBM Cloud Estimator

  Use IBM Cloud Estimator to forecast costs, configure resources, and compare pricing for self-managed and fully managed deployments. This tool ensures transparency and helps optimize budgets before implementation.

  Figure 3-13 shows IBM Cloud Estimator website with options to configure resources, select services, and view real-time pricing that is tailored to your requirements.

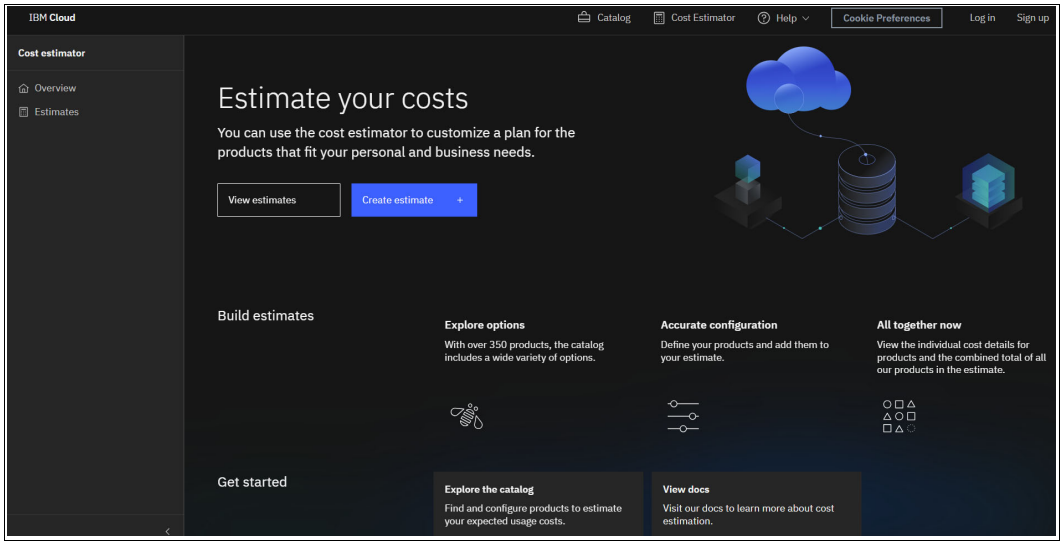


Figure 3-13 IBM Cloud Estimator

- No-cost trial self-managed:

  IBM offers a 60-day trial license for both Cloud Pak for Data and Red Hat OpenShift in a self-managed model. You provide your own underlying infrastructure (IaaS).

  The trial includes the following components:

  – Cloud Pak for Data: Follows IBM's standard software evaluation process
  – Red Hat OpenShift: Requires a Red Hat account and restricts usage to CP4D workloads

To start the trial, see the sign-up page. Provide basic account details such as email, name, and region. This trial is ideal if you want full control over infrastructure while evaluating CP4D capabilities, as shown in Figure 3-14.

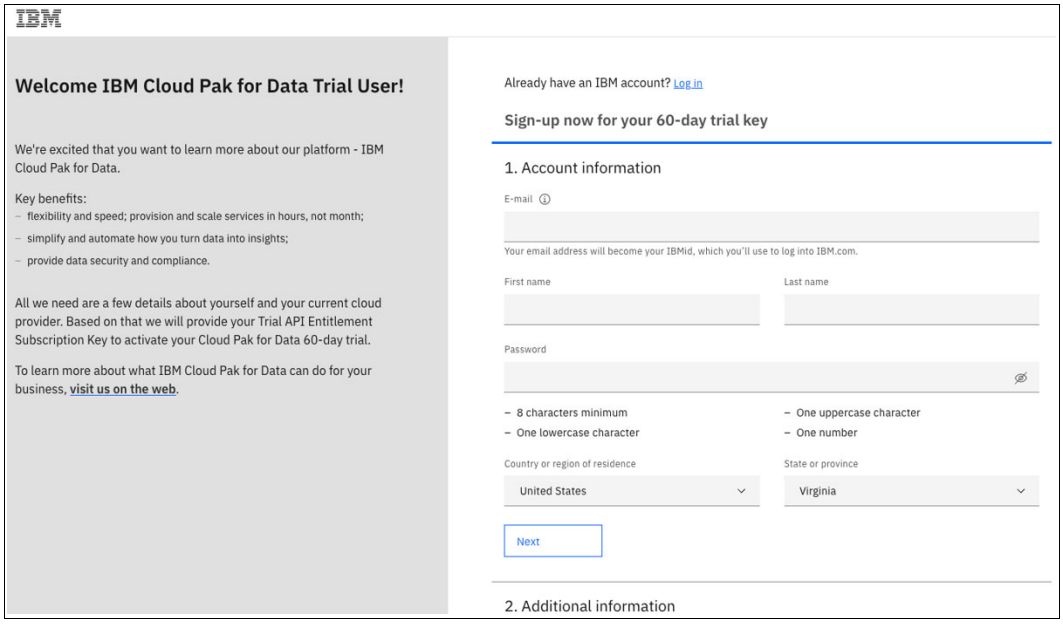


Figure 3-14 Shelf managed no-cost trial

- Free trial fully managed SaaS:

  IBM provides a fully managed SaaS trial for Cloud Pak for Data for organizations seeking simplicity. IBM handles infrastructure, installation, and maintenance, so you can focus on exploring data and AI services. For more information, see IBM Cloud Pak for Data for free. The trial has the following characteristics:

  – Available in Dallas, London, Frankfurt, and Tokyo data center regions
  – Requires an IBMid account for access
  – Includes core services such as DataStage, IBM Knowledge Catalog, Watson Machine Learning, Watson Studio, and more

To begin the trial, select a region, agree to the terms, and create an IBMid, as shown in Figure 3-15. This trial provides a fast, low-effort way to experience CP4D as a service.

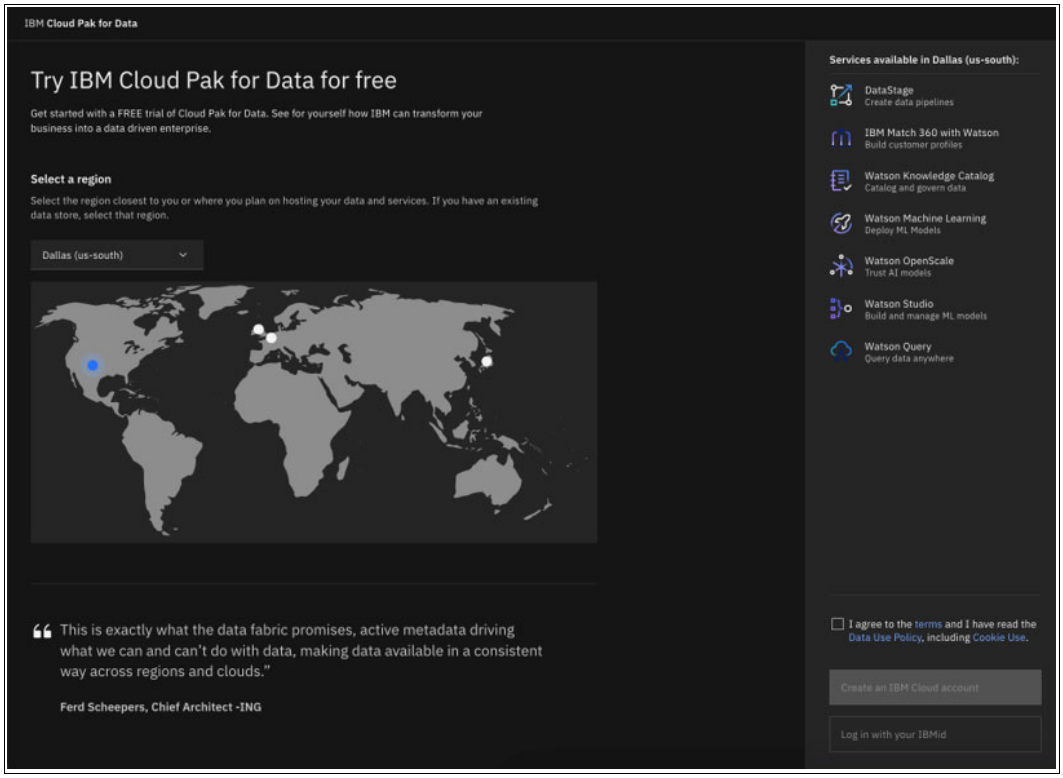


Figure 3-15 Fully managed SaaS

3.4 Terraform deployment for IBM Cloud Pak for Data

Terraform is an Infrastructure-as-Code (IaC) tool that automates the provisioning and configuration of infrastructure and applications. For IBM Cloud Pak for Data, Terraform simplifies deployment by enabling repeatable, consistent setups across IBM Cloud, AWS, Azure, and on-premises environments.

Terraform provides the following benefits for Cloud Pak for Data (CP4D) deployment:

- Automation: Reduces manual intervention and configuration errors by automating deployment tasks
- Consistency: Ensures repeatable and predictable deployments across development, test, and production environments
- Multi-cloud support: Enables deployment on IBM Cloud, AWS, Azure, and on-premises Red Hat OpenShift clusters, supporting hybrid and multi-cloud strategies
- Compliance: Facilitates adherence to governance and security policies, helping organizations maintain regulatory standards
- Faster deployment: Reduces setup time from days to hours
- Integration with GitOps: Supports version-controlled infrastructure
- Cost optimization: Automates resource cleanup with terraform destroy

3.4.1 Architecture

Figure 3-16 illustrates an architecture that delivers high availability, scalability, and resilience for enterprise workloads on IBM Cloud.

The multi-zone design removes single points of failure and maintains service continuity during zone-level disruptions. Portworx provides persistent storage and integrates with Red Hat OpenShift to support stateful applications and enterprise-grade data management. Combined with IBM Cloud infrastructure flexibility, this architecture enables dynamic scaling and consistent performance for mission-critical workloads across hybrid and multi-cloud environments.

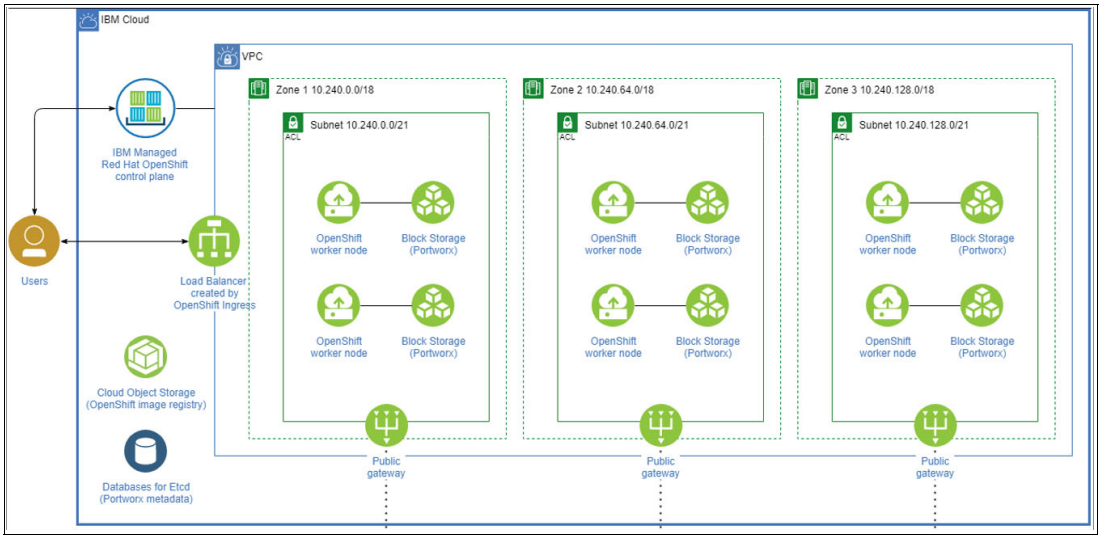


Figure 3-16 IBM Cloud environment running an IBM Managed Red Hat OpenShift cluster

Terraform modules commonly perform the following tasks:

- Provision the Red Hat OpenShift cluster: Deploys either IBM Cloud Red Hat OpenShift (ROKS) or a self-managed Red Hat OpenShift cluster
- Install Cloud Pak for Data Components: Sets up the CP4D control plane and selected services
- Configure infrastructure: Integrates persistent storage, networking resources, and cluster connectivity
- Apply security and compliance policies: Enforces governance, access controls, and security configurations

3.4.2 Automate service

Portworx Enterprise is the recommended storage solution for Terraform-based deployments of Cloud Pak for Data. Portworx delivers high availability, scalability, and advanced data management capabilities for containerized workloads, ensuring reliable performance for stateful applications.

Terraform automates the installation of the Cloud Pak for Data control plane and selected services, which reduces manual effort and improves consistency across environments. Terraform automates the installation of the following key CP4D services:

- Watson Knowledge Catalog (WKC)

- Watson Machine Learning (WML)
- Watson Studio Lite (WSL)
- Watson OpenScale (WOS)
- IBM Cognos® Dashboard Embedded (CDE)
- Data Virtualization (DV)
- Planning Analytics (PA)
- Watson Discovery (WD)
- IBM DataStage (DS)
- Cognos Analytics (CA)
- IBM OpenPages® (OP)
- IBM Db2
- Data Science and MLOps with IBM Data Fabric
- Multicloud Data Integration
- Data Governance and Privacy
- Express options for Data Science and Governance
- ELT Pushdown Express

These services are provisioned automatically during deployment, reducing complexity and accelerating time-to-value.

3.4.3 Deployment steps

This section outlines the complete sequence of tasks to deploy the solution end to end. Each step ensures accuracy, repeatability, and alignment with the reference architecture.

Complete the following steps to deploy Cloud Pak for Data with Terraform:

1. Install Terraform and IBM provider plug-in

   Install Terraform v0.13+ and terraform-provider-ibm.

2. Configure IBM Cloud credentials

   Set API keys and resource group in your Terraform configuration.

3. Provision the Red Hat OpenShift cluster

   Use IBM-provided Terraform modules for ROKS or specify an existing cluster.

4. Deploy the CP4D module

   Define variables for cluster name, region, storage type, and CP4D version.

Example 3-1 module detail

```
module "cp4d" {
  source           =
"git::https://github.com/ibm-build-lab/terraform-ibm-cloud-pak.git//modules/cp4d_4
.0"
  cluster_name_id  = var.cluster_name_id
  resource_group   = var.resource_group
  storage_type     = "portworx"
}
```

5. Run Terraform commands

   Run the following commands to deploy CP4D:

   a. **`terraform init`** to initialize modules
   b. **`terraform plan`** to review the configuration
   c. **`terraform apply`** to deploy CP4D

6. Access the CP4D console

   Retrieve the console URL and credentials from Terraform outputs.

For more information about installation, see terraform-ibm-cloudpak-dataon GitHub.

3.5 IBM Cloud catalog deployment

IBM Cloud catalog provides a simplified, automated method for deploying IBM Cloud Pak for Data (CP4D) on an existing Red Hat OpenShift cluster. This approach offers a streamlined purchasing experience and reduces operational complexity by using IBM-managed automation.

3.5.1 IBM Cloud catalog services

The IBM Cloud catalog provides the following advantages for deploying Cloud Pak for Data:

- Enables fully automated deployment
- Accelerates provisioning compared to manual or scripted installation methods
- Supports deployment on existing Red Hat OpenShift clusters with predefined resource requirements
- Provides direct access to IBM Cloud services for AI, analytics, and automation

The IBM Cloud catalog supports automatic provisioning of the following services:

- Analytics Engine
- BigSQL
- Cognos Analytics
- Cognos Dashboard
- Data Management Console
- DataStage
- Db2 and Db2 Warehouse
- Decision Optimization
- AI Factsheet
- Master Data Management
- OpenPages
- Planning Analytics
- RStudio
- IBM SPSS® Modeler
- Watson Query
- IBM Knowledge Catalog
- Watson Machine Learning
- Watson OpenScale
- Watson Pipelines
- Watson Studio
- Watson Assistant
- Watson Speech
- Watson Discovery
- MANTA

These services enable organizations to build a comprehensive data and AI platform without manual configuration.

3.5.2 Deployment architecture

IBM Cloud Catalog deployment requires an existing Red Hat OpenShift cluster that meets the following minimum requirements: three nodes, each with 16 cores, 64 GB memory, and 25 GB disk.

Red Hat OpenShift Data Foundation is the recommended storage option, ensuring high availability and scalability for containerized workloads.

Complete the following steps to deploy Cloud Pak for Data from the IBM Cloud catalog:

1. Access the IBM Cloud catalog

   From the IBM Cloud catalog, select **Cloud Pak for Data**.
2. Choose the deployment type

   Select **Existing OpenShift Cluster** as the target environment.
3. Verify cluster requirements

   Ensure the cluster meets minimum specifications.
4. Configure storage

   Select **OpenShift Data Foundation** for persistent storage.
5. Select services

   Select required services from the automated list.
6. Deploy

   Initiate deployment. IBM automation handles installation and configuration.
7. Access the CP4D console

   Retrieve the URL and credentials from the deployment summary.

4

Hybrid cloud use cases on Power

Hybrid cloud combines the resiliency of on-premises systems with the flexibility of cloud services. IBM Power Systems, integrated with IBM Power Virtual Server and powered by Power11, provides a secure and scalable foundation for hybrid cloud strategies.

This chapter includes key hybrid use cases:

- High availability and disaster recovery with placement groups and Global Replication Services (GRS)
- Development and testing acceleration
- PowerVS capabilities for agility and automation
- Application integration patterns
- AI workloads with watsonx and Power acceleration technologies
- Operational strategies for workload placement and observability

IBM Power combines advanced hardware, enterprise software, and cloud services to help organizations modernize applications and optimize operations across hybrid environments.

This chapter covers the following topics:

- “Hybrid cloud and business continuity with HA and DR”
- “Accelerating dev/test on cloud”
- “Using Power Virtual Server effectively”
- “Application integration in hybrid multicloud”
- “AI workloads in the cloud”
- “Operational strategies for hybrid environments”

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4.1 Hybrid cloud and business continuity with HA and DR

A hybrid cloud strategy enhances business continuity by distributing critical workloads across on-premises systems and cloud environments, ensuring operations remain resilient during localized failures or planned maintenance. High availability (HA) and disaster recovery (DR) capabilities extend into the hybrid cloud, enabling rapid failover, geographic redundancy, and workload mobility that protect mission-critical services.

Cloud-based automation and scalable infrastructure enable organizations to provision recovery environments, test replicas, or supplemental capacity in minutes. Hybrid architectures reduce infrastructure costs by eliminating the need for fully provisioned secondary sites through consumption-based cloud resource pricing.

IBM Power11 strengthens HA and DR in hybrid cloud environments through advanced capabilities for resiliency, automation, and integration with IBM Cloud services, enabling continuity strategies that span on-premises and cloud deployments.

4.1.1 Hybrid cloud resiliency on IBM Power Virtual Server

Hybrid cloud deployments on IBM Power Virtual Server (PowerVS) require a multi-layered resiliency strategy to protect workloads against local hardware failures and regional outages while meeting Recovery Point Objective (RPO) and Recovery Time Objective (RTO) targets. IBM provides the following capabilities:

- Placement groups: Local high availability (HA)
- Global Replication Services (GRS): Regional disaster recovery (DR)

Combined with automation, orchestration, and monitoring, these capabilities enable enterprises to design hybrid cloud architectures that minimize downtime and data loss.

Hybrid cloud provides the following benefits for HA and DR:

- Geographic resilience: Distributes workloads across multiple locations, reducing single points of failure
- Elastic recovery resources: Provisions cloud-based DR environments on demand, minimizing cost while ensuring rapid recovery
- Continuous operations: Enables active-active or active-passive deployments, supporting zero planned downtime during migrations or maintenance
- Automation and orchestration: Streamlines failover processes, reducing complexity and human error
- Compliance and data control: Maintains sensitive data on-premises while using cloud for resilience, meeting regulatory requirements

4.1.2 Integrated resiliency strategy

IBM Power Virtual Server provides an enterprise-grade resiliency model that combines Placement Groups for local high availability and Global Replication Services (GRS) for regional disaster recovery.

Core resiliency capabilities

IBM Power Virtual Server delivers the following resiliency capabilities:

Always-on operations: Placement Groups maintain workload availability by isolating workloads across physical hosts, reducing risk from hardware failures

Cross-region protection: GRS replicates data asynchronously across paired IBM Cloud regions, enabling rapid recovery from site-wide outages

Automation: Ansible and Terraform enable predictable, repeatable failover and fallback workflows

Observability: IBM Cloud Monitoring (Sysdig) and IBM Cloud Logs provide real-time insights and compliance-ready audit trails

These capabilities reduce recovery time from hours to minutes, maintain mission-critical application availability during disruptions, and meet stringent compliance standards through built-in security and encryption.

Advanced HA/DR capabilities

IBM offers the following advanced capabilities for enterprise workloads:

- IBM PowerHA® integration: Enterprise-grade clustering for AIX and IBM i workloads, with Geographic Logical Volume Manager (GLVM) for synchronous or asynchronous replication across regions
- IBM i geographic mirroring: Native OS-level replication for IBM i, delivering seamless failover without external appliances
- Resource Optimized High Availability (ROHA): Dynamically allocates CPU and memory during failover using Shared Processor Pools (SPP), reducing cost while maintaining performance
- Cyber resiliency: Provides immutable storage, retention policies, and FIPS 140-2 Level 4 encryption to protect against ransomware and data breaches
- Compliance and governance: Integrates identity and access management, audit capabilities, and encryption to meet regulatory requirements

These capabilities enable dynamic resource scaling with consumption-based pricing during DR events, protect data through advanced encryption and isolation, and support hybrid and multicloud strategies.

Business continuity value

Combining Placement Groups for local high availability with Global Replication Services for regional disaster recovery, which is enhanced with PowerHA, ROHA, isolated data, and encrypted replication, minimizes downtime, protects critical workloads, and maintains compliance. With the global average cost of a data breach at USD 4.88 million (up 10% from 2023), IBM Power Virtual Server helps organizations mitigate financial and operational risks in hybrid cloud environments.

Figure 4-1 shows average cost and frequency of data breaches[1].

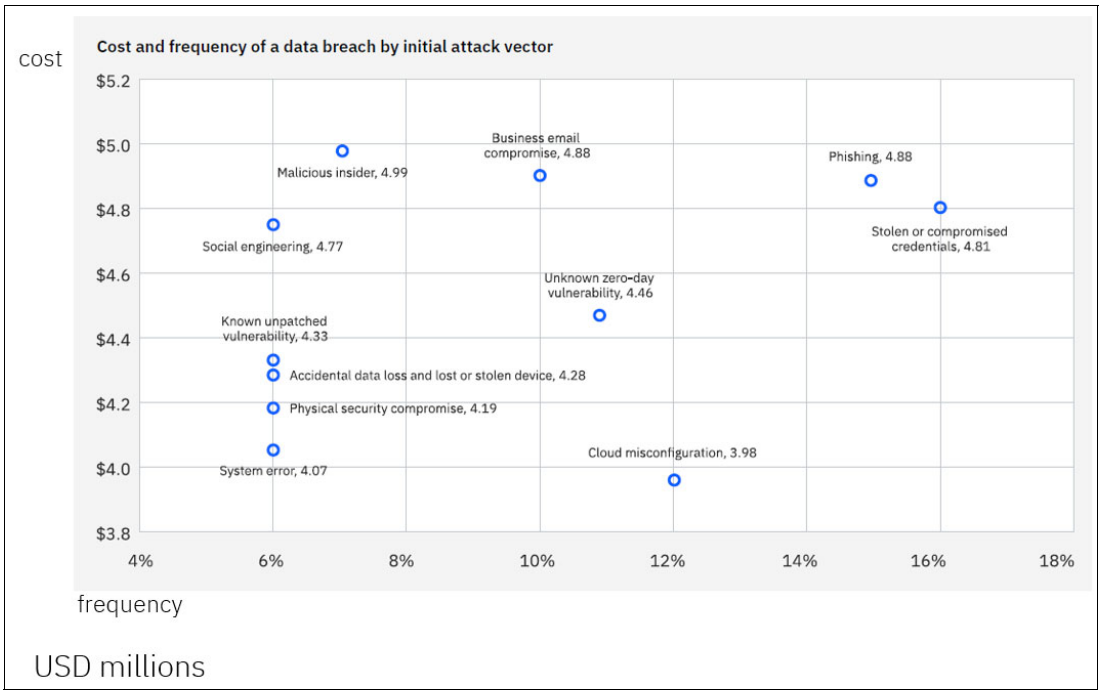


Figure 4-1 Cost of data breach

Placement Groups: Local high availability

Placement Groups are logical constructs that control how virtual server instances (VSIs) are distributed across physical hosts within an IBM Cloud zone. Placement Groups enforce anti-affinity policies to reduce the risk of simultaneous failures caused by hardware faults or maintenance events.

Placement Groups have the following characteristics:

- Scope: Operate within a single zone; do not span multiple zones or regions
- Policy: Anti-affinity ensures cluster nodes are placed on separate physical servers
- Supported Workloads: Support AIX, IBM i, and Linux clusters, including IBM PowerHA SystemMirror® and Red Hat HA Add-On

Configuration workflow

Complete the following steps to configure Placement Groups:

1. Create a placement group

   ```
   ibmcloud sl placement-group-create --name <pg-ha-cluster-name> --policy
   anti-affinity
   ```

2. Assign VSIs

   Specify the placement group during VSI creation or update.

3. Validate distribution

   Use the IBM Cloud CLI or API to confirm node placement.

[1] https://www.ibm.com/reports/data-breach

Best practices

Apply the following best practices when using Placement Groups:

- Combine Placement Groups with redundant VLANs and multipath I/O for storage.
- For HA clusters, distribute quorum nodes across different placement groups when possible.
- Monitor placement compliance using IBM Cloud Activity Tracker.

Global Replication Services (GRS) for regional disaster recovery

Global Replication Services (GRS) provides asynchronous block-level replication between paired IBM Cloud regions using IBM FlashSystem Global Mirror technology. GRS enables rapid recovery without third-party replication appliances.

Core components

GRS includes the following core components:

- Replication groups: Provide logical grouping of volumes for consistency
- Consistency groups: Backend construct that ensures write-order fidelity across volumes
- Auxiliary volumes: Remain read-only until failover; become active during DR events

Configuration

Complete the following steps to configure GRS:

1. Provision volumes in GRS-capable pools

   Select a storage pool that supports replication during volume creation.

2. Enable replication as in the following example:

   ```
   ibmcloud pi volume-create --name vol-db --size 500 --replication-enabled true
   ```

3. Create a volume group

   Group-related volumes for coordinated failover.

4. Onboard auxiliary volumes

   Attach replicated volumes to standby VSIs in the secondary region.

5. Validate replication state

   ```
   ibmcloud pi volume-group-get --id <group_id>
   ```

Failover process

Complete the following steps to fail over to the DR site:

1. Stop replication on the primary site.
2. Promote auxiliary volumes to read/write mode.
3. Boot standby VSIs and reconfigure IP addresses or DNS as needed.

Performance characteristics

Asynchronous replication introduces minimal latency impact on primary workloads. RPO typically ranges from seconds to minutes, depending on bandwidth and workload I/O profile.

4.2 Accelerating dev/test on cloud

Modern enterprises must deliver innovation quickly while maintaining reliability, security, and operational governance. Traditional on-premises provisioning creates bottlenecks through long lead times, manual processes, and limited resource elasticity, which slow delivery cycles and create friction between development and operations teams.

IBM Power Virtual Server combined with IBM Power11 systems modernizes and accelerates dev and test workflows. This hybrid cloud model enables rapid provisioning, automated lifecycle management, and Infrastructure-as-Code practices that provide consistent, repeatable environment creation. Developers gain on-demand access to Power-based resources. Operations teams maintain governance and visibility through unified management tools and policy-driven automation.

DevOps teams face the following challenges in cloud-enabled development models:

- Environment drift
- Inconsistent CI/CD pipelines across hybrid estates
- Integration complexity for Power-based workloads into automation frameworks
- Balancing security compliance with developer autonomy
- Cost governance as environments scale dynamically

PowerVS with Power11, integrated into a modern DevOps toolchain, addresses these challenges and enables faster, more reliable innovation across hybrid cloud platforms.

4.2.1 Considerations for hybrid cloud for dev and test

Development and testing environments are often constrained by on-premises hardware availability, manual provisioning, and rigid resource allocation. Hybrid cloud eliminates these bottlenecks by enabling on-demand scalability, automation, and consistent architecture across environments.

IBM Power Virtual Server (PowerVS) combined with Power11 systems provides the following capabilities:

- Architecture consistency: PowerVS runs AIX, IBM i, and Linux on the same Power architecture as on-premises systems, ensuring predictable performance and compatibility.
- Rapid provisioning: Infrastructure-as-Code (IaC) tools like Terraform and Ansible create production-like environments in minutes, not days.
- Dynamic resource allocation: Features such as Shared Processor Pools (SPP) and Live Partition Mobility (LPM) scale compute resources up or down without downtime.
- Integrated networking: Direct Link and VPN connectivity ensure secure, low-latency access between cloud and on-premises environments for hybrid workflows.
- Storage flexibility: Multiple tiers (Tier 0 for high IOPS, Tier 3 for cost efficiency) match workload requirements without overprovisioning.

Hybrid cloud dev/test environments can be automated, monitored, and optimized by using IBM Cloud Schematics, Sysdig monitoring, and PowerVM capabilities to deliver agility without sacrificing enterprise-grade reliability. For more information, see *High Availability and Disaster Recovery Options for IBM Power Cloud and On-Premises*, REDP-5656.

4.2.2 Observability and compliance for faster dev/test cycles

Accelerating development and testing requires more than speed. It demands control, cost efficiency, and risk reduction. IBM Cloud Observability services transform dev/test environments into intelligent, self-optimizing platforms that deliver measurable business value.

IBM Cloud Observability provides the following capabilities:

- Real-time insights: IBM Cloud Monitoring (powered by Sysdig) provides dashboards and alerts for CPU, memory, and network usage, enabling teams to optimize performance during stress tests.
- Cost control: Short-lived environments can become expensive if left unchecked. Observability tools track idle resources and trigger automated scale-down or teardown workflows, reducing unnecessary spending.
- Rapid issue resolution: Centralized logs and metrics accelerate troubleshooting, minimizing delays in iteration cycles and improving developer productivity.
- Compliance readiness: IBM Cloud Logs capture every provisioning and configuration change, ensuring audit readiness for regulated industries even in test environments.

Integrating observability into dev/test strategies delivers predictable costs, faster release cycles, and compliance assurance while maintaining enterprise-grade security, reducing operational risk and accelerating innovation.

4.3 Using Power Virtual Server effectively

IBM Power Virtual Server is more than a cloud extension, It is a strategic enabler for hybrid cloud agility, resilience, and cost optimization. As organizations advance their hybrid cloud strategies, integrating existing enterprise workloads with cloud-native services differentiates competitive IT operations. Traditional infrastructures often lack the agility, scalability, and financial flexibility that modern IT environments demand. PowerVS extends the capabilities of IBM Power into the cloud, enabling enterprises to run IBM i, AIX, and Linux workloads without refactoring. This seamless extension ensures smooth interoperability between on-premises systems and the cloud, supporting both lift-and-shift migrations and modernization strategies.

PowerVS unlocks value beyond conventional environments through on-demand scalability for seasonal workloads, rapid provisioning for development and testing, and integration with advanced IBM Cloud services such as AI, analytics, and automation. Beyond simple hosting, PowerVS enables strategic use cases including disaster recovery, hybrid application modernization, and containerized workloads through Red Hat OpenShift. With enterprise-grade security, compliance certifications, and high availability, PowerVS empowers organizations to modernize their IT landscape while preserving the reliability and performance of mission-critical systems.

4.3.1 Flexibility and agility

PowerVS provides flexibility and agility through the following capabilities:

- Rapid provisioning: Self-service provisioning delivers servers within 10–20 minutes, enabling organizations to quickly stand up test or development environments.
- Dynamic scaling: Dynamic Logical Partitioning (DLPAR) adjusts resources without shutting down the system. For example, an LPAR initially created with 2 cores can be

resized from 0.25 to 16 cores, subject to host capacity. (The DLPAR range is from one-eighth to eight times the initial allocation.)

- Host migration: If resources are unavailable on the current host, migration to another host with sufficient capacity can be requested. For requirements beyond the DLPAR range, additional changes can be applied after shutting down the LPAR.
- Storage tiering: Performance tiers (3, 10, 25 IOPS/GB, or a fixed tier providing 5,000 IOPS for volumes up to 200 GB) can be selected and modified later to meet evolving workload needs.
- Automation: API and CLI support enables scheduled resource scaling (for example, during nights or weekends) and restoration during business hours to optimize costs.

4.3.2 Automating, optimizing, and securing your PowerVS environment

PowerVS in hybrid cloud environments requires optimization across three dimensions: automation, performance, and security.

Automation

Infrastructure-as-Code (IaC) tools accelerate deployment and reduce operational impact. Terraform defines infrastructure in declarative scripts, ensuring consistency across environments. Ansible automates software installation and configuration. Together, these tools enable rapid deployment, updates, and teardown of environments, reducing provisioning time from hours to minutes while minimizing configuration errors.

Performance and placement

Performance optimization and strategic placement ensure reliability and cost control. PowerVS provides the following capabilities:

- Placement groups: Spread virtual servers across different physical hosts so that a single hardware failure does not affect the entire system.
- Global Replication Services (GRS): Protect data by replicating it to another region for disaster recovery.
- Processor compatibility modes: Support Power9 or Power10 compatibility modes to ensure legacy applications run smoothly while using newer hardware capabilities.

Security and compliance

PowerVS provides enterprise-grade security and compliance features:

- Encryption: All data is encrypted using FIPS 140-2 Level 4 standards, both at rest and in transit.
- Access control: Identity and Access Management (IAM) restricts changes to authorized personnel.
- Audit trails: IBM Cloud Logs record every action, simplifying compliance and providing transparency for audits.

Automation accelerates delivery; optimized placement improves reliability and cost efficiency; and integrated security ensures workloads remain protected and compliant.

4.3.3 Access to the latest hardware

IBM Power Virtual Server delivers instant access to the latest IBM Power hardware generations without waiting for on-premises refresh cycles. When IBM introduced Power11 for on-premises deployment, the same technology became available in multiple PowerVS data centers at launch. Organizations can immediately use the performance, scalability, and advanced features of Power11, even if their current infrastructure is based on older generations such as Power9 or Power10.

This capability is critical for hybrid modernization strategies. Instead of investing in new physical hardware upfront, enterprises can adopt Power11 in the cloud to accelerate transformation projects, test next-generation workloads, or run production systems with minimal disruption. PowerVS provides a pathway to modernization by delivering access to long-term support and advanced capabilities such as AI acceleration, enhanced security, and improved energy efficiency, all while maintaining operational continuity.

PowerVS offers processor compatibility modes (for example, Power9_Base or Power10) that allow administrators to run legacy applications alongside newer workloads. This flexibility ensures compatibility with an earlier version for existing software stacks and simplifies migration planning. For example, workloads that require older instruction sets can continue operating without modification, and new applications can take advantage of Power11 performance improvements.

PowerVS supports the following hybrid modernization strategies:

- Hybrid upgrade strategy: Validate workloads on Power11 in PowerVS before committing to on-premises upgrades.
- Performance testing: Benchmark applications in PowerVS to understand scaling behavior on newer hardware.
- Risk mitigation: Compatibility modes reduce migration complexity and avoid costly refactoring.
- Future-proofing: Immediate access to Power11 features such as AI acceleration and advanced security positions organizations for emerging workloads without delay.

PowerVS eliminates the traditional hardware refresh bottleneck, enabling enterprises to modernize at their own pace while using IBM Power technology in a secure, cloud-native environment.

4.3.4 Business benefits

PowerVS delivers clear business advantages compared to traditional deployments, which are organized across the following areas:

- Agility: Near-instant provisioning of development and test environments, typically within minutes rather than weeks, accelerates application release cycles, supports CI/CD, and enables cloud bursting for seasonal or unpredictable demand spikes without redesigning workloads.
- Financial flexibility: Consumption-based monthly subscription shifts IT spending from CapEx to OpEx, eliminating large upfront investments and improving cash flow predictability.
- Operational efficiency: IBM manages hardware, virtualization, and lifecycle operations, reducing internal impact for routine maintenance tasks such as firmware updates or hardware replacements.

- Cost optimization: On-demand scaling of compute and storage resources eliminates overprovisioning and wasted investment by aligning costs with actual usage.
- Risk mitigation: PowerVS supports hybrid disaster recovery strategies, allowing production workloads to remain on conventional systems while PowerVS serves as a remote backup or failover site. IBM Cloud Object Storage automatically replicates data across three geographically dispersed sites for high fault tolerance.
- Enterprise-grade security: Workloads hosted on IBM Cloud benefit from global compliance certifications (ISO, SOC) and industry-specific frameworks. The proven reliability and security-hardened architecture of IBM Power are preserved in PowerVS, combined with IBM Cloud resilient operations for mission-critical protection.

PowerVS combines agility, automation, and modernization with enterprise-grade security and cost control. By integrating PowerVS into hybrid cloud strategies, organizations can accelerate innovation, optimize resources, and maintain the trusted reliability of IBM Power while enabling new capabilities such as AI, analytics, and containerized workloads.

4.4 Application integration in hybrid multicloud

Hybrid multicloud architectures extend enterprise workloads across on-premises systems, IBM Power platforms, and public cloud services. Although network connectivity provides the foundation, application-level integration enables seamless business processes and real-time data exchange across these environments. Integration ensures interoperability, operational consistency, and agility without requiring extensive application rewrites.

IBM addresses these challenges through a comprehensive integration portfolio under IBM Cloud Pak for Integration, which includes IBM App Connect, API Connect, Event Streams, and IBM MQ. These components enable organizations to integrate applications, APIs, and data across hybrid environments securely and efficiently.

4.4.1 Integration patterns

Application integration in hybrid multicloud environments typically follows these patterns:

- API-driven integration

  APIs provide a standardized mechanism for connecting heterogeneous systems. RESTful APIs, GraphQL, and gRPC enable flexible, scalable communication between legacy workloads and modern cloud-native applications.

- Event-driven architecture

  Real-time event streaming allows applications to respond to changes instantly. Technologies such as Apache Kafka and IBM Event Streams support asynchronous, decoupled communication for high-performance integration.

- Messaging and queues

  Message-oriented middleware, such as IBM MQ, ensures reliable and secure message delivery between distributed components, even under network disruptions or system failures.

4.4.2 IBM integration solutions

IBM provides a modular integration portfolio under IBM Cloud Pak for Integration (see 3.2.3, “Integration with orchestration platforms” on page 78). For application integration in hybrid multicloud environments, three components play a critical role:

IBM App Connect

IBM App Connect is IBM's flagship integration platform designed to connect applications, APIs, and data across hybrid environments. It enables organizations to automate workflows, integrate systems without rewriting legacy applications, and deliver consistent operations across on-premises, cloud, and containerized deployments.

IBM App Connect provides the following key capabilities:

- Connect: Secure connectivity to on-premises and SaaS applications using 200+ prebuilt connectors.
- Build: Low-code and no-code flow design with AI-powered mapping and automation.
- Transform: High-performance data transformation engine supporting XML, JSON, CSV, EDI, SWIFT, HL7, and more.
- Operate: Deploy integrations as microservices or ESB with unified monitoring and governance.

Hybrid multicloud environments introduce complexity in connecting diverse applications and data sources. IBM App Connect addresses this complexity through the following capabilities:

- Bridging IBM Power workloads (AIX, IBM i, Linux) with SaaS and cloud-native services.
- Supporting API-first strategies for scalable, secure integration.
- Ensuring portability and operational consistency across multiple clouds.

Deployment options

IBM App Connect offers flexible deployment models to meet diverse enterprise needs:

- On-premises: Ideal for organizations with strict security or compliance requirements
- IBM Cloud (ACEaaS): A fully managed iPaaS for rapid integration without infrastructure overhead
- Cloud Pak for Integration: Enterprise-grade orchestration on Red Hat OpenShift for hybrid and multicloud deployments
- Certified containers: Optimized for Kubernetes/OpenShift with dynamic resource management

Deployment patterns

App Connect supports the following deployment patterns:

- Enterprise Service Bus (ESB): Typically deployed on bare metal or VMs, and can also run on cloud environments. ESB deployments support multiple HA/DR technologies for 24x7 operation but tend to be monolithic and more complex to update.
- Container-Native Microservices: Smaller, self-contained integrations managed by container orchestration platforms such as Red Hat OpenShift, Kubernetes, or Docker Swarm. These deployments are easier to update and scale horizontally for high availability.

App Connect provides the following container-native images:

- ACEcc for container deployments

- ACEccMT for managed cloud services

Supported operating systems include AIX, Linux on Power Systems, Linux on System x, Linux on System z, Windows, zCX, and IBM Integration Bus z/OS.

Elements for deployment

The solution requires the following foundational elements:

- Software (on-premises): Requires native hypervisor or virtualization with minimum specifications: 2 CPUs, 8 GB memory, and 4 GB disk. Capacity depends on message throughput, complexity of message formats, and protocols used.
- OCP/Kubernetes: Supports Red Hat OpenShift v4.12–4.17 and Kubernetes 1.25.x–1.29.x on Linux 64-bit. Recommended sizes per integration server:
  - Quick start: 0.3 vCPU, 5 GB storage, 368 MB memory
  - Small: 1.5 vCPU, 512 MB memory
  - Medium: 3 vCPU, 768 MB memory
  - Large: 6 vCPU, 1 GB memory
- ACE as a Service: IBM AWS-managed service with similar sizing options. For flow-based pricing, there is no minimum requirement.
- IBM App Connect Enterprise (ACE) v13 introduces the following enhancements:
- Local designer: No-code authoring tool is now available outside containers for laptop use.
- Unified runtime: One high-performance engine for APIs, events, and integrations.
- AI assistance: AI assistance includes the following capabilities:
  - Mapping assist: AI-powered field-mapping suggestions.
  - Data assist: Auto-generate transformations.
  - AI-generated flows: Create flows by using natural language.
- Expanded connectivity: Expanded connectivity includes the following features:
  - More than 30 new discovery connectors (Salesforce, SAP, Workday, IBM watsonx.ai®).
  - Connector Development Kit (CDK) for custom connectors.
- Advanced tools: Advanced tools include the following capabilities:
  - JSONata mapping node for flexible transformations.
  - Enhanced credential vault with external directory support.
- Observability and governance: Observability and governance capabilities include the following:
  - Business Transaction Monitoring (BTM).
  - Instana integration for end-to-end tracing.
  - Declarative configuration via YAML.
- Performance and security: Performance and security enhancements include the following:
  - Dynamic resource optimization for containers.
  - Default SSL for admin ports.
  - Java 17 support for modern workloads.

For more information, see What's new in Version 13.0?.

Installation packages in ACE v13

IBM App Connect Enterprise 13 provides two installation packages:

- Evaluation edition: Free, intended for evaluation purposes and development on macOS. There is no time limit on the evaluation period, but deployed message flows are restricted to a maximum of one message per second.

- IBM App Connect Enterprise 13.0.1.0: The paid version is available through IBM Passport Advantage® (PPA). After installation, you can configure the mode/edition for Development, Non-Production, or Production environments. For production, most customers choose the Advanced edition, which supports multiple servers and container deployments. The Standard edition is limited to one server per node and does not support containerized environments.

Both Development and Evaluation editions are without charge, making them ideal for initial exploration and testing. Production and Non-Production editions are designed for business-critical workloads. Edition selection depends on the specific use case and deployment strategy.

Figure 4-2 provides a visual overview of IBM App Connect Enterprise 13 installation options and editions. For more information and to register, see IBM App Connect Enterprise Evaluation Edition.

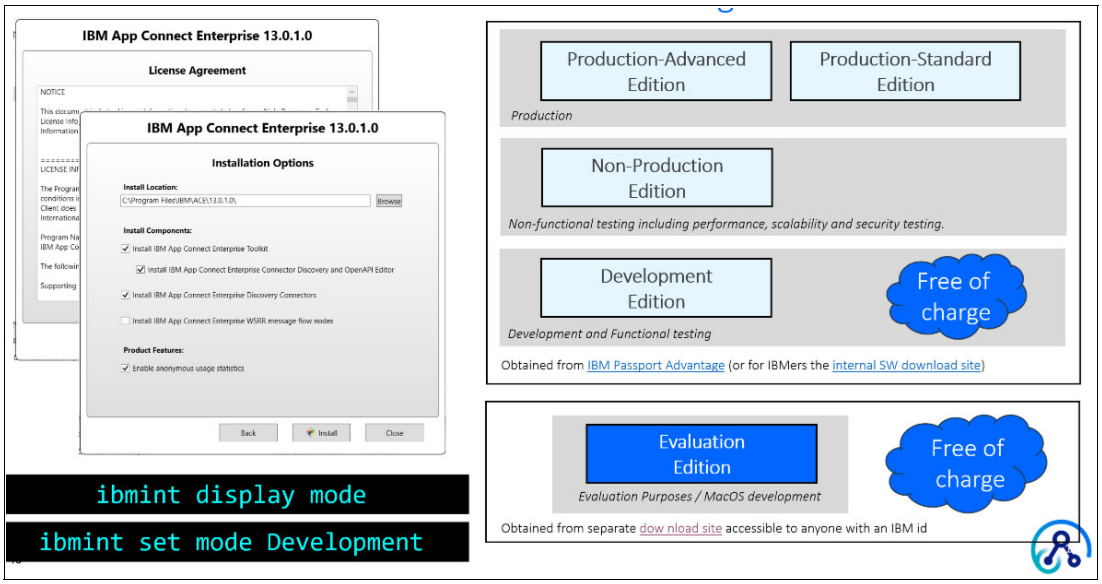


Figure 4-2 Installation option

Installation commands

Administrators can use the following commands to configure environments quickly:

- `ibmint display mode`
- `ibmint set mode Development`

IBM App Connect roadmap and market trends

The roadmap for IBM App Connect Enterprise addresses the most pressing trends shaping integration today. It focuses on four strategic pillars: Test and Automation, Hybrid Cloud Operations, Developer Productivity, and Extended Connectivity as shown in Figure 4-3 on page 102. These priorities align with the following market dynamics:

- Decentralization: As organizations adopt cloud-native architectures, integration is shifting from centralized models to decentralized approaches, empowering teams to manage their own services independently.
- Hybrid and multi-cloud: Enterprises increasingly operate across multiple clouds, introducing challenges around cost optimization, security, and architectural complexity.

- Cross-technology integration: Modern integration tools are becoming more accessible and intuitive, enabling a broader range of users, from business technologists to integration experts, to work efficiently across diverse platforms.
- AI-powered applications and integrations: Artificial intelligence is transforming integration by automating complex tasks such as data mapping and error detection, accelerating delivery and improving accuracy.

This roadmap ensures that IBM App Connect remains a future-ready platform, enabling organizations to integrate faster, smarter, and at scale across hybrid multicloud environments.

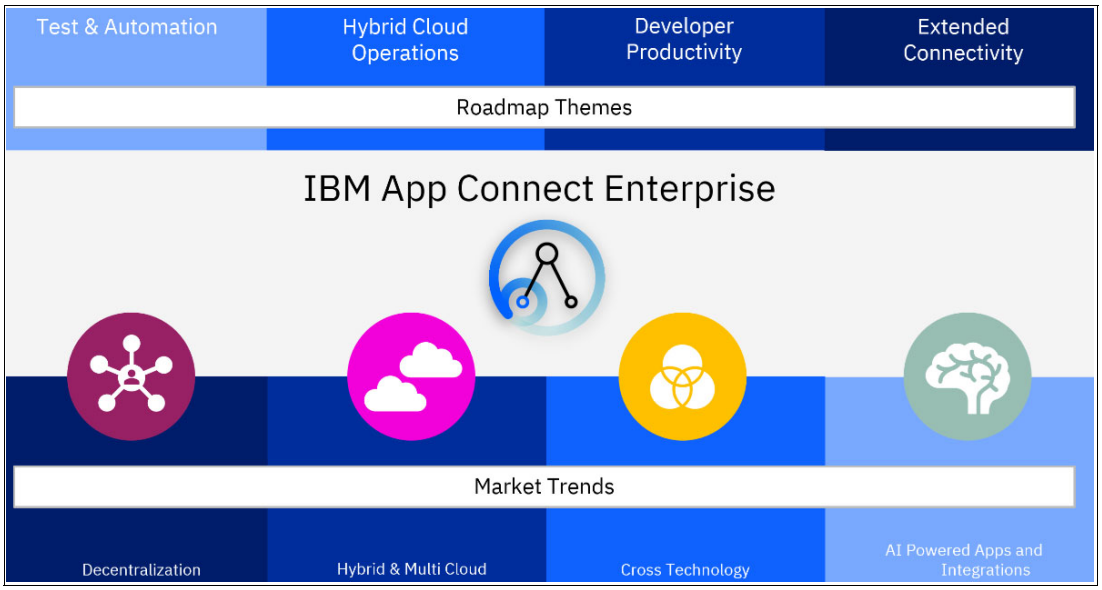


Figure 4-3 IBM App Connect Roadmap and Market Trends

Use cases and best practices

IBM App Connect supports the following use cases:

- Integrating IBM i ERP systems with Salesforce or SAP.
- Real-time event streaming between IBM Power workloads and public cloud services.
- API management for securely exposing IBM Power workloads to external systems.
- IoT and manufacturing integration using OPC-UA nodes.

Follow these best practices when implementing IBM App Connect:

- Use secure endpoints and encryption for all data flows.
- Standardize API governance across hybrid environments.
- Combine App Connect with IBM MQ for reliable messaging in mission-critical workflows.
- Use AI-powered mapping and templates to accelerate delivery.

IBM Event Streams

IBM Event Streams is IBM's enterprise-grade implementation of Apache Kafka, providing a robust and scalable foundation for event-driven architectures. It enables organizations to collect, store, and distribute streams of real-time business events across the enterprise, ensuring that critical data flows reliably between applications and services.

Figure 4-4 on page 103 shows Event Streams in IBM Cloud.

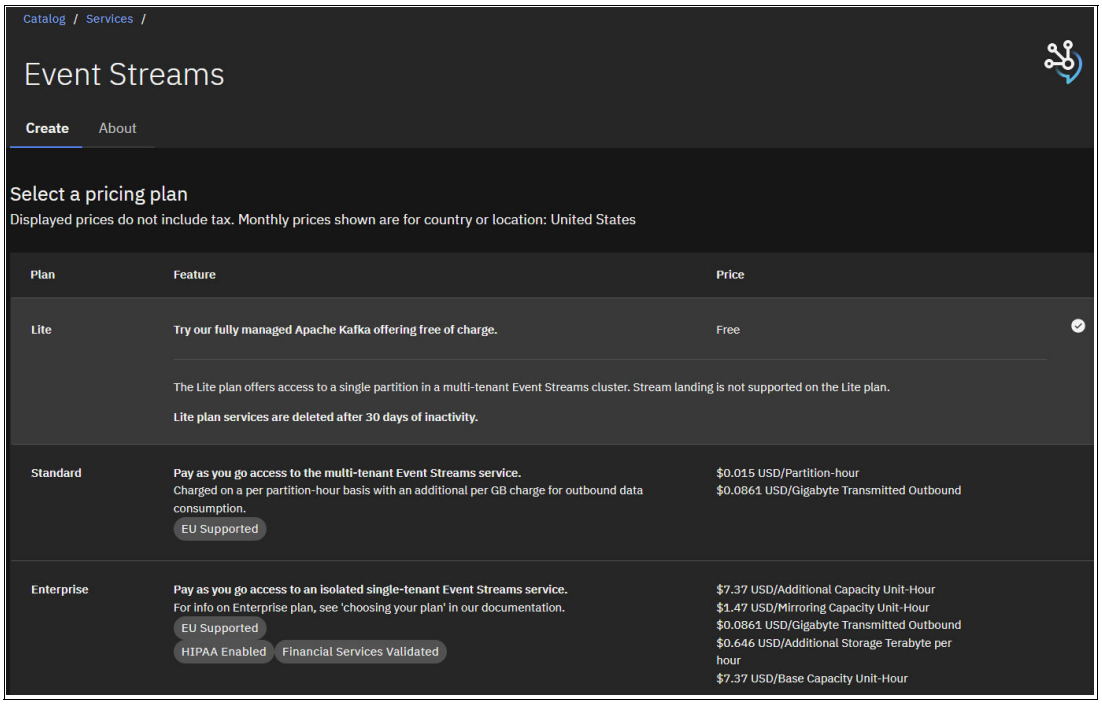


Figure 4-4 Event Streams

Event Streams and IBM Event Automation

Event Streams is a core component of the IBM Event Automation portfolio. Although Event Streams focuses on reliable event streaming, Event Automation adds intelligence and orchestration. Together, they enable organizations to perform the following actions:

- Detect business events as they occur
- Act in real time
- Automate decisions to improve efficiency and outcomes

By combining Event Streams with Event Processing and Event Endpoint Management, IBM delivers a complete solution for building event-driven enterprises.

How IBM Event Streams fits into hybrid cloud

Modern businesses operate across multiple environments including on-premises systems, private clouds, and public clouds. This creates challenges in moving data quickly and securely between these environments. IBM Event Streams solves this by providing a consistent, scalable, and secure event streaming platform that works seamlessly across hybrid cloud deployments, which includes the following examples:

- Container-native architecture

  Event Streams is built for Red Hat OpenShift and Kubernetes, making it easier to deploy in any cloud or on-premises environment. This flexibility ensures that organizations can run Kafka clusters wherever their data resides.

- Unified management across environments

  Operators simplify deployment and scaling of Kafka clusters in hybrid setups. Whether workloads run in a data center or in the cloud, Event Streams provides a single management experience.

- Resilience and high availability

  Event Streams supports multi-zone clusters and geo-replication, allowing data to be synchronized across regions and clouds. This ensures business continuity even during outages.

- Integration with IBM Cloud Pak for Integration

  Event Streams works as a core component of IBM Cloud Pak for Integration, enabling event-driven architectures alongside APIs, messaging, and application integration.

Capabilities and advantages of IBM Event Streams

IBM Event Streams delivers a comprehensive set of capabilities for building event-driven architectures in hybrid cloud environments. IBM Event Streams provides the following core capabilities:

- Enterprise-grade implementation of Apache Kafka with enhanced features
- Built-in schema registry for maintaining data consistency
- Rich connector catalog with more than 70 options for SaaS applications and IBM systems
- REST Producer API that helps to simplify event publishing without requiring a Kafka client
- Integration with IBM MQ, IBM Z, IBM DataPower®, and App Connect Enterprise for seamless two-way messaging
- Observability features powered by Instana and OpenTelemetry for real-time monitoring and troubleshooting

Recent innovations include the following enhancements:

- KRaft mode: Eliminates ZooKeeper, simplifying deployments and improving scalability
- Kafka node pools: Allow different broker configurations within the same cluster, optimizing resources for mixed workloads
- Tiered storage: Integrates with Amazon S3 to reduce local disk usage and extend data retention for compliance and analytics
- Advanced consumer rebalance protocols: Provide greater stability
- Share groups: Enable collaborative message consumption
- Improved topic operator: Enables efficient cluster management

IBM Event Streams delivers the following business benefits:

- Deployment flexibility: Deploy anywhere, on-premises, public cloud, or hybrid and increase in scale to handle millions of events per second.
- Enterprise-grade security: Encryption and role-based access control protect data and resources.
- Developer productivity: Intuitive tools, APIs, and starter templates accelerate development.
- Resilience: Multi-zone clusters and geo-replication ensure business continuity.

IBM Event Streams provides a secure, scalable, and future-ready platform for real-time data streaming across hybrid cloud environments.

For more information, see Event Streams.

4.4.3 IBM MQ

IBM MQ is an enterprise messaging middleware that is designed to provide secure, reliable, and exactly-once delivery of messages between applications. It is widely used in mission-critical environments where data integrity and guaranteed delivery are nonnegotiable. IBM MQ enables applications to communicate asynchronously, decoupling producers and consumers so that each can operate independently without affecting the other's availability or performance.

Message queuing is the foundation of this capability. Instead of requiring synchronous communication where both applications must be online at the same time, IBM MQ uses queues as a persistent buffer. A sending application places a message on a queue, and the receiving application retrieves it when ready, often described as *fire and forget*. This approach provides the following benefits:

- Resilience: Messages are not lost if a system or network fails.
- Isolation: Changes in one component do not disrupt others.
- Scalability: Workloads can be distributed across multiple consumers for load balancing.

This decoupling improves resilience and flexibility in distributed environments, as shown in Figure 4-5.

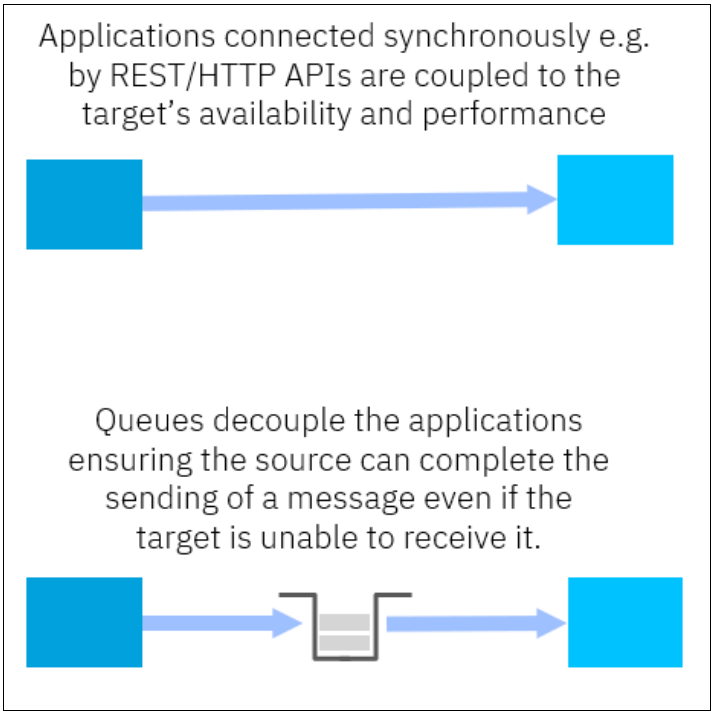


Figure 4-5 Couple and decouple queue example

IBM MQ supports advanced messaging patterns such as point-to-point, publish and subscribe, and request and response, along with features including transactional messaging, message persistence, priority handling, and dead-letter queues. These capabilities make IBM MQ a cornerstone for hybrid cloud integration, enabling consistent messaging across on-premises systems, IBM Power Virtual Server, public clouds, and containerized environments.

Figure 4-6 illustrates how IBM MQ extends beyond simple message queuing to provide a secure, reliable, and scalable messaging backbone that spans hybrid and multicloud environments, connecting applications across different platforms, operating systems, and geographies.

IBM MQ enables organizations to build resilient, event-driven architectures where applications communicate asynchronously with guaranteed delivery and transactional integrity, even in complex distributed systems. IBM MQ is the foundation for integrating mission-critical workloads running on-premises, in public clouds, and on container platforms such as Red Hat OpenShift, including deployments on IBM Power Virtual Server.

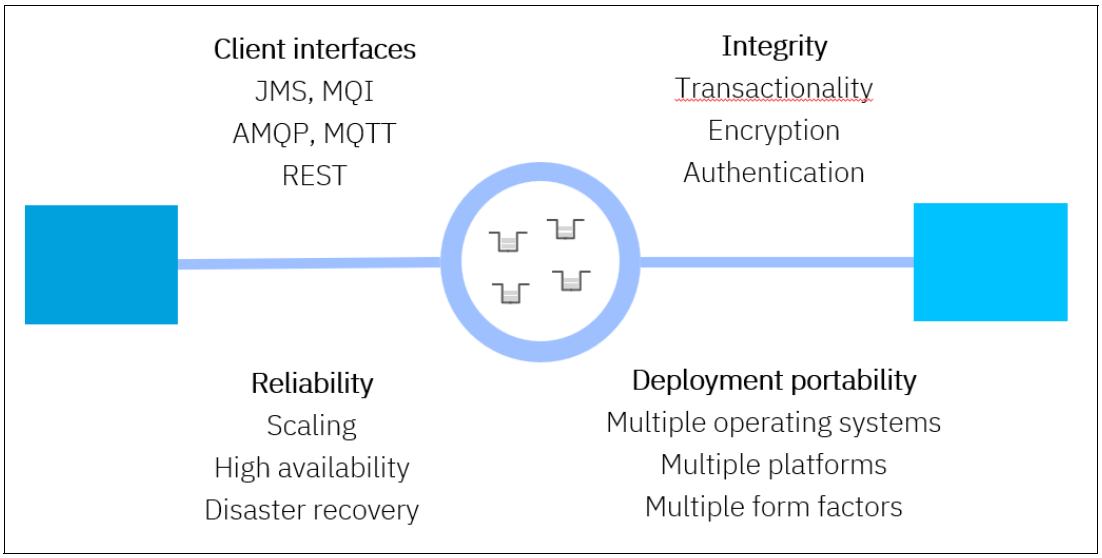


Figure 4-6 Queue manager architectural

IBM MQ in hybrid cloud environments

Modern businesses operate across distributed environments. Applications run on-premises, in public clouds, and on container platforms. IBM MQ provides the messaging backbone that connects these environments, ensuring data moves securely and reliably across all application deployments.

IBM MQ supports the following platforms:

- On-premises systems: AIX, Linux, Windows, IBM i, and IBM Z
- Public clouds: IBM Cloud, AWS, Azure, and Google Cloud
- Containers: Red Hat OpenShift and Kubernetes
- IBM Power Virtual Server: Mission-critical workloads in IBM Cloud remain connected to other systems

Applications on IBM Power Virtual Server integrate with cloud-native services and on-premises systems without code refactoring.

IBM MQ guarantees message delivery even during network failures. It ensures transactional integrity, which is critical for workloads such as banking or payment systems where losing or duplicating a message is unacceptable. IBM MQ supports the following integration scenarios:

- Connecting banking applications on IBM i with payment gateways
- Messaging between on-premises Power workloads and cloud services
- Supporting event-driven architectures in hybrid environments

IBM MQ runs on-premises, in the cloud, and in containers with the same capabilities. Organizations can choose the deployment model that fits their strategy and budget while preparing for multicloud and container adoption.

Figure 4-7 shows consistency to connect application wherever they reside.

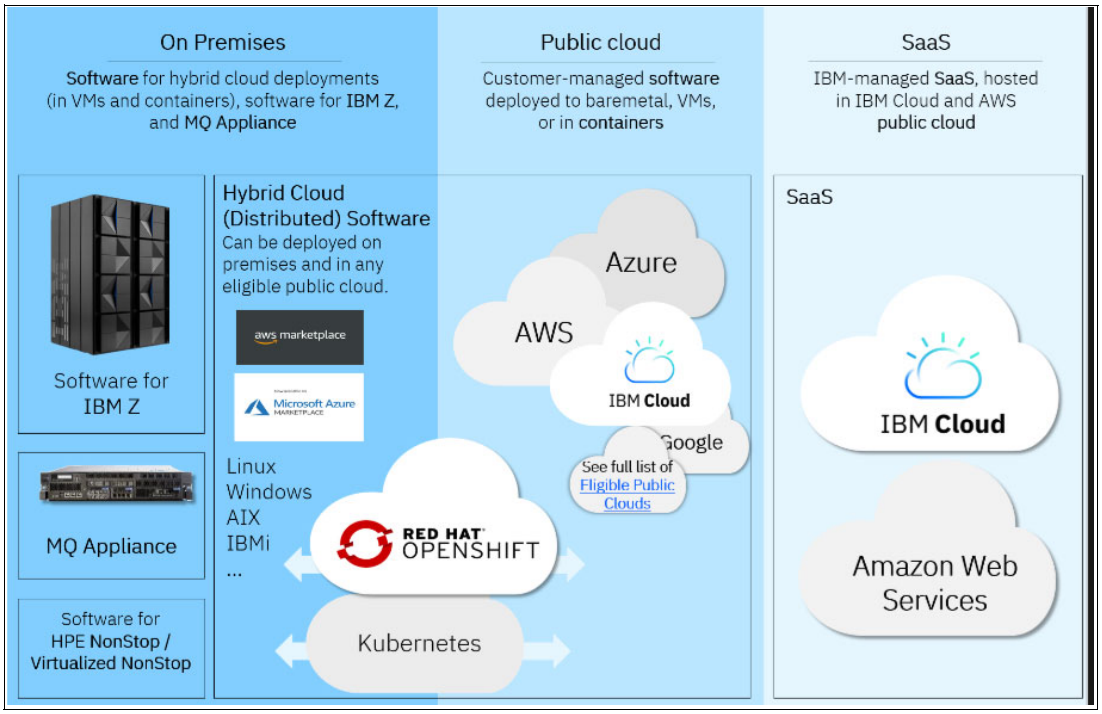


Figure 4-7 IBM MQ portfolio diagram

IBM MQ delivers reliability and control for hybrid cloud environments through the following benefits:

- Resilience: Built-in HA and DR for uninterrupted messaging
- Flexibility: Deploy anywhere, on-premises, IBM Cloud, or containers
- Security: End-to-end encryption and compliance-ready features

IBM MQ improves operational efficiency by enabling automation through DevOps tools and simplifying day-to-day management.

IBM MQ Advanced includes the following capabilities for advanced hybrid cloud scenarios:

- Managed File Transfer (MFT) for secure file movement
- Kafka connectors for event streaming integration
- Telemetry support through MQTT for IoT devices
- Enhanced HA/DR options and container orchestration

IBM MQ is available as a managed service on IBM Cloud and AWS, reducing operational impact. SaaS deployments provide the following features:

- Automatic upgrades and patching
- Built-in HA across multiple availability zones
- Private networking for secure connectivity
- Terraform support for automated provisioning

IBM MQ on IBM Power Virtual Server supports the following capabilities:

- Deploy AIX or Linux-based queue managers in IBM Cloud
- Connect securely to IBM MQ SaaS or containerized IBM MQ on Red Hat OpenShift
- Implement HA and DR strategies across regions by using IBM MQ clustering or CRR
- Modernize messaging without rewriting applications, by using the performance of IBM Power Systems in a hybrid cloud model

For more information, see the following sources:

- IBM MQ documentation
- IBM Developer: IBM MQ
- IBM Training

4.5 AI workloads in the cloud

Artificial Intelligence is transforming business operations, from predictive analytics to generative AI and automation. IBM Power Systems are designed to handle these demanding workloads with performance, scalability, and security. In hybrid cloud environments, IBM Power combines on-premises reliability with cloud flexibility, enabling organizations to run AI models where it makes the most sense, whether on IBM Power Virtual Server or integrated with SaaS offerings like IBM watsonx.

4.5.1 AI Workloads with watsonx SaaS and IBM Power Virtual Server

IBM watsonx SaaS on IBM Power Virtual Server (PowerVS) combines IBM's enterprise AI platform with the flexibility of Power-based virtual infrastructure in the cloud. This integration brings advanced AI capabilities, which include machine learning, generative AI, and AI governance, directly to mission-critical workloads in a secure hybrid cloud environment. Instead of moving sensitive data off-platform, organizations can keep their information within IBM Cloud while consuming watsonx services, ensuring compliance and reducing risk. This approach aligns with hybrid cloud principles: portability, security, and workload placement flexibility.

IBM watsonx SaaS offers the following core services:

- IBM watsonx.ai: Build and deploy machine learning and generative AI models
- watsonx.data: Access and analyze enterprise data efficiently
- watsonx.governance: Manage the AI lifecycle with transparency and accountability

When deployed on PowerVS, these services securely connect to enterprise data stored in IBM's virtual server infrastructure, enabling AI-driven insights without compromising performance or compliance. This is especially valuable for organizations that rely on IBM Power Systems for high-performance computing and need to integrate AI into existing workflows without migrating sensitive data.

The IBM Satellite® Connector provides secure communication between PowerVS and watsonx services. For example, in a banking scenario, fraud investigators can use natural language prompts to query transaction data stored on PowerVS without writing SQL or relying on database administrators. This capability accelerates decision-making and improves operational efficiency, making AI accessible to business users without deep technical expertise.

IBM watsonx SaaS on PowerVS supports the following use cases:

- Fraud detection and investigation: AI models analyze transaction patterns and flag anomalies.
- Customer experience enhancement: AI-powered chatbots and voice agents deliver personalized support.
- Data-driven marketing: AI insights identify high-value clients and tailor campaigns.
- AI governance: Models remain viewable, explainable, and compliant with regulations.

This integration delivers the following benefits:

- Securely run AI workloads without data migration
- Maintain consistent governance across environments
- Scale AI training and inference using the elasticity of PowerVS
- Modernize IT infrastructure while using existing investments in IBM Power technology

4.5.2 Power11 and advanced AI capabilities

IBM's Power11 servers, announced in July 2025, represent a major advancement for AI and hybrid cloud. Power11 includes the following innovations:

- IBM Spyre™ Accelerator: A system-on-a-chip designed to accelerate AI inference workloads, reducing latency and improving performance.
- Energy-efficient mode: Delivers up to 28% better efficiency compared to maximum performance settings.
- Advanced cyber resilience: Real-time ransomware detection and immutable snapshots.
- Combined with IBM watsonx Orchestrate®, which includes over 150 pre-built AI agents for HR, sales, procurement, and IT, these capabilities enable enterprises to automate operations and drive innovation at scale.

Power11 and watsonx on PowerVS support the following use cases:

- AI-driven IT automation: Use watsonx Orchestrate and IBM Concert to streamline patching, diagnostics, and workload mobility.
- Secure AI inference: Run locally with Spyre, reducing latency and improving compliance.
- Hybrid cloud modernization: Migrate legacy workloads to PowerVS while integrating AI and analytics.
- Cyber resilience: Use Power Cyber Vault for subminute ransomware detection and recovery.

IBM watsonx SaaS on PowerVS empowers organizations to harness the full potential of their data, modernize IT infrastructure, and drive innovation through AI. All within a secure and scalable hybrid cloud environment. By combining watsonx services with the performance and reliability of IBM Power Systems, enterprises can adopt AI without compromising compliance, security, or operational efficiency.

4.5.3 AI Acceleration on IBM Power

This section provides a brief overview of AI capabilities in the context of hybrid cloud. For comprehensive coverage of AI features, refer to the dedicated IBM Redbooks publication on AI acceleration.

AI workloads increasingly require high performance, low latency, and flexible deployment options across hybrid environments. IBM Power11 addresses these demands through integrated acceleration technologies and software environments that are designed for enterprise AI. Power11 introduces significant AI innovations, including enhanced Matrix Multiply Assist (MMA) for CPU-resident inference and the new IBM Spyre accelerator for scaling generative AI workloads across hybrid environments. This section examines how IBM Power enables AI inference using MMA within the CPU pipeline, scales to generative AI workloads with Spyre, and integrates with Red Hat OpenShift and watsonx services to deliver governed AI deployments across hybrid cloud environments.

AI and ML on Red Hat OpenShift for IBM Power

Red Hat OpenShift on IBM Power supports AI and machine learning workloads through optimized libraries including OpenBLAS, ONNX Runtime, PyTorch, and TensorFlow

Nodes can be labeled with `ai.inference.accelerator=mma` to ensure AI workloads are scheduled on hardware capable of MMA acceleration.

AI models can be deployed by using GitOps workflows, Kubeflow, or Watson OpenScale, integrated with ModelOps pipelines for drift detection and retraining.

IBM Cloud Pak solutions, including Cloud Pak for Data, are fully supported on Red Hat OpenShift running on IBM Power Virtual Server (PowerVS). This support enables AI workloads to run alongside traditional applications while using Power MMA capabilities for inference and analytics.

Acceleration technologies: MMA and Spyre

IBM Power11 integrates Matrix Multiply Assist (MMA) directly into the CPU pipeline to accelerate dense linear algebra operations that underpin AI and ML inference.

MMA provides the following capabilities:

- Supports mixed-precision floating point (FP16, bfloat16) and integer types for quantized inference, enabling low-latency "AI next to data" in application or database logical partitions (LPARs) without requiring external devices
- Delivers up to 80% higher on-chip inference performance compared to Power10 through expanded MMA units and optimized mixed-precision support (FP16, bfloat16)
- Enables low-latency AI processing directly within application or database partitions without requiring discrete GPUs

For workloads that exceed CPU-resident capabilities, IBM introduces Spyre, a PCIe-attached accelerator purpose-built for inference at scale. Spyre provides the following specifications and capabilities:

- 32 cores and 128 GB LPDDR5 memory
- Designed for transformer-based and generative AI models
- Up to 16 Spyre cards can be clustered in Power11 systems, providing scale-out capability for high-throughput inference workloads

MMA and Spyre work together: MMA handles lighter, in-place inference close to the data, and Spyre scales complex models for advanced AI scenarios.

Hybrid deployment models

IBM Power supports several deployment patterns to operationalize AI across hybrid environments.

In-place scoring with burst to PowerVS

MMA-optimized inference can run directly in application or database LPARs on-premises for low-latency, data-proximate scoring. When additional capacity is required for development, testing, regional proximity, or disaster recovery, the same container images can be replicated in IBM PowerVS. This ensures consistent governance, security, and operational processes across environments.

CPU-first with optional Spyre

Enterprises can begin with MMA-based inference for operational workloads and introduce Spyre accelerators as models grow in size or complexity. Spyre can be deployed on-premises first and later extended to PowerVS as cloud offerings mature, maintaining a consistent operational model across environments.

PowerVS with watsonx.ai Integration

The watsonx Toolkit on PowerVS, using Satellite Connector, provides a governed pattern to keep sensitive data on PowerVS while invoking watsonx.ai® services such as large language models and AI agents. Prompts and metadata are routed securely to watsonx.ai, and results are written back to Power workloads for auditability and compliance. This design enables advanced use cases like copilots and retrieval-augmented generation (RAG) without moving regulated datasets.

Design considerations

Consider the following design factors when deploying AI workloads on IBM Power:

- Placement decisions: Prioritize MMA when inference must be colocated with data, such as transactional paths or strict latency requirements. Spyre is recommended for high-throughput generative AI layers.
- Portability: Standardize on ONNX Runtime or MMA-enabled containers, allowing the same model artifacts to move between on-premises and PowerVS environments.
- Operational consistency: Power11 day-one availability in PowerVS enables hybrid billing and alignment with enterprise HA/DR and governance policies.
- Energy efficiency and resilience: Power11 introduces Energy Efficient mode, reducing power consumption by up to 28%, and autonomous IT features such as zero planned downtime and ransomware detection through Power Cyber Vault. These capabilities enhance operational resilience and sustainability in hybrid environments. For more information, see *Zero Downtime, Automation, and Energy Optimization,* SG24-8596.

Table 4-1 lists a brief description of AI-ready features.

Table 4-1 A hybrid AI-ready feature summary

Feature	Description
Integrated MMA	On-chip acceleration for AI inference directly on CPU cores, reducing latency.
External Spyre	PCIe-attached accelerator with 32 cores and 128 GB LPDDR5 for GenAI workloads.
PowerVS integration	Consistent hardware and OS experience across on-premises and IBM Cloud.

Feature	Description
Focus	Enterprise AI inference and ML modernization with CPU-first and scalable options.
watsonx Integration	Secure AI services with governed data residency.
Energy Efficient Mode	Optimized workload scheduling for sustainability.
Autonomous IT	Zero downtime and ransomware protection.

4.6 Operational strategies for hybrid environments

Hybrid environments introduce complexity in managing workloads, resources, and operational visibility across on-premises and cloud platforms. Effective strategies ensure performance, resilience, and cost optimization while maintaining governance and compliance. This section explores practical approaches for balancing workloads, making placement decisions, and achieving end-to-end observability.

4.6.1 Workload balancing between on-premises and PowerVS

In hybrid environments, workload balancing is a strategic enabler for agility, cost efficiency, and business continuity. IBM Power11 combined with IBM Power Virtual Server offers the ability to extend enterprise-grade Power Systems capabilities into IBM Cloud without sacrificing performance, security, or operational consistency.

Balancing workloads between on-premises and PowerVS starts with understanding the drivers for hybrid adoption. Organizations often face seasonal demand spikes, unpredictable workloads, or disaster recovery requirements. Instead of overprovisioning physical servers, PowerVS provides burst capacity or temporary workload offloading. For example, a retail customer running AIX or IBM i workloads on Power11 can maintain core transaction processing on-premises while shifting analytics or batch jobs to PowerVS during peak shopping seasons. This approach reduces capital expenditure while maintaining service-level agreements (SLAs).

Workload balancing involves policy-driven orchestration and automation frameworks. IBM Cloud Pak for Automation and Ansible enable dynamic scaling through the following policy examples:

- Latency-sensitive workloads remain on-premises for predictable performance.
- Compute-intensive but non-critical tasks migrate to PowerVS when usage exceeds defined thresholds.
- Disaster recovery workloads replicate to PowerVS using Global Replication Services (GRS) for rapid failover.

These policies can be enforced through IBM Turbonomic, which provides real-time resource optimization and placement recommendations based on performance, cost, and compliance metrics. For instance, if an SAP HANA workload running on Power11 approaches CPU saturation, Turbonomic can trigger an automated scale-out to PowerVS, ensuring uninterrupted operations without manual intervention.

Networking and data movement are critical considerations. IBM Direct Link offers secure, low-latency connectivity between on-premises and IBM Cloud, enabling seamless workload

migration. This connectivity minimizes performance degradation and supports compliance for industries like finance and healthcare. Additionally, PowerVS supports shared processor pools, enabling pay-per-use pricing models for operational flexibility.

As an example, A global manufacturing company running IBM i for ERP can keep production workloads on-premises while deploying development and testing environments in PowerVS. This accelerates innovation, isolates nonproduction activities from mission-critical systems, and reduces risk and cost.

Workload balancing between on-premises and PowerVS provides strategic flexibility, enabling organizations to optimize resources, control costs, and maintain resilience without compromising the enterprise-grade reliability of IBM Power Systems. IBM hybrid cloud solutions deliver the elasticity of cloud with the predictability of on-premises infrastructure, positioning Power11 and PowerVS as the foundation for modern, adaptive IT strategies.

4.6.2 Placement strategies and decision framework

In hybrid environments, workload placement is a structured decision that affects performance, cost, compliance, and business agility. Position IBM Power11 and IBM Power Virtual Server as complementary platforms that enable clients to make intelligent placement decisions based on real-time insights and policy-driven automation. The goal is to help clients answer a critical question: Where should each workload run to maximize value?

Begin with workload profiling. Guide clients to classify workloads by latency sensitivity, compute intensity, regulatory requirements, and interdependencies. For example, mission-critical SAP HANA or Oracle workloads with stringent latency SLAs often remain on-premises, and analytics, development, and seasonal batch jobs can be offloaded to Power Virtual Server for elasticity. Workload profiling is essential for avoiding performance degradation and ensuring compliance.

Consider the following placement drivers:

- Performance and latency: Applications requiring submillisecond response times typically stay on-premises. Power Virtual Server is ideal for workloads tolerant of network latency or those benefiting from IBM Cloud Direct Link for secure, low-latency connectivity.
- Cost optimization: Clients seeking to reduce capital expenses can use Power Virtual Server for temporary capacity or disaster recovery, paying only for consumed resources through shared processor pools.
- Compliance and data residency: Industries like finance and healthcare might require certain workloads to remain on-premises while using Power Virtual Server for nonsensitive operations.
- Scalability and innovation: Development and testing environments can be deployed in Power Virtual Server to accelerate time-to-market without affecting production systems.

Structure the decision framework in three phases:

1. Assessment: Use IBM Support Insights and workload analytics to evaluate resource usage and performance baselines.
2. Policy definition: Define placement rules using IBM Cloud Pak for Automation and Ansible, ensuring alignment with SLAs, compliance, and cost objectives.
3. Execution and optimization: Implement placement policies with orchestration tools and use IBM Turbonomic for continuous optimization. Turbonomic's AI-driven recommendations ensure workloads are placed where they deliver the best performance-to-cost ratio.

Consider the following example: A global bank running AIX workloads for transaction processing can keep core systems on-premises while deploying fraud detection analytics in Power Virtual Server. This approach isolates compute-intensive analytics from latency-sensitive transactions, improving performance and reducing infrastructure costs.

IBM hybrid solutions provide policy-driven placement intelligence, enabling clients to optimize resources dynamically. By combining Power11 reliability with Power Virtual Server elasticity, organizations gain a flexible, secure, and cost-effective platform for modern workloads.

4.6.3 Monitoring and observability across hybrid environments

Monitoring and observability in hybrid environments are foundational for operational resilience, compliance, and proactive optimization. When workloads span IBM Power11 on-premises systems and IBM Power Virtual Server in IBM Cloud, visibility must extend beyond traditional infrastructure metrics. Position IBM's observability stack as a differentiator that delivers unified insights across compute, storage, and network layers, enabling clients to manage complexity without sacrificing control.

Why observability matters in hybrid architectures

Hybrid environments introduce multiple failure domains and dynamic resource allocation. Without integrated observability, clients risk blind spots that lead to SLA breaches, security vulnerabilities, and inefficient resource usage. IBM addresses these challenges through a layered approach combining real-time telemetry, AI-driven analytics, and policy-based automation.

Core components of IBM's observability strategy

IBM's observability strategy includes the following components:

- IBM Instana: Provides full-stack observability for applications and infrastructure across on-premises and Power Virtual Server environments. Instana automatically discovers services, maps dependencies, and delivers real-time metrics with one-second granularity. For example, if an AIX workload in Power Virtual Server experiences latency due to network congestion, Instana pinpoints the root cause by correlating application traces with infrastructure metrics.
- IBM Turbonomic: Offers actionable optimization beyond monitoring. Turbonomic continuously analyzes resource consumption and recommends placement or scaling actions to maintain performance and cost efficiency. In hybrid environments, this means dynamically shifting workloads between Power11 and Power Virtual Server based on real-time demand.
- IBM Support Insights and AIOps: Integrates predictive analytics for proactive maintenance. By analyzing historical telemetry and anomaly patterns, AIOps forecasts potential failures and triggers automated remediation workflows through IBM Cloud Pak for Automation.
- Unified dashboards: IBM Cloud Monitoring consolidates metrics from Power Virtual Server and on-premises systems into a single interface. This eliminates the need for fragmented tools and supports compliance auditing with detailed logs of workload movements and configuration changes.

Technical best practices for hybrid observability

Apply the following best practices:

- Enable end-to-end tracing: Use Instana to trace transactions across hybrid boundaries, ensuring visibility from application layer to hardware.

- Use secure connectivity: IBM Cloud Direct Link ensures telemetry data flows securely between environments without introducing latency.
- Automate alerting and remediation: Configure policy-driven alerts for SLA breaches and integrate with Ansible playbooks for automated corrective actions.
- Maintain compliance and audit readiness: Maintain immutable logs of workload placement and configuration changes for regulatory audits.

Consider the following example: A global retailer running IBM i workloads on Power11 uses Power Virtual Server for seasonal analytics. Instana monitors application performance across both environments, and Turbonomic optimizes resource allocation during peak demand. When anomaly detection predicts potential CPU saturation in Power Virtual Server, AIOps triggers an automated scale-out policy, ensuring uninterrupted service without manual intervention.

IBM delivers observability with intelligence, enabling clients to monitor, optimize, and automate hybrid operations. This positions IBM Power11 and Power Virtual Server as the backbone of a resilient, adaptive IT strategy.

5

Modernization with Red Hat OpenShift on IBM Power and IBM Cloud

Modernizing enterprise workloads is critical for achieving agility, scalability, and operational efficiency in hybrid cloud environments. Red Hat OpenShift provides a Kubernetes-based platform that enables organizations to build, deploy, and manage containerized applications consistently across on-premises, IBM Power Virtual Server, and IBM Cloud. By using Red Hat OpenShift on IBM Power, you can modernize incrementally, preserving existing investments in AIX and IBM i and introducing cloud-native architectures.

Red Hat OpenShift accelerates digital transformation on IBM Power Systems and IBM Cloud through flexible deployment options, phased modernization strategies, and advanced capabilities such as AI integration and security enhancements. Red Hat OpenShift Platform Plus on IBM Power servers delivers a comprehensive set of services for multi-cluster security, compliance, and data management across hybrid environments.

This chapter covers the following topics:

- “Red Hat OpenShift on IBM Cloud”
- “Phased modernization with IBM PowerVS and Red Hat OpenShift”
- “Support for Red Hat OpenShift Platform Plus on IBM Power Servers”

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5.1 Red Hat OpenShift on IBM Cloud

Red Hat OpenShift Container Platform deploys cloud-native and legacy applications across hybrid cloud environments at scale. Built on Kubernetes, Red Hat OpenShift adds enterprise-grade features such as automated installation, lifecycle management, and integrated developer tools.

Beginning with Red Hat OpenShift release 3.10 in 2018, IBM Power hardware supports Red Hat OpenShift. IBM and Red Hat continue to strengthen this integration. Starting with Red Hat OpenShift version 4.19, deployment on IBM Power11 is fully supported, combining the reliability, security, and performance of IBM Power hardware with the flexibility of Kubernetes and Red Hat OpenShift.

Red Hat OpenShift on IBM Cloud (ROKS) is a fully managed cloud service that simplifies cluster lifecycle management and accelerates application delivery. ROKS is available in six IBM Cloud regions worldwide, spanning more than 35 data centers.

Figure 5-1 shows the IBM Cloud catalog page for creating a Red Hat OpenShift cluster.

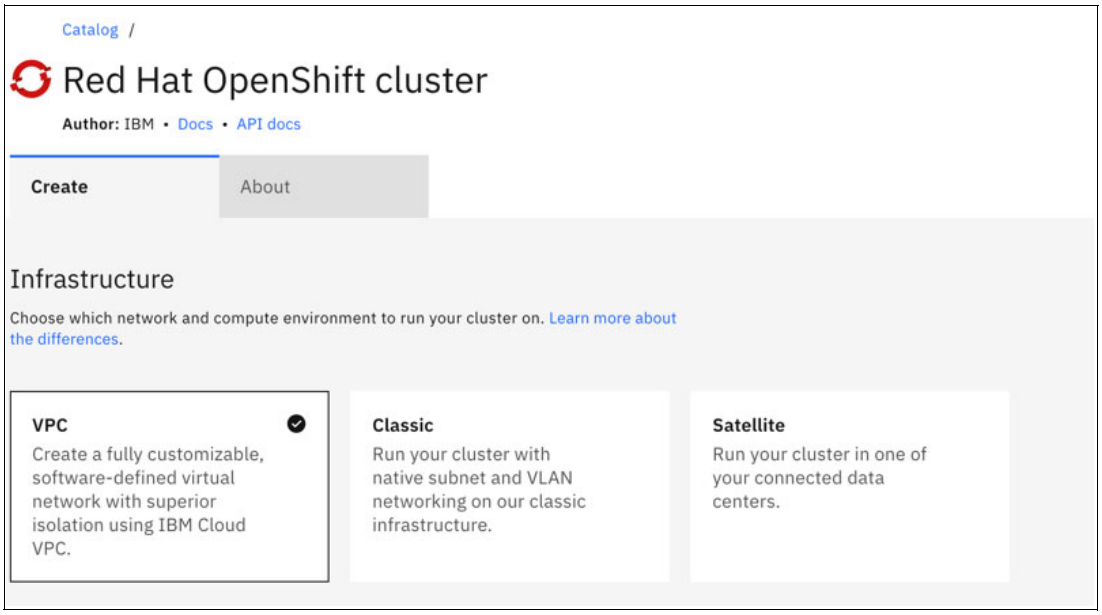


Figure 5-1 Red Hat OpenShift cluster

Red Hat OpenShift delivers a comprehensive foundation for operating secure, resilient, and fully managed Red Hat Enterprise Linux (RHEL) clusters in the cloud. The platform provides the following key capabilities:

- Cluster lifecycle automation: Single-click provisioning of RHEL clusters, automated installation, configuration, upgrades, and patch management
- Enterprise security and compliance: Built-in security hardening, image signing, vulnerability scanning, and compliance for HIPAA, PCI, SOC2, and ISO standards
- Resilient architecture: Automatic multi-zone deployments with failure recovery and highly available master nodes

- Integrated IBM Cloud Services: IBM Key Protect, KYOK with IBM Cloud Hyper Protect Crypto Services, IBM Cloud Identity and Access Management (IAM), IBM Cloud Activity Tracker, IBM Cloud Security Advisor, IBM Cloud Container Registry, and Vulnerability Advisor
- Observability and Storage: Built-in monitoring, logging, and storage options (File, Block, and IBM Cloud Object Storage).

Red Hat OpenShift supports multiple deployment models for flexibility:

- IPI (Installer-Provisioned Infrastructure)
- UPI (User-Provisioned Infrastructure)
- Agent-based Installation
- Assisted Installation

These models allow organizations to choose between fully automated provisioning or custom configurations for complex environments.

5.1.1 Installer-provisioned infrastructure

The installer-provisioned infrastructure (IPI) model automates the creation of Red Hat OpenShift clusters, including provisioning compute nodes, networking, and storage resources. Organizations can deploy production-ready clusters in less than two hours.

IPI uses the Red Hat OpenShift Container Platform installation program, also called the openshift-installer, to perform the following tasks:

- Generate manifests and ignition files
- Automate infrastructure provisioning
- Monitor deployment progress

This automation ensures consistency across environments and simplifies the operational lifecycle of Red Hat OpenShift clusters.

Currently, IPI deployment on IBM Power supports exclusively IBM Cloud Power Virtual Server (PowerVS). Plans for future releases include support for additional environments such as Power Virtualization Center (PowerVC).

Benefits of IPI deployment

The IPI model offers several advantages for organizations adopting Red Hat OpenShift on IBM Power:

- Minimal manual intervention: Infrastructure provisioning and cluster setup are fully automated.
- Rapid deployment: Clusters become operational in under two hours.
- Scalability: Compute resources scale up or down easily.
- Automated teardown: Clusters can be removed quickly when no longer needed.

Deployment workflow

Deploying an Red Hat OpenShift cluster by using IPI involves the following steps:

1. Prepare the configuration

   Create a YAML configuration file called `install-config.yaml` that defines the cluster layout, including networking, compute resources, and storage.

2. Run the following installation command:

   `openshift-install create cluster`

   The installer provisions the infrastructure and deploys the cluster automatically.

3. Monitor progress

   The installation program provides real-time feedback on deployment status.

Interactive configuration example

To deploy an Red Hat OpenShift cluster by using the Installer-Provisioned Infrastructure (IPI) method, you need an `install-config.yaml` file. This file defines the cluster's settings, such as platform, region, and domain. You create this file by using the Red Hat OpenShift installer's interactive process.

When you run the command **`openshift-install create install-config`**, the installer prompts you with questions. The installer uses your answers to generate the configuration file automatically.

Figure 5-2 shows the use of interactive prompts for creating `install-config.yaml` using openshift-installer on IBM PowerVS.

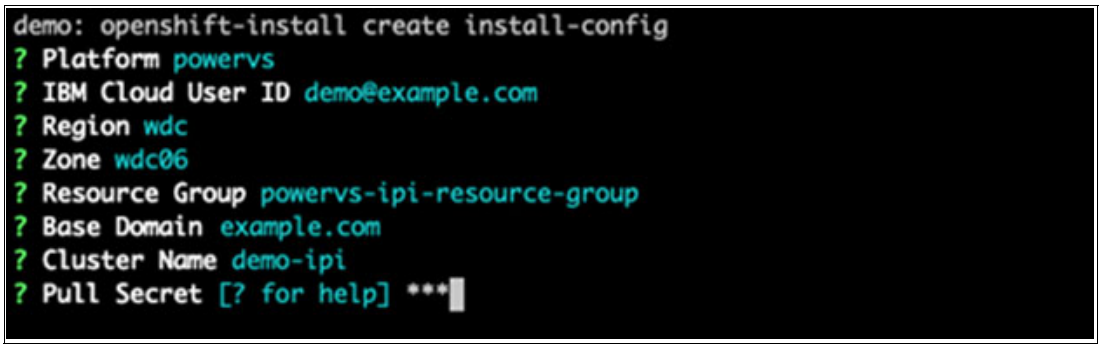


Figure 5-2 Red Hat OpenShift installer

The following list describes each field:

- Platform: Choose the environment where the cluster runs. For IBM Power, select IBM Power Virtual Server.
- IBM Cloud user ID: Your IBM Cloud account credentials.
- Region and Zone: The location for your cluster (for example, wdc for Washington, D.C., and wdc06 for the zone).
- Resource Group: A logical grouping of resources in IBM Cloud.
- Base Domain: The DNS domain for your cluster (for example, example.com).
- Cluster Name: A unique name for your Red Hat OpenShift cluster.
- Pull Secret: A secret key from Red Hat that provides access to container images.

The installation process requires the following key components:

- openshift-installer: The tool that creates manifests, ignition files, and manages the deployment process.
- Manifests and ignition files: Files that define cluster resources and bootstrap nodes.
- `install-config.yaml`: The main configuration file that describes cluster topology and settings.

Alternative method

Instead of answering prompts in the terminal, you can use the installer's survey mode, which asks similar questions and generates an `install-config.yaml` file that you can use immediately or customize later.

Figure 5-3 illustrates the survey-based generation of `install-config.yaml` by using openshift-installer.

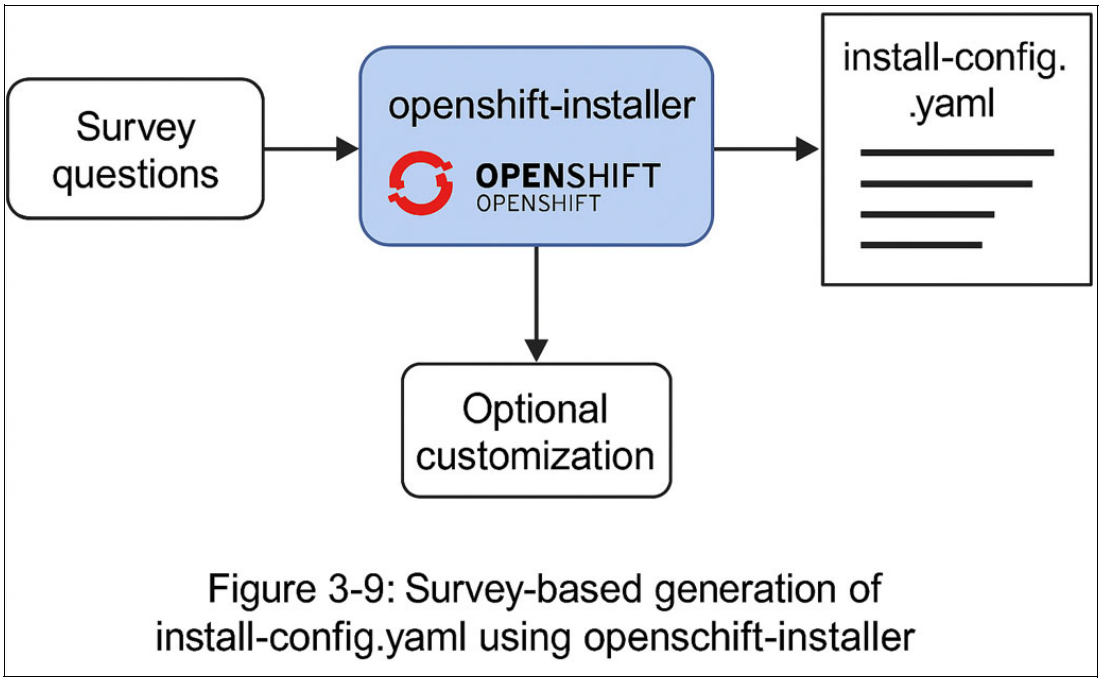


Figure 5-3 Survey-based configuration workflow

The openshift-installer is responsible for generating manifests, ignition files, and orchestrating the deployment process. These artifacts define cluster resources and bootstrap nodes. The `install-config.yaml` file serves as the foundation for cluster topology and settings.

Scaling and Teardown:

Clusters deployed with IPI include the Machine API, which makes scaling simple. To add or remove nodes, apply a manifest to the Red Hat OpenShift cluster. If a cluster is no longer needed, it can be removed with the following command:

```
openshift-install destroy cluster
```

For step by step instructions about using this deployment method, see Getting started with Red Hat OpenShift Container Platform on IBM Power Virtual Server.

5.1.2 User-provisioned infrastructure

The User-Provisioned Infrastructure (UPI) model offers maximum flexibility for organizations that require full control over their Red Hat OpenShift cluster deployment. Unlike IPI, where the installer automates infrastructure provisioning, UPI requires you to manually configure and prepare the environment before installing Red Hat OpenShift.

In a UPI deployment, the Red Hat OpenShift installer generates the necessary manifests and ignition files but does not create or manage the underlying infrastructure. You are responsible for provisioning compute nodes, networking, storage, and supporting services.

This approach is ideal for the following environments:

- On-premises IBM Power deployments
- Environments with strict compliance requirements
- Custom networking or advanced integration needs

UPI provides greater flexibility for environments that require custom DNS configurations, disconnected installations, or persistent storage provisioning by using Red Hat OpenShift Data Foundation or IBM Storage Scale.

The UPI model has the following characteristics:

- Greater infrastructure flexibility: Full control over architecture and configuration
- Broad platform support: Supports any IBM Power environment: PowerVS, PowerVM, or bare-metal Power servers
- Manual scaling: Adding or removing nodes requires manual steps or automation scripts
- Limited installer role: Generates manifests and ignition files but does not provision resources

Deployment Workflow

The UPI workflow includes the following steps:

1. Provision infrastructure manually

   Create virtual machines or physical servers for control plane and worker nodes. Configure networking, DNS, and load balancers.

2. Generate installation artifacts

   Use the following commands to create manifests and ignition files:

   ```
   openshift-install create manifests
   ```

   ```
   openshift-install create ignition-configs
   ```

3. Distribute ignition files

   Copy the ignition files to the appropriate nodes and configure bootstrapping.

4. Start the cluster

   Boot the nodes by using the ignition files and monitor the installation process.

Automation Options

Although you can perform UPI deployment manually, you can automate it using tools such as Terraform and Ansible. For more information, see the IBM-authored automation on GitHub that integrates with IBM PowerVC, PowerVM, and PowerVS:

- https://github.com/ocp-power-automation/ocp4-upi-powervm
- https://github.com/ocp-power-automation/ocp4-upi-powervs

UPI supports disconnected installations, custom DNS configurations, and persistent storage provisioning by using Red Hat OpenShift Data Foundation or IBM Storage Scale.

When deploying Red Hat OpenShift on IBM Power, organizations can select between two installation models: Installer Provisioned Infrastructure (IPI) and User Provisioned Infrastructure (UPI). Each model provides distinct benefits, depending on whether the priority

is rapid deployment with high automation or maximum flexibility and customization. As summarized in Table 5-1, these differences help decision makers determine the model that best aligns with their business objectives, operational readiness, and existing infrastructure strategy.

Table 5-1 Comparison of IPI and UPI

Feature	IPI	UPI
Automation Level	High (installer provisions infra)	Low to Medium (manual or scripted)
Supported Environment	IBM Cloud PowerVS	Any IBM Power environment
Flexibility	Limited (predefined workflow)	High (custom networking/storage)
Deployment Speed	Fast (< 2 hours)	Slower (depends on manual steps)
Scaling	Automated through the Machine API	Manual or automated by using scripts

5.1.3 Assisted Installer

The Assisted Installer provides a web-based interface for defining and deploying Red Hat OpenShift clusters. This method simplifies cluster setup by guiding you through configuration steps in an intuitive UI, while still allowing automation through APIs.

The Assisted Installer is technically similar to User-Provisioned Infrastructure. You create and manage the underlying infrastructure, but the Assisted Installer streamlines the process by enrolling infrastructure through its interface.

The Assisted Installer offers the following capabilities:

- Web-based interface for cluster definition and enrollment
- API-driven automation for advanced integration
- Flexibility to work with user-provisioned infrastructure

You create the infrastructure and then register it through the Assisted Installer UI, as shown in Figure 5-4 on page 124.

Deployment workflow

The Assisted Installer workflow includes the following steps:

1. Provision infrastructure manually

   Create the required nodes (control plane and workers) in your IBM Power environment (PowerVS, PowerVM, or on-premises).

2. Access Assisted Installer UI

   Define cluster parameters, which include the following parameters:

   - Cluster name
   - Base domain
   - Networking configuration
   - Node enrollment

3. Enroll nodes

   Upload discovery ISO or use the provided agent to register nodes with the Assisted Installer.

4. Generate and apply configuration

   The Assisted Installer creates manifests and ignition files, similar to UPI, and guides you through installation.

Advantages and considerations

The Assisted Installer's main advantage is ease of use. You can configure clusters visually without editing YAML files. Compared to IPI, it provides less automation but more flexibility for custom environments. However, you must still provision the infrastructure manually, and connectivity to the Assisted Installer service is required unless you use disconnected mode.

Automation options

You can automate deployments through the Assisted Installer API. Similar to the UPI method, IBM published automation to deploy a cluster with the Assisted Installer:

Additionally, automation is possible through its API. Similar to the UPI method, IBM provides automation to deploy a cluster with the Assisted installer. For more information, see https://github.com/ocp-power-automation/ocp4-ai-power/

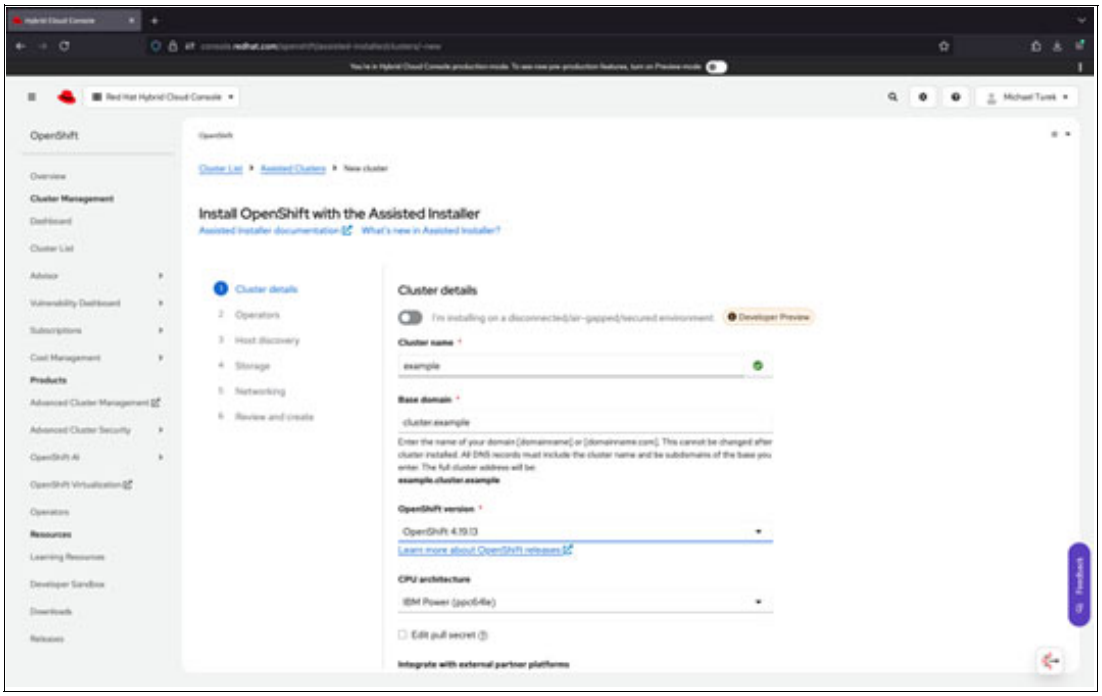


Figure 5-4 Install Red Hat OpenShift by Assistant Installer

5.1.4 Agent-based Installer

The agent-based installer is a variant of the assisted installer that is designed for disconnected or air-gapped environments. It provides the same user experience as the assisted installer but can be hosted locally, enabling deployments without requiring external connectivity.

This method is ideal for the following environments:

- Air-gapped environments with no internet access
- Highly secure deployments that require full control over installation artifacts
- Custom IBM Power environments where compliance and isolation are critical

The Agent Installer uses agents running on each node to coordinate installation locally. It eliminates dependency on Red Hat's cloud-hosted Assisted Installer service while maintaining flexibility and automation. The agent installer includes the following features:

- Disconnected installation: Works in fully offline environments
- Same UI experience as Assisted Installer: Hosted locally for cluster definition and node enrollment
- Agent-driven orchestration: Nodes communicate locally to complete installation
- Supports IBM Power platforms: PowerVS, PowerVM, and bare-metal Power servers

Deployment workflow

The Agent-Based Installer workflow includes the following steps:

1. Prepare installation artifacts

   Download the Agent Installer bundle and required images from Red Hat or a local mirror.

2. Host the Assisted Installer locally

   Run the service in your environment to provide the same UI experience offline.

3. Provision infrastructure

   Create nodes manually or using automation tools (Terraform, Ansible).

4. Deploy agents and enroll nodes

   Boot nodes with the agent ISO or container image. Nodes register locally and coordinate installation.

5. Start cluster installation

   The agents apply manifests and ignition files to configure the cluster.

Automation options

IBM provides automation scripts for Agent Installer deployments on IBM Power platforms. These scripts integrate with PowerVC, PowerVM, and PowerVS for streamlined provisioning and installation.

Advantages and considerations

The Agent Installer combines the simplicity of Assisted Installer with the ability to run in disconnected environments. However, it requires additional preparation for hosting the service locally and managing installation artifacts compared to the cloud-based Assisted Installer.

Figure 5-5 on page 126 illustrates how nodes participate in the installation process using agents in connected or disconnected environments.

For more information, see Understanding Agent-based Installer.

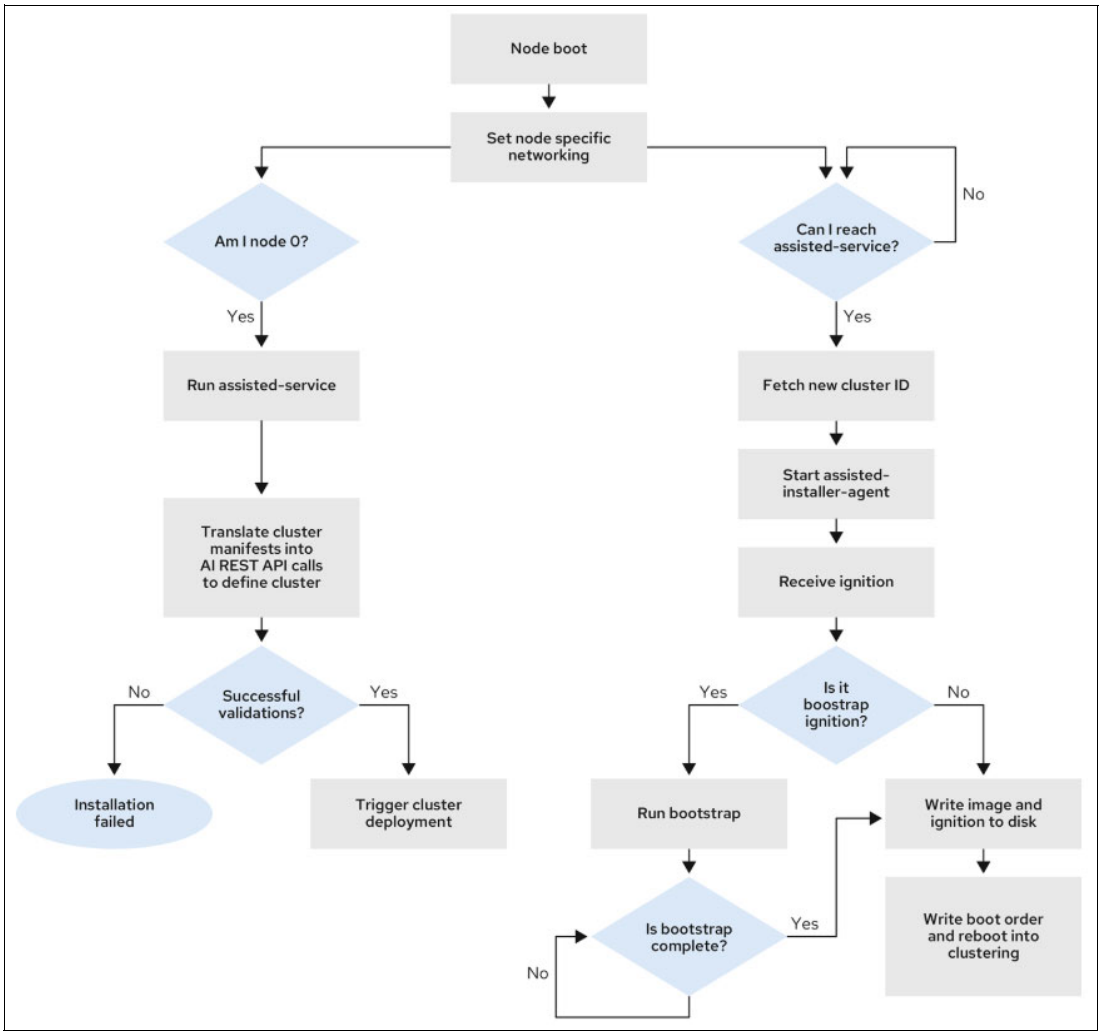


Figure 5-5

5.1.5 Factors to consider when choosing a deployment method

Choosing the right Red Hat OpenShift deployment method for IBM Power depends on technical requirements, operational constraints, and organizational priorities. Consider the following factors:

- Level of Automation:
  - IPI offers the highest automation, provisioning infrastructure and deploying clusters with minimal manual effort.
  - UPI, Assisted Installer, and Agent-Based Installer require manual provisioning but can be automated by using tools such as Terraform and Ansible.
- Infrastructure Control:
  - If you need full control over networking, storage, and node configuration, UPI or Agent-Based Installer are better suited.
  - IPI limits customization because infrastructure is provisioned automatically.
- Connectivity Requirements:
  - Assisted Installer requires connectivity to Red Hat's service unless hosted locally.

  - Agent-Based Installer is designed for disconnected or air-gapped environments.
- Deployment Environment:
  - IPI currently supports IBM Cloud PowerVS only.
  - UPI, Assisted Installer, and Agent-Based Installer support PowerVS, PowerVM, and on-premises Power servers.
- Compliance and Security

  For highly regulated environments or air-gapped deployments, UPI or Agent-Based Installer provide the most flexibility and offline capability.
- Skill Level and Operational Model
  - Choose IPI for simplicity and speed.
  - Select UPI, Assisted Installer, or Agent-Based Installer if your team has expertise in automation tools like Terraform and Ansible.
- Additional Considerations
  - All methods support disconnected installations, custom DNS configurations, and persistent storage provisioning using Red Hat OpenShift Data Foundation or IBM Storage Scale.
  - IBM provides automation scripts for UPI, Assisted Installer, and Agent-Based Installer on GitHub:
    - ocp4-upi-powervm
    - ocp4-upi-powervs
    - ocp4-ai-power

  Table 5-2 provides a comparison of these installation methods across supported environments.

Table 5-2 Compatibility of Red Hat OpenShift Deployment Methods for IBM Power Environments

Environment	IPI	UPI	Assisted	Agent
PowerVS	Supported	Supported	Not Supported	Not Supported
PowerVC	Not Supported	Supported	Supported	Supported
Bare metal	Not Supported	Supported	Supported	Supported

- Multi-Architecture Compute (Multi-Arch)

  Red Hat OpenShift 4.14 introduced Multi-Arch capability, expanded in 4.15, enabling clusters to span multiple CPU architectures such as ppc64le (IBM Power), amd64 (x86), arm64 (ARM), and s390x (IBM Z).

  Multi-Arch allows heterogeneous clusters that align workloads with architecture strengths, optimizing performance, cost, and resource usage.

  Multi-Arch containerization uses manifest lists to bundle architecture-specific images under a single tag. Tools like Docker Buildx, Buildah, and Podman create these images, and CI/CD pipelines (Tekton, GitLab) automate deployment.

  Scheduling is managed through node labels, affinity rules, and the Multiarch Tuning Operator, which automates architecture-aware workload placement.

  Multi-Arch supports the following use cases:
  - Legacy application support
  - Gradual hardware migration

- Unified DevOps pipelines
- Hybrid cloud deployments
- Compliance-driven environments

5.2 Phased modernization with IBM PowerVS and Red Hat OpenShift

Migrating workloads to IBM Power Virtual Server benefits from a phased approach. Organizations can start small, validate their strategies, and gradually modernize their environments by using two complementary strategies: incremental adoption and incremental modernization.

5.2.1 Starting in PowerVS before full migration

Migrating mission-critical workloads requires careful planning. Using PowerVS as a proving ground before committing to large-scale changes reduces risk, accelerates decision-making, and builds confidence in hybrid cloud operations.

Proof of concept and pilot projects

PowerVS provides an ideal environment for proof-of-concept initiatives or small-scale pilots. Enterprises can quickly provision instances to test applications, validate operating system versions, or benchmark performance without impacting production systems.

Rapid test and development environments

Self-service provisioning enables development and test environments to be created within minutes rather than weeks. Teams can provision resources for specific projects, use them only as needed, and then decommission them to avoid unnecessary costs. This capability aligns with DevOps and CI/CD pipelines.

Early evaluation of new hardware generations

When new hardware generations such as IBM Power11 are released, PowerVS allows organizations to validate application compatibility and performance before committing to on-premises upgrades. This early testing ensures smoother transitions and minimizes risk.

Phased modernization

Instead of a single, all-encompassing migration, enterprises can start by moving selected workloads or departmental systems into PowerVS. This incremental approach builds hybrid expertise and supports a broader modernization strategy over time.

5.2.2 Adding Red Hat OpenShift on PowerVS

Red Hat OpenShift on IBM PowerVS enables organizations to modernize workloads gradually by introducing containerized environments alongside existing systems. This approach supports hybrid deployments and cloud-native architectures without disrupting operations.

Red Hat OpenShift on PowerVS includes several key advantages:

- High performance: IBM Power Systems deliver up to 3.2× higher container density per core compared to x86.

- Cost efficiency: Internal benchmarks show up to 53% lower cost per transaction for OLTP workloads.
- Enterprise reliability: Features like Active Memory Mirroring and Chipkill ensure 99.999% uptime.
- Security: Supports secure VPC networking, IAM policies, and integration with IBM Cloud Internet Services.

Red Hat OpenShift on PowerVS is optimized for container workloads on Power11:

- Vector Scalar Matrix Engine v2: Doubles performance for INT8 and FP16 operations, ideal for AI/ML inference workloads.
- Adaptive frequency boost: Latency-sensitive containerized apps can reach turbo speeds of 4.8 GHz.
- DDR5 with open memory interface (OMI): Enhances memory bandwidth and efficiency for containerized microservices.
- IBM Spyre AI accelerator: Designed for high-throughput AI inference, which is integrated with Red Hat OpenShift AI.

Red Hat OpenShift on PowerVS includes modernization enablers:

- Infrastructure as Code (IaC): Tools like Terraform streamline automation and lifecycle management.
- Hybrid cloud flexibility: Power11 is available both on-premises and through PowerVS, enabling seamless hybrid deployments.

AI-ready infrastructure: Built-in AI acceleration and Red Hat OpenShift AI integration allow enterprises to deploy intelligent applications at scale.

5.3 Support for Red Hat OpenShift Platform Plus on IBM Power Servers

Red Hat OpenShift Platform Plus (OPP) is a unified platform that is designed to build, modernize, and deploy applications at scale. It provides multi-cluster security, compliance, application, and data management across infrastructures, ensuring consistency throughout the software supply chain. With a complete set of services for hybrid cloud environments, OPP helps organizations work smarter and faster by accelerating application delivery.

Red Hat OPP subscriptions include full support from Red Hat.

5.3.1 Description

As part of the Red Hat OpenShift portfolio, Red Hat OpenShift Platform Plus delivers a single, hybrid cloud platform for enterprises to build, deploy, run, manage, automate, and safeguard intelligent applications at scale. Building on the trusted Red Hat OpenShift Container Platform, OPP provides an enterprise-grade Kubernetes system for migrating workloads to the cloud, creating new digital experiences, and deploying applications to edge locations.

Enterprise IT operations teams deploy and manage OPP, which includes several integrated technology solutions:

- Red Hat OpenShift Container Platform

A complete set of services for rapid application development and efficient IT operations that includes advanced open source capabilities such as Red Hat OpenShift Serverless, Red Hat OpenShift Pipelines, and Red Hat OpenShift GitOps

- Red Hat Advanced Cluster Security for Kubernetes (ACS)

  Provides Kubernetes-native security across the entire application lifecycle, enhancing infrastructure and workload protection across clusters in data centers, clouds, and edge environments

- Red Hat Advanced Cluster Management for Kubernetes (ACM)

  Offers end-to-end visibility and control for cluster and application lifecycle management, improving security and compliance across multi-cluster domains

- Red Hat OpenShift Data Foundation Essentials

  Delivers data efficiency, resilience, and security for Red Hat OpenShift environments

- Red Hat Quay

  A secure private registry platform for managing container images across global data center and cloud environments, supporting cloud-native and DevSecOps models

- Red Hat OpenShift Lightspeed

  A generative AI-based virtual assistant integrated into the Red Hat OpenShift console, providing context-aware guidance to improve platform and application owner efficiency

5.3.2 Subscriptions and ordering information

Subscriptions

Red Hat OPP subscriptions are offered as 1-year terms with Red Hat support.

Confirmed Stateside Support (CSS) is available for US government customers that require specialized security and data handling. CSS ensures technical assistance exclusively from US citizens on US soil at designated support centers.

When Red Hat OpenShift products are ordered from IBM, a Product Information Card with activation details is shipped separately from hardware. To activate entitlements, contact the Red Hat OpenShift team by email: `openshift-orders@redhat.com`.

Ordering Information

Program number: 5639-OPP

Program name: Red Hat OpenShift Platform Plus for Power

Version: 1.0.0

The following Table 5-3 provides a comparison of program numbers, feature descriptions, billing feature codes, and supported regions for these subscription offerings.

Table 5-3 Feature Codes and Options

Program Number	Feature Description	Billing Feature Code	Regions
5639-OPP	RH OPP per Svr up to 2 cores w 1 yr PR Sub/CSS	0001	US
5639-OPP	RH OPP per Svr up to 2 cores w 1 yr ST Sub/CSS	0002	US

Program Number	Feature Description	Billing Feature Code	Regions
5639-OPP	RH OPP per Svr up to 2 cores w 1 yr PR Sub/Spt	0003	US, Canada, AP, Japan
5639-OPP	RH OPP per Svr up to 2 cores w 1 yr ST Sub/Spt	0004	US, Canada, AP, Japan

Use the Feature Code 5809 (Product Information Card) and Process Option Feature Code: 3439 (Subscription Term Registration).

For more information, see Red Hat OpenShift Platform Plus.

6

Security architecture for hybrid cloud on Power11

Hybrid cloud environments introduce unique security challenges that require a coordinated approach across on-premises and cloud resources. IBM Power11 security is designed to manage identity, protect data, and isolate workloads consistently, whether they run on local servers or in IBM Cloud. Multiple domains and technologies in a hybrid model can create complexity and blind spots if not properly managed. Identity is the new security perimeter in these distributed systems, and data must be encrypted and safeguarded both at rest and in motion.

This chapter unifies identity management, data protection, platform trust, compliance operations, integration strategies, and quantum-safe readiness into a single architecture for IBM Power11 and IBM Power Virtual Server

This chapter includes the following topics:

- Compliance, governance, and operational security
- Identity and access across hybrid environments
- Data protection
- Secure platform and workload isolation
- Integration and orchestration patterns
- Securing IBM hybrid cloud with IP-based controls

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6.1 Compliance, governance, and operational security

As organizations expand their hybrid and multi-cloud strategies, maintaining strong security, governance, and regulatory compliance becomes both more complex and more essential. Modern workloads span on-premises systems, public clouds, and distributed edge environments with each introducing unique risks, controls, and operational requirements. To meet these challenges, IBM provides an integrated set of security, compliance, and automation capabilities that are designed to protect data, enforce policy, and sustain operational integrity across the entire cloud estate.

This chapter explores how IBM technologies help organizations build a consistent and proactive security posture by automating policy enforcement, validating regulatory compliance, monitoring environments in real time, and rapidly responding to emerging threats. The chapter includes the following key focus areas:

- Automated Security and Compliance Enforcement

  IBM PowerSC provides automated security controls, policy enforcement, compliance reporting, and encryption key tracking for Power environments.

- Cloud Compliance Validation

  IBM Cloud Security and Compliance Center continuously evaluates IBM Power Virtual Server resources against regulatory benchmarks and organizational controls.

- Centralized Visibility and Monitoring

  Security events and operational logs from systems and cloud services can be integrated into a SIEM, such as IBM QRadar, for coordinated monitoring, threat detection, and forensic analysis.

- Data Protection and Auditing

  IBM Guardium® secures sensitive data with capabilities for access control, activity monitoring, and audit reporting.

- Ransomware Resilient Recovery

  IBM Power Cyber Vault creates isolated, immutable recovery points to accelerate restoration after ransomware or data corruption events.

- Multi-Cloud Governance and Policy Consistency

  Cloud Security Posture Management (CSPM), policy-as-code frameworks, regulatory mappings (PCI-DSS, HIPAA, GDPR), configuration drift detection, and centralized evidence collection help unify security across multiple clouds.

- AI-Driven Risk and Compliance Automation

  Solutions such as IBM Concert with watsonx.ai apply predictive analytics, automated remediation insights, and AI-driven governance to help security teams detect risks earlier and enforce controls more efficiently.

Together, these capabilities create a cohesive security and compliance framework that adapts to the speed and scale of hybrid cloud operations and reduce manual effort and operational risk. This chapter details how these tools and practices can be implemented to achieve consistent governance, strengthen security posture, and support regulatory readiness across diverse cloud environments.

6.2 Identity and access across hybrid environments

In a hybrid cloud environment that includes IBM Power11, strong identity and access management is essential to ensure that only authorized users and systems can access resources across on-premises and cloud components. These setups often involve separate identity systems, such as an on-premises Active Directory for enterprise users and a different identity service for cloud, which can create inconsistent policies and security gaps.

To address this challenge, organizations implement a centralized or federated identity and access management framework, enabling cohesive control of identities and access policies across all environments. Modern architectures treat identity as the primary security boundary. Zero trust principles require continuous verification of user identity and context for every access attempt.

6.2.1 Unified identity management on-premises

IBM Power11 strengthens unified identity management by aligning its security model with enterprise directory services and modern zero trust principles. All core operating systems on the platform (AIX, IBM i, and Linux) support centralized authentication and directory-based identity federation, enabling organizations to manage users and roles consistently across environments. IBM positions Power11 as secure-by-design, with identity, access, and compliance integrated into the full stack from firmware to operating system.

AIX and Linux systems can integrate directly with Microsoft Active Directory or other enterprise LDAP directories by using native LDAP client services. This integration allows administrators to centralize user lifecycle management, password policies, and authentication workflows. AIX also supports advanced role-based access control (RBAC), which can map system roles to LDAP-backed groups, ensuring a single source of truth for identity and privilege assignments. When you add a user to corporate groups such as DBAdmins or Auditors, Power11 systems can automatically inherit those role mappings without manual reconfiguration. This model reduces administrative requirements and improves compliance alignment.

IBM promotes zero trust capabilities across Power11 platforms, including encryption, RBAC, vTPM, and quantum-safe cryptography as part of its built-in security stack. These controls reinforce least-privilege enforcement by ensuring that users receive only the access required to perform their responsibilities.

Centralized identity also enhances operational resilience. When you modify user attributes or group memberships in the directory, such as revoking credentials or changing departmental roles, the updates propagate immediately across connected Power11 systems. This tight integration supports strong auditability, faster compliance responses, and reduced risk exposure.

IBM identity solutions, such as IBM Security Identity Manager, provide additional automation for AIX and Linux environments by enabling consistent provisioning and de-provisioning of accounts across UNIX and Linux hosts. This capability ensures that identity governance extends beyond authentication and into full lifecycle management, strengthening security hygiene and reducing manual workload.

6.2.2 IBM Cloud IAM for PowerVS

On the cloud side of a hybrid deployment, IBM provides identity management through IBM Cloud IAM, which is used by the IBM Power Virtual Server (PowerVS) service. IBM Cloud IAM enables you to securely authenticate users and control access to PowerVS resources by using fine-grained roles and policies.

In a PowerVS environment, all user and API access is governed by the IAM system of IBM Cloud. You can define custom roles, assign users to groups, and grant permissions scoped to specific resources or resource groups. By default, cloud user accounts have no access until they are granted roles, aligning with the least privilege model.

A key benefit of IBM Cloud IAM is its integration with enterprise identity providers through federation. Organizations can federate their corporate Identity Provider (IdP) with IBM Cloud, allowing corporate user accounts to log in to IBM Cloud services (including PowerVS) by using their regular company credentials.

This approach simplifies administration and enhances security by eliminating separate cloud logins. Employees use their existing IBMid or federated ID (for example, their Active Directory username) to access cloud resources. You can establish federation either through IBMid federation or an IBM Cloud App ID service by using SAML. After you enable federation, users are authenticated by the enterprise IdP and automatically granted access to the IBM Cloud account based on predefined mappings or trusted profiles. This approach preserves central identity control. For instance, if an employee leaves the company and their Active Directory account is disabled, their access to PowerVS is automatically revoked as part of the federation trust.

IBM Cloud IAM also supports organizational constructs such as resource groups and access groups to simplify managing permissions for multiple resources or users at the same time. You use these features to ensure that administrators, developers, and other roles have access only to the resources they require. For example, you can limit development teams to dev/test PowerVS instances while production environments remain restricted to operations staff.

Figure 6-1 on page 137 illustrates the high-level architecture of PowerVS Private Cloud in client locations, where federated IAM and enterprise directory integration are key to securing workloads.

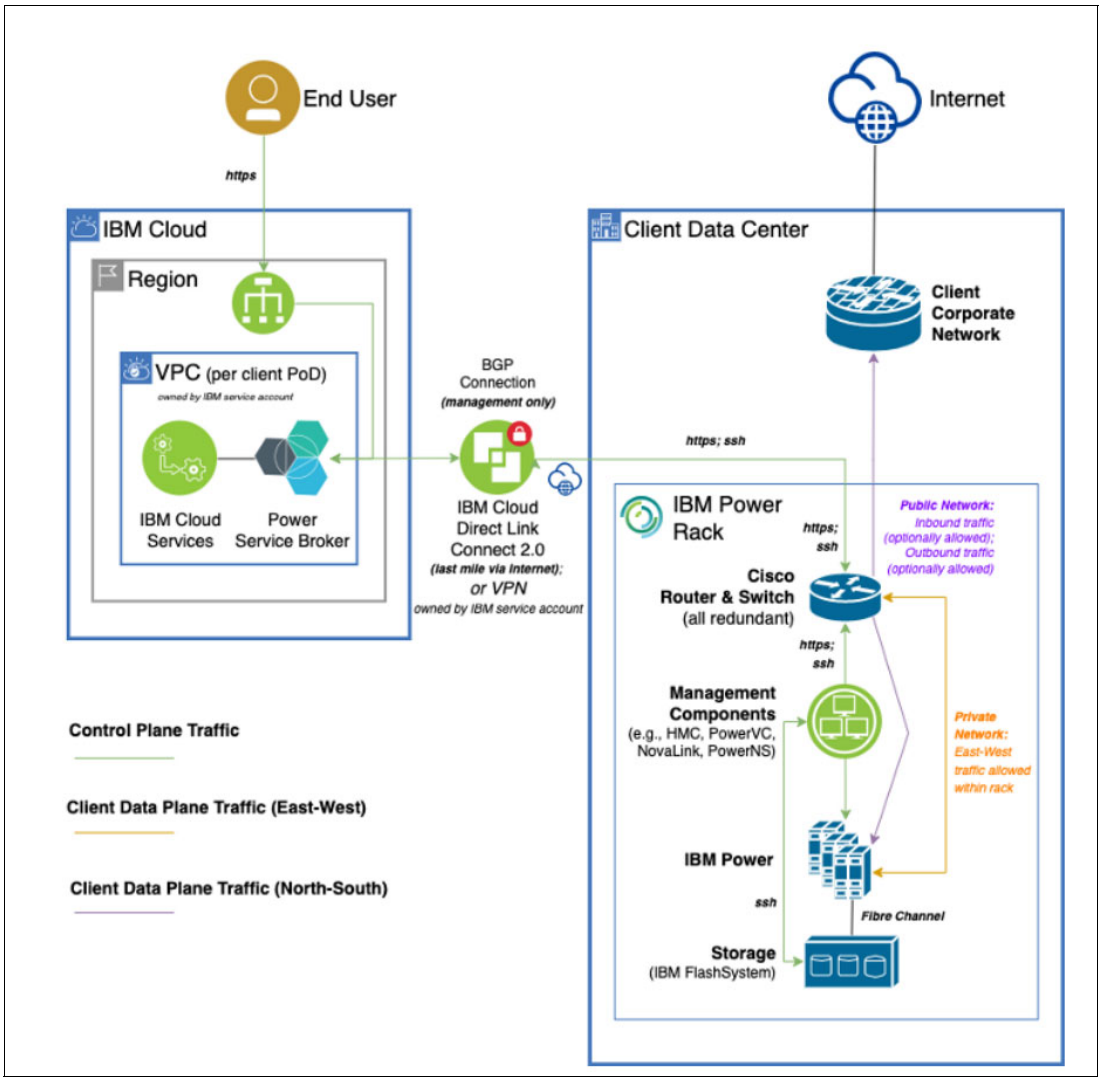


Figure 6-1 IBM Cloud and Client Data Center

6.2.3 Hybrid cloud enforcement

In hybrid cloud architectures, access control must extend beyond traditional identity-based permissions to include contextual and environmental factors. IBM hybrid cloud environments implement a layered security model that combines Identity and Access Management (IAM), Context-Based Restrictions (CBRs), and service-specific controls. This model enforces Zero Trust principles by ensuring that access decisions consider who is making a request, what they are allowed to do, and where the request originates.

Granular access control with IAM and CBRs

IBM Cloud IAM provides the foundational authorization model used across IBM Cloud services. IAM policies define the actions that users or service identities are permitted to perform, and you can scope these permissions to the account, resource group, or individual resource level. IAM roles serve as the primary mechanism for assigning identity-based access in IBM hybrid cloud environments.

CBRs add an additional enforcement layer by validating the conditions under which a request is made. CBRs examine network origin, endpoint type, and other request attributes to

determine whether access should be permitted. These restrictions apply along with IAM policies, providing protection even in cases where credentials are compromised or misconfigured. IBM Documentation specifies that CBRs evaluate IP ranges, virtual private cloud (VPC) identifiers, and endpoint types as part of their enforcement logic.

CBR rules operate independently of the user interface or API origin. Whether the request originates from the console, CLI, or automation, the access decision is based on network location and the configured contextual boundaries. This approach helps ensure consistent enforcement across hybrid environments.

Service-specific access controls

Beyond IAM and CBRs, many IBM Cloud services include the following additional security controls that help restrict access to sensitive resources:

- IP-allow lists for Cloud Databases
- Endpoint restrictions in API Connect
- Network-constrained administration interfaces for security services such as QRadar on Cloud

Service-level controls provide fine-grained protection even when broader IAM permissions apply.

For example, you can configure IP allowlists to ensure that only requests originating from approved corporate networks can reach a database instance. IBM Cloud documentation highlights the importance of combining RBAC, IP restrictions, geolocation enforcement, and resource-based policies to strengthen cloud access governance.

Hybrid cloud access enforcement example

The following example illustrates how hybrid cloud access controls work together to enforce secure operations across both cloud and on-premises environments:

- IAM Role: A PowerVS Administrator can start, stop, and manage virtual server instances.
- CBR Condition: These actions are allowed only when the request originates from a corporate IP address or designated VPC.
- Operational Context: The request is permitted only during authorized maintenance windows, with optional restrictions requiring the use of private endpoints.

In this scenario, IAM determines the operational capability, and CBRs and service-specific restrictions verify the trustworthiness of the request context before allowing the operation to proceed.

Importance of granular enforcement in hybrid cloud

Granular access control is essential for hybrid cloud security because it ensures that authorization decisions incorporate both identity and environmental context. IAM policies define permissions, but CBRs verify that access occurs only from trusted networks, devices, or service paths. This approach reduces the risk of unauthorized access, supports regulatory compliance, and maintains consistent governance across cloud and on-premises workloads. IBM guidance emphasizes that CBRs enhance IAM by enforcing restrictions that remain effective even in the event of compromised credentials.

By combining identity-based and context-aware controls, IBM hybrid cloud provides a strong security foundation for enterprises operating in complex, distributed environments.

6.2.4 Granular authorization in a hybrid cloud environment

Granular access control refers to the ability to define and enforce fine-grained permissions for users, applications, and services, rather than broad, all-or-nothing access. In the context of IBM Hybrid Cloud IAM, this capability means controlling who can do what, where, and under what conditions across both on-premises IBM Power Systems and cloud environments.

In a hybrid cloud, workloads span multiple environments, and security must adapt dynamically. Granular Access Control ensures the following security aspects:

- Least privilege by limiting permissions to specific actions and resources
- Context-aware enforcement based on location, device posture, and risk
- Compliance alignment with industry standards through policy-driven controls

Key characteristics of granular authorization include fine-grained permissions, policy-based access, dynamic and context-aware access, and integration with RBAC and ABAC.

Fine-grained permissions:

For fine-grained permissions, You can scope permissions beyond broad roles at the following levels:

- Resource Level: Specific VM, storage bucket, or API
- Action Level: Start instance, attach storage, view logs
- Condition Level: Time-based, IP-based, or device posture

In an example of fine-grained permissions, an operator can restart PowerVS instances only during scheduled maintenance windows.

Policy-based access

For policy-based access, define the following elements:

- Who: Identity or role (user, service ID)
- What: Allowed actions
- Where: Resource or environment
- When: Time or context

In an example of policy-based access, an AIX Operator role can restart servers only during maintenance windows and only from approved IP ranges.

Dynamic and context-aware access

Dynamic and context-aware access decisions can depend on the following factors:

- Device posture: Secure endpoint validation
- Location: On-premises versus remote network
- Risk score: Behavioral anomalies detected by SIEM

Integration with security models

Integration with RBAC and attribute-based access control (ABAC) can help with assigning permissions and adding access conditions:

RBAC: Assign permissions based on roles

ABAC: Add conditions such as department, project, compliance level, or network origin

6.2.5 Multi-factor authentication and zero trust

Multi-factor authentication (MFA) is recommended for enhanced IAM security across all environments. IBM Cloud IAM includes native support for MFA during user login, providing an additional level of verification alongside passwords. For on-premises environments, Power11 systems can incorporate MFA solutions to secure administrative access. IBM Power11 also supports integration with IBM Security Verify, allowing for unified MFA and single sign-on functionality across hybrid settings.

For example, an administrator logging in to the IBM Cloud console to manage PowerVS can be required to use a one-time passcode or biometric factor, the same as they do for accessing an on-premises application. This consistency supports the Zero Trust philosophy: every access request, whether from a user or an API, is verified with multiple factors and contextual policies before being allowed.

When you use fine-grained IAM controls on PowerVS, you can define the following permissions:

- Who can create or manage virtual servers
- Who can view or edit network configurations
- Who can perform sensitive actions with audit logging

6.2.6 Privileged access and compliance

Managing privileged access is another key aspect of IAM in a hybrid cloud. Organizations implement role separation so that highly privileged accounts, such as system administrators or security officers on IBM i, are distinct and closely monitored.

IBM PowerSC and related tools can be used to audit privileged actions on AIX, IBM i, and Linux, helping ensure that any use of an administrative credential is recorded and can be reviewed for compliance.

Third-party solutions are also available to enhance privileged access security. Vendors such as Fortra (PowerTech), Precisely, and Trinity Guard provide multi-factor authentication and privileged access management tools specifically for IBM i and AIX.

These solutions can enforce an extra authentication step (for example, a TOTP code or smart card) when signing on to a sensitive account or provide profile-switching capabilities (such as Fortra's Authority Broker, which allows controlled elevation to privileged profiles with full activity logging).

Implementing these practices and tools allows organizations to apply consistent multi-factor authentication and monitoring on traditional platforms within a hybrid cloud environment. Aligning identity and access management approaches across IBM Cloud services, AIX, Linux, and IBM i centralizes control and auditability of user access throughout the hybrid environment. Audit trails are created for both cloud and on-premises systems (through cloud audit logs and system logs forwarded to a SIEM), supporting unified visibility and compliance reporting.

6.3 Data protection

Data protection is an important aspect of hybrid cloud architecture. IBM Power11 provides multiple capabilities for safeguarding data, including encryption across various layers and

secure workload isolation to minimize unauthorized exposure between tenants or environments.

In a hybrid cloud environment on Power11, data is categorized into three states: at rest, in transit, and in use. Technologies are applied to secure each state. The platform also incorporates protections against emerging threats such as quantum decryption and includes mechanisms for isolating different workloads and cloud tenants.

6.3.1 Data at rest

Encryption of data at rest is supported end-to-end. Power11 systems offer encryption for disks, file systems, databases, and backups to ensure that sensitive information is not exposed even if storage media is accessed without authorization.

The following encryption capabilities are available:

- On AIX: JFS2 file systems can be encrypted, with keys managed by the OS or external key managers.
- On IBM i: Built-in field-level encryption is available for Db2 and for backup media encryption.
- Hardware-based encryption: Many organizations also deploy self-encrypting drives or IBM DS8900F storage systems.

At the processor level, Power11 includes cryptographic accelerators that handle encryption tasks efficiently, ensuring minimal performance impact. In IBM Power Virtual Server (PowerVS), all customer data that is stored in IBM Cloud data centers is encrypted by default, including volumes and object storage. You can perform key management with IBM Cloud Key Protect or with client-supplied keys, an important feature for compliance.

Additionally, PowerVS supports integration with secret management tools such as IBM HashiCorp® Vault. By centralizing credentials (API keys, database passwords, and certificates) in a secure vault, you avoid storing secrets in plain form on VMs and can enforce key rotation and auditing.

6.3.2 Data in transit

Encryption of data in transit is equally critical as information moves between on-premises Power Systems, cloud components, and external services. IBM Power11 supports industry-standard protocols such as TLS for applications and IPsec or VPN for network links.

For hybrid deployments, organizations often establish a secure VPN or Direct Link with encryption to IBM Cloud to ensure confidentiality of traffic. Within IBM Cloud, PowerVS supports the following features:

- Private networking with VLAN segmentation
- Integration with IBM Cloud Transit Gateway
- Use of security groups and firewall rules to restrict flows

All APIs and management interfaces are secured with HTTPS (TLS), and administrative access (for example, SSH into an AIX LPAR) can be tunneled through jump hosts or VPNs.

Organizations enforce the use of modern cipher suites such as TLS 1.3 with elliptic-curve or post-quantum key exchange. This approach ensures resilience against both present-day and future quantum-based attacks. IBM Cloud software-defined networking further isolates each tenant's traffic.

6.3.3 Data in use

Encryption of data in use is provided by Transparent Memory Encryption (TME) on Power11. TME is a hardware feature that encrypts all data stored in memory, transparently to applications and operating systems.

Introduced in Power10 and carried into Power11, TME ensures that data in DIMMs or on the memory bus is unreadable if accessed by attackers. This capability prevents cold-start or hardware snooping attacks that might otherwise expose secrets such as encryption keys or customer data. Importantly, TME operates inline, with negligible performance impact.

Table 6-1 compares memory encryption and isolation capabilities across Power9, Power10, and Power11. It shows how Power11 advances the design with quantum-safe cryptography, runtime validation, and trusted domain fencing, providing stronger isolation than prior generations.

Table 6-1 Features memory encryption over Power infrastructure

Feature	POWER9	POWER10	POWER11
Memory type	DDR4	DDR4/DDR5 through OMI	DDR5 DDIMM (up to 4800 MHz, 512 GB per module)
Encryption engines	Basic AES support	4 AES engines per core, transparent encryption	Quantum-safe + AES + Crystals Dilithium/SHA3-512
Secure boot chain	Firmware - based, RSA/SHA2	Hardware-rooted, ECDSA/SHA3	Immutable SROM + OTPROM + Quantum-safe boot
Memory isolation	Limited, firmware-level	SPPE introduced for service processor isolation	Full SPPE enforcement + Trust Domain fencing
Runtime validation	Minimal	Boot-time validation only	Continuous runtime integrity checks
Quantum-safe readiness	Not supported	Partial (experimental)	Full support: Crystals Dilithium, SHA3-512
Tamper resistance	Firmware-level	Cryptographic firmware validation	Hardware-enforced + firmware + runtime checks

With encryption at rest, in transit, and in use, Power11 offers comprehensive protection that is aligned with industry standards such as PCI-DSS and HIPAA.

6.3.4 Quantum-safe cryptography

Power11 uses quantum-safe cryptography alongside standard encryption, with Secure Boot verifying firmware and OS integrity through both classical and post-quantum signatures. This dual method maintains security even if traditional signatures are compromised by quantum computing.

Figure 5-3 illustrates the quantum-safe secure boot process, which uses dual-signature verification to protect system startup.

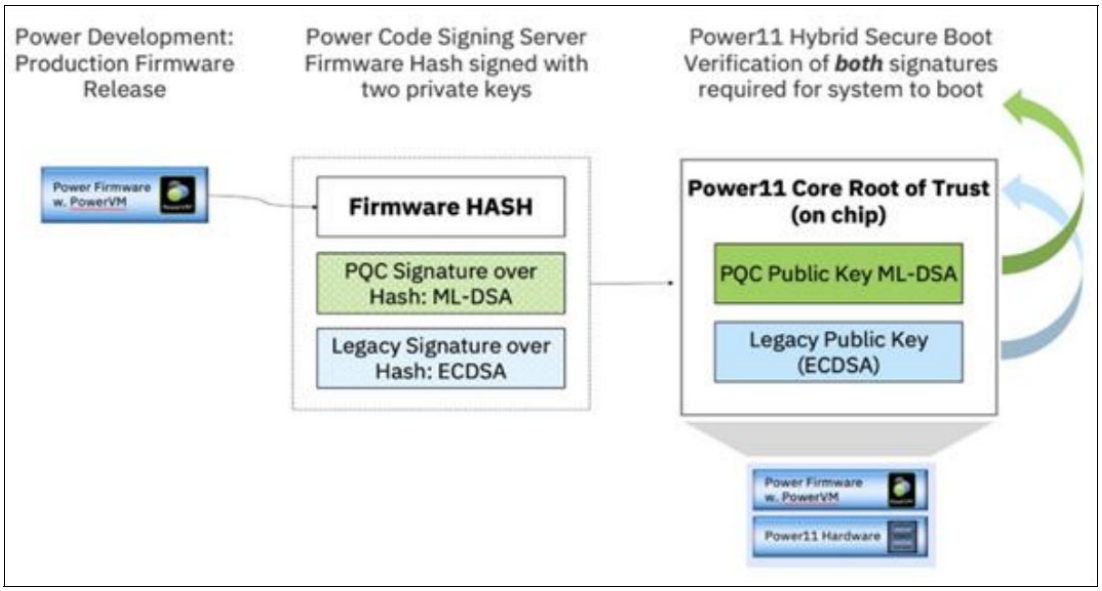


Figure 6-2 Flowchart of the Power11 Hybrid Secure Boot Verification process

IBM also secures Live Partition Mobility (LPM) with hybrid encryption. During migration, memory and state information are encrypted by using two key exchanges: one classical and one quantum-safe. This hybrid model ensures that if classical encryption is broken in the future, the quantum-safe component still protects confidentiality.

Figure 5-4 shows the quantum-safe LPM process, where dual-key negotiation mitigates “harvest-now, decrypt-later” attacks.

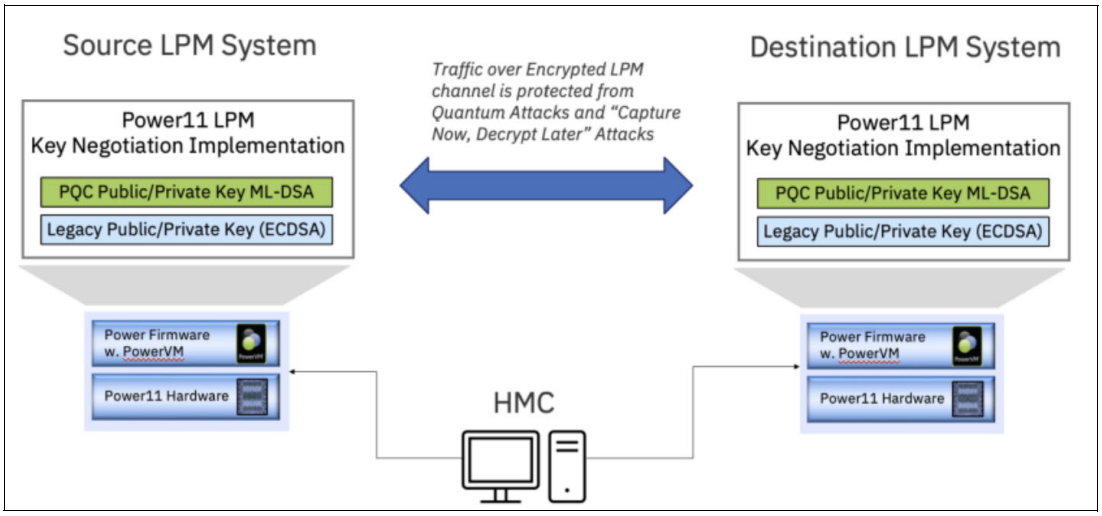


Figure 6-3 Key negotiation implement

These features align with NIST guidance on hybrid post-quantum cryptography, ensuring that Power11 is future-proof for sensitive workloads.

6.3.5 Workload isolation

Although encryption secures data, isolation ensures that each workload can access only its own resources.

At the hypervisor level, PowerVM enforces strong partitioning of CPU, memory, and I/O. Only signed firmware can run, and PowerVM has a long record of security assurance, including Common Criteria EAL4+ certification. This capability ensures that one LPAR cannot interfere with another, even in multi-tenant environments.

At the infrastructure level, PowerVS uses PowerVM and additional controls to enforce tenant isolation. Each client receives logically isolated compute, storage, and networking. IBM manages the hardware and hypervisor, and the client manages the OS and applications.

At the OS and application level, additional isolation is achieved through the following mechanisms:

- User access separation
- Containerization (for example, Red Hat OpenShift on Power)
- Network segmentation

6.3.6 Data integrity and availability

Power11 also addresses data integrity and resilience through the following capabilities:

- IBM PowerSC can monitor compliance, maintain a cryptographic inventory, and alert administrators to unauthorized changes.
- IBM Cloud Security and Compliance Center checks cloud-based PowerVS resources against regulatory benchmarks.
- IBM Power Cyber Vault isolates backup copies in a secure, immutable environment; these copies can be validated and restored quickly after a ransomware attack or data corruption event.

Together, these features ensure that organizations can encrypt and isolate data and can also recover it rapidly while maintaining compliance.

6.4 Secure platform and workload isolation

Running a hybrid cloud environment requires uncompromising security and data protection across both on-premises and cloud components. IBM Power11 addresses these requirements with a multi-layered security architecture, introducing enhancements in data encryption, cryptographic resilience, and workload isolation to protect sensitive workloads in any location.

This section explains how Power11 protects data at rest and in transit, uses advanced encryption with quantum-safe algorithms for future-proof security, and isolates workloads to block unauthorized access or interference.

6.4.1 Encryption and data protection

Power11 and PowerVS incorporate comprehensive encryption features to protect data both at rest and in transit. All data that is stored on PowerVS in IBM Cloud can be encrypted, and you

can secure network connections between on-premises Power11 systems and the cloud through VPN or encrypted tunnels.

The platform uses modern encryption standards such as AES-256, boosted by on-chip cryptographic engines for speed. On IBM Power Virtual Server, you can encrypt cloud storage and disk data with keys from IBM Cloud Key Protect or external managers.

Data traveling between the Power11 environment and IBM Cloud can use secure protocols (TLS, IPsec) to ensure confidentiality and integrity. These measures align with Zero Trust security principles by ensuring that data remains protected at all times, including on shared or external networks.

Identity and access management

Identity and access management is also integrated. Power11 systems and PowerVS support robust IAM controls, including integration with IBM Security Verify for single sign-on and multi-factor authentication. This integration means that whether a workload runs locally or in the cloud, the same strict authentication and authorization policies apply, greatly reducing the risk of breaches due to compromised credentials.

In practice, an administrator accessing a Power11 LPAR or a PowerVS instance can pass the same level of identity verification, and fine-grained role-based access control can limit their actions on each system. These controls help enforce a consistent security posture across hybrid cloud components, which is especially important for industries with strict regulatory requirements on data privacy and access logging.

Memory encryption

Power11 introduces memory encryption and other hardware-based data protection features. Building on the Power10 generation, Power11 implements memory-to-core encryption, which means data is automatically encrypted as it moves between memory and the processor caches. This capability guards against any snooping of the memory bus or installation of illicit hardware devices trying to capture sensitive information such as cryptographic keys or personal data while in use.

The memory encryption does not affect applications and does not require modifications to software. Additionally, the Power11 architecture uses DDR5 memory with an Open Memory Interface and includes integrity checks and redundancy to prevent data corruption or tampering in memory.

An important enhancement in Power11 is the extension of Secure Boot beyond just the processor firmware to also cover memory DIMM firmware. Each Power11 DDR5 DIMM module has an embedded controller that undergoes its own secure boot process: a hardware-anchored Secure ROM verifies the DIMM's firmware by using digital signatures before the memory is trusted by the system.

If the memory firmware fails verification or shows any sign of tampering, the system automatically unconfigures that memory module (isolating it from use) to maintain overall system integrity. This level of granular hardware validation ensures that data residing in memory is not only encrypted, but also that the memory hardware itself has not been maliciously modified.

6.4.2 Cryptographic resilience and quantum-safe encryption

Power11 introduces post-quantum cryptography, which is designed to address potential future security challenges. Traditional encryption and signature algorithms (such as RSA and

ECC) can become vulnerable to quantum computers. IBM has incorporated quantum-safe algorithms into essential system components.

Power11 encryption

The secure firmware in Power11 uses digital signatures based on CRYSTALS-Dilithium, a quantum-resistant, lattice-based signature algorithm, together with the traditional Elliptic Curve Digital Signature Algorithm (ECDSA) to verify boot code.

This dual-signature approach (both classical and quantum-safe) means that even if a powerful quantum computer in the future compromises ECDSA, the Dilithium signature remains unbreakable with quantum attacks, ensuring the firmware's authenticity remains trusted.

IBM uses CRYSTALS-Kyber, a quantum-resistant key encapsulation mechanism, to secure data in transit during key exchange and encryption. NIST and ETSI recommend these algorithms for post-quantum cryptography.

By adopting them early, Power11 provides long-term confidentiality for data. Data protected today remains secure even when quantum computing becomes mainstream.

Practical example of quantum-safe

A practical demonstration of the quantum-safe strategy in Power11 is its approach to securing Live Partition Mobility (LPM) operations within a hybrid cloud environment. With LPM, a running partition (a virtual machine) can be transferred from one physical server to another without service interruption.

During LPM, the system's memory state and processor state are transferred over the network, a sensitive process that must be encrypted. Power11 enhances LPM by using a hybrid key exchange that combines classical cryptography with quantum-safe cryptography to negotiate the encryption keys for migration.

During an LPM migration, Power11 generates a shared session key using both traditional algorithms (for example, an Elliptic Curve Diffie-Hellman component authenticated by classical signatures) and quantum-safe algorithms (such as CRYSTALS-Kyber for key exchange and Dilithium for authentication).

Even if an attacker records the encrypted migration traffic today with the intent to decrypt it in the future (the "harvest-now, decrypt-later" attack), the quantum-resistant portion of the encryption ensures that the data cannot be decrypted even with quantum computing power. IBM's implementation of this hybrid cryptography approach follows industry recommendations and serves as a vital defense against emerging threats.

Power11 employs quantum-safe methods in its Secure Boot process. The chain of trust that starts from hardware has been upgraded to use algorithms like SHA3-512 hashing and Dilithium signatures for verifying firmware and operating system bootloaders.

The entire platform establishes a root of trust that is robust against both today's attackers and tomorrow's quantum adversaries. This cryptographic agility and resilience give organizations confidence that their sensitive data (customer records, financial transactions, and so on) and cryptographic keys remain protected throughout the lifespan of the Power11 system, even as the threat landscape evolves.

6.4.3 Secure boot and workload isolation

Protecting data requires not only encryption but also ensuring that only authorized code and workloads can run on the system. IBM Power11 uses a Secure Boot mechanism to verify the integrity of firmware, operating systems, and hypervisor components every time the system starts.

Secure Boot on Power11 is anchored in hardware: an immutable boot ROM (the SROM) and one-time programmable fuses store IBM root cryptographic keys. At power-on, the system firmware (such as the Power Hypervisor and system management firmware) is cryptographically verified using these keys before execution.

If any code was tampered with or is not signed by a trusted key, the Secure Boot process detects this and prevents it from running. This prevents low-level malware or rootkits from inserting themselves into the platform.

Power11 extends this Secure Boot process to various components: not just the main processor firmware, but also adapter firmware, BMC (service processor) firmware, and even memory modules as noted earlier.

Trusted Boot complements Secure Boot by measuring each software component's cryptographic hash and storing those measurements (for example, in a TPM or another secure store). This creates an attestation log that you can use to verify the platform's state to external systems or for compliance audits. Together, Secure Boot and Trusted Boot ensure that the foundation on which workloads run is secure and trustworthy.

Secure workload isolation

With a trusted foundation in place, IBM Power11 provides secure workload isolation through its virtualization and partitioning features. Power Systems have a long history of strong logical partition (LPAR) isolation, which is certified to separation standards (previous Power hypervisors have been evaluated to Common Criteria EAL4+ or higher).

On Power11, each LPAR (or VM) is isolated by the Power Hypervisor such that it has dedicated or virtualized slices of processors, memory, and I/O, and cannot interfere with other partitions.

The hardware enforces memory separation, and the hypervisor controls ensure one partition cannot read or write another's memory. Power11 introduces Resource Groups and Trusted Execution features that further improve isolation and performance predictability for workloads (for example, isolating certain cores or memory regions for high-priority VMs).

Power11 introduces the Secure Partition Platform Engine (SPPE), a hardware mechanism that isolates the service processor (management controller) from the system's operational data paths.

SPPE acts as a gatekeeper so that even the privileged service processor cannot directly access memory or registers of running partitions, eliminating a potential attack vector and aligning with a Zero Trust approach internally. This means even if the management firmware were compromised, it cannot bypass partition isolation to leak sensitive information.

These measures ensure that workloads on Power11, whether multiple tenants in a cloud environment or multiple departments in a private data center, are walled off from each other.

IBM Power Virtual Server (PowerVS) uses that capability to provide secure multi-tenancy on the public cloud: the secure architecture of PowerVS ensures that one customer's VMs

cannot snoop on another's, with strict controls preventing any cross-tenant access or interference.

Network segmentation and virtual firewalls in PowerVS further contribute to isolating workloads, and integration with cloud security services (such as IBM Cloud Monitoring and Compliance) provides continuous oversight.

Operational security

In terms of operational security, Power11 supports a range of features to maintain a secure posture in hybrid environments. It includes Cryptographic Role-Based Access Control (RBAC) on the platform, meaning that administrators can be given fine-grained roles that limit what security-sensitive operations they can perform (for instance, separate roles for managing keys versus managing VMs).

You can use PowerSC, the IBM security and compliance tool for Power Systems, with Power11 to automate security policy enforcement and compliance audits across hybrid cloud deployments. Features such as trusted logging, security event forwarding to IBM QRadar (SIEM), and integration with enterprise directory services (LDAP/AD for unified identity across on-premises and cloud) are all part of the configuration that keeps the hybrid cloud secure.

Note: By using encryption, quantum-safe cryptography, secure boot, and strict isolation, IBM Power11 delivers layered security for hybrid cloud workloads without sacrificing performance or agility. Its next-generation cryptographic accelerators handle intensive encryption tasks with minimal impact, making always-on security practical for data replication across on-premises and cloud environments. With quantum-safe techniques, organizations can future-proof their data protection strategies. Consistent security models between Power11 and PowerVS enable unified governance and compliance reporting throughout the hybrid environment.

6.5 Integration and orchestration patterns

In hybrid environments, integration goes beyond individual tools. It requires cross-cloud interoperability and unified orchestration. This means aligning security policies across different providers, centralizing secrets and encryption key management, and ensuring end-to-end monitoring across IBM and non-IBM platforms.

IBM Power11 and IBM Power Virtual Server work seamlessly with key IBM security tools:

- IBM Security Verify for identity management
- IBM QRadar for threat detection
- IBM Guardium for data protection
- IBM PowerSC for compliance automation

Power Virtual Server also supports third-party solutions such as IBM HashiCorp Vault for managing secrets and Red Hat technologies for system hardening and automation.

6.5.1 IBM Verify Identity Governance

IBM Verify Identity Governance (IVIG), previously known as IBM Security Verify Governance, delivers identity and access capabilities for workforce and consumer users, with integrated governance to manage accounts, entitlements, and compliance across on-premises and cloud environments.

Figure 6-4 shows an overview of the IBM Verify Identity Fabric. It shows how Verify manages identity for employees and consumers (human users) and for applications, devices, and infrastructure (nonhuman identities), by using a layered approach that orchestrates identity flows and protects against risk.

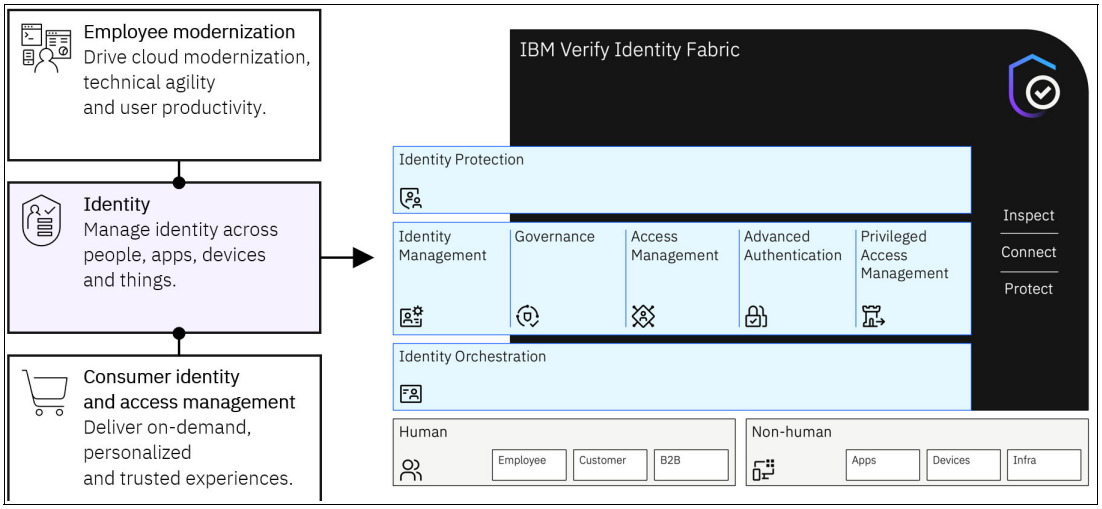


Figure 6-4 IBM Verify Identity Fabric

IVIG centralizes identity lifecycle, access requests, and access certifications, and enforces governance controls that reduce risk and support audits. It provides modern user and admin experiences, dashboards for operational visibility, and policy-driven controls aligned to separation of duties and sensitive access.

IVIG includes the following key capabilities:

- Identity governance and administration: Unified lifecycle and compliance, role optimization, and broad adapter coverage for mainframe, databases, software-as-a-service, and on-premises applications
- Access certification: Scheduled campaigns to review and validate user accounts and entitlements, with business-friendly interfaces and reporting
- Risk insights and operational visibility: Built-in dashboards for identity risks (such as orphaned accounts and separation of duties conflicts) and the status of provisioning and governance events
- User experiences: Streamlined request and approval flows, profile management, and low-code configuration for roles, policies, and campaigns

IVIG enables IAM administrators to gain better insights into currently running, failed, and successful provisioning and governance operations through an interactive dashboard, as shown in Figure 6-5 on page 150.



Figure 6-5 IBM Security Verify dashboard

In hybrid environments, IVIG harmonizes identity policies, centralizes entitlements, and enables end-to-end monitoring so that administrators can certify access, remediate misconfigurations, and demonstrate compliance across IBM and non-IBM platforms.

For more information, see IBM Verify Identity Governance 11.0.0 delivers the next-generation of IBM Verify Governance solution.

6.5.2 IBM QRadar

IBM QRadar does not run natively on IBM Power processors. You deploy IBM QRadar as an on-premises appliance or virtual machine on x86, or as a managed service, and then ingest security telemetry from your IBM Power11 servers and your IBM Power Virtual Server workloads. QRadar collects, normalizes, and correlates logs, events, and network flows from AIX, IBM i, and Linux on Power, turning them into actionable offenses for your Security Operations Center.

QRadar centralizes telemetry from logs, flows, endpoints, and applications; correlates events into offenses; and guides analysts through investigation and response with automation and artificial intelligence. It improves Security Operations Center efficiency with modern dashboards, workflow orchestration, and built-in compliance and retention features.

IBM QRadar includes the following core components:

- QRadar SIEM: Real-time analytics, offense management, and search across large data volumes. Recent capabilities include Rule Versioning (auditable changes and rollback), Tiered Storage (hot/warm data to balance speed and cost), Predictive Parsing, infographic-style visualizations, and an Investigation Assistant powered by artificial intelligence. User and Entity Behavior Analytics (UEBA) adds risk-aware detection for insider and behavioral threats.

- QRadar SOAR: Case management and playbook automation with low-code/no-code integrations, playbook lifecycle tracking, and an AI Response Assistant for incident summaries, context enrichment, and recommended actions.
- QRadar EDR: Endpoint protection with scalable on-premises deployments, expanded detection strategies for Linux and macOS, and tools to validate custom detections.

Architecture patterns

The following list provides example configurations and the resulting activity by QRadar:

1. On-premises QRadar with IBM Power11 (local data center):
   - Where QRadar runs: QRadar Console, Event Collector, and Event Processor run on x86 hardware or virtual machines in your data center.
   - Connectivity: Direct LAN, private VLANs, or segmented networks connect AIX, IBM i, and Linux on Power hosts to QRadar.
   - What you send:
     - AIX: syslog from syslogd/syslog-ng, audit records (AIX audit subsystem), SSH/auth logs, sudo, application logs.
     - IBM i: Security and operational messages (for example, from QAUDJRN and QSYSOPR) forwarded through syslog or a forwarding utility/agent.
     - Linux on Power: syslog (rsyslog/syslog-ng), auditd, PAM, kernel, and application logs.
     - Network flows: NetFlow/IPFIX/sFlow from Power-attached network devices (if applicable).
   - Outcome: QRadar correlates these sources to detect privilege abuse, policy violations, misconfigurations, and threat patterns across your IBM Power11 environment.
2. On-premises QRadar ingesting from IBM Power Virtual Server (hybrid):
   - Where QRadar runs: QRadar remains on-premises (x86).
   - Connectivity: Site-to-site VPN or IBM Cloud Direct Link provides private, encrypted pathways from IBM Power Virtual Server to your data center.
   - What you send: The same AIX, IBM i, and Linux on Power logs and events, plus cloud control-plane logs (where available) and security tool output running in IBM Power Virtual Server.
   - Outcome: A single QRadar view unifies on-premises IBM Power11 and IBM Power Virtual Server signals.
3. Distributed ingestion with edge collectors (scalability):
   - Where QRadar runs: QRadar Console and Event Processor on-premises; Event Collector (x86) can be placed close to the IBM Power Virtual Server workloads to buffer and compress logs before forwarding to your core QRadar.
   - Connectivity: TLS-secured forwarding from the edge collector to the central Event Processor.
   - Outcome: Reduced bandwidth usage and improved reliability for high-volume sources.

Detection, investigation, and response

You can use QRadar to assist you with detecting, analyzing, and remediating offenses:

- Correlation and offenses

  QRadar normalizes all IBM Power11 and IBM Power Virtual Server events and correlates them with rules, producing offenses (incidents) when behavior indicates risk (such as

brute-force login attempts, unauthorized profile changes, separation-of-duties violations, or anomalous access to system objects).

- Behavior analytics (optional)

  User and Entity Behavior Analytics scores risk across users and assets, highlighting insider threats or abnormal device behavior that might originate on AIX, IBM i, or Linux on Power.

- AI-assisted triage (optional)

  Investigation Assistant can summarize offenses in natural language, suggest next steps, and generate search queries, accelerating analyst response when incidents involve IBM Power workloads.

- SOAR and runbooks (optional)

  QRadar SOAR can orchestrate containment and remediation. For example, you can disable a suspicious IBM i profile, revoke a role on AIX, open a change in ServiceNow, or trigger a Guardium policy increase for a database running on IBM Power.

Use the following steps to configure QRadar in a hybrid environment:

1. Establish connectivity: Set up site-to-site VPN or Direct Link between your data center and IBM Power Virtual Server.
2. Create log destinations: On QRadar, define secured syslog listeners and appropriate DSMs for AIX, IBM i, Linux on Power, and key applications.
3. Configure sources:
   - AIX: Enable audit; set syslog-ng to forward through TCP/TLS.
   - IBM i: Configure journal export and a forwarding bridge to convert to syslog/LEEF; send through TCP/TLS.
   - Linux on Power: Forward rsyslog/auditd through TCP/TLS.
4. Validate parsing: Confirm fields (user, profile, IP, subsystem, command, object) map correctly in QRadar. Adjust parsing where needed.
5. Enable rules and dashboards: Turn on rules for authentication anomalies, privilege escalation, file integrity, and policy violations; publish SOC dashboards.
6. Automate response (optional): Integrate QRadar SOAR with ticketing and identity tools; create playbooks to quarantine accounts or require reauthentication based on offense severity.

6.5.3 IBM Guardium

Hybrid cloud environments amplify the need for crypto agility, which is the ability to adopt new cryptographic standards quickly and consistently across platforms. IBM Power11 addresses this challenge by introducing post-quantum cryptography (PQC) alongside classical algorithms, helping ensure resilience against emerging quantum threats while maintaining interoperability in multi-cloud configurations.

Power11 includes the following quantum-safe capabilities:

- Secure Boot with Dual Signatures: Firmware and boot loaders are verified by using both traditional algorithms (for example, ECDSA) and quantum-safe signatures (CRYSTALS-Dilithium), combined with SHA3-512 hashing for integrity.
- Quantum-Safe Key Exchange: Power11 adopts CRYSTALS-Kyber for secure key exchange, mitigating risks from quantum-based attacks.

- Hybrid Cryptography for Mobility: During Live Partition Mobility (LPM), memory and state migration use a hybrid approach (combining classical and quantum-safe algorithms) to prevent harvest-now, decrypt-later strategies.

Crypto-agility roadmap for hybrid cloud

A comprehensive crypto-agility strategy for hybrid cloud includes the following elements:

- Consistent PQ adoption for inter-cloud APIs, storage replication, and workload migrations
- Centralized key lifecycle governance across on-premises and cloud environments
- Compliance alignment with emerging PQ standards for regulated industries
- Operational orchestration by using AI-driven platforms for automation and resilience

IBM Guardium Cryptography Manager governance layer

To complement Power11's quantum-safe capabilities, IBM Guardium Cryptography Manager provides centralized cryptography governance across on-premises and cloud environments. Guardium performs the following functions:

- Discovers cryptographic endpoints and enforces policies
- Manages keys and certificates, including PQC assets
- Automates rotation and lifecycle management
- Prepares systems for quantum-safe resilience

In hybrid deployments, Guardium acts as a single control plane, helping ensure consistent cryptographic policies across IBM Power11 servers in your data center and workloads hosted on IBM Power Virtual Server (PowerVS). This integration delivers the following benefits:

- Visibility and compliance for PQC adoption
- Audit-ready reporting mapped to NIST and emerging PQ standards
- Operational consistency for secure workload portability across hybrid and multi-cloud environments

Figure 6-6 on page 154 shows the compliance posture management dashboard. Guardium Cryptography Manager continuously evaluates certificates, keys, and protocols, then classifies their PQC readiness. It rolls that assessment into an enterprise risk score using CVSS-style scoring plus business impact, so teams can focus on the riskiest violations first, not just the noisiest. This approach provides a single, consistent view of classical and PQC posture in one place.

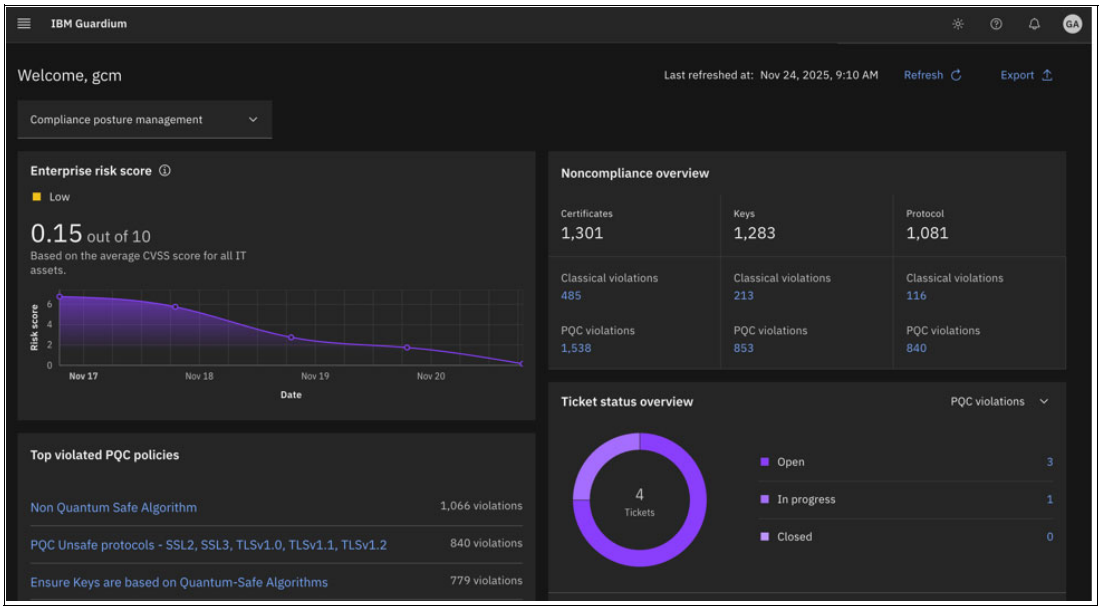


Figure 6-6 IBM Guardium dashboard

Prioritizing vulnerabilities by business risk

Policies act as the control plane for managing cryptographic compliance. IBM Guardium Cryptography Manager includes predefined policies, such as PQC-focused policies like Quantum-Safe Algorithms and Detection of RSA/ECC Use. You can enforce stricter PQC rules in production than in test environments. Each policy is linked to business impact ratings, so when a violation occurs, the system automatically reflects the criticality of the affected asset to the organization.

Figure 6-7 provides a centralized view of all cryptographic policies that are managed by IBM Guardium Cryptography Manager. The Create Policy button (highlighted in blue) allows administrators to define new policies that are tailored to organizational requirements, enabling flexibility for different environments such as production and test. This dashboard helps prioritize vulnerabilities based on business risk by connecting policy violations to compliance and impact ratings.

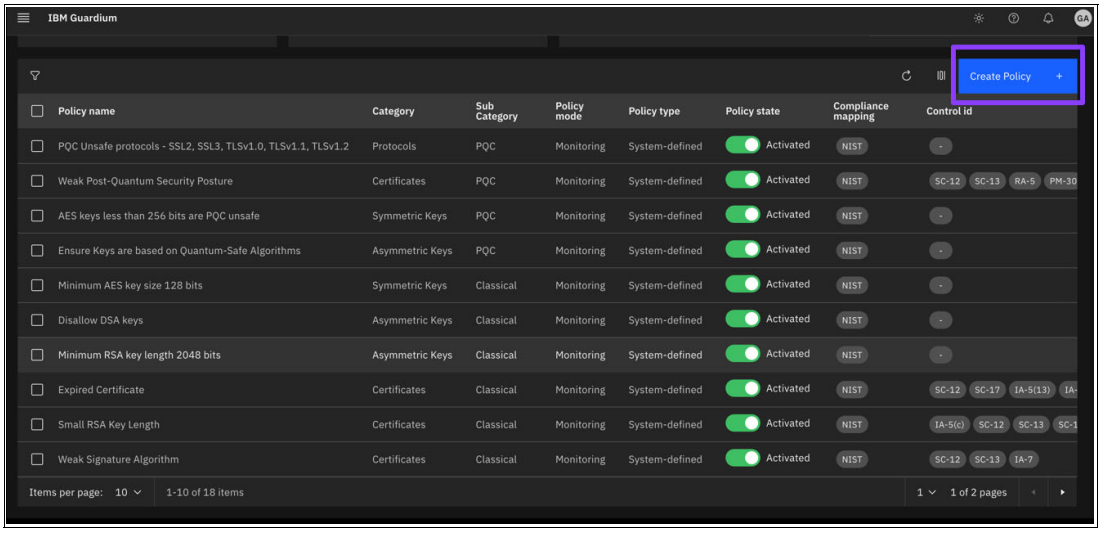


Figure 6-7 Policy Management Dashboard

Continuous compliance monitoring

IBM Guardium Cryptography Manager continuously evaluates every cryptographic object against active policies and automatically flags violations when assets fall out of compliance. These violations can range from weak RSA keys and deprecated TLS versions to poor post-quantum cryptography (PQC) posture or missing PQC algorithms.

Each violation record includes the following information:

- Severity level
- CVSS-based risk score
- Business impact rating
- Status (open, in remediation, resolved)

This approach provides teams with a dynamic, prioritized list of issues to address and a measurable roadmap toward PQC readiness.

Figure 6-8 provides a detailed view of an active cryptographic policy within IBM Guardium Cryptography Manager. The view combines compliance insights with specific policy configurations to help administrators monitor and enforce security standards.

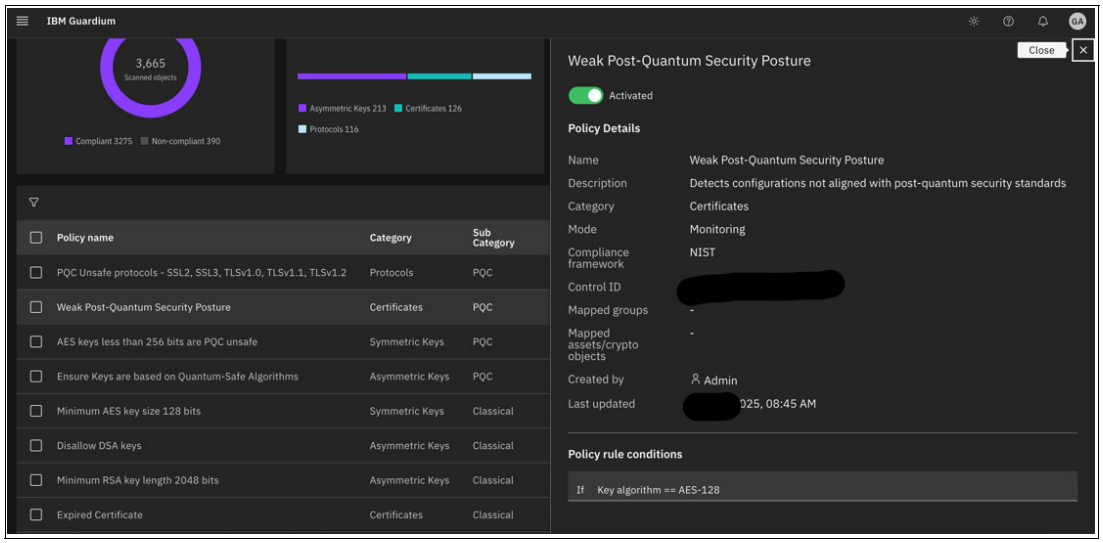


Figure 6-8 Compliance policy option

Automated enforcement and streamline compliance

IBM Guardium Cryptography Manager offers dedicated views for both classical and post-quantum cryptography (PQC) violations, enabling teams to filter by policy, object type, asset, or regulatory framework. You can link critical violations to ticket IDs, helping ensure seamless integration with existing workflows and accelerating remediation.

To support audits and regulatory reviews, the platform includes a one-click export feature. Security and compliance teams can instantly generate CSV or Excel reports mapped to NIST control IDs and other standards, providing ready-to-use evidence for compliance documentation.

Figure 6-9 shows the Violation Dashboard in IBM Guardium Cryptography Manager, focused on classical cryptographic violations. This dashboard provides the following capabilities:

- Monitor and prioritize classical cryptographic violations
- Identify the most common compliance gaps
- Drill down into specific violations and linked assets
- Export data for audits and regulatory evidence

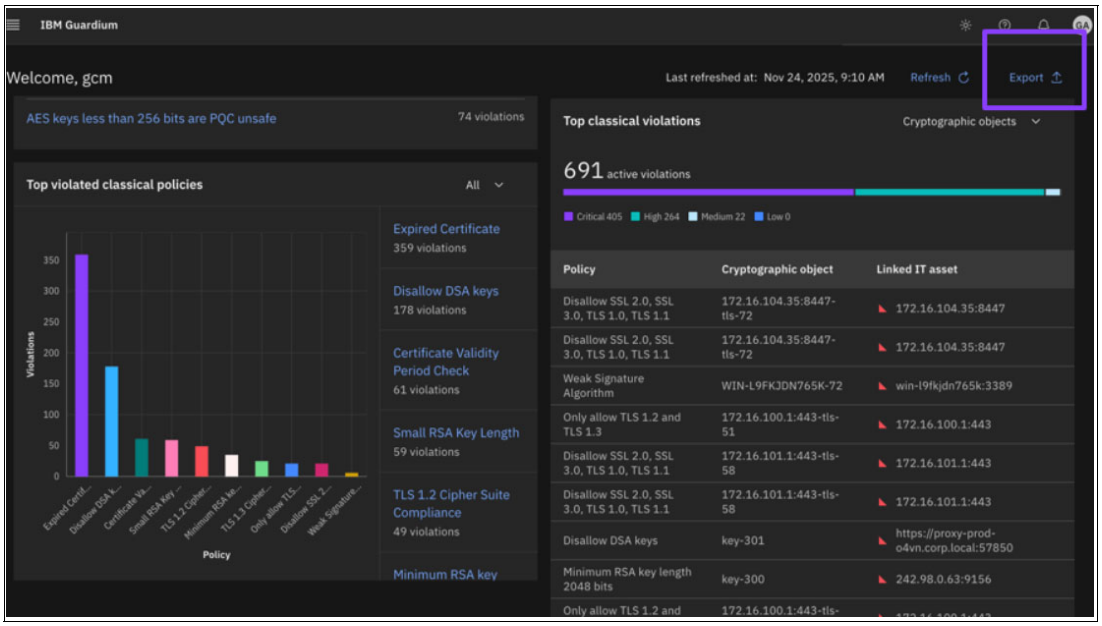


Figure 6-9 Guardium Cryptography Manager

Architecture patterns

The following architecture patterns support cryptographic management across hybrid environments:

- Centralized cryptography management with on-premises control
  - Where it runs: Deploy Guardium Cryptography Manager as software in your data center (x86).
  - What it manages: Cryptographic services and endpoints on IBM Power11 (AIX, IBM i, Linux on Power), plus associated applications and databases.
  - How it connects: Secure management channels (TLS) to discover endpoints, enroll certificates, manage keys, and automate rotations.
- Hybrid management for IBM Power Virtual Server
  - Where it runs: Keep the Guardium management server on-premises; manage IBM Power Virtual Server workloads remotely over a private, encrypted link (for example, site-to-site VPN or Direct Link).
  - What it manages: The same cryptographic assets: web services, SSH, application endpoints, and database encryption keys, which span both cloud-hosted and on-premises IBM Power environments.
  - Outcome: A single set of policies and dashboards for both locations, avoiding configuration drift.
- Distributed gateways for scale

- Edge placement: Optionally deploy lightweight collectors or proxies close to IBM Power Virtual Server workloads to cache discovery results and reduce bandwidth.
- Result: Faster scans, reliable certificate enrollment, and resilient key management across regions.

Why this matters in hybrid cloud

Unified cryptographic management across hybrid environments delivers the following benefits:

- One policy, many environments: Apply the same key and certificate lifecycle rules to AIX, IBM i, and Linux on Power on-premises, and to workloads running in IBM Power Virtual Server.
- Unified visibility: Maintain a consolidated cryptographic inventory (endpoints, keys, certificates) spanning local systems, virtual machines, databases, and cloud-exposed services.
- Crypto agility: Orchestrate algorithm upgrades, key rotation, and certificate renewals uniformly, including staged transitions to quantum-safe methods without application rewrites.
- Compliance and audit: Prove consistent control across multiple trust domains with centralized reporting and evidence collection.

6.5.4 IBM PowerSC

The cost of a data breach continues to rise, with the global average reaching $4.45 million in 2023. Industries such as healthcare and financial services face even higher risks. Organizations are increasing security investments, focusing on incident response planning, employee training, and threat detection. PowerSC addresses these challenges by automating compliance, monitoring security posture, and reducing operational complexity.

PowerSC uses two key principles of IBM Power Systems:

- Isolation: Strong separation between layers (chip, firmware, hypervisor, OS, VM).
- Integrity: Continuous validation of each layer to help ensure trustworthiness.

PowerSC sits on top of the OS stack and uses built-in security hooks, wrapping them in a GUI for centralized management.

Figure 6-10 on page 158 shows the high-level architecture of PowerSC and how the software interacts with the system.

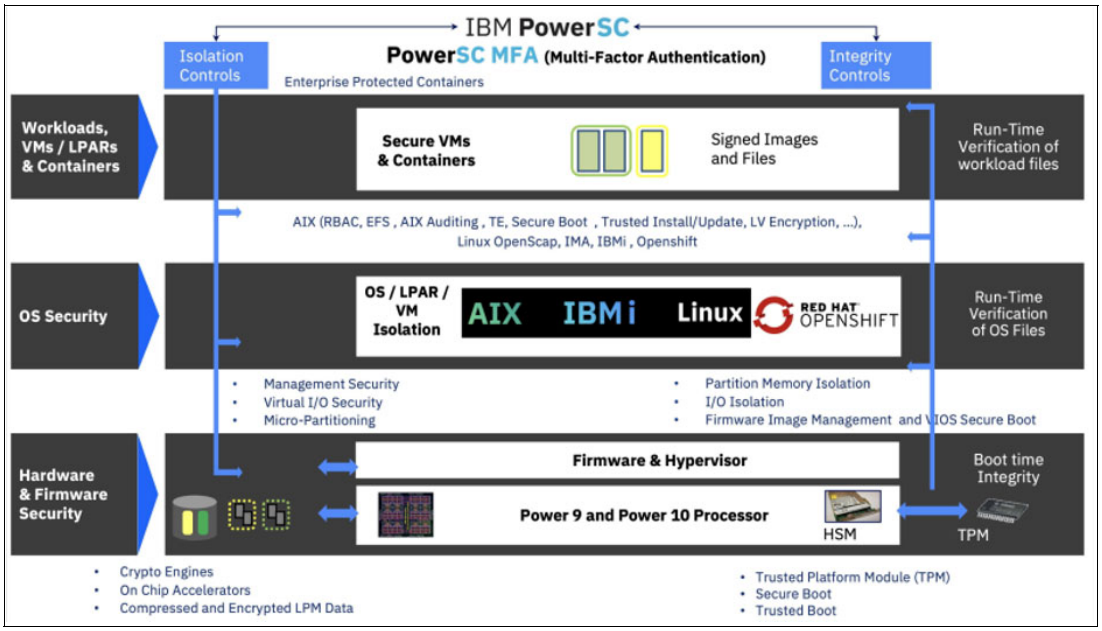


Figure 6-10 Power SC architecture

Hybrid cloud combines on-premises infrastructure with public and private cloud resources, creating a distributed environment that demands consistent security and compliance across all layers. IBM PowerSC extends these capabilities into hybrid environments, helping ensure that security policies and compliance controls remain uniform regardless of where workloads run.

PowerSC delivers the following capabilities for hybrid cloud security:

- Unified security across on-premises and cloud: PowerSC provides a single location to manage security and compliance for AIX, IBM i, and Linux workloads, whether they run on physical servers, virtualized environments, or cloud-based instances. This approach ensures that security policies and compliance profiles are applied consistently across hybrid environments.
- Compliance for regulated industries: Hybrid cloud often hosts sensitive workloads (finance, healthcare, energy). PowerSC automates enforcement of compliance profiles like HIPAA, PCI, and NERC, reducing risk when workloads move between on-premises and cloud.
- Integration with cloud security tools: PowerSC integrates with IBM QRadar SIEM and IBM Safeguarded Copy, extending threat detection and ransomware protection to hybrid environments. REST API and Swagger support enable automation and orchestration with cloud-native security tools.
- Patch management across distributed systems: Hybrid environments require timely patching across multiple platforms. PowerSC policy-based patch management ensures that both on-premises and cloud VMs remain secure and compliant.
- Zero-trust and MFA for cloud access: PowerSC Multifactor Authentication (MFA) strengthens identity assurance for users accessing hybrid cloud resources, aligning with zero-trust principles.
- Visibility and reporting: Hybrid cloud audits demand consolidated evidence. PowerSC exportable compliance reports (CSV/Excel) mapped to NIST controls provide ready-made documentation for regulatory reviews across hybrid deployments.

For more details about Power SC, see the following documentation:

- PowerSC 2.2.0.1 90-day free trial: PowerSC Try and Buy
- *IBM Power Security Catalog*, SG24-8568
- *Security and Cyber Resilience with IBM Power11*, SG24-8595
- *Simplify Management of IT Security and Compliance with IBM PowerSC in Cloud and Virtualized Environments*, SG24-8082

6.5.5 HarshiCorp Vault

As organizations adopt hybrid cloud, managing secrets and encryption consistently across environments becomes critical. HashiCorp Vault, integrated with IBM Cloud through IBM Secrets Manager, provides a secure and centralized way to store credentials, certificates, and encryption keys. When combined with IBM Power Virtual Server (PowerVS), Vault enables the following capabilities:

- Secure secrets management for AIX, IBM i, and Linux workloads running in PowerVS
- Dynamic secrets and key rotation, reducing risk from hardcoded credentials
- Encryption-as-a-service, allowing applications to encrypt and decrypt data without managing keys directly
- Policy-based access control and audit logging, supporting compliance for regulated industries

This integration helps simplify security operations and ensures consistent protection across on-premises and cloud environments.

IBM Secrets Manager, which is built on HashiCorp Vault, offers a 30-day free trial on IBM Cloud. This trial allows you to explore enterprise-grade secrets management capabilities, including the following features:

- Centralized secrets storage for API keys, passwords, and certificates
- Dynamic secrets and key rotation to reduce risk from static credentials
- Encryption-as-a-service for applications without manual key handling
- Policy-based access control and audit logging for compliance and governance

Figure 6-11 shows the IBM Cloud interface for selecting a pricing plan and configuring the resource.

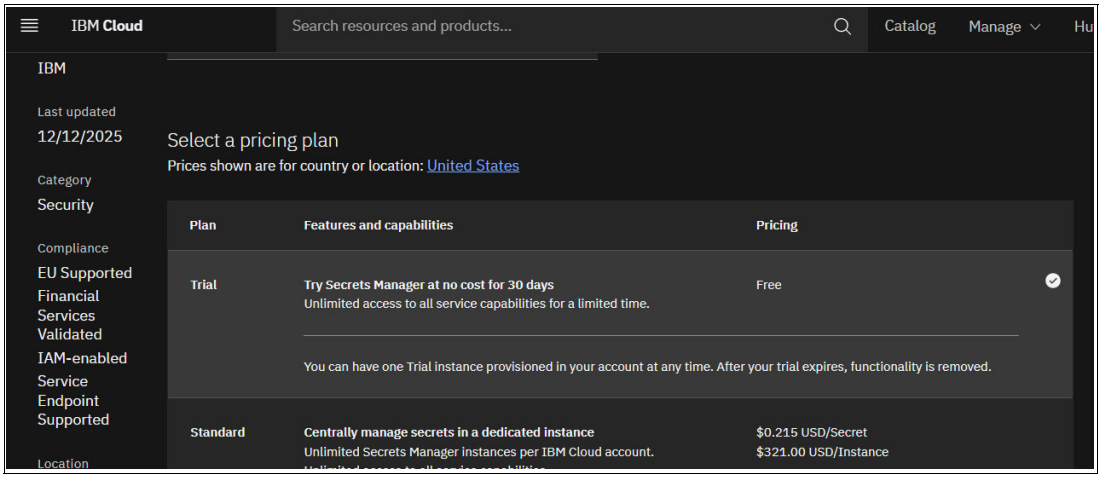


Figure 6-11 IBM Secrets Manager on IBM Cloud

6.6 Securing IBM hybrid cloud with IP-based controls

Hybrid cloud architectures rely heavily on consistent, predictable, and secure network connectivity. As discussed in Chapter 2, "Network Integration in Reference Architecture" on page 23, hybrid environments combine traditional on-premises networks with IBM Cloud Virtual Private Clouds (VPCs), VPN tunnels, Direct Link connectivity, transit gateways, and micro-segmented network zones. Building on that foundation, this section focuses on how IP-based access control becomes a critical security mechanism that spans both infrastructure and application layers.

Account-level IP restrictions

By default, all IP addresses can access IBM Cloud services. To strengthen security, you can restrict access to specific IP addresses or ranges at the account level:

- Navigate to **IBM Cloud Console** → **Manage** → **Access (IAM)** → **Settings**.
- Enable Restrict IP address access.
- Enter allowed IP addresses (single IP, range, or CIDR).
- Only requests from these IP addresses can log in or use API keys.

Important: If both account-level and user-level restrictions exist, the IP must satisfy both conditions.

Limitation: IBM Cloud Shell cannot be accessed when IP restrictions are enabled because it uses shared infrastructure.

User-level IP restrictions

For more granular control, you can restrict access for individual users:

- Navigate to **Manage** → **Access (IAM)** → **Users** → **User Details** → **IP address restrictions**.
- Add allowed IP addresses for Cloud platform and Classic infrastructure.
- The service supports IPv4 and IPv6, ranges, and subnets.

Context-based restrictions (CBRs)

For fine-grained, policy-driven control, IBM Cloud provides Context-Based Restrictions:

- Define network zones with allowed IP addresses, ranges, or VPC IDs.
- Apply these zones to specific services or resources.
- Ideal for hybrid setups, where you want to restrict access based on network origin.
- Manage by using a UI or CLI (ibmcloud plug-in install cbr).

Service-specific IP allowlists

Certain IBM Cloud services allow IP listing for additional protection:

- Cloud Databases: Add up to 100 IP entries per deployment.
- API Connect: Configure allowlists by using CLI commands.
- QRadar on Cloud: Supports IP-based restrictions for secure integration.

Important considerations:

When implementing IP-based restrictions, consider the following points:

- Only public IP addresses are supported for account-level restrictions.
- IP restrictions apply to login and API key usage, not necessarily to all service endpoints.
- Combine IP restrictions with IAM policies and CBRs for layered security.

These IP-based controls help enforce network-origin security policies across hybrid environments, ensuring that only trusted networks can access IBM Cloud resources. When combined with IAM and context-based restrictions, they provide a strong foundation for securing workloads that span on-premises and cloud.

Abbreviations and acronyms

ABAC	Attribute-Based Access Control
ACE	App Connect Enterprise
ACM	Advanced Cluster Management
ALB	Application Load Balancer
ARO	Application Resource Optimization
BAW	Business Automation Workflow
BGP	Border Gateway Protocol
BTM	Business Transaction Monitoring
CA	Cognos Analytics
CBRs	Context-Based Restrictions
CDE	Cognos Dashboard Embedded
CDK	Connector Development Kit
CDN	Content Delivery Network
CIO	Chief Information Officer
CIS	Cloud Internet Services
CMC	Cloud Management Console
CP4D	Cloud Pak for Data
CRN	Cloud Resource Name
CSPM	Cloud Security Posture Management
CSS	Confirmed Stateside Support
DIB	Defense Industrial Base
DLPAR	Dynamic LPAR
DPD	Dead Peer Detection
DR	Disaster Recovery
DRA	Disaster Recovery Automation
DS	DataStage
DV	Data Virtualization
DoD	Department of Defense
ECDSA	Elliptic Curve Digital Signature Algorithm
ECS	Enterprise Cloud Services
ESB	Enterprise Service Bus
ESS	Entitled Systems Support
EUS	Extended Update Support
FMEC	Fog and Mobile Edge Computing
FQDNs	Fully Qualified Domain Names
FVT	Functional Verification Test
GA	General Availability
GLB	Global Load Balancer
GLVM	Geographic Logical Volume Manager
GRE	Generic Routing Encapsulation
GRS	Global Replication Services
HA	High Availability
HMC	Hardware Management Console
IAM	Identity and Access Management
IBM	International Business Machines Corporation
IPI	Installer-Provisioned Infrastructure
IPsec	Internet Protocol security
IVIG	IBM Verify Identity Governance
IaC	Infrastructure-as-Code
IaaS	Infrastructure-as-a-Service
IdP	Identity Provider
KLP	Kernel Live Patching
LKU	Live Kernel Update
LLU	Live Library Update
LPAR	Logical Partition
LPM	Live Partition Mobility
LPPs	Licensed Program Products
MFA	Multi-Factor authentication
MFT	Managed File Transfer
MMA	Matrix Multiply Assist
MZR	Multi-Zone Regions
NAG	Network Address Group
NIC	network interface
NLB	Network Load Balancer
NNI	network-to-Network interfaces
NSA	National Security Agency
NSG	Network Security Group
NSS	National Security Systems
ODM	Operational Decision Manager
OMI	Open Memory Interface
OP	OpenPages
OPP	OpenShift Platform Plus
PA	Planning Analytics
PER	Power Edge Router
POD	Performance Optimized Datacenter
PPA	Passport Advantage
PQC	Post-Quantum Cryptography

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PSK	Pre-Shared Key
PVS1	Package feature code
PoPs	Points of Presence
RAG	Retrieval-Augmented Generation
RAS	Reliability, Availability, and Serviceability
RBAC	Role-Based Access Control
RHEL	Red Hat Enterprise Linux
ROHA	Resource Optimized High Availability
RPA	Robotic Process Automation
RPO	Recovery Point Objective
RTO	Recovery Time Objective
SDN	Software-Defined Networking
SLAs	Service Level Agreements
SLES	SUSE Linux Enterprise Server
SPP	Shared Processor Pools
SPPE	Secure Partition Platform Engine
SRE	Site Reliability Engineering
SREs	Site Reliability Engineers
TGW	Transit Gateway
TME	Transparent Memory Encryption
UEBA	User and Entity Behavior Analytics
UPI	User-Provisioned Infrastructure
VM	Virtual Machine
VNF	Virtual Network Function
VPC	Virtual Private Cloud
VPE	Virtual Private Endpoint
VSI	Virtual Server Instance
VSN	Virtual Serial Number
WAF	Web Application Firewall
WAN	Wide Area Network
WD	Watson Discovery
WKC	Watson Knowledge Catalog
WML	Watson Machine Learning
WOS	Watson OpenScale
WSL	Watson Studio Lite
ZPD	Zero Planned Downtime

Related publications

The publications listed in this section are considered particularly suitable for a more detailed discussion of the topics covered in this book.

IBM Redbooks

The following IBM Redbooks publications provide additional information about the topic in this document. Note that some publications referenced in this list might be available in softcopy only.

- *High Availability and Disaster Recovery Options for IBM Power Cloud and On-Premises*, REDP-5656
- *Simplify Management of IT Security and Compliance with IBM PowerSC in Cloud and Virtualized Environments*, SG24-8082
- *IBM Power Security Catalog*, SG24-8568
- *Migrating SAP Workloads to IBM Power Virtual Server: A Practical Guide for Hybrid Cloud Transformation*, SG24-8577
- *IBM Power E1180: Introduction and Overview*, SG24-8587
- *IBM Power11 E1150 Introduction and Technical Overview*, SG24-8589
- *IBM Power11 Scale-Out Servers: Introduction and Overview*, SG24-8590
- *Implementing AI on Power11: Introducing the IBM Spyre Accelerator*, SG24-8592
- *Security and Cyber Resilience with IBM Power11*, SG24-8595
- *Zero Downtime, Automation, and Energy Optimization,* SG24-8596
- *High Availability and Disaster Recovery Solutions on IBM Power Virtual Server*, SG24-8603

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